



Thesis Project by Group CLS

THESIS 1-3

Thesis Project

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Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1-F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1-F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid,

and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from $\{2, \dots, NL\}$ on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e_1, e_2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL-k)/NL$ decays to zero as $k \rightarrow NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_{CR}	0.35
Max FES	$10,000 \times D$
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an

algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

Library used: `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power (<i>Deprecated</i>)	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800

No.	Function Name	Global Optimum
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300

No.	Component Functions	Global Optimum
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
<code>python</code>	Core programming language	3.10+
<code>numpy</code>	Numerical computing and array operations	2.0.2
<code>matplotlib</code>	Plotting and saving convergence graphs	3.10.0
<code>cec2017</code>	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
<code>cec2017.simple</code>	Simple benchmark functions F1-F10	—
<code>cec2017.hybrid</code>	Hybrid benchmark functions F11-F20	—
<code>cec2017.composition</code>	Composition benchmark functions F21-F30	—
<code>dplldc</code>	DP-LLDE optimization algorithm (uploaded locally)	Custom
<code>google.colab.files</code>	File upload utility for loading <code>dplldc.py</code> into the Colab environment	Built-in
<code>sys</code>	Clears cached module before re-importing <code>dplldc.py</code>	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
In [ ]: !pip install cec2017 --quiet
```

ERROR: Could not find a version that satisfies the requirement cec2017 (from versions: none)
ERROR: No matching distribution found for cec2017

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
In [ ]: from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

cec2017 loaded!

Import `sys` control over warning messages for verification that `lblde.py` exists

```
In [20]: from google.colab import files
import sys

uploaded = files.upload()

if 'lblde_mutation_rule.py' in uploaded:
    if 'lblde_mutation_rule' in sys.modules:
        del sys.modules['lblde']
    import lblde_mutation_rule
    from lblde_mutation_rule import LBLDE_MutationRule
    print('lblde_mutation_rule.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named lblde.py')
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving `lblde_mutation_rule.py` to `lblde_mutation_rule.py`
`lblde_mutation_rule.py` uploaded and imported successfully!

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
In [21]: import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
In [22]: D = 30 # Change this when testing different dimensions
```

```
In [23]: bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

```
In [24]: all_results = {}
```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

```
In [25]: def make_wrapper(fn):
    """Wrap a cec2017 function (expects (1,D) input) to accept (D,) input."""
    def wrapper(x):
        val = fn(x.reshape(1, -1))
        if hasattr(val, '__len__'):
            val = float(val[0])
        else:
            val = float(val)
        return val if np.isfinite(val) else 1e18
    return wrapper
```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```
In [26]: def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run LBLDE_m on a function and plot convergence + diversity graphs."""
    print('=' * 80)
    print(f'LBLDE_m on CEC 2017 {fn_name} – Single Run')
    print('=' * 80)
    print(f'Function      : {fn_label} ({D}D)')
    print(f'Bounds        : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = LBLDE_MutationRule(
        objective_func=fn_eval,
        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    # ← Unpack 4 values now
    best_sol, best_fit, history, diversity_history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
    print(f'  Best Fitness      : {best_fit:.6e}')
    print(f'  Global Optimum    : {f_global:.6e}')
    print(f'  Error (|best-optimum|) : {error:.6e}')
    print(f'  Best Solution (5 dim) : {best_sol[:5]}')
    print(f'  D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'{"="*80}')

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    fes_axis = np.linspace(0, MAX_FES, len(history))
    gen_axis = np.arange(len(diversity_history)) # Generations (0, 1, 2, ...)

    # — Two-panel figure —
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 10))

    # Convergence curve (top) – FEs on x-axis
    ax1.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    ax1.set_ylabel('Best Fitness', fontsize=12)
    ax1.set_xlabel('Function Evaluations (FEs)', fontsize=12)
    ax1.set_title(
        f'LBLDE_m – CEC 2017 {fn_name}: {fn_label} ({D}D)',
```

```

        fontsize=12
    )
    ax1.grid(True, which='both', linestyle='--', alpha=0.6)

    # Diversity curve (bottom) – Generations on x-axis
    ax2.plot(gen_axis, diversity_history, color='darkorange', linewidth=1.5)
    ax2.set_xlabel('Generation', fontsize=12)
    ax2.set_ylabel('Population Diversity\n(Mean Std Dev)', fontsize=12)
    ax2.set_title(f'Diversity Across Generations – {fn_name}', fontsize=12)
    ax2.grid(True, which='both', linestyle='--', alpha=0.6)

    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'
    plt.savefig(fname, dpi=150)
    plt.show()
    print(f'Graph saved as {fname}\n')

```

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

```

In [27]: # =====
# F1 – Shifted and Rotated Bent Cigar
# =====
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and Rotated Bent C

```

=====

=

LBLDE_m on CEC 2017 F1 – Single Run

=====

=

Function : Shifted and Rotated Bent Cigar (30D)
Bounds : [-100, 100]
Global optimum : 100.0
Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 1.189140e+12 | Diversity = 5.7976e+01

Generation	10:	Best = 2.454394e+11	FES = 1100	Diversity = 1.9062e+01
Generation	20:	Best = 1.311194e+11	FES = 2100	Diversity = 6.5093e+00
Generation	30:	Best = 8.245890e+10	FES = 3100	Diversity = 3.3234e+00
Generation	40:	Best = 6.499477e+10	FES = 4100	Diversity = 1.8773e+00
Generation	50:	Best = 5.398721e+10	FES = 5100	Diversity = 1.3791e+00
Generation	60:	Best = 4.659503e+10	FES = 6100	Diversity = 9.5435e-01
Generation	70:	Best = 4.237127e+10	FES = 7100	Diversity = 7.9418e-01
Generation	80:	Best = 3.992337e+10	FES = 8100	Diversity = 6.3613e-01
Generation	90:	Best = 3.745207e+10	FES = 9100	Diversity = 6.6447e-01
Generation	100:	Best = 3.555185e+10	FES = 10100	Diversity = 6.6067e-01
Generation	110:	Best = 3.381748e+10	FES = 11100	Diversity = 5.8055e-01
Generation	120:	Best = 3.235464e+10	FES = 12100	Diversity = 6.6064e-01
Generation	130:	Best = 3.082050e+10	FES = 13100	Diversity = 6.6164e-01
Generation	140:	Best = 2.891902e+10	FES = 14100	Diversity = 8.5398e-01
Generation	150:	Best = 2.734717e+10	FES = 15100	Diversity = 7.6177e-01
Generation	160:	Best = 2.624519e+10	FES = 16100	Diversity = 6.7516e-01
Generation	170:	Best = 2.483848e+10	FES = 17100	Diversity = 7.0053e-01
Generation	180:	Best = 2.329884e+10	FES = 18100	Diversity = 7.2748e-01
Generation	190:	Best = 2.172749e+10	FES = 19100	Diversity = 5.2436e-01
Generation	200:	Best = 2.040037e+10	FES = 20100	Diversity = 6.0459e-01
Generation	210:	Best = 1.961262e+10	FES = 21100	Diversity = 5.9522e-01
Generation	220:	Best = 1.895884e+10	FES = 22100	Diversity = 6.1925e-01
Generation	230:	Best = 1.852365e+10	FES = 23100	Diversity = 5.0225e-01
Generation	240:	Best = 1.818653e+10	FES = 24100	Diversity = 3.8108e-01
Generation	250:	Best = 1.793919e+10	FES = 25100	Diversity = 3.2723e-01
Generation	260:	Best = 1.773343e+10	FES = 26100	Diversity = 3.3747e-01
Generation	270:	Best = 1.755952e+10	FES = 27100	Diversity = 3.0785e-01
Generation	280:	Best = 1.741003e+10	FES = 28100	Diversity = 3.0655e-01
Generation	290:	Best = 1.730705e+10	FES = 29100	Diversity = 3.2673e-01
Generation	300:	Best = 1.722805e+10	FES = 30100	Diversity = 2.7317e-01
Generation	310:	Best = 1.714093e+10	FES = 31100	Diversity = 2.5478e-01
Generation	320:	Best = 1.706041e+10	FES = 32100	Diversity = 2.4986e-01
Generation	330:	Best = 1.695988e+10	FES = 33100	Diversity = 3.8254e-01
Generation	340:	Best = 1.684837e+10	FES = 34100	Diversity = 4.0005e-01
Generation	350:	Best = 1.674146e+10	FES = 35100	Diversity = 4.0623e-01
Generation	360:	Best = 1.667892e+10	FES = 36100	Diversity = 5.7680e-01
Generation	370:	Best = 1.660575e+10	FES = 37100	Diversity = 4.6495e-01
Generation	380:	Best = 1.655311e+10	FES = 38100	Diversity = 3.3489e-01
Generation	390:	Best = 1.651265e+10	FES = 39100	Diversity = 4.5350e-01
Generation	400:	Best = 1.642528e+10	FES = 40100	Diversity = 4.0262e-01
Generation	410:	Best = 1.636919e+10	FES = 41100	Diversity = 4.2729e-01
Generation	420:	Best = 1.632355e+10	FES = 42100	Diversity = 4.0876e-01
Generation	430:	Best = 1.629911e+10	FES = 43100	Diversity = 3.1242e-01
Generation	440:	Best = 1.627429e+10	FES = 44100	Diversity = 2.7478e-01

Generation	450:	Best = 1.625765e+10	FES = 45100	Diversity = 2.8384e-01
Generation	460:	Best = 1.624219e+10	FES = 46100	Diversity = 2.8563e-01
Generation	470:	Best = 1.623170e+10	FES = 47100	Diversity = 2.2474e-01
Generation	480:	Best = 1.622388e+10	FES = 48100	Diversity = 1.8390e-01
Generation	490:	Best = 1.621984e+10	FES = 49100	Diversity = 1.4804e-01
Generation	500:	Best = 1.621613e+10	FES = 50100	Diversity = 1.3348e-01
Generation	510:	Best = 1.621193e+10	FES = 51100	Diversity = 1.1790e-01
Generation	520:	Best = 1.620842e+10	FES = 52100	Diversity = 1.1648e-01
Generation	530:	Best = 1.620745e+10	FES = 53100	Diversity = 6.3566e-02
Generation	540:	Best = 1.620661e+10	FES = 54100	Diversity = 5.4810e-02
Generation	550:	Best = 1.620601e+10	FES = 55100	Diversity = 4.5017e-02
Generation	560:	Best = 1.620556e+10	FES = 56100	Diversity = 2.9132e-02
Generation	570:	Best = 1.620529e+10	FES = 57100	Diversity = 1.9986e-02
Generation	580:	Best = 1.620509e+10	FES = 58100	Diversity = 2.4373e-02
Generation	590:	Best = 1.620485e+10	FES = 59100	Diversity = 2.5041e-02
Generation	600:	Best = 1.620475e+10	FES = 60100	Diversity = 1.5141e-02
Generation	610:	Best = 1.620470e+10	FES = 61100	Diversity = 5.9176e-03
Generation	620:	Best = 1.620468e+10	FES = 62100	Diversity = 6.8670e-03
Generation	630:	Best = 1.620465e+10	FES = 63100	Diversity = 6.9972e-03
Generation	640:	Best = 1.620463e+10	FES = 64100	Diversity = 6.4124e-03
Generation	650:	Best = 1.620462e+10	FES = 65100	Diversity = 4.2956e-03
Generation	660:	Best = 1.620461e+10	FES = 66100	Diversity = 3.0632e-03
Generation	670:	Best = 1.620461e+10	FES = 67100	Diversity = 1.7651e-03
Generation	680:	Best = 1.620461e+10	FES = 68100	Diversity = 1.7132e-03
Generation	690:	Best = 1.620461e+10	FES = 69100	Diversity = 1.4860e-03
Generation	700:	Best = 1.620461e+10	FES = 70100	Diversity = 3.9374e-04
Generation	710:	Best = 1.620461e+10	FES = 71100	Diversity = 2.8195e-04
Generation	720:	Best = 1.620461e+10	FES = 72100	Diversity = 3.4163e-04
Generation	730:	Best = 1.620461e+10	FES = 73100	Diversity = 1.3505e-04
Generation	740:	Best = 1.620461e+10	FES = 74100	Diversity = 1.7151e-04
Generation	750:	Best = 1.620461e+10	FES = 75100	Diversity = 1.6700e-04
Generation	760:	Best = 1.620461e+10	FES = 76100	Diversity = 1.3254e-04
Generation	770:	Best = 1.620461e+10	FES = 77100	Diversity = 2.0319e-04
Generation	780:	Best = 1.620461e+10	FES = 78100	Diversity = 5.6024e-05
Generation	790:	Best = 1.620461e+10	FES = 79100	Diversity = 6.1713e-05
Generation	800:	Best = 1.620461e+10	FES = 80100	Diversity = 4.3312e-05
Generation	810:	Best = 1.620461e+10	FES = 81100	Diversity = 4.4876e-05
Generation	820:	Best = 1.620461e+10	FES = 82100	Diversity = 5.1070e-05
Generation	830:	Best = 1.620461e+10	FES = 83100	Diversity = 3.6643e-05
Generation	840:	Best = 1.620461e+10	FES = 84100	Diversity = 2.5470e-05
Generation	850:	Best = 1.620461e+10	FES = 85100	Diversity = 2.6599e-05
Generation	860:	Best = 1.620461e+10	FES = 86100	Diversity = 1.6309e-05
Generation	870:	Best = 1.620461e+10	FES = 87100	Diversity = 2.6597e-05
Generation	880:	Best = 1.620461e+10	FES = 88100	Diversity = 9.5942e-06
Generation	890:	Best = 1.620461e+10	FES = 89100	Diversity = 1.0572e-05
Generation	900:	Best = 1.620461e+10	FES = 90100	Diversity = 4.8340e-06
Generation	910:	Best = 1.620461e+10	FES = 91100	Diversity = 4.4030e-06
Generation	920:	Best = 1.620461e+10	FES = 92100	Diversity = 5.6568e-06
Generation	930:	Best = 1.620461e+10	FES = 93100	Diversity = 2.0142e-06
Generation	940:	Best = 1.620461e+10	FES = 94100	Diversity = 8.8663e-07
Generation	950:	Best = 1.620461e+10	FES = 95100	Diversity = 3.5921e-07
Generation	960:	Best = 1.620461e+10	FES = 96100	Diversity = 4.8430e-07
Generation	970:	Best = 1.620461e+10	FES = 97100	Diversity = 4.6015e-07
Generation	980:	Best = 1.620461e+10	FES = 98100	Diversity = 3.4307e-07

[illegible]

[illegible]

[illegible]

Generation 2610:	Best = 1.620461e+10	FES = 261100	Diversity = 3.7037e-14
Generation 2620:	Best = 1.620461e+10	FES = 262100	Diversity = 3.7037e-14
Generation 2630:	Best = 1.620461e+10	FES = 263100	Diversity = 3.7037e-14
Generation 2640:	Best = 1.620461e+10	FES = 264100	Diversity = 3.7037e-14
Generation 2650:	Best = 1.620461e+10	FES = 265100	Diversity = 3.7037e-14
Generation 2660:	Best = 1.620461e+10	FES = 266100	Diversity = 3.7037e-14
Generation 2670:	Best = 1.620461e+10	FES = 267100	Diversity = 3.7037e-14
Generation 2680:	Best = 1.620461e+10	FES = 268100	Diversity = 3.7037e-14
Generation 2690:	Best = 1.620461e+10	FES = 269100	Diversity = 3.7037e-14
Generation 2700:	Best = 1.620461e+10	FES = 270100	Diversity = 3.7037e-14
Generation 2710:	Best = 1.620461e+10	FES = 271100	Diversity = 3.7037e-14
Generation 2720:	Best = 1.620461e+10	FES = 272100	Diversity = 3.7037e-14
Generation 2730:	Best = 1.620461e+10	FES = 273100	Diversity = 3.7037e-14
Generation 2740:	Best = 1.620461e+10	FES = 274100	Diversity = 3.7037e-14
Generation 2750:	Best = 1.620461e+10	FES = 275100	Diversity = 3.7037e-14
Generation 2760:	Best = 1.620461e+10	FES = 276100	Diversity = 3.7037e-14
Generation 2770:	Best = 1.620461e+10	FES = 277100	Diversity = 3.7037e-14
Generation 2780:	Best = 1.620461e+10	FES = 278100	Diversity = 3.7037e-14
Generation 2790:	Best = 1.620461e+10	FES = 279100	Diversity = 3.7037e-14
Generation 2800:	Best = 1.620461e+10	FES = 280100	Diversity = 3.7037e-14
Generation 2810:	Best = 1.620461e+10	FES = 281100	Diversity = 3.7037e-14
Generation 2820:	Best = 1.620461e+10	FES = 282100	Diversity = 3.7037e-14
Generation 2830:	Best = 1.620461e+10	FES = 283100	Diversity = 3.7037e-14
Generation 2840:	Best = 1.620461e+10	FES = 284100	Diversity = 3.7037e-14
Generation 2850:	Best = 1.620461e+10	FES = 285100	Diversity = 3.7037e-14
Generation 2860:	Best = 1.620461e+10	FES = 286100	Diversity = 3.7037e-14
Generation 2870:	Best = 1.620461e+10	FES = 287100	Diversity = 3.7037e-14
Generation 2880:	Best = 1.620461e+10	FES = 288100	Diversity = 3.7037e-14
Generation 2890:	Best = 1.620461e+10	FES = 289100	Diversity = 3.7037e-14
Generation 2900:	Best = 1.620461e+10	FES = 290100	Diversity = 3.7037e-14
Generation 2910:	Best = 1.620461e+10	FES = 291100	Diversity = 3.7037e-14
Generation 2920:	Best = 1.620461e+10	FES = 292100	Diversity = 3.7037e-14
Generation 2930:	Best = 1.620461e+10	FES = 293100	Diversity = 3.7037e-14
Generation 2940:	Best = 1.620461e+10	FES = 294100	Diversity = 3.7037e-14
Generation 2950:	Best = 1.620461e+10	FES = 295100	Diversity = 3.7037e-14
Generation 2960:	Best = 1.620461e+10	FES = 296100	Diversity = 3.7037e-14
Generation 2970:	Best = 1.620461e+10	FES = 297100	Diversity = 3.7037e-14
Generation 2980:	Best = 1.620461e+10	FES = 298100	Diversity = 3.7037e-14
Generation 2990:	Best = 1.620461e+10	FES = 299100	Diversity = 3.7037e-14

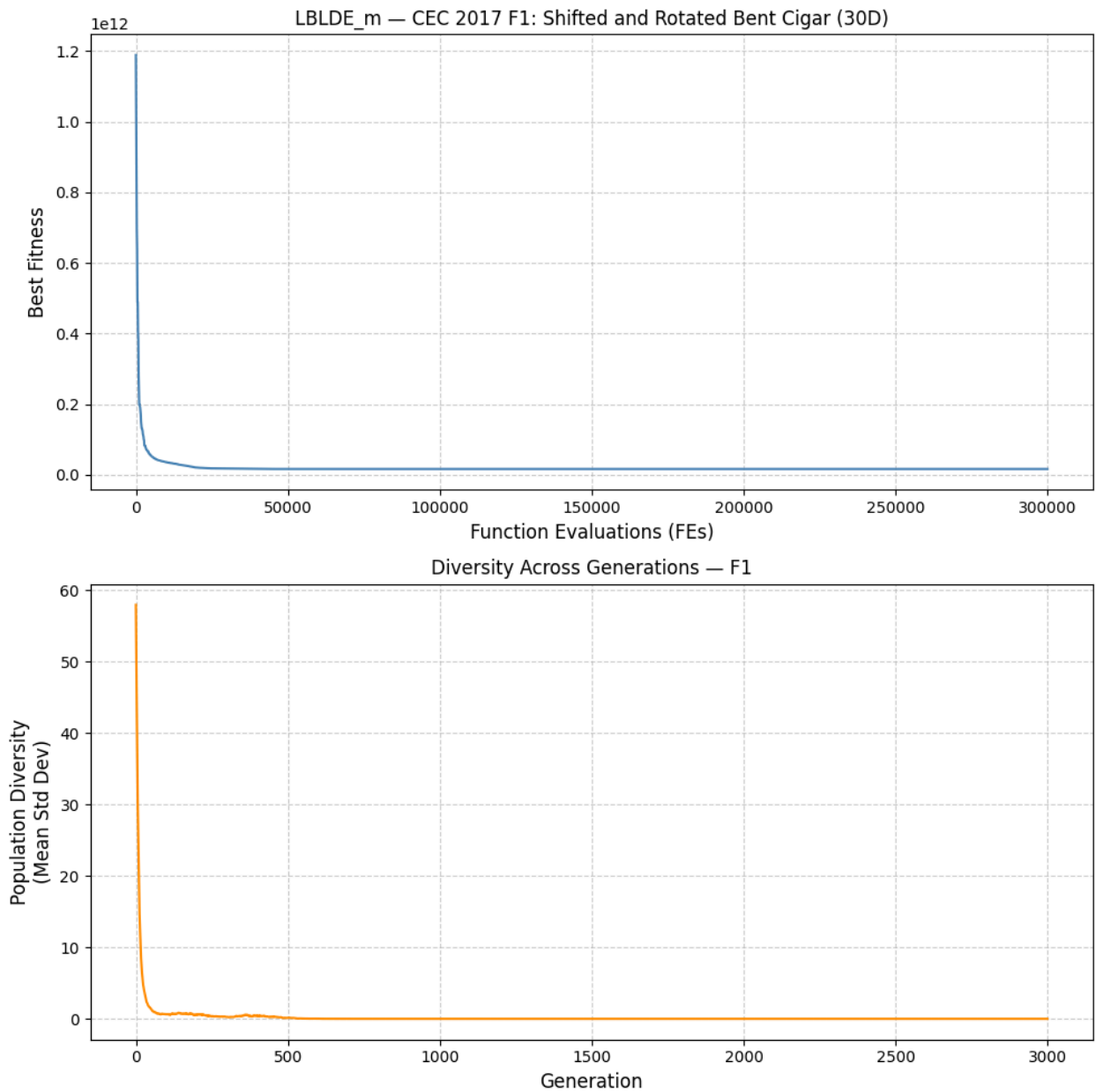
Optimization completed!

Final Best Fitness = 1.620461e+10

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness      : 1.620461e+10
  Global Optimum    : 1.000000e+02
  Error (|best-optimum|) : 1.620461e+10
  Best Solution (5 dim) : [-43.38234137 -63.85845013 -16.36256786 -58.29151073
20.70419427]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F1.png

```
In [28]: # =====  
# F2 – Shifted and Rotated Sum of Different Power  
# =====  
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and Rotated Sum of
```

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LBLDE_m on CEC 2017 F2 – Single Run

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Function : Shifted and Rotated Sum of Different Power (Deprecated) (30D)
Bounds : [-100, 100]
Global optimum : 200.0

WARNING: f2 has been deprecated from the CEC 2017 benchmark suite

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.044577e+48 | Diversity = 5.7976e+01

Generation	10:	Best = 2.174379e+37	FES = 1100	Diversity = 2.7867e+01
Generation	20:	Best = 3.286194e+33	FES = 2100	Diversity = 1.3334e+01
Generation	30:	Best = 1.376940e+31	FES = 3100	Diversity = 6.7758e+00
Generation	40:	Best = 2.248104e+30	FES = 4100	Diversity = 4.6390e+00
Generation	50:	Best = 6.424008e+29	FES = 5100	Diversity = 3.2630e+00
Generation	60:	Best = 2.329359e+29	FES = 6100	Diversity = 2.5295e+00
Generation	70:	Best = 1.439539e+29	FES = 7100	Diversity = 2.3176e+00
Generation	80:	Best = 8.731954e+28	FES = 8100	Diversity = 1.8672e+00
Generation	90:	Best = 6.780543e+28	FES = 9100	Diversity = 1.5786e+00
Generation	100:	Best = 5.530385e+28	FES = 10100	Diversity = 1.4896e+00
Generation	110:	Best = 5.048715e+28	FES = 11100	Diversity = 1.3496e+00
Generation	120:	Best = 4.671708e+28	FES = 12100	Diversity = 1.4513e+00
Generation	130:	Best = 4.433628e+28	FES = 13100	Diversity = 1.5430e+00
Generation	140:	Best = 4.208325e+28	FES = 14100	Diversity = 1.6045e+00
Generation	150:	Best = 4.070762e+28	FES = 15100	Diversity = 1.5531e+00
Generation	160:	Best = 3.994083e+28	FES = 16100	Diversity = 1.3560e+00
Generation	170:	Best = 3.941118e+28	FES = 17100	Diversity = 1.6608e+00
Generation	180:	Best = 3.899625e+28	FES = 18100	Diversity = 1.8218e+00
Generation	190:	Best = 3.865374e+28	FES = 19100	Diversity = 1.5305e+00
Generation	200:	Best = 3.841217e+28	FES = 20100	Diversity = 1.2706e+00
Generation	210:	Best = 3.826034e+28	FES = 21100	Diversity = 1.5630e+00
Generation	220:	Best = 3.812707e+28	FES = 22100	Diversity = 1.6094e+00
Generation	230:	Best = 3.799820e+28	FES = 23100	Diversity = 1.6263e+00
Generation	240:	Best = 3.789336e+28	FES = 24100	Diversity = 1.4624e+00
Generation	250:	Best = 3.777774e+28	FES = 25100	Diversity = 1.6930e+00
Generation	260:	Best = 3.767673e+28	FES = 26100	Diversity = 1.7517e+00
Generation	270:	Best = 3.757706e+28	FES = 27100	Diversity = 1.7521e+00
Generation	280:	Best = 3.750940e+28	FES = 28100	Diversity = 1.6475e+00
Generation	290:	Best = 3.746519e+28	FES = 29100	Diversity = 1.6317e+00
Generation	300:	Best = 3.742698e+28	FES = 30100	Diversity = 1.3330e+00
Generation	310:	Best = 3.740216e+28	FES = 31100	Diversity = 1.4159e+00
Generation	320:	Best = 3.737814e+28	FES = 32100	Diversity = 1.4432e+00
Generation	330:	Best = 3.735540e+28	FES = 33100	Diversity = 1.3545e+00
Generation	340:	Best = 3.733981e+28	FES = 34100	Diversity = 1.5180e+00
Generation	350:	Best = 3.732724e+28	FES = 35100	Diversity = 1.6432e+00
Generation	360:	Best = 3.730943e+28	FES = 36100	Diversity = 1.6834e+00
Generation	370:	Best = 3.729481e+28	FES = 37100	Diversity = 1.3697e+00
Generation	380:	Best = 3.728019e+28	FES = 38100	Diversity = 1.6448e+00
Generation	390:	Best = 3.726530e+28	FES = 39100	Diversity = 1.6330e+00
Generation	400:	Best = 3.725611e+28	FES = 40100	Diversity = 1.4829e+00
Generation	410:	Best = 3.724975e+28	FES = 41100	Diversity = 1.5079e+00
Generation	420:	Best = 3.724659e+28	FES = 42100	Diversity = 1.3858e+00
Generation	430:	Best = 3.724391e+28	FES = 43100	Diversity = 1.2581e+00

Generation	440:	Best = 3.724246e+28	FES = 44100	Diversity = 1.1188e+00
Generation	450:	Best = 3.724109e+28	FES = 45100	Diversity = 1.1002e+00
Generation	460:	Best = 3.724003e+28	FES = 46100	Diversity = 1.0102e+00
Generation	470:	Best = 3.723918e+28	FES = 47100	Diversity = 1.1457e+00
Generation	480:	Best = 3.723852e+28	FES = 48100	Diversity = 9.8290e-01
Generation	490:	Best = 3.723799e+28	FES = 49100	Diversity = 9.4043e-01
Generation	500:	Best = 3.723746e+28	FES = 50100	Diversity = 8.7201e-01
Generation	510:	Best = 3.723713e+28	FES = 51100	Diversity = 8.8935e-01
Generation	520:	Best = 3.723687e+28	FES = 52100	Diversity = 7.9160e-01
Generation	530:	Best = 3.723651e+28	FES = 53100	Diversity = 9.1233e-01
Generation	540:	Best = 3.723620e+28	FES = 54100	Diversity = 9.4762e-01
Generation	550:	Best = 3.723598e+28	FES = 55100	Diversity = 1.0111e+00
Generation	560:	Best = 3.723587e+28	FES = 56100	Diversity = 6.9309e-01
Generation	570:	Best = 3.723582e+28	FES = 57100	Diversity = 7.0202e-01
Generation	580:	Best = 3.723578e+28	FES = 58100	Diversity = 6.6868e-01
Generation	590:	Best = 3.723576e+28	FES = 59100	Diversity = 7.0993e-01
Generation	600:	Best = 3.723574e+28	FES = 60100	Diversity = 7.1284e-01
Generation	610:	Best = 3.723572e+28	FES = 61100	Diversity = 6.0873e-01
Generation	620:	Best = 3.723571e+28	FES = 62100	Diversity = 5.5183e-01
Generation	630:	Best = 3.723570e+28	FES = 63100	Diversity = 5.2070e-01
Generation	640:	Best = 3.723570e+28	FES = 64100	Diversity = 6.1946e-01
Generation	650:	Best = 3.723570e+28	FES = 65100	Diversity = 3.9441e-01
Generation	660:	Best = 3.723569e+28	FES = 66100	Diversity = 3.8412e-01
Generation	670:	Best = 3.723569e+28	FES = 67100	Diversity = 3.8000e-01
Generation	680:	Best = 3.723569e+28	FES = 68100	Diversity = 3.5217e-01
Generation	690:	Best = 3.723569e+28	FES = 69100	Diversity = 3.4685e-01
Generation	700:	Best = 3.723569e+28	FES = 70100	Diversity = 3.4915e-01
Generation	710:	Best = 3.723569e+28	FES = 71100	Diversity = 3.9467e-01
Generation	720:	Best = 3.723569e+28	FES = 72100	Diversity = 4.0012e-01
Generation	730:	Best = 3.723569e+28	FES = 73100	Diversity = 3.7591e-01
Generation	740:	Best = 3.723569e+28	FES = 74100	Diversity = 3.8384e-01
Generation	750:	Best = 3.723569e+28	FES = 75100	Diversity = 3.3789e-01
Generation	760:	Best = 3.723569e+28	FES = 76100	Diversity = 2.6765e-01
Generation	770:	Best = 3.723569e+28	FES = 77100	Diversity = 2.6978e-01
Generation	780:	Best = 3.723569e+28	FES = 78100	Diversity = 2.6403e-01
Generation	790:	Best = 3.723569e+28	FES = 79100	Diversity = 2.0988e-01
Generation	800:	Best = 3.723569e+28	FES = 80100	Diversity = 1.7750e-01
Generation	810:	Best = 3.723569e+28	FES = 81100	Diversity = 2.0499e-01
Generation	820:	Best = 3.723569e+28	FES = 82100	Diversity = 1.7539e-01
Generation	830:	Best = 3.723569e+28	FES = 83100	Diversity = 2.2447e-01
Generation	840:	Best = 3.723569e+28	FES = 84100	Diversity = 2.0422e-01
Generation	850:	Best = 3.723569e+28	FES = 85100	Diversity = 8.8892e-02
Generation	860:	Best = 3.723569e+28	FES = 86100	Diversity = 7.6269e-02
Generation	870:	Best = 3.723569e+28	FES = 87100	Diversity = 5.0721e-02
Generation	880:	Best = 3.723569e+28	FES = 88100	Diversity = 1.3182e-01
Generation	890:	Best = 3.723569e+28	FES = 89100	Diversity = 2.9224e-02
Generation	900:	Best = 3.723569e+28	FES = 90100	Diversity = 1.3487e-02
Generation	910:	Best = 3.723569e+28	FES = 91100	Diversity = 8.2305e-03
Generation	920:	Best = 3.723569e+28	FES = 92100	Diversity = 6.2260e-03
Generation	930:	Best = 3.723569e+28	FES = 93100	Diversity = 3.8966e-03
Generation	940:	Best = 3.723569e+28	FES = 94100	Diversity = 3.5089e-03
Generation	950:	Best = 3.723569e+28	FES = 95100	Diversity = 2.1714e-03
Generation	960:	Best = 3.723569e+28	FES = 96100	Diversity = 2.6807e-03
Generation	970:	Best = 3.723569e+28	FES = 97100	Diversity = 2.1428e-03

[illegible]

[illegible]

[illegible]

Generation 2600:	Best = 3.723569e+28	FES = 260100	Diversity = 3.0405e-14
Generation 2610:	Best = 3.723569e+28	FES = 261100	Diversity = 3.0405e-14
Generation 2620:	Best = 3.723569e+28	FES = 262100	Diversity = 3.0405e-14
Generation 2630:	Best = 3.723569e+28	FES = 263100	Diversity = 3.0405e-14
Generation 2640:	Best = 3.723569e+28	FES = 264100	Diversity = 3.0405e-14
Generation 2650:	Best = 3.723569e+28	FES = 265100	Diversity = 3.0405e-14
Generation 2660:	Best = 3.723569e+28	FES = 266100	Diversity = 3.0405e-14
Generation 2670:	Best = 3.723569e+28	FES = 267100	Diversity = 3.0405e-14
Generation 2680:	Best = 3.723569e+28	FES = 268100	Diversity = 3.0405e-14
Generation 2690:	Best = 3.723569e+28	FES = 269100	Diversity = 3.0405e-14
Generation 2700:	Best = 3.723569e+28	FES = 270100	Diversity = 3.0405e-14
Generation 2710:	Best = 3.723569e+28	FES = 271100	Diversity = 3.0405e-14
Generation 2720:	Best = 3.723569e+28	FES = 272100	Diversity = 3.0405e-14
Generation 2730:	Best = 3.723569e+28	FES = 273100	Diversity = 3.0405e-14
Generation 2740:	Best = 3.723569e+28	FES = 274100	Diversity = 3.0405e-14
Generation 2750:	Best = 3.723569e+28	FES = 275100	Diversity = 3.0405e-14
Generation 2760:	Best = 3.723569e+28	FES = 276100	Diversity = 3.0405e-14
Generation 2770:	Best = 3.723569e+28	FES = 277100	Diversity = 3.0405e-14
Generation 2780:	Best = 3.723569e+28	FES = 278100	Diversity = 3.0405e-14
Generation 2790:	Best = 3.723569e+28	FES = 279100	Diversity = 3.0405e-14
Generation 2800:	Best = 3.723569e+28	FES = 280100	Diversity = 3.0405e-14
Generation 2810:	Best = 3.723569e+28	FES = 281100	Diversity = 3.0405e-14
Generation 2820:	Best = 3.723569e+28	FES = 282100	Diversity = 3.0405e-14
Generation 2830:	Best = 3.723569e+28	FES = 283100	Diversity = 3.0405e-14
Generation 2840:	Best = 3.723569e+28	FES = 284100	Diversity = 3.0405e-14
Generation 2850:	Best = 3.723569e+28	FES = 285100	Diversity = 3.0405e-14
Generation 2860:	Best = 3.723569e+28	FES = 286100	Diversity = 3.0405e-14
Generation 2870:	Best = 3.723569e+28	FES = 287100	Diversity = 3.0405e-14
Generation 2880:	Best = 3.723569e+28	FES = 288100	Diversity = 3.0405e-14
Generation 2890:	Best = 3.723569e+28	FES = 289100	Diversity = 3.0405e-14
Generation 2900:	Best = 3.723569e+28	FES = 290100	Diversity = 3.0405e-14
Generation 2910:	Best = 3.723569e+28	FES = 291100	Diversity = 3.0405e-14
Generation 2920:	Best = 3.723569e+28	FES = 292100	Diversity = 3.0405e-14
Generation 2930:	Best = 3.723569e+28	FES = 293100	Diversity = 3.0405e-14
Generation 2940:	Best = 3.723569e+28	FES = 294100	Diversity = 3.0405e-14
Generation 2950:	Best = 3.723569e+28	FES = 295100	Diversity = 3.0405e-14
Generation 2960:	Best = 3.723569e+28	FES = 296100	Diversity = 3.0405e-14
Generation 2970:	Best = 3.723569e+28	FES = 297100	Diversity = 3.0405e-14
Generation 2980:	Best = 3.723569e+28	FES = 298100	Diversity = 3.0405e-14
Generation 2990:	Best = 3.723569e+28	FES = 299100	Diversity = 3.0405e-14

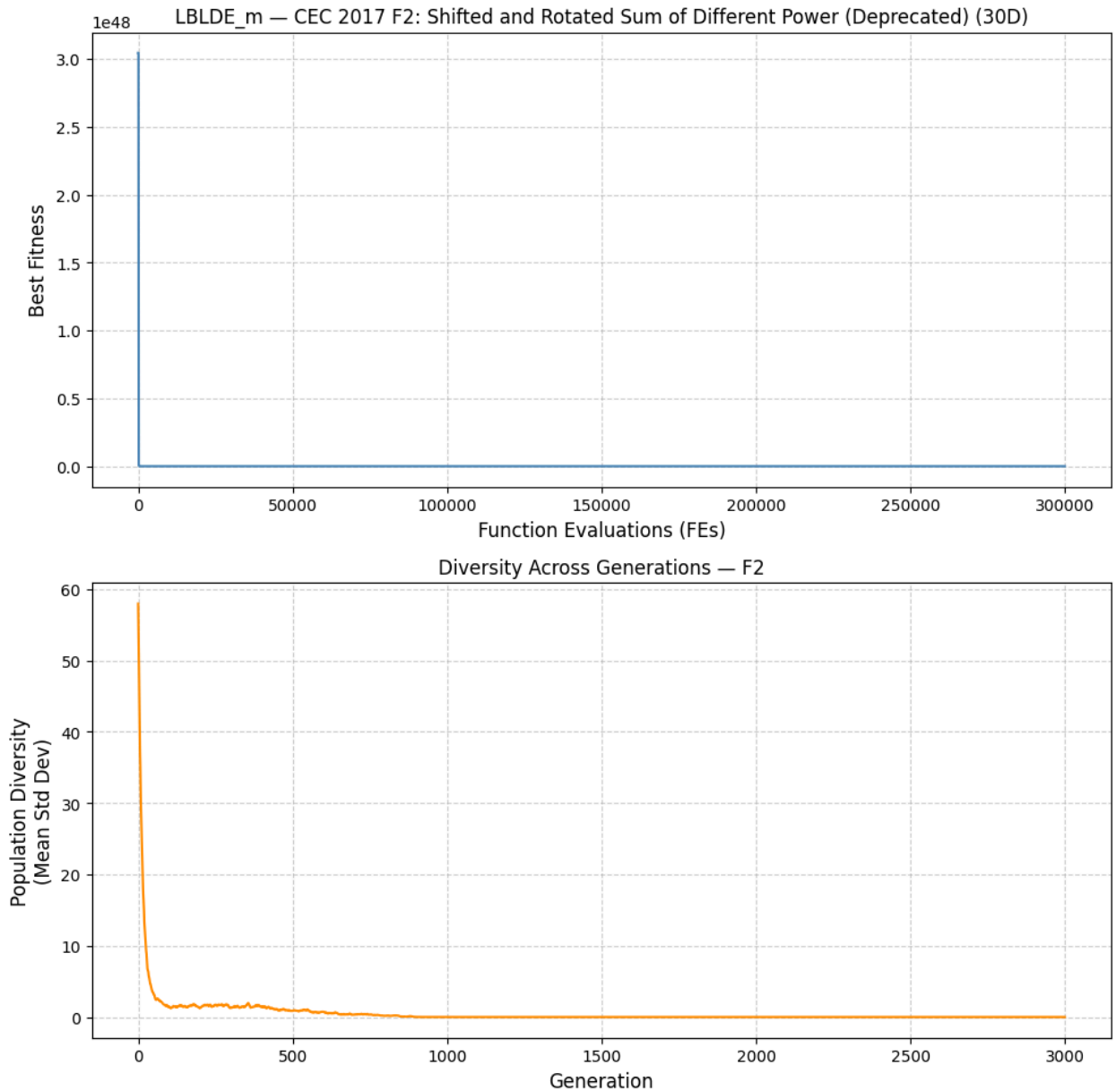
Optimization completed!

Final Best Fitness = 3.723569e+28

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 3.723569e+28
  Global Optimum         : 2.000000e+02
  Error (|best-optimum|) : 3.723569e+28
  Best Solution (5 dim)  : [ -9.32355291 -10.85330212 -58.53214502  34.03260747
-9.13296352]
  D^(0) (init. diversity): 5.7976e+01
```

II



Graph saved as convergence_F2.png

```
In [29]: # =====  
# F3 — Shifted and Rotated Zakharov  
# =====  
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and Rotated Zakharov')
```

=====

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LBLDE_m on CEC 2017 F3 – Single Run

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Function : Shifted and Rotated Zakharov (30D)

Bounds : [-100, 100]

Global optimum : 300.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 4.672334e+07 | Diversity = 5.7976e+01

Generation	10:	Best = 8.970981e+04	FES = 1100	Diversity = 4.4121e+01
Generation	20:	Best = 8.957687e+04	FES = 2100	Diversity = 3.5551e+01
Generation	30:	Best = 6.397746e+04	FES = 3100	Diversity = 2.6945e+01
Generation	40:	Best = 5.991229e+04	FES = 4100	Diversity = 1.9336e+01
Generation	50:	Best = 4.828121e+04	FES = 5100	Diversity = 1.2278e+01
Generation	60:	Best = 4.591958e+04	FES = 6100	Diversity = 8.6307e+00
Generation	70:	Best = 4.053551e+04	FES = 7100	Diversity = 6.9842e+00
Generation	80:	Best = 4.053094e+04	FES = 8100	Diversity = 5.8759e+00
Generation	90:	Best = 3.891697e+04	FES = 9100	Diversity = 4.7936e+00
Generation	100:	Best = 3.801346e+04	FES = 10100	Diversity = 3.6707e+00
Generation	110:	Best = 3.682313e+04	FES = 11100	Diversity = 3.7008e+00
Generation	120:	Best = 3.540896e+04	FES = 12100	Diversity = 3.2831e+00
Generation	130:	Best = 3.512085e+04	FES = 13100	Diversity = 3.1685e+00
Generation	140:	Best = 3.314362e+04	FES = 14100	Diversity = 2.7505e+00
Generation	150:	Best = 3.237051e+04	FES = 15100	Diversity = 2.6636e+00
Generation	160:	Best = 3.121428e+04	FES = 16100	Diversity = 2.4899e+00
Generation	170:	Best = 2.998527e+04	FES = 17100	Diversity = 2.4212e+00
Generation	180:	Best = 2.852982e+04	FES = 18100	Diversity = 2.5138e+00
Generation	190:	Best = 2.808877e+04	FES = 19100	Diversity = 2.7332e+00
Generation	200:	Best = 2.707263e+04	FES = 20100	Diversity = 2.7226e+00
Generation	210:	Best = 2.608293e+04	FES = 21100	Diversity = 2.9253e+00
Generation	220:	Best = 2.498356e+04	FES = 22100	Diversity = 2.9546e+00
Generation	230:	Best = 2.417993e+04	FES = 23100	Diversity = 2.6063e+00
Generation	240:	Best = 2.388838e+04	FES = 24100	Diversity = 2.3034e+00
Generation	250:	Best = 2.323700e+04	FES = 25100	Diversity = 2.1704e+00
Generation	260:	Best = 2.268149e+04	FES = 26100	Diversity = 1.9329e+00
Generation	270:	Best = 2.224161e+04	FES = 27100	Diversity = 2.1483e+00
Generation	280:	Best = 2.086092e+04	FES = 28100	Diversity = 2.0451e+00
Generation	290:	Best = 2.052844e+04	FES = 29100	Diversity = 2.3534e+00
Generation	300:	Best = 1.999406e+04	FES = 30100	Diversity = 2.3173e+00
Generation	310:	Best = 1.938639e+04	FES = 31100	Diversity = 2.2718e+00
Generation	320:	Best = 1.865177e+04	FES = 32100	Diversity = 2.0414e+00
Generation	330:	Best = 1.817757e+04	FES = 33100	Diversity = 2.2766e+00
Generation	340:	Best = 1.784663e+04	FES = 34100	Diversity = 2.1851e+00
Generation	350:	Best = 1.705638e+04	FES = 35100	Diversity = 2.0074e+00
Generation	360:	Best = 1.582632e+04	FES = 36100	Diversity = 2.1325e+00
Generation	370:	Best = 1.537390e+04	FES = 37100	Diversity = 2.2278e+00
Generation	380:	Best = 1.435965e+04	FES = 38100	Diversity = 1.9722e+00
Generation	390:	Best = 1.388977e+04	FES = 39100	Diversity = 1.2128e+00
Generation	400:	Best = 1.356290e+04	FES = 40100	Diversity = 8.0040e-01
Generation	410:	Best = 1.329866e+04	FES = 41100	Diversity = 3.6154e-01
Generation	420:	Best = 1.318031e+04	FES = 42100	Diversity = 1.2032e-01
Generation	430:	Best = 1.311942e+04	FES = 43100	Diversity = 6.8891e-02
Generation	440:	Best = 1.307543e+04	FES = 44100	Diversity = 7.1824e-02

Generation	450:	Best = 1.304637e+04	FES = 45100	Diversity = 4.9224e-02
Generation	460:	Best = 1.303725e+04	FES = 46100	Diversity = 1.9003e-02
Generation	470:	Best = 1.303388e+04	FES = 47100	Diversity = 1.2181e-02
Generation	480:	Best = 1.303225e+04	FES = 48100	Diversity = 7.2016e-03
Generation	490:	Best = 1.303091e+04	FES = 49100	Diversity = 9.2540e-03
Generation	500:	Best = 1.303028e+04	FES = 50100	Diversity = 6.1405e-03
Generation	510:	Best = 1.303001e+04	FES = 51100	Diversity = 2.7542e-03
Generation	520:	Best = 1.302991e+04	FES = 52100	Diversity = 2.4934e-03
Generation	530:	Best = 1.302989e+04	FES = 53100	Diversity = 5.2696e-04
Generation	540:	Best = 1.302989e+04	FES = 54100	Diversity = 3.7618e-04
Generation	550:	Best = 1.302988e+04	FES = 55100	Diversity = 2.2307e-04
Generation	560:	Best = 1.302988e+04	FES = 56100	Diversity = 2.1709e-04
Generation	570:	Best = 1.302988e+04	FES = 57100	Diversity = 1.7597e-04
Generation	580:	Best = 1.302988e+04	FES = 58100	Diversity = 1.2920e-04
Generation	590:	Best = 1.302988e+04	FES = 59100	Diversity = 5.8372e-05
Generation	600:	Best = 1.302988e+04	FES = 60100	Diversity = 4.6873e-05
Generation	610:	Best = 1.302988e+04	FES = 61100	Diversity = 2.1185e-05
Generation	620:	Best = 1.302988e+04	FES = 62100	Diversity = 1.4037e-05
Generation	630:	Best = 1.302988e+04	FES = 63100	Diversity = 1.5947e-05
Generation	640:	Best = 1.302988e+04	FES = 64100	Diversity = 1.5950e-05
Generation	650:	Best = 1.302988e+04	FES = 65100	Diversity = 1.1393e-05
Generation	660:	Best = 1.302988e+04	FES = 66100	Diversity = 7.0880e-06
Generation	670:	Best = 1.302988e+04	FES = 67100	Diversity = 2.8354e-06
Generation	680:	Best = 1.302988e+04	FES = 68100	Diversity = 8.7354e-07
Generation	690:	Best = 1.302988e+04	FES = 69100	Diversity = 8.5741e-07
Generation	700:	Best = 1.302988e+04	FES = 70100	Diversity = 7.4729e-07
Generation	710:	Best = 1.302988e+04	FES = 71100	Diversity = 4.4922e-07
Generation	720:	Best = 1.302988e+04	FES = 72100	Diversity = 5.2078e-07
Generation	730:	Best = 1.302988e+04	FES = 73100	Diversity = 7.3828e-07
Generation	740:	Best = 1.302988e+04	FES = 74100	Diversity = 6.6846e-07
Generation	750:	Best = 1.302988e+04	FES = 75100	Diversity = 2.4671e-07
Generation	760:	Best = 1.302988e+04	FES = 76100	Diversity = 1.6199e-07
Generation	770:	Best = 1.302988e+04	FES = 77100	Diversity = 7.4513e-08
Generation	780:	Best = 1.302988e+04	FES = 78100	Diversity = 8.1753e-08
Generation	790:	Best = 1.302988e+04	FES = 79100	Diversity = 7.4978e-08
Generation	800:	Best = 1.302988e+04	FES = 80100	Diversity = 8.2119e-08
Generation	810:	Best = 1.302988e+04	FES = 81100	Diversity = 7.3482e-08
Generation	820:	Best = 1.302988e+04	FES = 82100	Diversity = 5.1667e-08
Generation	830:	Best = 1.302988e+04	FES = 83100	Diversity = 2.3423e-08
Generation	840:	Best = 1.302988e+04	FES = 84100	Diversity = 3.1782e-08
Generation	850:	Best = 1.302988e+04	FES = 85100	Diversity = 3.2159e-08
Generation	860:	Best = 1.302988e+04	FES = 86100	Diversity = 3.2680e-08
Generation	870:	Best = 1.302988e+04	FES = 87100	Diversity = 3.3707e-08
Generation	880:	Best = 1.302988e+04	FES = 88100	Diversity = 2.4320e-08
Generation	890:	Best = 1.302988e+04	FES = 89100	Diversity = 1.1450e-08
Generation	900:	Best = 1.302988e+04	FES = 90100	Diversity = 9.3055e-09
Generation	910:	Best = 1.302988e+04	FES = 91100	Diversity = 9.2368e-09
Generation	920:	Best = 1.302988e+04	FES = 92100	Diversity = 8.9160e-09
Generation	930:	Best = 1.302988e+04	FES = 93100	Diversity = 7.6514e-09
Generation	940:	Best = 1.302988e+04	FES = 94100	Diversity = 4.8350e-09
Generation	950:	Best = 1.302988e+04	FES = 95100	Diversity = 1.9625e-09
Generation	960:	Best = 1.302988e+04	FES = 96100	Diversity = 2.3635e-10
Generation	970:	Best = 1.302988e+04	FES = 97100	Diversity = 3.2951e-14
Generation	980:	Best = 1.302988e+04	FES = 98100	Diversity = 3.2951e-14

[illegible]

[illegible]

[illegible]

Generation 2610:	Best = 1.302988e+04	FES = 261100	Diversity = 3.2951e-14
Generation 2620:	Best = 1.302988e+04	FES = 262100	Diversity = 3.2951e-14
Generation 2630:	Best = 1.302988e+04	FES = 263100	Diversity = 3.2951e-14
Generation 2640:	Best = 1.302988e+04	FES = 264100	Diversity = 3.2951e-14
Generation 2650:	Best = 1.302988e+04	FES = 265100	Diversity = 3.2951e-14
Generation 2660:	Best = 1.302988e+04	FES = 266100	Diversity = 3.2951e-14
Generation 2670:	Best = 1.302988e+04	FES = 267100	Diversity = 3.2951e-14
Generation 2680:	Best = 1.302988e+04	FES = 268100	Diversity = 3.2951e-14
Generation 2690:	Best = 1.302988e+04	FES = 269100	Diversity = 3.2951e-14
Generation 2700:	Best = 1.302988e+04	FES = 270100	Diversity = 3.2951e-14
Generation 2710:	Best = 1.302988e+04	FES = 271100	Diversity = 3.2951e-14
Generation 2720:	Best = 1.302988e+04	FES = 272100	Diversity = 3.2951e-14
Generation 2730:	Best = 1.302988e+04	FES = 273100	Diversity = 3.2951e-14
Generation 2740:	Best = 1.302988e+04	FES = 274100	Diversity = 3.2951e-14
Generation 2750:	Best = 1.302988e+04	FES = 275100	Diversity = 3.2951e-14
Generation 2760:	Best = 1.302988e+04	FES = 276100	Diversity = 3.2951e-14
Generation 2770:	Best = 1.302988e+04	FES = 277100	Diversity = 3.2951e-14
Generation 2780:	Best = 1.302988e+04	FES = 278100	Diversity = 3.2951e-14
Generation 2790:	Best = 1.302988e+04	FES = 279100	Diversity = 3.2951e-14
Generation 2800:	Best = 1.302988e+04	FES = 280100	Diversity = 3.2951e-14
Generation 2810:	Best = 1.302988e+04	FES = 281100	Diversity = 3.2951e-14
Generation 2820:	Best = 1.302988e+04	FES = 282100	Diversity = 3.2951e-14
Generation 2830:	Best = 1.302988e+04	FES = 283100	Diversity = 3.2951e-14
Generation 2840:	Best = 1.302988e+04	FES = 284100	Diversity = 3.2951e-14
Generation 2850:	Best = 1.302988e+04	FES = 285100	Diversity = 3.2951e-14
Generation 2860:	Best = 1.302988e+04	FES = 286100	Diversity = 3.2951e-14
Generation 2870:	Best = 1.302988e+04	FES = 287100	Diversity = 3.2951e-14
Generation 2880:	Best = 1.302988e+04	FES = 288100	Diversity = 3.2951e-14
Generation 2890:	Best = 1.302988e+04	FES = 289100	Diversity = 3.2951e-14
Generation 2900:	Best = 1.302988e+04	FES = 290100	Diversity = 3.2951e-14
Generation 2910:	Best = 1.302988e+04	FES = 291100	Diversity = 3.2951e-14
Generation 2920:	Best = 1.302988e+04	FES = 292100	Diversity = 3.2951e-14
Generation 2930:	Best = 1.302988e+04	FES = 293100	Diversity = 3.2951e-14
Generation 2940:	Best = 1.302988e+04	FES = 294100	Diversity = 3.2951e-14
Generation 2950:	Best = 1.302988e+04	FES = 295100	Diversity = 3.2951e-14
Generation 2960:	Best = 1.302988e+04	FES = 296100	Diversity = 3.2951e-14
Generation 2970:	Best = 1.302988e+04	FES = 297100	Diversity = 3.2951e-14
Generation 2980:	Best = 1.302988e+04	FES = 298100	Diversity = 3.2951e-14
Generation 2990:	Best = 1.302988e+04	FES = 299100	Diversity = 3.2951e-14

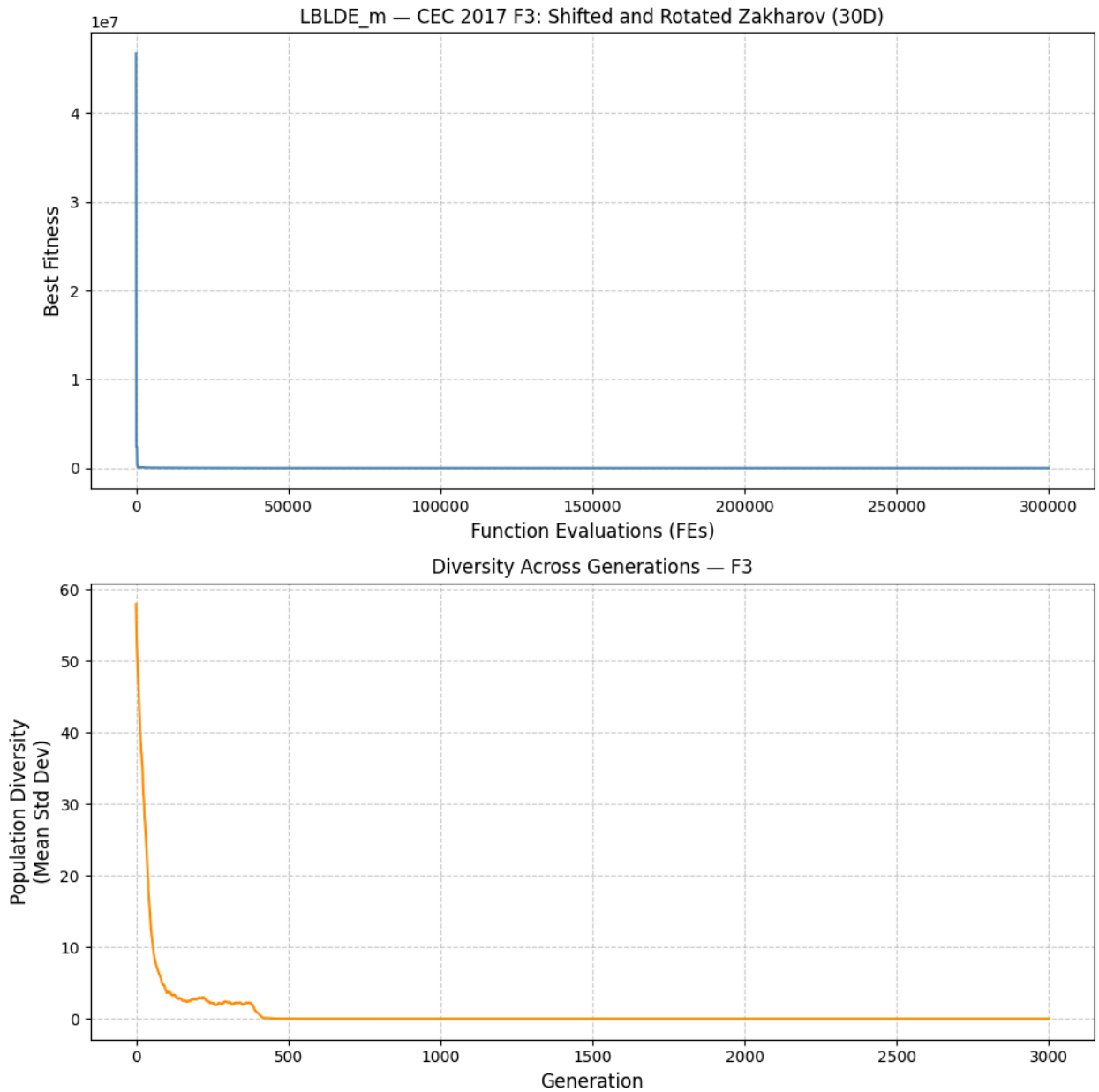
Optimization completed!

Final Best Fitness = 1.302988e+04

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 1.302988e+04
  Global Optimum         : 3.000000e+02
  Error (|best-optimum|) : 1.272988e+04
  Best Solution (5 dim)  : [-40.25521233   9.60054679  24.53725452  13.04803381
-49.25669306]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F3.png

6.2 Multimodal Functions (F4-F10)

```
In [30]: # =====  
# F4 — Shifted and Rotated Rosenbrock  
# =====  
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and Rotated Rosent
```

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LBLDE_m on CEC 2017 F4 – Single Run

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Function : Shifted and Rotated Rosenbrock's (30D)

Bounds : [-100, 100]

Global optimum : 400.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.421845e+04 | Diversity = 5.7976e+01

Generation	10:	Best = 4.543432e+03	FES = 1100	Diversity = 1.9164e+01
Generation	20:	Best = 3.009860e+03	FES = 2100	Diversity = 6.7044e+00
Generation	30:	Best = 2.261403e+03	FES = 3100	Diversity = 3.1525e+00
Generation	40:	Best = 2.012558e+03	FES = 4100	Diversity = 1.6989e+00
Generation	50:	Best = 1.866927e+03	FES = 5100	Diversity = 1.2392e+00
Generation	60:	Best = 1.737234e+03	FES = 6100	Diversity = 9.8200e-01
Generation	70:	Best = 1.632907e+03	FES = 7100	Diversity = 8.5755e-01
Generation	80:	Best = 1.576182e+03	FES = 8100	Diversity = 8.0897e-01
Generation	90:	Best = 1.513961e+03	FES = 9100	Diversity = 8.3578e-01
Generation	100:	Best = 1.459415e+03	FES = 10100	Diversity = 7.5307e-01
Generation	110:	Best = 1.404963e+03	FES = 11100	Diversity = 7.6248e-01
Generation	120:	Best = 1.353686e+03	FES = 12100	Diversity = 6.8248e-01
Generation	130:	Best = 1.322138e+03	FES = 13100	Diversity = 6.9255e-01
Generation	140:	Best = 1.294934e+03	FES = 14100	Diversity = 5.9783e-01
Generation	150:	Best = 1.276359e+03	FES = 15100	Diversity = 5.2836e-01
Generation	160:	Best = 1.259505e+03	FES = 16100	Diversity = 5.6929e-01
Generation	170:	Best = 1.243386e+03	FES = 17100	Diversity = 4.5077e-01
Generation	180:	Best = 1.231438e+03	FES = 18100	Diversity = 3.1947e-01
Generation	190:	Best = 1.222702e+03	FES = 19100	Diversity = 2.2644e-01
Generation	200:	Best = 1.214231e+03	FES = 20100	Diversity = 2.8090e-01
Generation	210:	Best = 1.206289e+03	FES = 21100	Diversity = 2.6979e-01
Generation	220:	Best = 1.200290e+03	FES = 22100	Diversity = 2.7038e-01
Generation	230:	Best = 1.194378e+03	FES = 23100	Diversity = 2.5586e-01
Generation	240:	Best = 1.188299e+03	FES = 24100	Diversity = 1.7955e-01
Generation	250:	Best = 1.185745e+03	FES = 25100	Diversity = 1.2084e-01
Generation	260:	Best = 1.184148e+03	FES = 26100	Diversity = 9.2077e-02
Generation	270:	Best = 1.182790e+03	FES = 27100	Diversity = 7.8742e-02
Generation	280:	Best = 1.181268e+03	FES = 28100	Diversity = 7.2487e-02
Generation	290:	Best = 1.180486e+03	FES = 29100	Diversity = 5.7949e-02
Generation	300:	Best = 1.179922e+03	FES = 30100	Diversity = 4.5349e-02
Generation	310:	Best = 1.179510e+03	FES = 31100	Diversity = 3.2904e-02
Generation	320:	Best = 1.179227e+03	FES = 32100	Diversity = 3.1300e-02
Generation	330:	Best = 1.179036e+03	FES = 33100	Diversity = 2.4551e-02
Generation	340:	Best = 1.178895e+03	FES = 34100	Diversity = 2.1055e-02
Generation	350:	Best = 1.178813e+03	FES = 35100	Diversity = 1.7174e-02
Generation	360:	Best = 1.178763e+03	FES = 36100	Diversity = 1.2954e-02
Generation	370:	Best = 1.178736e+03	FES = 37100	Diversity = 6.9435e-03
Generation	380:	Best = 1.178693e+03	FES = 38100	Diversity = 8.1976e-03
Generation	390:	Best = 1.178656e+03	FES = 39100	Diversity = 8.8505e-03
Generation	400:	Best = 1.178623e+03	FES = 40100	Diversity = 6.6396e-03
Generation	410:	Best = 1.178608e+03	FES = 41100	Diversity = 4.6973e-03
Generation	420:	Best = 1.178593e+03	FES = 42100	Diversity = 5.8363e-03
Generation	430:	Best = 1.178581e+03	FES = 43100	Diversity = 4.6064e-03
Generation	440:	Best = 1.178574e+03	FES = 44100	Diversity = 2.9311e-03

Generation	450:	Best = 1.178570e+03	FES = 45100	Diversity = 1.9827e-03
Generation	460:	Best = 1.178567e+03	FES = 46100	Diversity = 1.7963e-03
Generation	470:	Best = 1.178564e+03	FES = 47100	Diversity = 1.7663e-03
Generation	480:	Best = 1.178562e+03	FES = 48100	Diversity = 1.1286e-03
Generation	490:	Best = 1.178560e+03	FES = 49100	Diversity = 1.1159e-03
Generation	500:	Best = 1.178558e+03	FES = 50100	Diversity = 9.0272e-04
Generation	510:	Best = 1.178557e+03	FES = 51100	Diversity = 7.5678e-04
Generation	520:	Best = 1.178557e+03	FES = 52100	Diversity = 6.5926e-04
Generation	530:	Best = 1.178556e+03	FES = 53100	Diversity = 5.4442e-04
Generation	540:	Best = 1.178556e+03	FES = 54100	Diversity = 4.2556e-04
Generation	550:	Best = 1.178556e+03	FES = 55100	Diversity = 2.7996e-04
Generation	560:	Best = 1.178556e+03	FES = 56100	Diversity = 2.4842e-04
Generation	570:	Best = 1.178556e+03	FES = 57100	Diversity = 1.9310e-04
Generation	580:	Best = 1.178556e+03	FES = 58100	Diversity = 1.4779e-04
Generation	590:	Best = 1.178556e+03	FES = 59100	Diversity = 1.2653e-04
Generation	600:	Best = 1.178556e+03	FES = 60100	Diversity = 9.8202e-05
Generation	610:	Best = 1.178556e+03	FES = 61100	Diversity = 6.4521e-05
Generation	620:	Best = 1.178556e+03	FES = 62100	Diversity = 7.2699e-05
Generation	630:	Best = 1.178556e+03	FES = 63100	Diversity = 5.0202e-05
Generation	640:	Best = 1.178556e+03	FES = 64100	Diversity = 4.6178e-05
Generation	650:	Best = 1.178556e+03	FES = 65100	Diversity = 2.7074e-05
Generation	660:	Best = 1.178556e+03	FES = 66100	Diversity = 2.2791e-05
Generation	670:	Best = 1.178556e+03	FES = 67100	Diversity = 1.0141e-05
Generation	680:	Best = 1.178556e+03	FES = 68100	Diversity = 1.2342e-05
Generation	690:	Best = 1.178556e+03	FES = 69100	Diversity = 6.7762e-06
Generation	700:	Best = 1.178556e+03	FES = 70100	Diversity = 5.1342e-06
Generation	710:	Best = 1.178556e+03	FES = 71100	Diversity = 4.0946e-06
Generation	720:	Best = 1.178556e+03	FES = 72100	Diversity = 3.6939e-06
Generation	730:	Best = 1.178556e+03	FES = 73100	Diversity = 2.0578e-06
Generation	740:	Best = 1.178556e+03	FES = 74100	Diversity = 1.4240e-06
Generation	750:	Best = 1.178556e+03	FES = 75100	Diversity = 7.8325e-07
Generation	760:	Best = 1.178556e+03	FES = 76100	Diversity = 4.3459e-07
Generation	770:	Best = 1.178556e+03	FES = 77100	Diversity = 2.4233e-07
Generation	780:	Best = 1.178556e+03	FES = 78100	Diversity = 2.4000e-07
Generation	790:	Best = 1.178556e+03	FES = 79100	Diversity = 1.6986e-07
Generation	800:	Best = 1.178556e+03	FES = 80100	Diversity = 1.1079e-07
Generation	810:	Best = 1.178556e+03	FES = 81100	Diversity = 5.6515e-08
Generation	820:	Best = 1.178556e+03	FES = 82100	Diversity = 3.9894e-08
Generation	830:	Best = 1.178556e+03	FES = 83100	Diversity = 2.4937e-08
Generation	840:	Best = 1.178556e+03	FES = 84100	Diversity = 1.5027e-08
Generation	850:	Best = 1.178556e+03	FES = 85100	Diversity = 1.0908e-08
Generation	860:	Best = 1.178556e+03	FES = 86100	Diversity = 9.8201e-11
Generation	870:	Best = 1.178556e+03	FES = 87100	Diversity = 1.1782e-12
Generation	880:	Best = 1.178556e+03	FES = 88100	Diversity = 1.0639e-13
Generation	890:	Best = 1.178556e+03	FES = 89100	Diversity = 4.0296e-14
Generation	900:	Best = 1.178556e+03	FES = 90100	Diversity = 4.0296e-14
Generation	910:	Best = 1.178556e+03	FES = 91100	Diversity = 4.0296e-14
Generation	920:	Best = 1.178556e+03	FES = 92100	Diversity = 4.0296e-14
Generation	930:	Best = 1.178556e+03	FES = 93100	Diversity = 4.0296e-14
Generation	940:	Best = 1.178556e+03	FES = 94100	Diversity = 4.0296e-14
Generation	950:	Best = 1.178556e+03	FES = 95100	Diversity = 4.0296e-14
Generation	960:	Best = 1.178556e+03	FES = 96100	Diversity = 4.0296e-14
Generation	970:	Best = 1.178556e+03	FES = 97100	Diversity = 4.0296e-14
Generation	980:	Best = 1.178556e+03	FES = 98100	Diversity = 4.0296e-14

[illegible]

[illegible]

[illegible]

Generation 2610:	Best = 1.178556e+03	FES = 261100	Diversity = 4.0296e-14
Generation 2620:	Best = 1.178556e+03	FES = 262100	Diversity = 4.0296e-14
Generation 2630:	Best = 1.178556e+03	FES = 263100	Diversity = 4.0296e-14
Generation 2640:	Best = 1.178556e+03	FES = 264100	Diversity = 4.0296e-14
Generation 2650:	Best = 1.178556e+03	FES = 265100	Diversity = 4.0296e-14
Generation 2660:	Best = 1.178556e+03	FES = 266100	Diversity = 4.0296e-14
Generation 2670:	Best = 1.178556e+03	FES = 267100	Diversity = 4.0296e-14
Generation 2680:	Best = 1.178556e+03	FES = 268100	Diversity = 4.0296e-14
Generation 2690:	Best = 1.178556e+03	FES = 269100	Diversity = 4.0296e-14
Generation 2700:	Best = 1.178556e+03	FES = 270100	Diversity = 4.0296e-14
Generation 2710:	Best = 1.178556e+03	FES = 271100	Diversity = 4.0296e-14
Generation 2720:	Best = 1.178556e+03	FES = 272100	Diversity = 4.0296e-14
Generation 2730:	Best = 1.178556e+03	FES = 273100	Diversity = 4.0296e-14
Generation 2740:	Best = 1.178556e+03	FES = 274100	Diversity = 4.0296e-14
Generation 2750:	Best = 1.178556e+03	FES = 275100	Diversity = 4.0296e-14
Generation 2760:	Best = 1.178556e+03	FES = 276100	Diversity = 4.0296e-14
Generation 2770:	Best = 1.178556e+03	FES = 277100	Diversity = 4.0296e-14
Generation 2780:	Best = 1.178556e+03	FES = 278100	Diversity = 4.0296e-14
Generation 2790:	Best = 1.178556e+03	FES = 279100	Diversity = 4.0296e-14
Generation 2800:	Best = 1.178556e+03	FES = 280100	Diversity = 4.0296e-14
Generation 2810:	Best = 1.178556e+03	FES = 281100	Diversity = 4.0296e-14
Generation 2820:	Best = 1.178556e+03	FES = 282100	Diversity = 4.0296e-14
Generation 2830:	Best = 1.178556e+03	FES = 283100	Diversity = 4.0296e-14
Generation 2840:	Best = 1.178556e+03	FES = 284100	Diversity = 4.0296e-14
Generation 2850:	Best = 1.178556e+03	FES = 285100	Diversity = 4.0296e-14
Generation 2860:	Best = 1.178556e+03	FES = 286100	Diversity = 4.0296e-14
Generation 2870:	Best = 1.178556e+03	FES = 287100	Diversity = 4.0296e-14
Generation 2880:	Best = 1.178556e+03	FES = 288100	Diversity = 4.0296e-14
Generation 2890:	Best = 1.178556e+03	FES = 289100	Diversity = 4.0296e-14
Generation 2900:	Best = 1.178556e+03	FES = 290100	Diversity = 4.0296e-14
Generation 2910:	Best = 1.178556e+03	FES = 291100	Diversity = 4.0296e-14
Generation 2920:	Best = 1.178556e+03	FES = 292100	Diversity = 4.0296e-14
Generation 2930:	Best = 1.178556e+03	FES = 293100	Diversity = 4.0296e-14
Generation 2940:	Best = 1.178556e+03	FES = 294100	Diversity = 4.0296e-14
Generation 2950:	Best = 1.178556e+03	FES = 295100	Diversity = 4.0296e-14
Generation 2960:	Best = 1.178556e+03	FES = 296100	Diversity = 4.0296e-14
Generation 2970:	Best = 1.178556e+03	FES = 297100	Diversity = 4.0296e-14
Generation 2980:	Best = 1.178556e+03	FES = 298100	Diversity = 4.0296e-14
Generation 2990:	Best = 1.178556e+03	FES = 299100	Diversity = 4.0296e-14

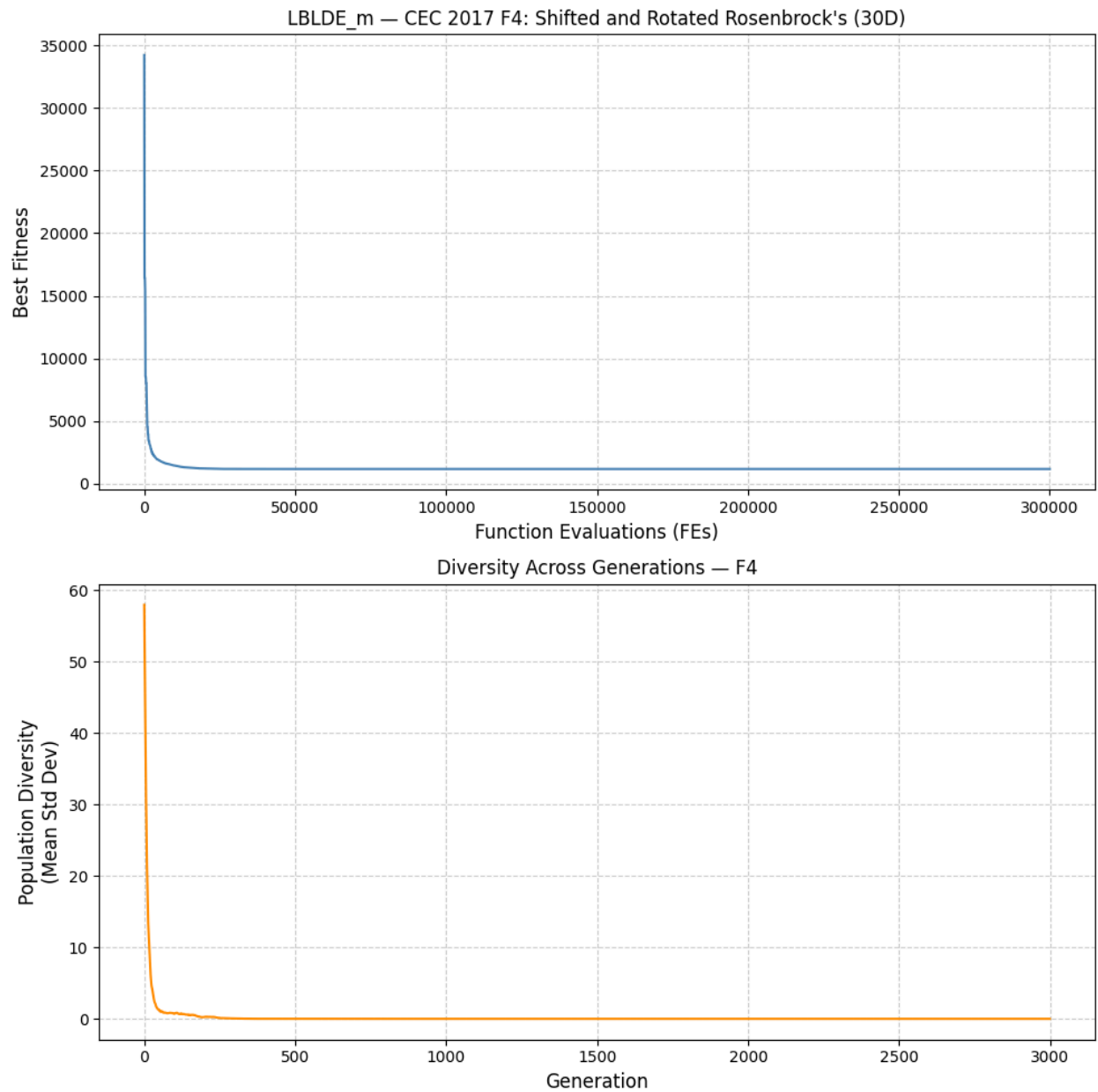
Optimization completed!

Final Best Fitness = 1.178556e+03

Total Function Evaluations = 300000

```
=====
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Results:
  Best Fitness           : 1.178556e+03
  Global Optimum         : 4.000000e+02
  Error (|best-optimum|) : 7.785557e+02
  Best Solution (5 dim)  : [ 66.25205697  10.73696167 -14.31671988  91.21437002
12.10001223]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F4.png

```
In [31]: # =====  
# F5 – Shifted and Rotated Rastrigin  
# =====  
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and Rotated Rastrigin")
```

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LBLDE_m on CEC 2017 F5 – Single Run

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Function : Shifted and Rotated Rastrigin's (30D)
Bounds : [-100, 100]
Global optimum : 500.0
Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 1.177228e+03 | Diversity = 5.7976e+01

Generation	10:	Best = 8.394134e+02	FES = 1100	Diversity = 2.4438e+01
Generation	20:	Best = 7.765545e+02	FES = 2100	Diversity = 1.6556e+01
Generation	30:	Best = 7.499913e+02	FES = 3100	Diversity = 1.4726e+01
Generation	40:	Best = 7.499913e+02	FES = 4100	Diversity = 1.3978e+01
Generation	50:	Best = 7.253643e+02	FES = 5100	Diversity = 1.3469e+01
Generation	60:	Best = 7.221003e+02	FES = 6100	Diversity = 1.2818e+01
Generation	70:	Best = 7.221003e+02	FES = 7100	Diversity = 1.2419e+01
Generation	80:	Best = 6.979875e+02	FES = 8100	Diversity = 1.2214e+01
Generation	90:	Best = 6.765403e+02	FES = 9100	Diversity = 1.2001e+01
Generation	100:	Best = 6.656849e+02	FES = 10100	Diversity = 1.2108e+01
Generation	110:	Best = 6.618657e+02	FES = 11100	Diversity = 1.1796e+01
Generation	120:	Best = 6.475971e+02	FES = 12100	Diversity = 1.1850e+01
Generation	130:	Best = 6.473881e+02	FES = 13100	Diversity = 1.1966e+01
Generation	140:	Best = 6.124424e+02	FES = 14100	Diversity = 1.1988e+01
Generation	150:	Best = 6.122026e+02	FES = 15100	Diversity = 1.1862e+01
Generation	160:	Best = 6.075251e+02	FES = 16100	Diversity = 1.1827e+01
Generation	170:	Best = 6.075251e+02	FES = 17100	Diversity = 1.1683e+01
Generation	180:	Best = 6.022241e+02	FES = 18100	Diversity = 1.1649e+01
Generation	190:	Best = 6.022241e+02	FES = 19100	Diversity = 1.1462e+01
Generation	200:	Best = 5.995478e+02	FES = 20100	Diversity = 1.1359e+01
Generation	210:	Best = 5.991523e+02	FES = 21100	Diversity = 1.1332e+01
Generation	220:	Best = 5.989286e+02	FES = 22100	Diversity = 1.1348e+01
Generation	230:	Best = 5.989286e+02	FES = 23100	Diversity = 1.1408e+01
Generation	240:	Best = 5.988941e+02	FES = 24100	Diversity = 1.1350e+01
Generation	250:	Best = 5.988941e+02	FES = 25100	Diversity = 1.1321e+01
Generation	260:	Best = 5.988941e+02	FES = 26100	Diversity = 1.1282e+01
Generation	270:	Best = 5.866901e+02	FES = 27100	Diversity = 1.1199e+01
Generation	280:	Best = 5.866901e+02	FES = 28100	Diversity = 1.1070e+01
Generation	290:	Best = 5.866901e+02	FES = 29100	Diversity = 1.1079e+01
Generation	300:	Best = 5.866901e+02	FES = 30100	Diversity = 1.0972e+01
Generation	310:	Best = 5.859427e+02	FES = 31100	Diversity = 1.0910e+01
Generation	320:	Best = 5.859427e+02	FES = 32100	Diversity = 1.0886e+01
Generation	330:	Best = 5.859427e+02	FES = 33100	Diversity = 1.0827e+01
Generation	340:	Best = 5.859164e+02	FES = 34100	Diversity = 1.0790e+01
Generation	350:	Best = 5.859164e+02	FES = 35100	Diversity = 1.0783e+01
Generation	360:	Best = 5.859164e+02	FES = 36100	Diversity = 1.0670e+01
Generation	370:	Best = 5.859164e+02	FES = 37100	Diversity = 1.0753e+01
Generation	380:	Best = 5.859164e+02	FES = 38100	Diversity = 1.0762e+01
Generation	390:	Best = 5.859164e+02	FES = 39100	Diversity = 1.0763e+01
Generation	400:	Best = 5.858741e+02	FES = 40100	Diversity = 1.0620e+01
Generation	410:	Best = 5.857655e+02	FES = 41100	Diversity = 1.0526e+01
Generation	420:	Best = 5.857655e+02	FES = 42100	Diversity = 1.0513e+01
Generation	430:	Best = 5.857655e+02	FES = 43100	Diversity = 1.0437e+01
Generation	440:	Best = 5.854311e+02	FES = 44100	Diversity = 1.0414e+01

Generation	450:	Best = 5.854311e+02	FES = 45100	Diversity = 1.0457e+01
Generation	460:	Best = 5.854311e+02	FES = 46100	Diversity = 1.0450e+01
Generation	470:	Best = 5.854311e+02	FES = 47100	Diversity = 1.0502e+01
Generation	480:	Best = 5.854311e+02	FES = 48100	Diversity = 1.0404e+01
Generation	490:	Best = 5.849888e+02	FES = 49100	Diversity = 1.0384e+01
Generation	500:	Best = 5.849397e+02	FES = 50100	Diversity = 1.0413e+01
Generation	510:	Best = 5.848544e+02	FES = 51100	Diversity = 1.0412e+01
Generation	520:	Best = 5.848502e+02	FES = 52100	Diversity = 1.0407e+01
Generation	530:	Best = 5.845492e+02	FES = 53100	Diversity = 1.0391e+01
Generation	540:	Best = 5.838680e+02	FES = 54100	Diversity = 1.0376e+01
Generation	550:	Best = 5.838680e+02	FES = 55100	Diversity = 1.0344e+01
Generation	560:	Best = 5.824979e+02	FES = 56100	Diversity = 1.0343e+01
Generation	570:	Best = 5.822825e+02	FES = 57100	Diversity = 1.0290e+01
Generation	580:	Best = 5.821379e+02	FES = 58100	Diversity = 1.0189e+01
Generation	590:	Best = 5.821379e+02	FES = 59100	Diversity = 1.0217e+01
Generation	600:	Best = 5.821379e+02	FES = 60100	Diversity = 1.0207e+01
Generation	610:	Best = 5.821379e+02	FES = 61100	Diversity = 1.0275e+01
Generation	620:	Best = 5.821379e+02	FES = 62100	Diversity = 1.0263e+01
Generation	630:	Best = 5.821379e+02	FES = 63100	Diversity = 1.0372e+01
Generation	640:	Best = 5.819913e+02	FES = 64100	Diversity = 1.0374e+01
Generation	650:	Best = 5.819608e+02	FES = 65100	Diversity = 1.0374e+01
Generation	660:	Best = 5.819395e+02	FES = 66100	Diversity = 1.0372e+01
Generation	670:	Best = 5.819379e+02	FES = 67100	Diversity = 1.0214e+01
Generation	680:	Best = 5.819284e+02	FES = 68100	Diversity = 1.0264e+01
Generation	690:	Best = 5.819227e+02	FES = 69100	Diversity = 1.0157e+01
Generation	700:	Best = 5.819072e+02	FES = 70100	Diversity = 1.0159e+01
Generation	710:	Best = 5.819070e+02	FES = 71100	Diversity = 1.0163e+01
Generation	720:	Best = 5.818954e+02	FES = 72100	Diversity = 1.0158e+01
Generation	730:	Best = 5.818954e+02	FES = 73100	Diversity = 1.0136e+01
Generation	740:	Best = 5.818935e+02	FES = 74100	Diversity = 1.0150e+01
Generation	750:	Best = 5.818935e+02	FES = 75100	Diversity = 1.0110e+01
Generation	760:	Best = 5.818679e+02	FES = 76100	Diversity = 1.0110e+01
Generation	770:	Best = 5.818679e+02	FES = 77100	Diversity = 1.0131e+01
Generation	780:	Best = 5.818637e+02	FES = 78100	Diversity = 1.0117e+01
Generation	790:	Best = 5.818622e+02	FES = 79100	Diversity = 1.0081e+01
Generation	800:	Best = 5.818430e+02	FES = 80100	Diversity = 1.0133e+01
Generation	810:	Best = 5.818430e+02	FES = 81100	Diversity = 1.0091e+01
Generation	820:	Best = 5.818157e+02	FES = 82100	Diversity = 1.0093e+01
Generation	830:	Best = 5.818157e+02	FES = 83100	Diversity = 1.0095e+01
Generation	840:	Best = 5.818149e+02	FES = 84100	Diversity = 1.0039e+01
Generation	850:	Best = 5.818147e+02	FES = 85100	Diversity = 1.0073e+01
Generation	860:	Best = 5.817973e+02	FES = 86100	Diversity = 1.0045e+01
Generation	870:	Best = 5.810935e+02	FES = 87100	Diversity = 1.0052e+01
Generation	880:	Best = 5.810895e+02	FES = 88100	Diversity = 9.9571e+00
Generation	890:	Best = 5.810648e+02	FES = 89100	Diversity = 9.9533e+00
Generation	900:	Best = 5.810499e+02	FES = 90100	Diversity = 9.9343e+00
Generation	910:	Best = 5.810497e+02	FES = 91100	Diversity = 9.9369e+00
Generation	920:	Best = 5.810497e+02	FES = 92100	Diversity = 9.9343e+00
Generation	930:	Best = 5.810465e+02	FES = 93100	Diversity = 9.9161e+00
Generation	940:	Best = 5.806870e+02	FES = 94100	Diversity = 9.9150e+00
Generation	950:	Best = 5.806868e+02	FES = 95100	Diversity = 9.9409e+00
Generation	960:	Best = 5.806868e+02	FES = 96100	Diversity = 9.9876e+00
Generation	970:	Best = 5.806865e+02	FES = 97100	Diversity = 1.0003e+01
Generation	980:	Best = 5.806863e+02	FES = 98100	Diversity = 9.9521e+00

Generation 990:	Best = 5.762837e+02	FES = 99100	Diversity = 9.8958e+00
Generation 1000:	Best = 5.755070e+02	FES = 100100	Diversity = 9.8933e+00
Generation 1010:	Best = 5.752463e+02	FES = 101100	Diversity = 9.8346e+00
Generation 1020:	Best = 5.752461e+02	FES = 102100	Diversity = 9.8603e+00
Generation 1030:	Best = 5.752207e+02	FES = 103100	Diversity = 9.8382e+00
Generation 1040:	Best = 5.752137e+02	FES = 104100	Diversity = 9.7972e+00
Generation 1050:	Best = 5.752131e+02	FES = 105100	Diversity = 9.7547e+00
Generation 1060:	Best = 5.752123e+02	FES = 106100	Diversity = 9.7576e+00
Generation 1070:	Best = 5.752116e+02	FES = 107100	Diversity = 9.7382e+00
Generation 1080:	Best = 5.752114e+02	FES = 108100	Diversity = 9.7420e+00
Generation 1090:	Best = 5.752114e+02	FES = 109100	Diversity = 9.6561e+00
Generation 1100:	Best = 5.752114e+02	FES = 110100	Diversity = 9.6607e+00
Generation 1110:	Best = 5.748217e+02	FES = 111100	Diversity = 9.6260e+00
Generation 1120:	Best = 5.743987e+02	FES = 112100	Diversity = 9.6779e+00
Generation 1130:	Best = 5.743920e+02	FES = 113100	Diversity = 9.6620e+00
Generation 1140:	Best = 5.743795e+02	FES = 114100	Diversity = 9.6951e+00
Generation 1150:	Best = 5.743777e+02	FES = 115100	Diversity = 9.6903e+00
Generation 1160:	Best = 5.743776e+02	FES = 116100	Diversity = 9.6684e+00
Generation 1170:	Best = 5.743773e+02	FES = 117100	Diversity = 9.6682e+00
Generation 1180:	Best = 5.743772e+02	FES = 118100	Diversity = 9.6631e+00
Generation 1190:	Best = 5.743769e+02	FES = 119100	Diversity = 9.6212e+00
Generation 1200:	Best = 5.743769e+02	FES = 120100	Diversity = 9.6196e+00
Generation 1210:	Best = 5.743769e+02	FES = 121100	Diversity = 9.6150e+00
Generation 1220:	Best = 5.743769e+02	FES = 122100	Diversity = 9.6035e+00
Generation 1230:	Best = 5.743768e+02	FES = 123100	Diversity = 9.5849e+00
Generation 1240:	Best = 5.743768e+02	FES = 124100	Diversity = 9.5813e+00
Generation 1250:	Best = 5.743768e+02	FES = 125100	Diversity = 9.5623e+00
Generation 1260:	Best = 5.743768e+02	FES = 126100	Diversity = 9.5908e+00
Generation 1270:	Best = 5.743768e+02	FES = 127100	Diversity = 9.5703e+00
Generation 1280:	Best = 5.743768e+02	FES = 128100	Diversity = 9.6330e+00
Generation 1290:	Best = 5.743768e+02	FES = 129100	Diversity = 9.6274e+00
Generation 1300:	Best = 5.743768e+02	FES = 130100	Diversity = 9.6272e+00
Generation 1310:	Best = 5.743768e+02	FES = 131100	Diversity = 9.6133e+00
Generation 1320:	Best = 5.743768e+02	FES = 132100	Diversity = 9.5605e+00
Generation 1330:	Best = 5.743768e+02	FES = 133100	Diversity = 9.4747e+00
Generation 1340:	Best = 5.743768e+02	FES = 134100	Diversity = 9.3937e+00
Generation 1350:	Best = 5.743768e+02	FES = 135100	Diversity = 9.4415e+00
Generation 1360:	Best = 5.743768e+02	FES = 136100	Diversity = 9.4027e+00
Generation 1370:	Best = 5.743768e+02	FES = 137100	Diversity = 9.4038e+00
Generation 1380:	Best = 5.743768e+02	FES = 138100	Diversity = 9.3342e+00
Generation 1390:	Best = 5.730334e+02	FES = 139100	Diversity = 9.2947e+00
Generation 1400:	Best = 5.730334e+02	FES = 140100	Diversity = 9.2968e+00
Generation 1410:	Best = 5.730334e+02	FES = 141100	Diversity = 9.3072e+00
Generation 1420:	Best = 5.678484e+02	FES = 142100	Diversity = 9.3036e+00
Generation 1430:	Best = 5.678484e+02	FES = 143100	Diversity = 9.3011e+00
Generation 1440:	Best = 5.677616e+02	FES = 144100	Diversity = 9.2684e+00
Generation 1450:	Best = 5.677616e+02	FES = 145100	Diversity = 9.2652e+00
Generation 1460:	Best = 5.677616e+02	FES = 146100	Diversity = 9.2257e+00
Generation 1470:	Best = 5.672747e+02	FES = 147100	Diversity = 9.1783e+00
Generation 1480:	Best = 5.661832e+02	FES = 148100	Diversity = 9.1746e+00
Generation 1490:	Best = 5.661652e+02	FES = 149100	Diversity = 9.1759e+00
Generation 1500:	Best = 5.648378e+02	FES = 150100	Diversity = 9.1731e+00
Generation 1510:	Best = 5.614877e+02	FES = 151100	Diversity = 9.0208e+00
Generation 1520:	Best = 5.607400e+02	FES = 152100	Diversity = 9.0454e+00

Generation 1530:	Best = 5.607380e+02	FES = 153100	Diversity = 9.0389e+00
Generation 1540:	Best = 5.597064e+02	FES = 154100	Diversity = 9.0761e+00
Generation 1550:	Best = 5.596998e+02	FES = 155100	Diversity = 9.0262e+00
Generation 1560:	Best = 5.594524e+02	FES = 156100	Diversity = 9.0181e+00
Generation 1570:	Best = 5.592573e+02	FES = 157100	Diversity = 8.9594e+00
Generation 1580:	Best = 5.592573e+02	FES = 158100	Diversity = 8.9567e+00
Generation 1590:	Best = 5.591702e+02	FES = 159100	Diversity = 8.8642e+00
Generation 1600:	Best = 5.591643e+02	FES = 160100	Diversity = 8.8381e+00
Generation 1610:	Best = 5.591527e+02	FES = 161100	Diversity = 8.8409e+00
Generation 1620:	Best = 5.591465e+02	FES = 162100	Diversity = 8.8403e+00
Generation 1630:	Best = 5.591463e+02	FES = 163100	Diversity = 8.7821e+00
Generation 1640:	Best = 5.591463e+02	FES = 164100	Diversity = 8.7479e+00
Generation 1650:	Best = 5.591427e+02	FES = 165100	Diversity = 8.6935e+00
Generation 1660:	Best = 5.586650e+02	FES = 166100	Diversity = 8.6409e+00
Generation 1670:	Best = 5.586649e+02	FES = 167100	Diversity = 8.6390e+00
Generation 1680:	Best = 5.586649e+02	FES = 168100	Diversity = 8.6089e+00
Generation 1690:	Best = 5.586315e+02	FES = 169100	Diversity = 8.5148e+00
Generation 1700:	Best = 5.584828e+02	FES = 170100	Diversity = 8.4388e+00
Generation 1710:	Best = 5.583969e+02	FES = 171100	Diversity = 8.3883e+00
Generation 1720:	Best = 5.583560e+02	FES = 172100	Diversity = 8.3466e+00
Generation 1730:	Best = 5.583209e+02	FES = 173100	Diversity = 8.3396e+00
Generation 1740:	Best = 5.583205e+02	FES = 174100	Diversity = 8.2913e+00
Generation 1750:	Best = 5.580654e+02	FES = 175100	Diversity = 8.2911e+00
Generation 1760:	Best = 5.580654e+02	FES = 176100	Diversity = 8.2839e+00
Generation 1770:	Best = 5.580654e+02	FES = 177100	Diversity = 8.2626e+00
Generation 1780:	Best = 5.580621e+02	FES = 178100	Diversity = 8.3303e+00
Generation 1790:	Best = 5.580610e+02	FES = 179100	Diversity = 8.2536e+00
Generation 1800:	Best = 5.580601e+02	FES = 180100	Diversity = 8.2716e+00
Generation 1810:	Best = 5.579633e+02	FES = 181100	Diversity = 8.1223e+00
Generation 1820:	Best = 5.565766e+02	FES = 182100	Diversity = 8.1533e+00
Generation 1830:	Best = 5.563515e+02	FES = 183100	Diversity = 8.0741e+00
Generation 1840:	Best = 5.559892e+02	FES = 184100	Diversity = 7.9362e+00
Generation 1850:	Best = 5.556048e+02	FES = 185100	Diversity = 7.8739e+00
Generation 1860:	Best = 5.556016e+02	FES = 186100	Diversity = 7.8780e+00
Generation 1870:	Best = 5.554913e+02	FES = 187100	Diversity = 7.9333e+00
Generation 1880:	Best = 5.554059e+02	FES = 188100	Diversity = 7.7005e+00
Generation 1890:	Best = 5.554013e+02	FES = 189100	Diversity = 7.5324e+00
Generation 1900:	Best = 5.553993e+02	FES = 190100	Diversity = 7.4736e+00
Generation 1910:	Best = 5.553930e+02	FES = 191100	Diversity = 7.3148e+00
Generation 1920:	Best = 5.553812e+02	FES = 192100	Diversity = 7.1058e+00
Generation 1930:	Best = 5.550548e+02	FES = 193100	Diversity = 7.0778e+00
Generation 1940:	Best = 5.537328e+02	FES = 194100	Diversity = 6.8337e+00
Generation 1950:	Best = 5.535478e+02	FES = 195100	Diversity = 6.6451e+00
Generation 1960:	Best = 5.531470e+02	FES = 196100	Diversity = 6.2524e+00
Generation 1970:	Best = 5.514756e+02	FES = 197100	Diversity = 6.2136e+00
Generation 1980:	Best = 5.510033e+02	FES = 198100	Diversity = 5.8189e+00
Generation 1990:	Best = 5.507697e+02	FES = 199100	Diversity = 5.2499e+00
Generation 2000:	Best = 5.504915e+02	FES = 200100	Diversity = 4.9261e+00
Generation 2010:	Best = 5.503765e+02	FES = 201100	Diversity = 4.2795e+00
Generation 2020:	Best = 5.501920e+02	FES = 202100	Diversity = 3.6244e+00
Generation 2030:	Best = 5.501457e+02	FES = 203100	Diversity = 3.1830e+00
Generation 2040:	Best = 5.501196e+02	FES = 204100	Diversity = 2.6880e+00
Generation 2050:	Best = 5.499751e+02	FES = 205100	Diversity = 2.2782e+00
Generation 2060:	Best = 5.498845e+02	FES = 206100	Diversity = 2.1420e+00

Generation 2070:	Best = 5.472781e+02	FES = 207100	Diversity = 2.0292e+00
Generation 2080:	Best = 5.471569e+02	FES = 208100	Diversity = 1.9494e+00
Generation 2090:	Best = 5.470682e+02	FES = 209100	Diversity = 1.8489e+00
Generation 2100:	Best = 5.470235e+02	FES = 210100	Diversity = 1.7909e+00
Generation 2110:	Best = 5.470138e+02	FES = 211100	Diversity = 1.6453e+00
Generation 2120:	Best = 5.470132e+02	FES = 212100	Diversity = 1.4787e+00
Generation 2130:	Best = 5.468100e+02	FES = 213100	Diversity = 1.2364e+00
Generation 2140:	Best = 5.468067e+02	FES = 214100	Diversity = 7.6425e-01
Generation 2150:	Best = 5.468047e+02	FES = 215100	Diversity = 7.0234e-01
Generation 2160:	Best = 5.468024e+02	FES = 216100	Diversity = 6.8201e-01
Generation 2170:	Best = 5.467788e+02	FES = 217100	Diversity = 7.0246e-01
Generation 2180:	Best = 5.467761e+02	FES = 218100	Diversity = 6.0944e-01
Generation 2190:	Best = 5.467755e+02	FES = 219100	Diversity = 5.8482e-01
Generation 2200:	Best = 5.467646e+02	FES = 220100	Diversity = 5.6039e-01
Generation 2210:	Best = 5.467623e+02	FES = 221100	Diversity = 5.0701e-01
Generation 2220:	Best = 5.467623e+02	FES = 222100	Diversity = 4.6443e-01
Generation 2230:	Best = 5.467622e+02	FES = 223100	Diversity = 4.6445e-01
Generation 2240:	Best = 5.467387e+02	FES = 224100	Diversity = 4.6425e-01
Generation 2250:	Best = 5.467383e+02	FES = 225100	Diversity = 4.4331e-01
Generation 2260:	Best = 5.467382e+02	FES = 226100	Diversity = 4.4334e-01
Generation 2270:	Best = 5.467371e+02	FES = 227100	Diversity = 4.0159e-01
Generation 2280:	Best = 5.467363e+02	FES = 228100	Diversity = 4.0159e-01
Generation 2290:	Best = 5.467358e+02	FES = 229100	Diversity = 4.0122e-01
Generation 2300:	Best = 5.467358e+02	FES = 230100	Diversity = 4.0123e-01
Generation 2310:	Best = 5.467358e+02	FES = 231100	Diversity = 4.0109e-01
Generation 2320:	Best = 5.467358e+02	FES = 232100	Diversity = 4.0109e-01
Generation 2330:	Best = 5.467358e+02	FES = 233100	Diversity = 4.4362e-01
Generation 2340:	Best = 5.467358e+02	FES = 234100	Diversity = 3.7833e-01
Generation 2350:	Best = 5.467358e+02	FES = 235100	Diversity = 3.7566e-01
Generation 2360:	Best = 5.467348e+02	FES = 236100	Diversity = 3.7481e-01
Generation 2370:	Best = 5.467346e+02	FES = 237100	Diversity = 3.7477e-01
Generation 2380:	Best = 5.467346e+02	FES = 238100	Diversity = 3.7476e-01
Generation 2390:	Best = 5.467345e+02	FES = 239100	Diversity = 3.7466e-01
Generation 2400:	Best = 5.467345e+02	FES = 240100	Diversity = 3.7473e-01
Generation 2410:	Best = 5.467345e+02	FES = 241100	Diversity = 3.7469e-01
Generation 2420:	Best = 5.467345e+02	FES = 242100	Diversity = 3.7468e-01
Generation 2430:	Best = 5.467345e+02	FES = 243100	Diversity = 3.7467e-01
Generation 2440:	Best = 5.467345e+02	FES = 244100	Diversity = 3.7467e-01
Generation 2450:	Best = 5.467345e+02	FES = 245100	Diversity = 5.4662e-01
Generation 2460:	Best = 5.467345e+02	FES = 246100	Diversity = 5.4666e-01
Generation 2470:	Best = 5.467345e+02	FES = 247100	Diversity = 5.4658e-01
Generation 2480:	Best = 5.467345e+02	FES = 248100	Diversity = 5.4654e-01
Generation 2490:	Best = 5.467345e+02	FES = 249100	Diversity = 5.4636e-01
Generation 2500:	Best = 5.467338e+02	FES = 250100	Diversity = 5.4641e-01
Generation 2510:	Best = 5.467338e+02	FES = 251100	Diversity = 5.4644e-01
Generation 2520:	Best = 5.467338e+02	FES = 252100	Diversity = 5.4645e-01
Generation 2530:	Best = 5.467338e+02	FES = 253100	Diversity = 5.1682e-01
Generation 2540:	Best = 5.467338e+02	FES = 254100	Diversity = 5.1680e-01
Generation 2550:	Best = 5.446365e+02	FES = 255100	Diversity = 5.8642e-01
Generation 2560:	Best = 5.446365e+02	FES = 256100	Diversity = 5.8814e-01
Generation 2570:	Best = 5.443465e+02	FES = 257100	Diversity = 5.8211e-01
Generation 2580:	Best = 5.439779e+02	FES = 258100	Diversity = 5.8361e-01
Generation 2590:	Best = 5.439710e+02	FES = 259100	Diversity = 6.1748e-01
Generation 2600:	Best = 5.439705e+02	FES = 260100	Diversity = 6.1910e-01

Generation 2610:	Best = 5.439270e+02	FES = 261100	Diversity = 6.5116e-01
Generation 2620:	Best = 5.439162e+02	FES = 262100	Diversity = 6.4949e-01
Generation 2630:	Best = 5.438981e+02	FES = 263100	Diversity = 6.4903e-01
Generation 2640:	Best = 5.438973e+02	FES = 264100	Diversity = 6.4863e-01
Generation 2650:	Best = 5.438969e+02	FES = 265100	Diversity = 7.7321e-01
Generation 2660:	Best = 5.438969e+02	FES = 266100	Diversity = 7.7251e-01
Generation 2670:	Best = 5.438969e+02	FES = 267100	Diversity = 7.9442e-01
Generation 2680:	Best = 5.438969e+02	FES = 268100	Diversity = 7.9465e-01
Generation 2690:	Best = 5.437609e+02	FES = 269100	Diversity = 7.9519e-01
Generation 2700:	Best = 5.437600e+02	FES = 270100	Diversity = 7.0642e-01
Generation 2710:	Best = 5.436770e+02	FES = 271100	Diversity = 8.3649e-01
Generation 2720:	Best = 5.436645e+02	FES = 272100	Diversity = 8.6431e-01
Generation 2730:	Best = 5.436335e+02	FES = 273100	Diversity = 8.8156e-01
Generation 2740:	Best = 5.434778e+02	FES = 274100	Diversity = 8.2294e-01
Generation 2750:	Best = 5.434132e+02	FES = 275100	Diversity = 8.2520e-01
Generation 2760:	Best = 5.434076e+02	FES = 276100	Diversity = 8.4638e-01
Generation 2770:	Best = 5.434060e+02	FES = 277100	Diversity = 8.4560e-01
Generation 2780:	Best = 5.433805e+02	FES = 278100	Diversity = 8.7667e-01
Generation 2790:	Best = 5.433799e+02	FES = 279100	Diversity = 9.0574e-01
Generation 2800:	Best = 5.433795e+02	FES = 280100	Diversity = 1.0208e+00
Generation 2810:	Best = 5.433478e+02	FES = 281100	Diversity = 1.0287e+00
Generation 2820:	Best = 5.433399e+02	FES = 282100	Diversity = 1.0536e+00
Generation 2830:	Best = 5.420835e+02	FES = 283100	Diversity = 1.3009e+00
Generation 2840:	Best = 5.415387e+02	FES = 284100	Diversity = 1.5051e+00
Generation 2850:	Best = 5.415159e+02	FES = 285100	Diversity = 1.6090e+00
Generation 2860:	Best = 5.414182e+02	FES = 286100	Diversity = 1.5901e+00
Generation 2870:	Best = 5.413843e+02	FES = 287100	Diversity = 1.3214e+00
Generation 2880:	Best = 5.413423e+02	FES = 288100	Diversity = 1.0158e+00
Generation 2890:	Best = 5.413396e+02	FES = 289100	Diversity = 9.7801e-01
Generation 2900:	Best = 5.413223e+02	FES = 290100	Diversity = 9.8264e-01
Generation 2910:	Best = 5.413191e+02	FES = 291100	Diversity = 9.8297e-01
Generation 2920:	Best = 5.413177e+02	FES = 292100	Diversity = 9.8853e-01
Generation 2930:	Best = 5.413167e+02	FES = 293100	Diversity = 9.9346e-01
Generation 2940:	Best = 5.413145e+02	FES = 294100	Diversity = 9.9019e-01
Generation 2950:	Best = 5.413137e+02	FES = 295100	Diversity = 9.4351e-01
Generation 2960:	Best = 5.412866e+02	FES = 296100	Diversity = 9.5422e-01
Generation 2970:	Best = 5.412687e+02	FES = 297100	Diversity = 9.6820e-01
Generation 2980:	Best = 5.412635e+02	FES = 298100	Diversity = 9.7243e-01
Generation 2990:	Best = 5.412626e+02	FES = 299100	Diversity = 9.7412e-01

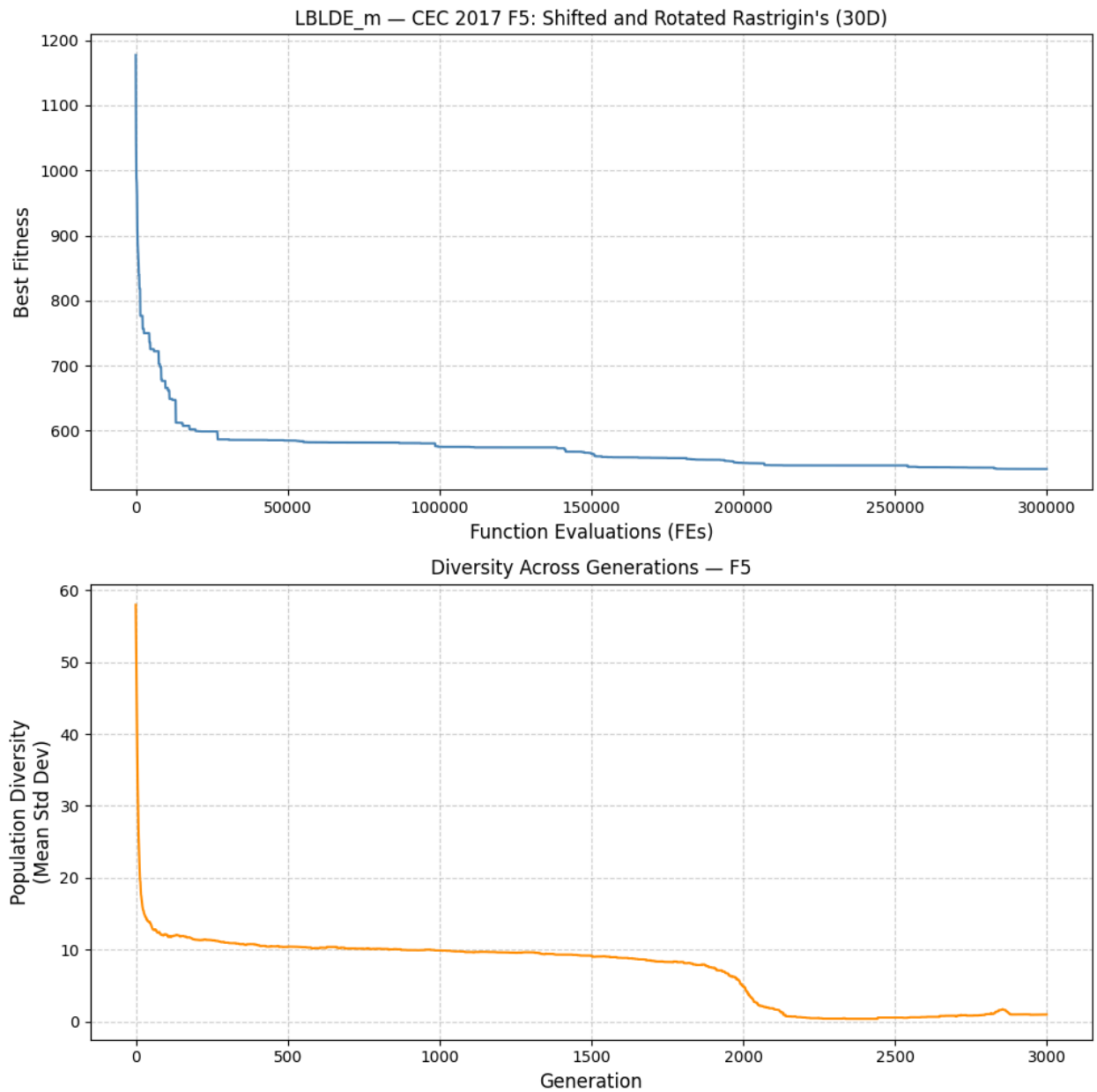
Optimization completed!

Final Best Fitness = 5.412613e+02

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness      : 5.412613e+02
  Global Optimum    : 5.000000e+02
  Error (|best-optimum|) : 4.126126e+01
  Best Solution (5 dim) : [-44.02840126  79.89618215 -24.43560054 -49.91043455
8.31286824]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F5.png

```
In [32]: # =====  
# F6 – Shifted and Rotated Schaffer's F7  
# =====  
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and Rotated Schaff
```

=====

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LBLDE_m on CEC 2017 F6 – Single Run

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Function : Shifted and Rotated Schaffer's F7 (30D)

Bounds : [-100, 100]

Global optimum : 600.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 7.816536e+02 | Diversity = 5.7976e+01

Generation	10:	Best = 6.812170e+02	FES = 1100	Diversity = 2.8321e+01
Generation	20:	Best = 6.598563e+02	FES = 2100	Diversity = 1.6095e+01
Generation	30:	Best = 6.505999e+02	FES = 3100	Diversity = 1.1366e+01
Generation	40:	Best = 6.427825e+02	FES = 4100	Diversity = 9.5780e+00
Generation	50:	Best = 6.407149e+02	FES = 5100	Diversity = 8.4014e+00
Generation	60:	Best = 6.362907e+02	FES = 6100	Diversity = 7.1166e+00
Generation	70:	Best = 6.275621e+02	FES = 7100	Diversity = 5.9468e+00
Generation	80:	Best = 6.252430e+02	FES = 8100	Diversity = 4.8325e+00
Generation	90:	Best = 6.220518e+02	FES = 9100	Diversity = 4.2221e+00
Generation	100:	Best = 6.183392e+02	FES = 10100	Diversity = 3.9426e+00
Generation	110:	Best = 6.166295e+02	FES = 11100	Diversity = 3.0303e+00
Generation	120:	Best = 6.154427e+02	FES = 12100	Diversity = 2.0599e+00
Generation	130:	Best = 6.137650e+02	FES = 13100	Diversity = 1.6827e+00
Generation	140:	Best = 6.122106e+02	FES = 14100	Diversity = 1.3220e+00
Generation	150:	Best = 6.122106e+02	FES = 15100	Diversity = 1.1064e+00
Generation	160:	Best = 6.115462e+02	FES = 16100	Diversity = 1.0305e+00
Generation	170:	Best = 6.114270e+02	FES = 17100	Diversity = 9.4913e-01
Generation	180:	Best = 6.111815e+02	FES = 18100	Diversity = 9.0290e-01
Generation	190:	Best = 6.111231e+02	FES = 19100	Diversity = 8.4196e-01
Generation	200:	Best = 6.110536e+02	FES = 20100	Diversity = 8.3723e-01
Generation	210:	Best = 6.105193e+02	FES = 21100	Diversity = 8.6670e-01
Generation	220:	Best = 6.103094e+02	FES = 22100	Diversity = 8.7585e-01
Generation	230:	Best = 6.101126e+02	FES = 23100	Diversity = 8.5289e-01
Generation	240:	Best = 6.094672e+02	FES = 24100	Diversity = 8.3755e-01
Generation	250:	Best = 6.088096e+02	FES = 25100	Diversity = 8.3262e-01
Generation	260:	Best = 6.088096e+02	FES = 26100	Diversity = 8.1269e-01
Generation	270:	Best = 6.087606e+02	FES = 27100	Diversity = 7.9223e-01
Generation	280:	Best = 6.086523e+02	FES = 28100	Diversity = 7.7503e-01
Generation	290:	Best = 6.085434e+02	FES = 29100	Diversity = 7.5872e-01
Generation	300:	Best = 6.083024e+02	FES = 30100	Diversity = 7.5651e-01
Generation	310:	Best = 6.083024e+02	FES = 31100	Diversity = 7.4996e-01
Generation	320:	Best = 6.082611e+02	FES = 32100	Diversity = 7.6227e-01
Generation	330:	Best = 6.077581e+02	FES = 33100	Diversity = 8.0564e-01
Generation	340:	Best = 6.077581e+02	FES = 34100	Diversity = 8.2312e-01
Generation	350:	Best = 6.069416e+02	FES = 35100	Diversity = 9.1595e-01
Generation	360:	Best = 6.069414e+02	FES = 36100	Diversity = 8.6192e-01
Generation	370:	Best = 6.062995e+02	FES = 37100	Diversity = 7.6455e-01
Generation	380:	Best = 6.061333e+02	FES = 38100	Diversity = 6.7355e-01
Generation	390:	Best = 6.061333e+02	FES = 39100	Diversity = 5.3356e-01
Generation	400:	Best = 6.061333e+02	FES = 40100	Diversity = 4.9043e-01
Generation	410:	Best = 6.061333e+02	FES = 41100	Diversity = 4.6099e-01
Generation	420:	Best = 6.061262e+02	FES = 42100	Diversity = 4.4551e-01
Generation	430:	Best = 6.057568e+02	FES = 43100	Diversity = 4.2723e-01
Generation	440:	Best = 6.057568e+02	FES = 44100	Diversity = 4.3295e-01

Generation	450:	Best = 6.057440e+02	FES = 45100	Diversity = 4.2340e-01
Generation	460:	Best = 6.057440e+02	FES = 46100	Diversity = 4.2185e-01
Generation	470:	Best = 6.056551e+02	FES = 47100	Diversity = 4.1925e-01
Generation	480:	Best = 6.055431e+02	FES = 48100	Diversity = 3.9562e-01
Generation	490:	Best = 6.053882e+02	FES = 49100	Diversity = 3.8693e-01
Generation	500:	Best = 6.052725e+02	FES = 50100	Diversity = 3.7995e-01
Generation	510:	Best = 6.050623e+02	FES = 51100	Diversity = 3.7134e-01
Generation	520:	Best = 6.050623e+02	FES = 52100	Diversity = 3.6240e-01
Generation	530:	Best = 6.049364e+02	FES = 53100	Diversity = 3.4550e-01
Generation	540:	Best = 6.049364e+02	FES = 54100	Diversity = 3.4333e-01
Generation	550:	Best = 6.049364e+02	FES = 55100	Diversity = 3.3435e-01
Generation	560:	Best = 6.049364e+02	FES = 56100	Diversity = 3.2743e-01
Generation	570:	Best = 6.048648e+02	FES = 57100	Diversity = 3.3177e-01
Generation	580:	Best = 6.048288e+02	FES = 58100	Diversity = 3.4415e-01
Generation	590:	Best = 6.047979e+02	FES = 59100	Diversity = 3.2525e-01
Generation	600:	Best = 6.047493e+02	FES = 60100	Diversity = 3.3189e-01
Generation	610:	Best = 6.047074e+02	FES = 61100	Diversity = 3.3240e-01
Generation	620:	Best = 6.047074e+02	FES = 62100	Diversity = 3.2775e-01
Generation	630:	Best = 6.046911e+02	FES = 63100	Diversity = 3.2723e-01
Generation	640:	Best = 6.046911e+02	FES = 64100	Diversity = 3.2003e-01
Generation	650:	Best = 6.046837e+02	FES = 65100	Diversity = 3.2146e-01
Generation	660:	Best = 6.046837e+02	FES = 66100	Diversity = 3.1777e-01
Generation	670:	Best = 6.046805e+02	FES = 67100	Diversity = 3.0574e-01
Generation	680:	Best = 6.046390e+02	FES = 68100	Diversity = 3.0557e-01
Generation	690:	Best = 6.046390e+02	FES = 69100	Diversity = 2.9822e-01
Generation	700:	Best = 6.046350e+02	FES = 70100	Diversity = 2.9292e-01
Generation	710:	Best = 6.046350e+02	FES = 71100	Diversity = 2.9621e-01
Generation	720:	Best = 6.046320e+02	FES = 72100	Diversity = 2.8536e-01
Generation	730:	Best = 6.046320e+02	FES = 73100	Diversity = 2.8830e-01
Generation	740:	Best = 6.046001e+02	FES = 74100	Diversity = 2.7926e-01
Generation	750:	Best = 6.046001e+02	FES = 75100	Diversity = 2.6525e-01
Generation	760:	Best = 6.046001e+02	FES = 76100	Diversity = 2.5306e-01
Generation	770:	Best = 6.045981e+02	FES = 77100	Diversity = 2.5248e-01
Generation	780:	Best = 6.045606e+02	FES = 78100	Diversity = 2.6317e-01
Generation	790:	Best = 6.045317e+02	FES = 79100	Diversity = 2.6444e-01
Generation	800:	Best = 6.045303e+02	FES = 80100	Diversity = 2.5826e-01
Generation	810:	Best = 6.045303e+02	FES = 81100	Diversity = 2.5591e-01
Generation	820:	Best = 6.044905e+02	FES = 82100	Diversity = 2.4867e-01
Generation	830:	Best = 6.044905e+02	FES = 83100	Diversity = 2.4797e-01
Generation	840:	Best = 6.044482e+02	FES = 84100	Diversity = 2.4103e-01
Generation	850:	Best = 6.044402e+02	FES = 85100	Diversity = 2.4127e-01
Generation	860:	Best = 6.044402e+02	FES = 86100	Diversity = 2.4315e-01
Generation	870:	Best = 6.044167e+02	FES = 87100	Diversity = 2.4241e-01
Generation	880:	Best = 6.044154e+02	FES = 88100	Diversity = 2.4128e-01
Generation	890:	Best = 6.043945e+02	FES = 89100	Diversity = 2.3605e-01
Generation	900:	Best = 6.043546e+02	FES = 90100	Diversity = 2.3365e-01
Generation	910:	Best = 6.043546e+02	FES = 91100	Diversity = 2.3852e-01
Generation	920:	Best = 6.043532e+02	FES = 92100	Diversity = 2.3680e-01
Generation	930:	Best = 6.043351e+02	FES = 93100	Diversity = 2.3675e-01
Generation	940:	Best = 6.043304e+02	FES = 94100	Diversity = 2.3751e-01
Generation	950:	Best = 6.043304e+02	FES = 95100	Diversity = 2.3291e-01
Generation	960:	Best = 6.043286e+02	FES = 96100	Diversity = 2.3116e-01
Generation	970:	Best = 6.043140e+02	FES = 97100	Diversity = 2.3125e-01
Generation	980:	Best = 6.043099e+02	FES = 98100	Diversity = 2.2858e-01

Generation 990:	Best = 6.043094e+02	FES = 99100	Diversity = 2.2708e-01
Generation 1000:	Best = 6.043012e+02	FES = 100100	Diversity = 2.1886e-01
Generation 1010:	Best = 6.042876e+02	FES = 101100	Diversity = 2.1460e-01
Generation 1020:	Best = 6.042856e+02	FES = 102100	Diversity = 2.1697e-01
Generation 1030:	Best = 6.042716e+02	FES = 103100	Diversity = 2.2338e-01
Generation 1040:	Best = 6.042437e+02	FES = 104100	Diversity = 2.1687e-01
Generation 1050:	Best = 6.042362e+02	FES = 105100	Diversity = 2.1543e-01
Generation 1060:	Best = 6.042226e+02	FES = 106100	Diversity = 2.1296e-01
Generation 1070:	Best = 6.042226e+02	FES = 107100	Diversity = 2.1361e-01
Generation 1080:	Best = 6.042176e+02	FES = 108100	Diversity = 2.1041e-01
Generation 1090:	Best = 6.042176e+02	FES = 109100	Diversity = 2.1135e-01
Generation 1100:	Best = 6.042142e+02	FES = 110100	Diversity = 2.0222e-01
Generation 1110:	Best = 6.042139e+02	FES = 111100	Diversity = 2.0276e-01
Generation 1120:	Best = 6.042139e+02	FES = 112100	Diversity = 2.0419e-01
Generation 1130:	Best = 6.042139e+02	FES = 113100	Diversity = 2.0467e-01
Generation 1140:	Best = 6.042139e+02	FES = 114100	Diversity = 2.0204e-01
Generation 1150:	Best = 6.042049e+02	FES = 115100	Diversity = 2.0064e-01
Generation 1160:	Best = 6.042049e+02	FES = 116100	Diversity = 1.9773e-01
Generation 1170:	Best = 6.041888e+02	FES = 117100	Diversity = 1.9764e-01
Generation 1180:	Best = 6.041888e+02	FES = 118100	Diversity = 2.0256e-01
Generation 1190:	Best = 6.041888e+02	FES = 119100	Diversity = 2.0446e-01
Generation 1200:	Best = 6.041888e+02	FES = 120100	Diversity = 2.0032e-01
Generation 1210:	Best = 6.041865e+02	FES = 121100	Diversity = 1.9510e-01
Generation 1220:	Best = 6.041822e+02	FES = 122100	Diversity = 1.9551e-01
Generation 1230:	Best = 6.041822e+02	FES = 123100	Diversity = 1.9243e-01
Generation 1240:	Best = 6.041822e+02	FES = 124100	Diversity = 1.8202e-01
Generation 1250:	Best = 6.041819e+02	FES = 125100	Diversity = 1.8455e-01
Generation 1260:	Best = 6.041819e+02	FES = 126100	Diversity = 1.8458e-01
Generation 1270:	Best = 6.041819e+02	FES = 127100	Diversity = 1.8476e-01
Generation 1280:	Best = 6.041819e+02	FES = 128100	Diversity = 1.8065e-01
Generation 1290:	Best = 6.041819e+02	FES = 129100	Diversity = 1.8056e-01
Generation 1300:	Best = 6.041819e+02	FES = 130100	Diversity = 1.7981e-01
Generation 1310:	Best = 6.041814e+02	FES = 131100	Diversity = 1.7913e-01
Generation 1320:	Best = 6.041814e+02	FES = 132100	Diversity = 1.8103e-01
Generation 1330:	Best = 6.041814e+02	FES = 133100	Diversity = 1.7489e-01
Generation 1340:	Best = 6.041814e+02	FES = 134100	Diversity = 1.7519e-01
Generation 1350:	Best = 6.041807e+02	FES = 135100	Diversity = 1.7252e-01
Generation 1360:	Best = 6.041762e+02	FES = 136100	Diversity = 1.7455e-01
Generation 1370:	Best = 6.041604e+02	FES = 137100	Diversity = 1.7449e-01
Generation 1380:	Best = 6.041564e+02	FES = 138100	Diversity = 1.6873e-01
Generation 1390:	Best = 6.041564e+02	FES = 139100	Diversity = 1.6843e-01
Generation 1400:	Best = 6.041469e+02	FES = 140100	Diversity = 1.6757e-01
Generation 1410:	Best = 6.041469e+02	FES = 141100	Diversity = 1.6696e-01
Generation 1420:	Best = 6.041469e+02	FES = 142100	Diversity = 1.6651e-01
Generation 1430:	Best = 6.041469e+02	FES = 143100	Diversity = 1.6542e-01
Generation 1440:	Best = 6.041469e+02	FES = 144100	Diversity = 1.6517e-01
Generation 1450:	Best = 6.041287e+02	FES = 145100	Diversity = 1.6524e-01
Generation 1460:	Best = 6.041252e+02	FES = 146100	Diversity = 1.6498e-01
Generation 1470:	Best = 6.041252e+02	FES = 147100	Diversity = 1.6324e-01
Generation 1480:	Best = 6.041252e+02	FES = 148100	Diversity = 1.6473e-01
Generation 1490:	Best = 6.041251e+02	FES = 149100	Diversity = 1.6681e-01
Generation 1500:	Best = 6.041251e+02	FES = 150100	Diversity = 1.6676e-01
Generation 1510:	Best = 6.041177e+02	FES = 151100	Diversity = 1.6624e-01
Generation 1520:	Best = 6.041129e+02	FES = 152100	Diversity = 1.6684e-01

Generation 1530:	Best = 6.041129e+02	FES = 153100	Diversity = 1.6539e-01
Generation 1540:	Best = 6.041129e+02	FES = 154100	Diversity = 1.6412e-01
Generation 1550:	Best = 6.041101e+02	FES = 155100	Diversity = 1.6459e-01
Generation 1560:	Best = 6.041080e+02	FES = 156100	Diversity = 1.6323e-01
Generation 1570:	Best = 6.041080e+02	FES = 157100	Diversity = 1.6351e-01
Generation 1580:	Best = 6.040968e+02	FES = 158100	Diversity = 1.6372e-01
Generation 1590:	Best = 6.040950e+02	FES = 159100	Diversity = 1.6448e-01
Generation 1600:	Best = 6.040932e+02	FES = 160100	Diversity = 1.6410e-01
Generation 1610:	Best = 6.040932e+02	FES = 161100	Diversity = 1.6238e-01
Generation 1620:	Best = 6.040658e+02	FES = 162100	Diversity = 1.6212e-01
Generation 1630:	Best = 6.040658e+02	FES = 163100	Diversity = 1.6148e-01
Generation 1640:	Best = 6.040658e+02	FES = 164100	Diversity = 1.6137e-01
Generation 1650:	Best = 6.040658e+02	FES = 165100	Diversity = 1.5154e-01
Generation 1660:	Best = 6.040658e+02	FES = 166100	Diversity = 1.5146e-01
Generation 1670:	Best = 6.040658e+02	FES = 167100	Diversity = 1.4181e-01
Generation 1680:	Best = 6.040658e+02	FES = 168100	Diversity = 1.4076e-01
Generation 1690:	Best = 6.040658e+02	FES = 169100	Diversity = 1.4123e-01
Generation 1700:	Best = 6.040624e+02	FES = 170100	Diversity = 1.4070e-01
Generation 1710:	Best = 6.040609e+02	FES = 171100	Diversity = 1.4013e-01
Generation 1720:	Best = 6.040609e+02	FES = 172100	Diversity = 1.3598e-01
Generation 1730:	Best = 6.040606e+02	FES = 173100	Diversity = 1.3660e-01
Generation 1740:	Best = 6.040481e+02	FES = 174100	Diversity = 1.3032e-01
Generation 1750:	Best = 6.040375e+02	FES = 175100	Diversity = 1.2894e-01
Generation 1760:	Best = 6.040375e+02	FES = 176100	Diversity = 1.2670e-01
Generation 1770:	Best = 6.040370e+02	FES = 177100	Diversity = 1.2503e-01
Generation 1780:	Best = 6.040370e+02	FES = 178100	Diversity = 1.2476e-01
Generation 1790:	Best = 6.040107e+02	FES = 179100	Diversity = 1.2265e-01
Generation 1800:	Best = 6.040107e+02	FES = 180100	Diversity = 1.2382e-01
Generation 1810:	Best = 6.039945e+02	FES = 181100	Diversity = 1.2249e-01
Generation 1820:	Best = 6.039840e+02	FES = 182100	Diversity = 1.2313e-01
Generation 1830:	Best = 6.039722e+02	FES = 183100	Diversity = 1.1595e-01
Generation 1840:	Best = 6.039722e+02	FES = 184100	Diversity = 1.0306e-01
Generation 1850:	Best = 6.039688e+02	FES = 185100	Diversity = 1.0407e-01
Generation 1860:	Best = 6.039560e+02	FES = 186100	Diversity = 9.3162e-02
Generation 1870:	Best = 6.039560e+02	FES = 187100	Diversity = 9.0529e-02
Generation 1880:	Best = 6.039526e+02	FES = 188100	Diversity = 8.7304e-02
Generation 1890:	Best = 6.039523e+02	FES = 189100	Diversity = 8.2195e-02
Generation 1900:	Best = 6.039436e+02	FES = 190100	Diversity = 8.2861e-02
Generation 1910:	Best = 6.039436e+02	FES = 191100	Diversity = 8.2607e-02
Generation 1920:	Best = 6.039436e+02	FES = 192100	Diversity = 8.3596e-02
Generation 1930:	Best = 6.039436e+02	FES = 193100	Diversity = 8.1859e-02
Generation 1940:	Best = 6.039435e+02	FES = 194100	Diversity = 8.1702e-02
Generation 1950:	Best = 6.039433e+02	FES = 195100	Diversity = 7.9777e-02
Generation 1960:	Best = 6.039433e+02	FES = 196100	Diversity = 7.7264e-02
Generation 1970:	Best = 6.039406e+02	FES = 197100	Diversity = 7.6470e-02
Generation 1980:	Best = 6.039406e+02	FES = 198100	Diversity = 7.2822e-02
Generation 1990:	Best = 6.039406e+02	FES = 199100	Diversity = 6.9530e-02
Generation 2000:	Best = 6.039406e+02	FES = 200100	Diversity = 6.9529e-02
Generation 2010:	Best = 6.039406e+02	FES = 201100	Diversity = 7.0522e-02
Generation 2020:	Best = 6.039384e+02	FES = 202100	Diversity = 6.4214e-02
Generation 2030:	Best = 6.039384e+02	FES = 203100	Diversity = 6.2808e-02
Generation 2040:	Best = 6.039384e+02	FES = 204100	Diversity = 6.5194e-02
Generation 2050:	Best = 6.039384e+02	FES = 205100	Diversity = 6.2412e-02
Generation 2060:	Best = 6.039384e+02	FES = 206100	Diversity = 5.7290e-02

Generation 2070:	Best = 6.039384e+02	FES = 207100	Diversity = 5.5448e-02
Generation 2080:	Best = 6.039384e+02	FES = 208100	Diversity = 5.4607e-02
Generation 2090:	Best = 6.039344e+02	FES = 209100	Diversity = 5.2116e-02
Generation 2100:	Best = 6.039312e+02	FES = 210100	Diversity = 5.0380e-02
Generation 2110:	Best = 6.039312e+02	FES = 211100	Diversity = 4.8576e-02
Generation 2120:	Best = 6.039307e+02	FES = 212100	Diversity = 4.8063e-02
Generation 2130:	Best = 6.039191e+02	FES = 213100	Diversity = 4.6318e-02
Generation 2140:	Best = 6.039191e+02	FES = 214100	Diversity = 4.4602e-02
Generation 2150:	Best = 6.039191e+02	FES = 215100	Diversity = 4.2594e-02
Generation 2160:	Best = 6.039191e+02	FES = 216100	Diversity = 4.1995e-02
Generation 2170:	Best = 6.039170e+02	FES = 217100	Diversity = 4.1003e-02
Generation 2180:	Best = 6.039170e+02	FES = 218100	Diversity = 4.1266e-02
Generation 2190:	Best = 6.039170e+02	FES = 219100	Diversity = 3.9903e-02
Generation 2200:	Best = 6.039170e+02	FES = 220100	Diversity = 3.8201e-02
Generation 2210:	Best = 6.039170e+02	FES = 221100	Diversity = 3.8738e-02
Generation 2220:	Best = 6.039170e+02	FES = 222100	Diversity = 3.7096e-02
Generation 2230:	Best = 6.039170e+02	FES = 223100	Diversity = 3.6451e-02
Generation 2240:	Best = 6.039170e+02	FES = 224100	Diversity = 3.2640e-02
Generation 2250:	Best = 6.039170e+02	FES = 225100	Diversity = 3.3281e-02
Generation 2260:	Best = 6.039170e+02	FES = 226100	Diversity = 3.2896e-02
Generation 2270:	Best = 6.039131e+02	FES = 227100	Diversity = 3.2084e-02
Generation 2280:	Best = 6.038984e+02	FES = 228100	Diversity = 3.0027e-02
Generation 2290:	Best = 6.038984e+02	FES = 229100	Diversity = 2.8974e-02
Generation 2300:	Best = 6.038984e+02	FES = 230100	Diversity = 2.9105e-02
Generation 2310:	Best = 6.038984e+02	FES = 231100	Diversity = 2.6812e-02
Generation 2320:	Best = 6.038970e+02	FES = 232100	Diversity = 2.6111e-02
Generation 2330:	Best = 6.038970e+02	FES = 233100	Diversity = 2.4409e-02
Generation 2340:	Best = 6.038969e+02	FES = 234100	Diversity = 2.2540e-02
Generation 2350:	Best = 6.038963e+02	FES = 235100	Diversity = 2.1575e-02
Generation 2360:	Best = 6.038963e+02	FES = 236100	Diversity = 2.1550e-02
Generation 2370:	Best = 6.038963e+02	FES = 237100	Diversity = 2.1511e-02
Generation 2380:	Best = 6.038963e+02	FES = 238100	Diversity = 2.1373e-02
Generation 2390:	Best = 6.038942e+02	FES = 239100	Diversity = 2.1136e-02
Generation 2400:	Best = 6.038942e+02	FES = 240100	Diversity = 2.0994e-02
Generation 2410:	Best = 6.038938e+02	FES = 241100	Diversity = 2.0789e-02
Generation 2420:	Best = 6.038938e+02	FES = 242100	Diversity = 2.0676e-02
Generation 2430:	Best = 6.038938e+02	FES = 243100	Diversity = 2.0569e-02
Generation 2440:	Best = 6.038938e+02	FES = 244100	Diversity = 2.0568e-02
Generation 2450:	Best = 6.038938e+02	FES = 245100	Diversity = 2.0496e-02
Generation 2460:	Best = 6.038938e+02	FES = 246100	Diversity = 2.0497e-02
Generation 2470:	Best = 6.038938e+02	FES = 247100	Diversity = 2.0496e-02
Generation 2480:	Best = 6.038938e+02	FES = 248100	Diversity = 2.0484e-02
Generation 2490:	Best = 6.038935e+02	FES = 249100	Diversity = 2.0447e-02
Generation 2500:	Best = 6.038935e+02	FES = 250100	Diversity = 2.0394e-02
Generation 2510:	Best = 6.038916e+02	FES = 251100	Diversity = 2.0390e-02
Generation 2520:	Best = 6.038916e+02	FES = 252100	Diversity = 2.0329e-02
Generation 2530:	Best = 6.038916e+02	FES = 253100	Diversity = 2.0220e-02
Generation 2540:	Best = 6.038916e+02	FES = 254100	Diversity = 2.0241e-02
Generation 2550:	Best = 6.038916e+02	FES = 255100	Diversity = 2.0192e-02
Generation 2560:	Best = 6.038916e+02	FES = 256100	Diversity = 2.0119e-02
Generation 2570:	Best = 6.038914e+02	FES = 257100	Diversity = 1.9991e-02
Generation 2580:	Best = 6.038914e+02	FES = 258100	Diversity = 1.9956e-02
Generation 2590:	Best = 6.038911e+02	FES = 259100	Diversity = 2.1091e-02
Generation 2600:	Best = 6.038911e+02	FES = 260100	Diversity = 2.1146e-02

Generation 2610:	Best = 6.038911e+02	FES = 261100	Diversity = 2.1115e-02
Generation 2620:	Best = 6.038882e+02	FES = 262100	Diversity = 2.1051e-02
Generation 2630:	Best = 6.038882e+02	FES = 263100	Diversity = 2.0969e-02
Generation 2640:	Best = 6.038882e+02	FES = 264100	Diversity = 2.0856e-02
Generation 2650:	Best = 6.038881e+02	FES = 265100	Diversity = 2.0699e-02
Generation 2660:	Best = 6.038881e+02	FES = 266100	Diversity = 1.9792e-02
Generation 2670:	Best = 6.038880e+02	FES = 267100	Diversity = 1.9834e-02
Generation 2680:	Best = 6.038880e+02	FES = 268100	Diversity = 1.9833e-02
Generation 2690:	Best = 6.038880e+02	FES = 269100	Diversity = 1.8657e-02
Generation 2700:	Best = 6.038880e+02	FES = 270100	Diversity = 1.8721e-02
Generation 2710:	Best = 6.038879e+02	FES = 271100	Diversity = 1.7679e-02
Generation 2720:	Best = 6.038879e+02	FES = 272100	Diversity = 1.7604e-02
Generation 2730:	Best = 6.038879e+02	FES = 273100	Diversity = 1.7559e-02
Generation 2740:	Best = 6.038873e+02	FES = 274100	Diversity = 1.7561e-02
Generation 2750:	Best = 6.038873e+02	FES = 275100	Diversity = 1.7503e-02
Generation 2760:	Best = 6.038873e+02	FES = 276100	Diversity = 1.6396e-02
Generation 2770:	Best = 6.038873e+02	FES = 277100	Diversity = 1.6391e-02
Generation 2780:	Best = 6.038873e+02	FES = 278100	Diversity = 1.4836e-02
Generation 2790:	Best = 6.038873e+02	FES = 279100	Diversity = 1.4691e-02
Generation 2800:	Best = 6.038873e+02	FES = 280100	Diversity = 1.4741e-02
Generation 2810:	Best = 6.038830e+02	FES = 281100	Diversity = 1.4751e-02
Generation 2820:	Best = 6.038830e+02	FES = 282100	Diversity = 1.4710e-02
Generation 2830:	Best = 6.038780e+02	FES = 283100	Diversity = 1.4709e-02
Generation 2840:	Best = 6.038780e+02	FES = 284100	Diversity = 1.4688e-02
Generation 2850:	Best = 6.038779e+02	FES = 285100	Diversity = 1.4692e-02
Generation 2860:	Best = 6.038779e+02	FES = 286100	Diversity = 1.4800e-02
Generation 2870:	Best = 6.038778e+02	FES = 287100	Diversity = 1.4829e-02
Generation 2880:	Best = 6.038778e+02	FES = 288100	Diversity = 1.4827e-02
Generation 2890:	Best = 6.038771e+02	FES = 289100	Diversity = 1.4804e-02
Generation 2900:	Best = 6.038771e+02	FES = 290100	Diversity = 1.4664e-02
Generation 2910:	Best = 6.038767e+02	FES = 291100	Diversity = 1.4587e-02
Generation 2920:	Best = 6.038767e+02	FES = 292100	Diversity = 1.4584e-02
Generation 2930:	Best = 6.038767e+02	FES = 293100	Diversity = 1.4555e-02
Generation 2940:	Best = 6.038767e+02	FES = 294100	Diversity = 1.4550e-02
Generation 2950:	Best = 6.038767e+02	FES = 295100	Diversity = 1.4551e-02
Generation 2960:	Best = 6.038767e+02	FES = 296100	Diversity = 1.4548e-02
Generation 2970:	Best = 6.038767e+02	FES = 297100	Diversity = 1.4675e-02
Generation 2980:	Best = 6.038764e+02	FES = 298100	Diversity = 1.4676e-02
Generation 2990:	Best = 6.038764e+02	FES = 299100	Diversity = 1.4676e-02

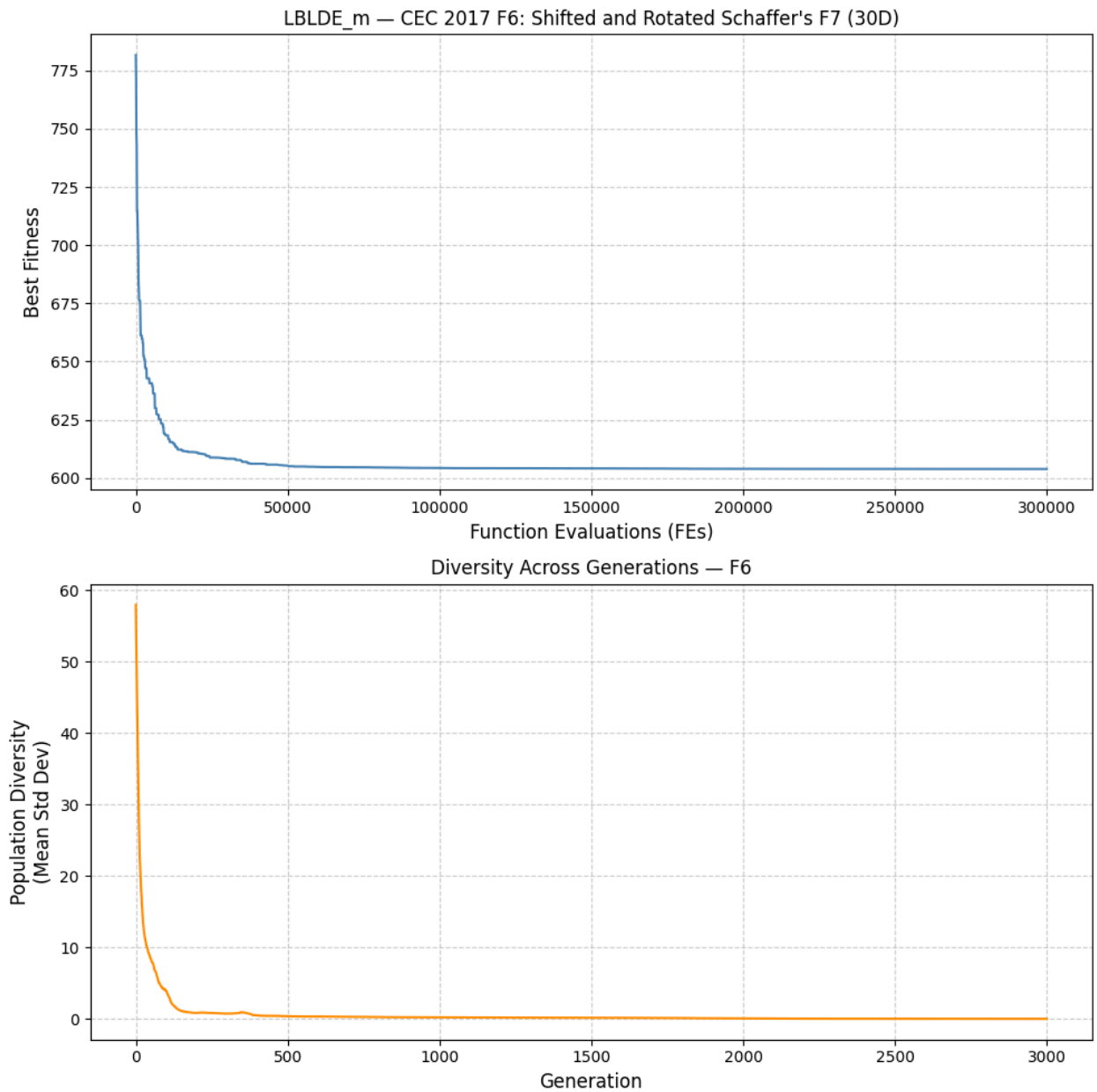
Optimization completed!

Final Best Fitness = 6.038764e+02

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 6.038764e+02
  Global Optimum         : 6.000000e+02
  Error (|best-optimum|) : 3.876356e+00
  Best Solution (5 dim)  : [ 65.37176425 -24.59323377 -37.09904591 -28.70441911
78.52328655]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F6.png

```
In [33]: # =====  
# F7 – Shifted and Rotated Lunacek Bi-Rastrigin  
# =====  
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and Rotated Lunacek Bi-Rastrigin")
```

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LBLDE_m on CEC 2017 F7 – Single Run

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Function : Shifted and Rotated Lunacek Bi-Rastrigin's (30D)

Bounds : [-100, 100]

Global optimum : 700.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.268478e+03 | Diversity = 5.7976e+01

Generation	10:	Best = 1.313745e+03	FES = 1100	Diversity = 1.6747e+01
Generation	20:	Best = 1.152319e+03	FES = 2100	Diversity = 8.0721e+00
Generation	30:	Best = 1.040928e+03	FES = 3100	Diversity = 5.9389e+00
Generation	40:	Best = 1.006438e+03	FES = 4100	Diversity = 4.8862e+00
Generation	50:	Best = 9.922333e+02	FES = 5100	Diversity = 4.5611e+00
Generation	60:	Best = 9.922333e+02	FES = 6100	Diversity = 4.2994e+00
Generation	70:	Best = 9.922333e+02	FES = 7100	Diversity = 4.2059e+00
Generation	80:	Best = 9.847245e+02	FES = 8100	Diversity = 4.1443e+00
Generation	90:	Best = 9.572929e+02	FES = 9100	Diversity = 4.0357e+00
Generation	100:	Best = 9.532596e+02	FES = 10100	Diversity = 4.0286e+00
Generation	110:	Best = 9.310926e+02	FES = 11100	Diversity = 4.0457e+00
Generation	120:	Best = 9.310669e+02	FES = 12100	Diversity = 3.9936e+00
Generation	130:	Best = 9.310669e+02	FES = 13100	Diversity = 3.9710e+00
Generation	140:	Best = 9.286195e+02	FES = 14100	Diversity = 3.8997e+00
Generation	150:	Best = 9.179778e+02	FES = 15100	Diversity = 3.8654e+00
Generation	160:	Best = 9.179778e+02	FES = 16100	Diversity = 3.8206e+00
Generation	170:	Best = 9.050951e+02	FES = 17100	Diversity = 3.8086e+00
Generation	180:	Best = 9.050951e+02	FES = 18100	Diversity = 3.7524e+00
Generation	190:	Best = 9.050951e+02	FES = 19100	Diversity = 3.7450e+00
Generation	200:	Best = 9.039062e+02	FES = 20100	Diversity = 3.7347e+00
Generation	210:	Best = 9.039062e+02	FES = 21100	Diversity = 3.6936e+00
Generation	220:	Best = 8.945611e+02	FES = 22100	Diversity = 3.6778e+00
Generation	230:	Best = 8.945611e+02	FES = 23100	Diversity = 3.6754e+00
Generation	240:	Best = 8.945611e+02	FES = 24100	Diversity = 3.6807e+00
Generation	250:	Best = 8.926616e+02	FES = 25100	Diversity = 3.6779e+00
Generation	260:	Best = 8.926504e+02	FES = 26100	Diversity = 3.6526e+00
Generation	270:	Best = 8.863876e+02	FES = 27100	Diversity = 3.6388e+00
Generation	280:	Best = 8.761699e+02	FES = 28100	Diversity = 3.6184e+00
Generation	290:	Best = 8.761699e+02	FES = 29100	Diversity = 3.6645e+00
Generation	300:	Best = 8.761699e+02	FES = 30100	Diversity = 3.6655e+00
Generation	310:	Best = 8.761699e+02	FES = 31100	Diversity = 3.6585e+00
Generation	320:	Best = 8.761699e+02	FES = 32100	Diversity = 3.6302e+00
Generation	330:	Best = 8.761699e+02	FES = 33100	Diversity = 3.5806e+00
Generation	340:	Best = 8.754842e+02	FES = 34100	Diversity = 3.5616e+00
Generation	350:	Best = 8.754842e+02	FES = 35100	Diversity = 3.5274e+00
Generation	360:	Best = 8.754842e+02	FES = 36100	Diversity = 3.4738e+00
Generation	370:	Best = 8.734571e+02	FES = 37100	Diversity = 3.4525e+00
Generation	380:	Best = 8.734571e+02	FES = 38100	Diversity = 3.4310e+00
Generation	390:	Best = 8.662737e+02	FES = 39100	Diversity = 3.4159e+00
Generation	400:	Best = 8.662737e+02	FES = 40100	Diversity = 3.4095e+00
Generation	410:	Best = 8.662737e+02	FES = 41100	Diversity = 3.4252e+00
Generation	420:	Best = 8.662737e+02	FES = 42100	Diversity = 3.4353e+00
Generation	430:	Best = 8.662737e+02	FES = 43100	Diversity = 3.4286e+00
Generation	440:	Best = 8.662737e+02	FES = 44100	Diversity = 3.4200e+00

Generation	450:	Best = 8.662737e+02	FES = 45100	Diversity = 3.4181e+00
Generation	460:	Best = 8.662737e+02	FES = 46100	Diversity = 3.4298e+00
Generation	470:	Best = 8.662737e+02	FES = 47100	Diversity = 3.4288e+00
Generation	480:	Best = 8.662737e+02	FES = 48100	Diversity = 3.4364e+00
Generation	490:	Best = 8.662737e+02	FES = 49100	Diversity = 3.4344e+00
Generation	500:	Best = 8.662737e+02	FES = 50100	Diversity = 3.4310e+00
Generation	510:	Best = 8.650474e+02	FES = 51100	Diversity = 3.4357e+00
Generation	520:	Best = 8.650474e+02	FES = 52100	Diversity = 3.4331e+00
Generation	530:	Best = 8.635574e+02	FES = 53100	Diversity = 3.4408e+00
Generation	540:	Best = 8.635574e+02	FES = 54100	Diversity = 3.4385e+00
Generation	550:	Best = 8.616008e+02	FES = 55100	Diversity = 3.4326e+00
Generation	560:	Best = 8.616008e+02	FES = 56100	Diversity = 3.4235e+00
Generation	570:	Best = 8.616008e+02	FES = 57100	Diversity = 3.4194e+00
Generation	580:	Best = 8.616008e+02	FES = 58100	Diversity = 3.4194e+00
Generation	590:	Best = 8.572705e+02	FES = 59100	Diversity = 3.4193e+00
Generation	600:	Best = 8.572705e+02	FES = 60100	Diversity = 3.4184e+00
Generation	610:	Best = 8.550677e+02	FES = 61100	Diversity = 3.4089e+00
Generation	620:	Best = 8.550677e+02	FES = 62100	Diversity = 3.4160e+00
Generation	630:	Best = 8.550677e+02	FES = 63100	Diversity = 3.4144e+00
Generation	640:	Best = 8.550677e+02	FES = 64100	Diversity = 3.4153e+00
Generation	650:	Best = 8.505422e+02	FES = 65100	Diversity = 3.4143e+00
Generation	660:	Best = 8.505422e+02	FES = 66100	Diversity = 3.4112e+00
Generation	670:	Best = 8.504310e+02	FES = 67100	Diversity = 3.4106e+00
Generation	680:	Best = 8.504310e+02	FES = 68100	Diversity = 3.4030e+00
Generation	690:	Best = 8.487966e+02	FES = 69100	Diversity = 3.3938e+00
Generation	700:	Best = 8.487966e+02	FES = 70100	Diversity = 3.3923e+00
Generation	710:	Best = 8.487875e+02	FES = 71100	Diversity = 3.3919e+00
Generation	720:	Best = 8.487875e+02	FES = 72100	Diversity = 3.3886e+00
Generation	730:	Best = 8.487875e+02	FES = 73100	Diversity = 3.3886e+00
Generation	740:	Best = 8.487875e+02	FES = 74100	Diversity = 3.3846e+00
Generation	750:	Best = 8.487875e+02	FES = 75100	Diversity = 3.3844e+00
Generation	760:	Best = 8.487875e+02	FES = 76100	Diversity = 3.3898e+00
Generation	770:	Best = 8.487875e+02	FES = 77100	Diversity = 3.3875e+00
Generation	780:	Best = 8.480092e+02	FES = 78100	Diversity = 3.3868e+00
Generation	790:	Best = 8.480092e+02	FES = 79100	Diversity = 3.3866e+00
Generation	800:	Best = 8.480092e+02	FES = 80100	Diversity = 3.3840e+00
Generation	810:	Best = 8.480092e+02	FES = 81100	Diversity = 3.3814e+00
Generation	820:	Best = 8.445818e+02	FES = 82100	Diversity = 3.3809e+00
Generation	830:	Best = 8.445818e+02	FES = 83100	Diversity = 3.3783e+00
Generation	840:	Best = 8.445818e+02	FES = 84100	Diversity = 3.3763e+00
Generation	850:	Best = 8.443150e+02	FES = 85100	Diversity = 3.3719e+00
Generation	860:	Best = 8.443150e+02	FES = 86100	Diversity = 3.3741e+00
Generation	870:	Best = 8.443150e+02	FES = 87100	Diversity = 3.3685e+00
Generation	880:	Best = 8.435070e+02	FES = 88100	Diversity = 3.3661e+00
Generation	890:	Best = 8.435070e+02	FES = 89100	Diversity = 3.3649e+00
Generation	900:	Best = 8.435070e+02	FES = 90100	Diversity = 3.3660e+00
Generation	910:	Best = 8.390803e+02	FES = 91100	Diversity = 3.3654e+00
Generation	920:	Best = 8.390803e+02	FES = 92100	Diversity = 3.3756e+00
Generation	930:	Best = 8.390803e+02	FES = 93100	Diversity = 3.3838e+00
Generation	940:	Best = 8.390803e+02	FES = 94100	Diversity = 3.3837e+00
Generation	950:	Best = 8.388707e+02	FES = 95100	Diversity = 3.3832e+00
Generation	960:	Best = 8.379135e+02	FES = 96100	Diversity = 3.3639e+00
Generation	970:	Best = 8.379135e+02	FES = 97100	Diversity = 3.3575e+00
Generation	980:	Best = 8.378858e+02	FES = 98100	Diversity = 3.3569e+00

Generation 990:	Best = 8.376292e+02	FES = 99100	Diversity = 3.3562e+00
Generation 1000:	Best = 8.325678e+02	FES = 100100	Diversity = 3.3569e+00
Generation 1010:	Best = 8.325272e+02	FES = 101100	Diversity = 3.3406e+00
Generation 1020:	Best = 8.325272e+02	FES = 102100	Diversity = 3.3462e+00
Generation 1030:	Best = 8.325272e+02	FES = 103100	Diversity = 3.3362e+00
Generation 1040:	Best = 8.292674e+02	FES = 104100	Diversity = 3.3419e+00
Generation 1050:	Best = 8.292674e+02	FES = 105100	Diversity = 3.3418e+00
Generation 1060:	Best = 8.292674e+02	FES = 106100	Diversity = 3.3558e+00
Generation 1070:	Best = 8.292674e+02	FES = 107100	Diversity = 3.3554e+00
Generation 1080:	Best = 8.292674e+02	FES = 108100	Diversity = 3.3550e+00
Generation 1090:	Best = 8.277887e+02	FES = 109100	Diversity = 3.3491e+00
Generation 1100:	Best = 8.277887e+02	FES = 110100	Diversity = 3.3471e+00
Generation 1110:	Best = 8.260867e+02	FES = 111100	Diversity = 3.3430e+00
Generation 1120:	Best = 8.257841e+02	FES = 112100	Diversity = 3.3391e+00
Generation 1130:	Best = 8.257841e+02	FES = 113100	Diversity = 3.3361e+00
Generation 1140:	Best = 8.257841e+02	FES = 114100	Diversity = 3.3351e+00
Generation 1150:	Best = 8.243246e+02	FES = 115100	Diversity = 3.3357e+00
Generation 1160:	Best = 8.243246e+02	FES = 116100	Diversity = 3.3334e+00
Generation 1170:	Best = 8.243246e+02	FES = 117100	Diversity = 3.3324e+00
Generation 1180:	Best = 8.243246e+02	FES = 118100	Diversity = 3.3318e+00
Generation 1190:	Best = 8.243246e+02	FES = 119100	Diversity = 3.3313e+00
Generation 1200:	Best = 8.241708e+02	FES = 120100	Diversity = 3.3323e+00
Generation 1210:	Best = 8.241708e+02	FES = 121100	Diversity = 3.3338e+00
Generation 1220:	Best = 8.241708e+02	FES = 122100	Diversity = 3.3328e+00
Generation 1230:	Best = 8.183355e+02	FES = 123100	Diversity = 3.3351e+00
Generation 1240:	Best = 8.183355e+02	FES = 124100	Diversity = 3.3342e+00
Generation 1250:	Best = 8.183355e+02	FES = 125100	Diversity = 3.3342e+00
Generation 1260:	Best = 8.183355e+02	FES = 126100	Diversity = 3.3334e+00
Generation 1270:	Best = 8.183355e+02	FES = 127100	Diversity = 3.3327e+00
Generation 1280:	Best = 8.183355e+02	FES = 128100	Diversity = 3.3321e+00
Generation 1290:	Best = 8.183355e+02	FES = 129100	Diversity = 3.3314e+00
Generation 1300:	Best = 8.183355e+02	FES = 130100	Diversity = 3.3302e+00
Generation 1310:	Best = 8.183355e+02	FES = 131100	Diversity = 3.3264e+00
Generation 1320:	Best = 8.183355e+02	FES = 132100	Diversity = 3.3238e+00
Generation 1330:	Best = 8.183355e+02	FES = 133100	Diversity = 3.3240e+00
Generation 1340:	Best = 8.183355e+02	FES = 134100	Diversity = 3.3248e+00
Generation 1350:	Best = 8.177033e+02	FES = 135100	Diversity = 3.3119e+00
Generation 1360:	Best = 8.177033e+02	FES = 136100	Diversity = 3.3102e+00
Generation 1370:	Best = 8.177033e+02	FES = 137100	Diversity = 3.3101e+00
Generation 1380:	Best = 8.177033e+02	FES = 138100	Diversity = 3.3087e+00
Generation 1390:	Best = 8.177033e+02	FES = 139100	Diversity = 3.3063e+00
Generation 1400:	Best = 8.177033e+02	FES = 140100	Diversity = 3.3056e+00
Generation 1410:	Best = 8.177033e+02	FES = 141100	Diversity = 3.3050e+00
Generation 1420:	Best = 8.177033e+02	FES = 142100	Diversity = 3.3281e+00
Generation 1430:	Best = 8.177033e+02	FES = 143100	Diversity = 3.3285e+00
Generation 1440:	Best = 8.175677e+02	FES = 144100	Diversity = 3.3291e+00
Generation 1450:	Best = 8.175677e+02	FES = 145100	Diversity = 3.3276e+00
Generation 1460:	Best = 8.175473e+02	FES = 146100	Diversity = 3.3308e+00
Generation 1470:	Best = 8.175473e+02	FES = 147100	Diversity = 3.3309e+00
Generation 1480:	Best = 8.167434e+02	FES = 148100	Diversity = 3.3307e+00
Generation 1490:	Best = 8.167434e+02	FES = 149100	Diversity = 3.3289e+00
Generation 1500:	Best = 8.167434e+02	FES = 150100	Diversity = 3.3290e+00
Generation 1510:	Best = 8.167434e+02	FES = 151100	Diversity = 3.3289e+00
Generation 1520:	Best = 8.167434e+02	FES = 152100	Diversity = 3.3291e+00

Generation 1530:	Best = 8.167434e+02	FES = 153100	Diversity = 3.3289e+00
Generation 1540:	Best = 8.167434e+02	FES = 154100	Diversity = 3.3288e+00
Generation 1550:	Best = 8.167434e+02	FES = 155100	Diversity = 3.3283e+00
Generation 1560:	Best = 8.167434e+02	FES = 156100	Diversity = 3.3289e+00
Generation 1570:	Best = 8.167157e+02	FES = 157100	Diversity = 3.3288e+00
Generation 1580:	Best = 8.166902e+02	FES = 158100	Diversity = 3.3287e+00
Generation 1590:	Best = 8.166902e+02	FES = 159100	Diversity = 3.3289e+00
Generation 1600:	Best = 8.166902e+02	FES = 160100	Diversity = 3.3298e+00
Generation 1610:	Best = 8.166902e+02	FES = 161100	Diversity = 3.3215e+00
Generation 1620:	Best = 8.166902e+02	FES = 162100	Diversity = 3.3216e+00
Generation 1630:	Best = 8.166902e+02	FES = 163100	Diversity = 3.3221e+00
Generation 1640:	Best = 8.166902e+02	FES = 164100	Diversity = 3.3221e+00
Generation 1650:	Best = 8.166902e+02	FES = 165100	Diversity = 3.3384e+00
Generation 1660:	Best = 8.166902e+02	FES = 166100	Diversity = 3.3380e+00
Generation 1670:	Best = 8.166902e+02	FES = 167100	Diversity = 3.3374e+00
Generation 1680:	Best = 8.166902e+02	FES = 168100	Diversity = 3.3366e+00
Generation 1690:	Best = 8.166902e+02	FES = 169100	Diversity = 3.3363e+00
Generation 1700:	Best = 8.166902e+02	FES = 170100	Diversity = 3.3364e+00
Generation 1710:	Best = 8.166902e+02	FES = 171100	Diversity = 3.3367e+00
Generation 1720:	Best = 8.166902e+02	FES = 172100	Diversity = 3.3257e+00
Generation 1730:	Best = 8.166902e+02	FES = 173100	Diversity = 3.3255e+00
Generation 1740:	Best = 8.166902e+02	FES = 174100	Diversity = 3.3198e+00
Generation 1750:	Best = 8.150721e+02	FES = 175100	Diversity = 3.3205e+00
Generation 1760:	Best = 8.150721e+02	FES = 176100	Diversity = 3.3208e+00
Generation 1770:	Best = 8.150721e+02	FES = 177100	Diversity = 3.3202e+00
Generation 1780:	Best = 8.150721e+02	FES = 178100	Diversity = 3.3206e+00
Generation 1790:	Best = 8.150721e+02	FES = 179100	Diversity = 3.3211e+00
Generation 1800:	Best = 8.150721e+02	FES = 180100	Diversity = 3.3208e+00
Generation 1810:	Best = 8.146124e+02	FES = 181100	Diversity = 3.3196e+00
Generation 1820:	Best = 8.146124e+02	FES = 182100	Diversity = 3.3191e+00
Generation 1830:	Best = 8.144293e+02	FES = 183100	Diversity = 3.3191e+00
Generation 1840:	Best = 8.144293e+02	FES = 184100	Diversity = 3.3178e+00
Generation 1850:	Best = 8.127301e+02	FES = 185100	Diversity = 3.3152e+00
Generation 1860:	Best = 8.127240e+02	FES = 186100	Diversity = 3.3150e+00
Generation 1870:	Best = 8.127240e+02	FES = 187100	Diversity = 3.3228e+00
Generation 1880:	Best = 8.127240e+02	FES = 188100	Diversity = 3.3313e+00
Generation 1890:	Best = 8.126582e+02	FES = 189100	Diversity = 3.3308e+00
Generation 1900:	Best = 8.126582e+02	FES = 190100	Diversity = 3.3307e+00
Generation 1910:	Best = 8.126579e+02	FES = 191100	Diversity = 3.3291e+00
Generation 1920:	Best = 8.126579e+02	FES = 192100	Diversity = 3.3306e+00
Generation 1930:	Best = 8.126579e+02	FES = 193100	Diversity = 3.3304e+00
Generation 1940:	Best = 8.126579e+02	FES = 194100	Diversity = 3.3305e+00
Generation 1950:	Best = 8.126579e+02	FES = 195100	Diversity = 3.3302e+00
Generation 1960:	Best = 8.126579e+02	FES = 196100	Diversity = 3.3302e+00
Generation 1970:	Best = 8.126579e+02	FES = 197100	Diversity = 3.3303e+00
Generation 1980:	Best = 8.108620e+02	FES = 198100	Diversity = 3.3302e+00
Generation 1990:	Best = 8.108620e+02	FES = 199100	Diversity = 3.3319e+00
Generation 2000:	Best = 8.108620e+02	FES = 200100	Diversity = 3.3319e+00
Generation 2010:	Best = 8.108620e+02	FES = 201100	Diversity = 3.3321e+00
Generation 2020:	Best = 8.108620e+02	FES = 202100	Diversity = 3.3304e+00
Generation 2030:	Best = 8.108620e+02	FES = 203100	Diversity = 3.3222e+00
Generation 2040:	Best = 8.108620e+02	FES = 204100	Diversity = 3.3385e+00
Generation 2050:	Best = 8.108620e+02	FES = 205100	Diversity = 3.3388e+00
Generation 2060:	Best = 8.108620e+02	FES = 206100	Diversity = 3.3327e+00

Generation 2070:	Best = 8.108620e+02	FES = 207100	Diversity = 3.3332e+00
Generation 2080:	Best = 8.108620e+02	FES = 208100	Diversity = 3.3241e+00
Generation 2090:	Best = 8.108454e+02	FES = 209100	Diversity = 3.3245e+00
Generation 2100:	Best = 8.108454e+02	FES = 210100	Diversity = 3.3246e+00
Generation 2110:	Best = 8.108454e+02	FES = 211100	Diversity = 3.3242e+00
Generation 2120:	Best = 8.108454e+02	FES = 212100	Diversity = 3.3249e+00
Generation 2130:	Best = 8.108454e+02	FES = 213100	Diversity = 3.3238e+00
Generation 2140:	Best = 8.108454e+02	FES = 214100	Diversity = 3.3239e+00
Generation 2150:	Best = 8.102297e+02	FES = 215100	Diversity = 3.3235e+00
Generation 2160:	Best = 8.102297e+02	FES = 216100	Diversity = 3.3108e+00
Generation 2170:	Best = 8.102297e+02	FES = 217100	Diversity = 3.3110e+00
Generation 2180:	Best = 8.102297e+02	FES = 218100	Diversity = 3.3112e+00
Generation 2190:	Best = 8.102297e+02	FES = 219100	Diversity = 3.3112e+00
Generation 2200:	Best = 8.102296e+02	FES = 220100	Diversity = 3.3099e+00
Generation 2210:	Best = 8.101630e+02	FES = 221100	Diversity = 3.3107e+00
Generation 2220:	Best = 8.101630e+02	FES = 222100	Diversity = 3.3106e+00
Generation 2230:	Best = 8.094714e+02	FES = 223100	Diversity = 3.3106e+00
Generation 2240:	Best = 8.094714e+02	FES = 224100	Diversity = 3.3099e+00
Generation 2250:	Best = 8.094209e+02	FES = 225100	Diversity = 3.3094e+00
Generation 2260:	Best = 8.094209e+02	FES = 226100	Diversity = 3.3093e+00
Generation 2270:	Best = 8.094209e+02	FES = 227100	Diversity = 3.3089e+00
Generation 2280:	Best = 8.093055e+02	FES = 228100	Diversity = 3.3190e+00
Generation 2290:	Best = 8.093055e+02	FES = 229100	Diversity = 3.3190e+00
Generation 2300:	Best = 8.093055e+02	FES = 230100	Diversity = 3.3187e+00
Generation 2310:	Best = 8.090731e+02	FES = 231100	Diversity = 3.3187e+00
Generation 2320:	Best = 8.090731e+02	FES = 232100	Diversity = 3.3259e+00
Generation 2330:	Best = 8.090721e+02	FES = 233100	Diversity = 3.3252e+00
Generation 2340:	Best = 8.084451e+02	FES = 234100	Diversity = 3.3217e+00
Generation 2350:	Best = 8.084451e+02	FES = 235100	Diversity = 3.3220e+00
Generation 2360:	Best = 8.084451e+02	FES = 236100	Diversity = 3.3219e+00
Generation 2370:	Best = 8.084451e+02	FES = 237100	Diversity = 3.3250e+00
Generation 2380:	Best = 8.084451e+02	FES = 238100	Diversity = 3.3248e+00
Generation 2390:	Best = 8.084451e+02	FES = 239100	Diversity = 3.3248e+00
Generation 2400:	Best = 8.084180e+02	FES = 240100	Diversity = 3.3245e+00
Generation 2410:	Best = 8.084180e+02	FES = 241100	Diversity = 3.3241e+00
Generation 2420:	Best = 8.084136e+02	FES = 242100	Diversity = 3.3181e+00
Generation 2430:	Best = 8.084136e+02	FES = 243100	Diversity = 3.3176e+00
Generation 2440:	Best = 8.084136e+02	FES = 244100	Diversity = 3.3177e+00
Generation 2450:	Best = 8.084045e+02	FES = 245100	Diversity = 3.3179e+00
Generation 2460:	Best = 8.081785e+02	FES = 246100	Diversity = 3.3181e+00
Generation 2470:	Best = 8.081785e+02	FES = 247100	Diversity = 3.3148e+00
Generation 2480:	Best = 8.081785e+02	FES = 248100	Diversity = 3.3152e+00
Generation 2490:	Best = 8.081785e+02	FES = 249100	Diversity = 3.3289e+00
Generation 2500:	Best = 8.081785e+02	FES = 250100	Diversity = 3.3286e+00
Generation 2510:	Best = 8.081785e+02	FES = 251100	Diversity = 3.3287e+00
Generation 2520:	Best = 8.081785e+02	FES = 252100	Diversity = 3.3282e+00
Generation 2530:	Best = 8.081785e+02	FES = 253100	Diversity = 3.3273e+00
Generation 2540:	Best = 8.081785e+02	FES = 254100	Diversity = 3.3265e+00
Generation 2550:	Best = 8.081785e+02	FES = 255100	Diversity = 3.3118e+00
Generation 2560:	Best = 8.081785e+02	FES = 256100	Diversity = 3.3121e+00
Generation 2570:	Best = 8.081785e+02	FES = 257100	Diversity = 3.3118e+00
Generation 2580:	Best = 8.081785e+02	FES = 258100	Diversity = 3.3121e+00
Generation 2590:	Best = 8.081785e+02	FES = 259100	Diversity = 3.3121e+00
Generation 2600:	Best = 8.081785e+02	FES = 260100	Diversity = 3.3115e+00

Generation 2610:	Best = 8.081785e+02	FES = 261100	Diversity = 3.3116e+00
Generation 2620:	Best = 8.081785e+02	FES = 262100	Diversity = 3.3110e+00
Generation 2630:	Best = 8.081785e+02	FES = 263100	Diversity = 3.3099e+00
Generation 2640:	Best = 8.081785e+02	FES = 264100	Diversity = 3.3086e+00
Generation 2650:	Best = 8.081785e+02	FES = 265100	Diversity = 3.3085e+00
Generation 2660:	Best = 8.081785e+02	FES = 266100	Diversity = 3.3085e+00
Generation 2670:	Best = 8.081785e+02	FES = 267100	Diversity = 3.3083e+00
Generation 2680:	Best = 8.081785e+02	FES = 268100	Diversity = 3.3086e+00
Generation 2690:	Best = 8.081785e+02	FES = 269100	Diversity = 3.3089e+00
Generation 2700:	Best = 8.081785e+02	FES = 270100	Diversity = 3.3091e+00
Generation 2710:	Best = 8.081785e+02	FES = 271100	Diversity = 3.3096e+00
Generation 2720:	Best = 8.081785e+02	FES = 272100	Diversity = 3.3097e+00
Generation 2730:	Best = 8.081785e+02	FES = 273100	Diversity = 3.3098e+00
Generation 2740:	Best = 8.081785e+02	FES = 274100	Diversity = 3.3080e+00
Generation 2750:	Best = 8.081785e+02	FES = 275100	Diversity = 3.3080e+00
Generation 2760:	Best = 8.081785e+02	FES = 276100	Diversity = 3.3083e+00
Generation 2770:	Best = 8.079105e+02	FES = 277100	Diversity = 3.3087e+00
Generation 2780:	Best = 8.079105e+02	FES = 278100	Diversity = 3.3070e+00
Generation 2790:	Best = 8.079105e+02	FES = 279100	Diversity = 3.3071e+00
Generation 2800:	Best = 8.079105e+02	FES = 280100	Diversity = 3.3059e+00
Generation 2810:	Best = 8.079105e+02	FES = 281100	Diversity = 3.3063e+00
Generation 2820:	Best = 8.079105e+02	FES = 282100	Diversity = 3.3060e+00
Generation 2830:	Best = 8.079105e+02	FES = 283100	Diversity = 3.3061e+00
Generation 2840:	Best = 8.079105e+02	FES = 284100	Diversity = 3.3059e+00
Generation 2850:	Best = 8.079105e+02	FES = 285100	Diversity = 3.3059e+00
Generation 2860:	Best = 8.079105e+02	FES = 286100	Diversity = 3.3059e+00
Generation 2870:	Best = 8.079105e+02	FES = 287100	Diversity = 3.3061e+00
Generation 2880:	Best = 8.079105e+02	FES = 288100	Diversity = 3.3054e+00
Generation 2890:	Best = 8.079105e+02	FES = 289100	Diversity = 3.3046e+00
Generation 2900:	Best = 8.079105e+02	FES = 290100	Diversity = 3.3046e+00
Generation 2910:	Best = 8.079105e+02	FES = 291100	Diversity = 3.3352e+00
Generation 2920:	Best = 8.079105e+02	FES = 292100	Diversity = 3.3352e+00
Generation 2930:	Best = 8.079105e+02	FES = 293100	Diversity = 3.3360e+00
Generation 2940:	Best = 8.079105e+02	FES = 294100	Diversity = 3.3462e+00
Generation 2950:	Best = 8.076178e+02	FES = 295100	Diversity = 3.3469e+00
Generation 2960:	Best = 8.076178e+02	FES = 296100	Diversity = 3.3467e+00
Generation 2970:	Best = 8.075230e+02	FES = 297100	Diversity = 3.3472e+00
Generation 2980:	Best = 8.075230e+02	FES = 298100	Diversity = 3.3471e+00
Generation 2990:	Best = 8.075230e+02	FES = 299100	Diversity = 3.3472e+00

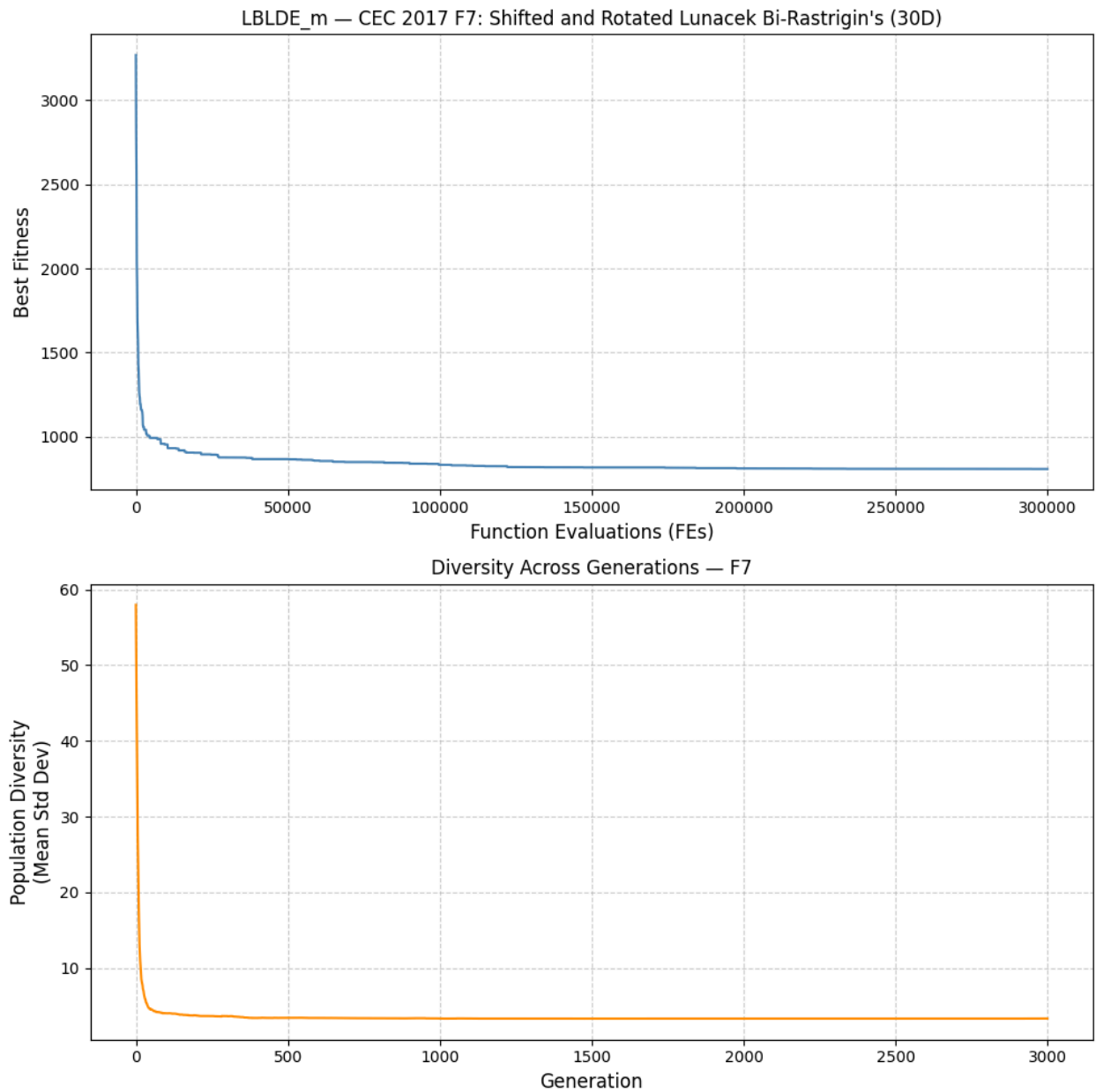
Optimization completed!

Final Best Fitness = 8.075230e+02

Total Function Evaluations = 300000

```
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=
Results:
  Best Fitness           : 8.075230e+02
  Global Optimum         : 7.000000e+02
  Error (|best-optimum|) : 1.075230e+02
  Best Solution (5 dim)  : [-20.37613768  13.22105098 -15.2299402   0.5319321
10.95671605]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F7.png

```
In [34]: # =====  
# F8 — Shifted and Rotated Non-Continuous Rastrigin  
# =====  
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and Rotated Non-Co
```

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LBLDE_m on CEC 2017 F8 – Single Run

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Function : Shifted and Rotated Non-Continuous Rastrigin's (30D)
Bounds : [-100, 100]
Global optimum : 800.0
Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 1.442304e+03 | Diversity = 5.7976e+01

Generation	10:	Best = 1.130617e+03	FES = 1100	Diversity = 2.7454e+01
Generation	20:	Best = 1.050565e+03	FES = 2100	Diversity = 1.8914e+01
Generation	30:	Best = 1.036677e+03	FES = 3100	Diversity = 1.5822e+01
Generation	40:	Best = 1.036677e+03	FES = 4100	Diversity = 1.4917e+01
Generation	50:	Best = 9.743870e+02	FES = 5100	Diversity = 1.4097e+01
Generation	60:	Best = 9.708037e+02	FES = 6100	Diversity = 1.3849e+01
Generation	70:	Best = 9.708037e+02	FES = 7100	Diversity = 1.3461e+01
Generation	80:	Best = 9.708037e+02	FES = 8100	Diversity = 1.3135e+01
Generation	90:	Best = 9.708037e+02	FES = 9100	Diversity = 1.2882e+01
Generation	100:	Best = 9.708037e+02	FES = 10100	Diversity = 1.2777e+01
Generation	110:	Best = 9.603278e+02	FES = 11100	Diversity = 1.2666e+01
Generation	120:	Best = 9.603278e+02	FES = 12100	Diversity = 1.2442e+01
Generation	130:	Best = 9.603211e+02	FES = 13100	Diversity = 1.2295e+01
Generation	140:	Best = 9.573172e+02	FES = 14100	Diversity = 1.2344e+01
Generation	150:	Best = 9.489392e+02	FES = 15100	Diversity = 1.2203e+01
Generation	160:	Best = 9.430235e+02	FES = 16100	Diversity = 1.2165e+01
Generation	170:	Best = 9.430235e+02	FES = 17100	Diversity = 1.2041e+01
Generation	180:	Best = 9.430235e+02	FES = 18100	Diversity = 1.1980e+01
Generation	190:	Best = 9.430235e+02	FES = 19100	Diversity = 1.1906e+01
Generation	200:	Best = 9.291259e+02	FES = 20100	Diversity = 1.1842e+01
Generation	210:	Best = 9.291259e+02	FES = 21100	Diversity = 1.1812e+01
Generation	220:	Best = 9.215726e+02	FES = 22100	Diversity = 1.1744e+01
Generation	230:	Best = 9.215726e+02	FES = 23100	Diversity = 1.1685e+01
Generation	240:	Best = 9.215726e+02	FES = 24100	Diversity = 1.1517e+01
Generation	250:	Best = 9.211189e+02	FES = 25100	Diversity = 1.1503e+01
Generation	260:	Best = 9.211189e+02	FES = 26100	Diversity = 1.1470e+01
Generation	270:	Best = 9.185799e+02	FES = 27100	Diversity = 1.1436e+01
Generation	280:	Best = 9.149215e+02	FES = 28100	Diversity = 1.1359e+01
Generation	290:	Best = 9.149215e+02	FES = 29100	Diversity = 1.1229e+01
Generation	300:	Best = 9.149215e+02	FES = 30100	Diversity = 1.1315e+01
Generation	310:	Best = 9.145232e+02	FES = 31100	Diversity = 1.1243e+01
Generation	320:	Best = 9.136050e+02	FES = 32100	Diversity = 1.1215e+01
Generation	330:	Best = 9.087475e+02	FES = 33100	Diversity = 1.1035e+01
Generation	340:	Best = 9.039868e+02	FES = 34100	Diversity = 1.1035e+01
Generation	350:	Best = 9.027780e+02	FES = 35100	Diversity = 1.1043e+01
Generation	360:	Best = 9.027780e+02	FES = 36100	Diversity = 1.1042e+01
Generation	370:	Best = 9.027780e+02	FES = 37100	Diversity = 1.1029e+01
Generation	380:	Best = 9.027780e+02	FES = 38100	Diversity = 1.1097e+01
Generation	390:	Best = 8.931093e+02	FES = 39100	Diversity = 1.1047e+01
Generation	400:	Best = 8.931093e+02	FES = 40100	Diversity = 1.1036e+01
Generation	410:	Best = 8.931093e+02	FES = 41100	Diversity = 1.1028e+01
Generation	420:	Best = 8.931093e+02	FES = 42100	Diversity = 1.0992e+01
Generation	430:	Best = 8.931093e+02	FES = 43100	Diversity = 1.0999e+01
Generation	440:	Best = 8.886031e+02	FES = 44100	Diversity = 1.0912e+01

Generation	450:	Best = 8.885757e+02	FES = 45100	Diversity = 1.0892e+01
Generation	460:	Best = 8.885757e+02	FES = 46100	Diversity = 1.0894e+01
Generation	470:	Best = 8.885757e+02	FES = 47100	Diversity = 1.0902e+01
Generation	480:	Best = 8.885757e+02	FES = 48100	Diversity = 1.0892e+01
Generation	490:	Best = 8.884259e+02	FES = 49100	Diversity = 1.0791e+01
Generation	500:	Best = 8.884259e+02	FES = 50100	Diversity = 1.0779e+01
Generation	510:	Best = 8.884259e+02	FES = 51100	Diversity = 1.0768e+01
Generation	520:	Best = 8.883571e+02	FES = 52100	Diversity = 1.0787e+01
Generation	530:	Best = 8.842340e+02	FES = 53100	Diversity = 1.0761e+01
Generation	540:	Best = 8.842340e+02	FES = 54100	Diversity = 1.0764e+01
Generation	550:	Best = 8.842340e+02	FES = 55100	Diversity = 1.0714e+01
Generation	560:	Best = 8.842340e+02	FES = 56100	Diversity = 1.0703e+01
Generation	570:	Best = 8.842340e+02	FES = 57100	Diversity = 1.0705e+01
Generation	580:	Best = 8.842340e+02	FES = 58100	Diversity = 1.0722e+01
Generation	590:	Best = 8.842340e+02	FES = 59100	Diversity = 1.0739e+01
Generation	600:	Best = 8.791338e+02	FES = 60100	Diversity = 1.0739e+01
Generation	610:	Best = 8.791338e+02	FES = 61100	Diversity = 1.0677e+01
Generation	620:	Best = 8.791338e+02	FES = 62100	Diversity = 1.0644e+01
Generation	630:	Best = 8.750928e+02	FES = 63100	Diversity = 1.0603e+01
Generation	640:	Best = 8.750928e+02	FES = 64100	Diversity = 1.0612e+01
Generation	650:	Best = 8.750928e+02	FES = 65100	Diversity = 1.0613e+01
Generation	660:	Best = 8.750928e+02	FES = 66100	Diversity = 1.0509e+01
Generation	670:	Best = 8.750928e+02	FES = 67100	Diversity = 1.0480e+01
Generation	680:	Best = 8.750928e+02	FES = 68100	Diversity = 1.0536e+01
Generation	690:	Best = 8.750928e+02	FES = 69100	Diversity = 1.0508e+01
Generation	700:	Best = 8.750928e+02	FES = 70100	Diversity = 1.0497e+01
Generation	710:	Best = 8.683799e+02	FES = 71100	Diversity = 1.0477e+01
Generation	720:	Best = 8.683799e+02	FES = 72100	Diversity = 1.0459e+01
Generation	730:	Best = 8.683799e+02	FES = 73100	Diversity = 1.0441e+01
Generation	740:	Best = 8.666340e+02	FES = 74100	Diversity = 1.0451e+01
Generation	750:	Best = 8.651672e+02	FES = 75100	Diversity = 1.0447e+01
Generation	760:	Best = 8.645404e+02	FES = 76100	Diversity = 1.0442e+01
Generation	770:	Best = 8.645404e+02	FES = 77100	Diversity = 1.0435e+01
Generation	780:	Best = 8.628458e+02	FES = 78100	Diversity = 1.0453e+01
Generation	790:	Best = 8.628458e+02	FES = 79100	Diversity = 1.0447e+01
Generation	800:	Best = 8.627076e+02	FES = 80100	Diversity = 1.0429e+01
Generation	810:	Best = 8.627076e+02	FES = 81100	Diversity = 1.0435e+01
Generation	820:	Best = 8.621419e+02	FES = 82100	Diversity = 1.0433e+01
Generation	830:	Best = 8.621419e+02	FES = 83100	Diversity = 1.0384e+01
Generation	840:	Best = 8.596170e+02	FES = 84100	Diversity = 1.0386e+01
Generation	850:	Best = 8.596170e+02	FES = 85100	Diversity = 1.0418e+01
Generation	860:	Best = 8.596170e+02	FES = 86100	Diversity = 1.0412e+01
Generation	870:	Best = 8.594808e+02	FES = 87100	Diversity = 1.0360e+01
Generation	880:	Best = 8.581480e+02	FES = 88100	Diversity = 1.0327e+01
Generation	890:	Best = 8.581480e+02	FES = 89100	Diversity = 1.0319e+01
Generation	900:	Best = 8.581480e+02	FES = 90100	Diversity = 1.0379e+01
Generation	910:	Best = 8.580422e+02	FES = 91100	Diversity = 1.0364e+01
Generation	920:	Best = 8.580422e+02	FES = 92100	Diversity = 1.0364e+01
Generation	930:	Best = 8.580422e+02	FES = 93100	Diversity = 1.0398e+01
Generation	940:	Best = 8.580422e+02	FES = 94100	Diversity = 1.0391e+01
Generation	950:	Best = 8.549387e+02	FES = 95100	Diversity = 1.0342e+01
Generation	960:	Best = 8.549386e+02	FES = 96100	Diversity = 1.0343e+01
Generation	970:	Best = 8.546704e+02	FES = 97100	Diversity = 1.0267e+01
Generation	980:	Best = 8.540879e+02	FES = 98100	Diversity = 1.0272e+01

Generation 990:	Best = 8.540786e+02	FES = 99100	Diversity = 1.0271e+01
Generation 1000:	Best = 8.530177e+02	FES = 100100	Diversity = 1.0271e+01
Generation 1010:	Best = 8.516447e+02	FES = 101100	Diversity = 1.0283e+01
Generation 1020:	Best = 8.513797e+02	FES = 102100	Diversity = 1.0242e+01
Generation 1030:	Best = 8.513337e+02	FES = 103100	Diversity = 1.0231e+01
Generation 1040:	Best = 8.513337e+02	FES = 104100	Diversity = 1.0224e+01
Generation 1050:	Best = 8.512733e+02	FES = 105100	Diversity = 1.0220e+01
Generation 1060:	Best = 8.512733e+02	FES = 106100	Diversity = 1.0247e+01
Generation 1070:	Best = 8.512733e+02	FES = 107100	Diversity = 1.0247e+01
Generation 1080:	Best = 8.508794e+02	FES = 108100	Diversity = 1.0309e+01
Generation 1090:	Best = 8.508794e+02	FES = 109100	Diversity = 1.0310e+01
Generation 1100:	Best = 8.507652e+02	FES = 110100	Diversity = 1.0297e+01
Generation 1110:	Best = 8.507652e+02	FES = 111100	Diversity = 1.0352e+01
Generation 1120:	Best = 8.507354e+02	FES = 112100	Diversity = 1.0354e+01
Generation 1130:	Best = 8.507354e+02	FES = 113100	Diversity = 1.0359e+01
Generation 1140:	Best = 8.506917e+02	FES = 114100	Diversity = 1.0354e+01
Generation 1150:	Best = 8.506917e+02	FES = 115100	Diversity = 1.0353e+01
Generation 1160:	Best = 8.506917e+02	FES = 116100	Diversity = 1.0350e+01
Generation 1170:	Best = 8.506471e+02	FES = 117100	Diversity = 1.0336e+01
Generation 1180:	Best = 8.505468e+02	FES = 118100	Diversity = 1.0336e+01
Generation 1190:	Best = 8.501405e+02	FES = 119100	Diversity = 1.0326e+01
Generation 1200:	Best = 8.501405e+02	FES = 120100	Diversity = 1.0364e+01
Generation 1210:	Best = 8.499979e+02	FES = 121100	Diversity = 1.0386e+01
Generation 1220:	Best = 8.499506e+02	FES = 122100	Diversity = 1.0386e+01
Generation 1230:	Best = 8.499506e+02	FES = 123100	Diversity = 1.0386e+01
Generation 1240:	Best = 8.496370e+02	FES = 124100	Diversity = 1.0345e+01
Generation 1250:	Best = 8.486296e+02	FES = 125100	Diversity = 1.0346e+01
Generation 1260:	Best = 8.485968e+02	FES = 126100	Diversity = 1.0345e+01
Generation 1270:	Best = 8.484132e+02	FES = 127100	Diversity = 1.0339e+01
Generation 1280:	Best = 8.482163e+02	FES = 128100	Diversity = 1.0298e+01
Generation 1290:	Best = 8.479269e+02	FES = 129100	Diversity = 1.0287e+01
Generation 1300:	Best = 8.479269e+02	FES = 130100	Diversity = 1.0289e+01
Generation 1310:	Best = 8.479269e+02	FES = 131100	Diversity = 1.0275e+01
Generation 1320:	Best = 8.478399e+02	FES = 132100	Diversity = 1.0277e+01
Generation 1330:	Best = 8.478399e+02	FES = 133100	Diversity = 1.0279e+01
Generation 1340:	Best = 8.478399e+02	FES = 134100	Diversity = 1.0274e+01
Generation 1350:	Best = 8.476898e+02	FES = 135100	Diversity = 1.0245e+01
Generation 1360:	Best = 8.476880e+02	FES = 136100	Diversity = 1.0246e+01
Generation 1370:	Best = 8.476737e+02	FES = 137100	Diversity = 1.0232e+01
Generation 1380:	Best = 8.476522e+02	FES = 138100	Diversity = 1.0255e+01
Generation 1390:	Best = 8.476507e+02	FES = 139100	Diversity = 1.0253e+01
Generation 1400:	Best = 8.476459e+02	FES = 140100	Diversity = 1.0237e+01
Generation 1410:	Best = 8.476446e+02	FES = 141100	Diversity = 1.0228e+01
Generation 1420:	Best = 8.476446e+02	FES = 142100	Diversity = 1.0230e+01
Generation 1430:	Best = 8.476446e+02	FES = 143100	Diversity = 1.0306e+01
Generation 1440:	Best = 8.476446e+02	FES = 144100	Diversity = 1.0309e+01
Generation 1450:	Best = 8.476289e+02	FES = 145100	Diversity = 1.0307e+01
Generation 1460:	Best = 8.476289e+02	FES = 146100	Diversity = 1.0189e+01
Generation 1470:	Best = 8.476271e+02	FES = 147100	Diversity = 1.0169e+01
Generation 1480:	Best = 8.476271e+02	FES = 148100	Diversity = 1.0199e+01
Generation 1490:	Best = 8.476270e+02	FES = 149100	Diversity = 1.0201e+01
Generation 1500:	Best = 8.476270e+02	FES = 150100	Diversity = 1.0147e+01
Generation 1510:	Best = 8.476185e+02	FES = 151100	Diversity = 1.0145e+01
Generation 1520:	Best = 8.476184e+02	FES = 152100	Diversity = 1.0156e+01

Generation 1530:	Best = 8.476184e+02	FES = 153100	Diversity = 1.0185e+01
Generation 1540:	Best = 8.476184e+02	FES = 154100	Diversity = 1.0185e+01
Generation 1550:	Best = 8.476184e+02	FES = 155100	Diversity = 1.0139e+01
Generation 1560:	Best = 8.476182e+02	FES = 156100	Diversity = 1.0142e+01
Generation 1570:	Best = 8.476182e+02	FES = 157100	Diversity = 1.0143e+01
Generation 1580:	Best = 8.476182e+02	FES = 158100	Diversity = 1.0139e+01
Generation 1590:	Best = 8.460930e+02	FES = 159100	Diversity = 1.0136e+01
Generation 1600:	Best = 8.460364e+02	FES = 160100	Diversity = 1.0124e+01
Generation 1610:	Best = 8.458128e+02	FES = 161100	Diversity = 1.0077e+01
Generation 1620:	Best = 8.458109e+02	FES = 162100	Diversity = 1.0038e+01
Generation 1630:	Best = 8.458109e+02	FES = 163100	Diversity = 1.0065e+01
Generation 1640:	Best = 8.458109e+02	FES = 164100	Diversity = 1.0067e+01
Generation 1650:	Best = 8.457997e+02	FES = 165100	Diversity = 1.0084e+01
Generation 1660:	Best = 8.457979e+02	FES = 166100	Diversity = 1.0020e+01
Generation 1670:	Best = 8.457955e+02	FES = 167100	Diversity = 1.0033e+01
Generation 1680:	Best = 8.457893e+02	FES = 168100	Diversity = 1.0006e+01
Generation 1690:	Best = 8.457893e+02	FES = 169100	Diversity = 9.9999e+00
Generation 1700:	Best = 8.457889e+02	FES = 170100	Diversity = 9.9976e+00
Generation 1710:	Best = 8.457886e+02	FES = 171100	Diversity = 9.9978e+00
Generation 1720:	Best = 8.457886e+02	FES = 172100	Diversity = 9.9855e+00
Generation 1730:	Best = 8.457772e+02	FES = 173100	Diversity = 9.9523e+00
Generation 1740:	Best = 8.457772e+02	FES = 174100	Diversity = 9.9561e+00
Generation 1750:	Best = 8.457771e+02	FES = 175100	Diversity = 9.9559e+00
Generation 1760:	Best = 8.457771e+02	FES = 176100	Diversity = 9.9552e+00
Generation 1770:	Best = 8.457771e+02	FES = 177100	Diversity = 9.9610e+00
Generation 1780:	Best = 8.457771e+02	FES = 178100	Diversity = 9.9673e+00
Generation 1790:	Best = 8.457771e+02	FES = 179100	Diversity = 9.9665e+00
Generation 1800:	Best = 8.457771e+02	FES = 180100	Diversity = 9.9331e+00
Generation 1810:	Best = 8.457771e+02	FES = 181100	Diversity = 9.9887e+00
Generation 1820:	Best = 8.457771e+02	FES = 182100	Diversity = 9.9861e+00
Generation 1830:	Best = 8.439627e+02	FES = 183100	Diversity = 9.9844e+00
Generation 1840:	Best = 8.439627e+02	FES = 184100	Diversity = 9.9875e+00
Generation 1850:	Best = 8.438092e+02	FES = 185100	Diversity = 9.9790e+00
Generation 1860:	Best = 8.438092e+02	FES = 186100	Diversity = 9.9412e+00
Generation 1870:	Best = 8.438083e+02	FES = 187100	Diversity = 9.9385e+00
Generation 1880:	Best = 8.437605e+02	FES = 188100	Diversity = 9.9367e+00
Generation 1890:	Best = 8.437409e+02	FES = 189100	Diversity = 9.9371e+00
Generation 1900:	Best = 8.433852e+02	FES = 190100	Diversity = 9.9356e+00
Generation 1910:	Best = 8.433852e+02	FES = 191100	Diversity = 9.9393e+00
Generation 1920:	Best = 8.433826e+02	FES = 192100	Diversity = 9.9433e+00
Generation 1930:	Best = 8.433712e+02	FES = 193100	Diversity = 9.9421e+00
Generation 1940:	Best = 8.433712e+02	FES = 194100	Diversity = 9.8594e+00
Generation 1950:	Best = 8.433712e+02	FES = 195100	Diversity = 9.8570e+00
Generation 1960:	Best = 8.433705e+02	FES = 196100	Diversity = 9.8586e+00
Generation 1970:	Best = 8.433690e+02	FES = 197100	Diversity = 9.8569e+00
Generation 1980:	Best = 8.433682e+02	FES = 198100	Diversity = 9.8586e+00
Generation 1990:	Best = 8.433681e+02	FES = 199100	Diversity = 9.8395e+00
Generation 2000:	Best = 8.433681e+02	FES = 200100	Diversity = 9.8064e+00
Generation 2010:	Best = 8.433679e+02	FES = 201100	Diversity = 9.8006e+00
Generation 2020:	Best = 8.433679e+02	FES = 202100	Diversity = 9.8018e+00
Generation 2030:	Best = 8.433678e+02	FES = 203100	Diversity = 9.8097e+00
Generation 2040:	Best = 8.433678e+02	FES = 204100	Diversity = 9.8438e+00
Generation 2050:	Best = 8.433678e+02	FES = 205100	Diversity = 9.8344e+00
Generation 2060:	Best = 8.433678e+02	FES = 206100	Diversity = 9.8578e+00

Generation 2070:	Best = 8.433677e+02	FES = 207100	Diversity = 9.8548e+00
Generation 2080:	Best = 8.433677e+02	FES = 208100	Diversity = 9.8542e+00
Generation 2090:	Best = 8.433677e+02	FES = 209100	Diversity = 9.8303e+00
Generation 2100:	Best = 8.433677e+02	FES = 210100	Diversity = 9.8309e+00
Generation 2110:	Best = 8.433677e+02	FES = 211100	Diversity = 9.8310e+00
Generation 2120:	Best = 8.433677e+02	FES = 212100	Diversity = 9.8332e+00
Generation 2130:	Best = 8.433677e+02	FES = 213100	Diversity = 9.7963e+00
Generation 2140:	Best = 8.433677e+02	FES = 214100	Diversity = 9.7944e+00
Generation 2150:	Best = 8.433677e+02	FES = 215100	Diversity = 9.7942e+00
Generation 2160:	Best = 8.433677e+02	FES = 216100	Diversity = 9.7963e+00
Generation 2170:	Best = 8.433677e+02	FES = 217100	Diversity = 9.7740e+00
Generation 2180:	Best = 8.433677e+02	FES = 218100	Diversity = 9.7733e+00
Generation 2190:	Best = 8.433677e+02	FES = 219100	Diversity = 9.8107e+00
Generation 2200:	Best = 8.433677e+02	FES = 220100	Diversity = 9.8113e+00
Generation 2210:	Best = 8.433677e+02	FES = 221100	Diversity = 9.8115e+00
Generation 2220:	Best = 8.433677e+02	FES = 222100	Diversity = 9.8115e+00
Generation 2230:	Best = 8.433677e+02	FES = 223100	Diversity = 9.8119e+00
Generation 2240:	Best = 8.433677e+02	FES = 224100	Diversity = 9.7839e+00
Generation 2250:	Best = 8.433677e+02	FES = 225100	Diversity = 9.7834e+00
Generation 2260:	Best = 8.433677e+02	FES = 226100	Diversity = 9.7832e+00
Generation 2270:	Best = 8.433677e+02	FES = 227100	Diversity = 9.7813e+00
Generation 2280:	Best = 8.433677e+02	FES = 228100	Diversity = 9.7812e+00
Generation 2290:	Best = 8.433677e+02	FES = 229100	Diversity = 9.7806e+00
Generation 2300:	Best = 8.433677e+02	FES = 230100	Diversity = 9.8218e+00
Generation 2310:	Best = 8.433677e+02	FES = 231100	Diversity = 9.8064e+00
Generation 2320:	Best = 8.433677e+02	FES = 232100	Diversity = 9.8077e+00
Generation 2330:	Best = 8.433677e+02	FES = 233100	Diversity = 9.8047e+00
Generation 2340:	Best = 8.433677e+02	FES = 234100	Diversity = 9.8029e+00
Generation 2350:	Best = 8.433677e+02	FES = 235100	Diversity = 9.8027e+00
Generation 2360:	Best = 8.433677e+02	FES = 236100	Diversity = 9.8011e+00
Generation 2370:	Best = 8.433677e+02	FES = 237100	Diversity = 9.7998e+00
Generation 2380:	Best = 8.433677e+02	FES = 238100	Diversity = 9.7992e+00
Generation 2390:	Best = 8.433677e+02	FES = 239100	Diversity = 9.8001e+00
Generation 2400:	Best = 8.433677e+02	FES = 240100	Diversity = 9.7991e+00
Generation 2410:	Best = 8.433677e+02	FES = 241100	Diversity = 9.7450e+00
Generation 2420:	Best = 8.433677e+02	FES = 242100	Diversity = 9.7419e+00
Generation 2430:	Best = 8.433677e+02	FES = 243100	Diversity = 9.7422e+00
Generation 2440:	Best = 8.433677e+02	FES = 244100	Diversity = 9.7434e+00
Generation 2450:	Best = 8.433677e+02	FES = 245100	Diversity = 9.7416e+00
Generation 2460:	Best = 8.433677e+02	FES = 246100	Diversity = 9.7420e+00
Generation 2470:	Best = 8.433677e+02	FES = 247100	Diversity = 9.7786e+00
Generation 2480:	Best = 8.433677e+02	FES = 248100	Diversity = 9.7833e+00
Generation 2490:	Best = 8.433677e+02	FES = 249100	Diversity = 9.7044e+00
Generation 2500:	Best = 8.433677e+02	FES = 250100	Diversity = 9.7039e+00
Generation 2510:	Best = 8.433677e+02	FES = 251100	Diversity = 9.7042e+00
Generation 2520:	Best = 8.433677e+02	FES = 252100	Diversity = 9.6647e+00
Generation 2530:	Best = 8.433677e+02	FES = 253100	Diversity = 9.6420e+00
Generation 2540:	Best = 8.433677e+02	FES = 254100	Diversity = 9.6408e+00
Generation 2550:	Best = 8.433677e+02	FES = 255100	Diversity = 9.6073e+00
Generation 2560:	Best = 8.410945e+02	FES = 256100	Diversity = 9.5302e+00
Generation 2570:	Best = 8.379163e+02	FES = 257100	Diversity = 9.5299e+00
Generation 2580:	Best = 8.379083e+02	FES = 258100	Diversity = 9.5318e+00
Generation 2590:	Best = 8.379016e+02	FES = 259100	Diversity = 9.5132e+00
Generation 2600:	Best = 8.379015e+02	FES = 260100	Diversity = 9.5147e+00

Generation 2610:	Best = 8.379014e+02	FES = 261100	Diversity = 9.5112e+00
Generation 2620:	Best = 8.379009e+02	FES = 262100	Diversity = 9.4701e+00
Generation 2630:	Best = 8.379009e+02	FES = 263100	Diversity = 9.4362e+00
Generation 2640:	Best = 8.379009e+02	FES = 264100	Diversity = 9.4090e+00
Generation 2650:	Best = 8.379009e+02	FES = 265100	Diversity = 9.4577e+00
Generation 2660:	Best = 8.379009e+02	FES = 266100	Diversity = 9.4574e+00
Generation 2670:	Best = 8.379009e+02	FES = 267100	Diversity = 9.3453e+00
Generation 2680:	Best = 8.379009e+02	FES = 268100	Diversity = 9.3456e+00
Generation 2690:	Best = 8.379009e+02	FES = 269100	Diversity = 9.3047e+00
Generation 2700:	Best = 8.378834e+02	FES = 270100	Diversity = 9.3008e+00
Generation 2710:	Best = 8.378827e+02	FES = 271100	Diversity = 9.3249e+00
Generation 2720:	Best = 8.378827e+02	FES = 272100	Diversity = 9.3251e+00
Generation 2730:	Best = 8.378823e+02	FES = 273100	Diversity = 9.3265e+00
Generation 2740:	Best = 8.378818e+02	FES = 274100	Diversity = 9.3727e+00
Generation 2750:	Best = 8.378818e+02	FES = 275100	Diversity = 9.3759e+00
Generation 2760:	Best = 8.378818e+02	FES = 276100	Diversity = 9.3756e+00
Generation 2770:	Best = 8.378818e+02	FES = 277100	Diversity = 9.3756e+00
Generation 2780:	Best = 8.378818e+02	FES = 278100	Diversity = 9.3751e+00
Generation 2790:	Best = 8.378817e+02	FES = 279100	Diversity = 9.3765e+00
Generation 2800:	Best = 8.378817e+02	FES = 280100	Diversity = 9.3728e+00
Generation 2810:	Best = 8.378817e+02	FES = 281100	Diversity = 9.3225e+00
Generation 2820:	Best = 8.378817e+02	FES = 282100	Diversity = 9.3225e+00
Generation 2830:	Best = 8.378817e+02	FES = 283100	Diversity = 9.3240e+00
Generation 2840:	Best = 8.378817e+02	FES = 284100	Diversity = 9.3124e+00
Generation 2850:	Best = 8.378817e+02	FES = 285100	Diversity = 9.2911e+00
Generation 2860:	Best = 8.378817e+02	FES = 286100	Diversity = 9.2928e+00
Generation 2870:	Best = 8.378817e+02	FES = 287100	Diversity = 9.2247e+00
Generation 2880:	Best = 8.378817e+02	FES = 288100	Diversity = 9.2026e+00
Generation 2890:	Best = 8.378817e+02	FES = 289100	Diversity = 9.2042e+00
Generation 2900:	Best = 8.378627e+02	FES = 290100	Diversity = 9.2046e+00
Generation 2910:	Best = 8.378620e+02	FES = 291100	Diversity = 9.1958e+00
Generation 2920:	Best = 8.378620e+02	FES = 292100	Diversity = 9.1950e+00
Generation 2930:	Best = 8.378620e+02	FES = 293100	Diversity = 9.1950e+00
Generation 2940:	Best = 8.378620e+02	FES = 294100	Diversity = 9.1946e+00
Generation 2950:	Best = 8.378620e+02	FES = 295100	Diversity = 9.1608e+00
Generation 2960:	Best = 8.378620e+02	FES = 296100	Diversity = 9.0809e+00
Generation 2970:	Best = 8.378620e+02	FES = 297100	Diversity = 9.0810e+00
Generation 2980:	Best = 8.378620e+02	FES = 298100	Diversity = 9.0812e+00
Generation 2990:	Best = 8.378620e+02	FES = 299100	Diversity = 9.0815e+00

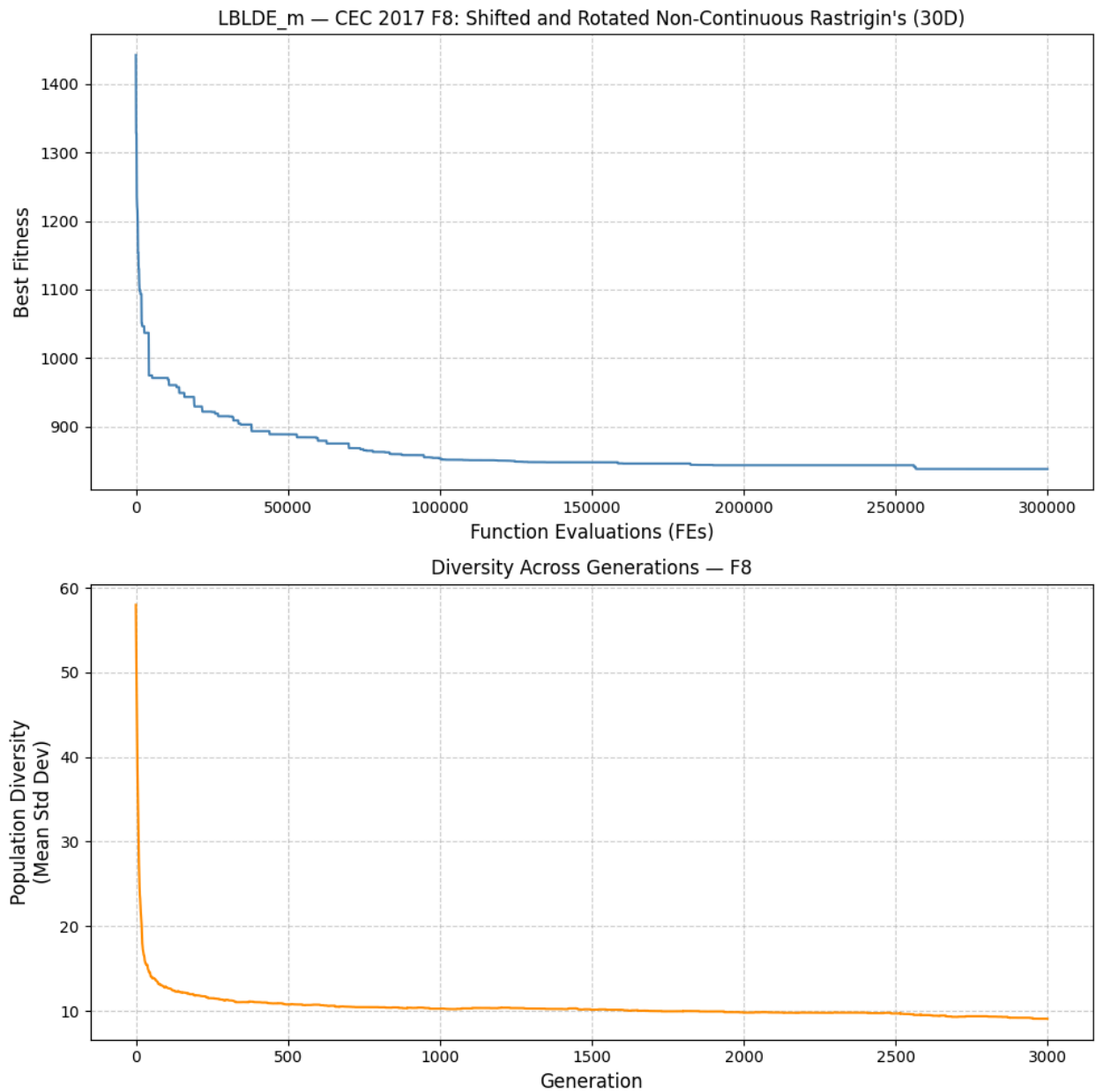
Optimization completed!

Final Best Fitness = 8.378620e+02

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 8.378620e+02
  Global Optimum         : 8.000000e+02
  Error (|best-optimum|) : 3.786196e+01
  Best Solution (5 dim)  : [ 13.54200637 -16.75755602  18.55573835 -20.5914708
-26.65394988]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F8.png

```
In [35]: # =====  
# F9 — Shifted and Rotated Levy  
# =====  
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and Rotated Levy')
```

=====

=

LBLDE_m on CEC 2017 F9 – Single Run

=====

=

Function : Shifted and Rotated Levy (30D)
Bounds : [-100, 100]
Global optimum : 900.0
Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 2.774722e+04 | Diversity = 5.7976e+01

Generation	10:	Best = 8.961470e+03	FES = 1100	Diversity = 2.7004e+01
Generation	20:	Best = 3.608951e+03	FES = 2100	Diversity = 1.3395e+01
Generation	30:	Best = 2.891153e+03	FES = 3100	Diversity = 9.5068e+00
Generation	40:	Best = 2.891153e+03	FES = 4100	Diversity = 8.3537e+00
Generation	50:	Best = 2.891153e+03	FES = 5100	Diversity = 7.6320e+00
Generation	60:	Best = 2.511211e+03	FES = 6100	Diversity = 6.9735e+00
Generation	70:	Best = 2.511211e+03	FES = 7100	Diversity = 6.8685e+00
Generation	80:	Best = 2.345968e+03	FES = 8100	Diversity = 6.5303e+00
Generation	90:	Best = 2.340171e+03	FES = 9100	Diversity = 6.3828e+00
Generation	100:	Best = 1.990622e+03	FES = 10100	Diversity = 6.2912e+00
Generation	110:	Best = 1.990622e+03	FES = 11100	Diversity = 6.2929e+00
Generation	120:	Best = 1.926507e+03	FES = 12100	Diversity = 6.2733e+00
Generation	130:	Best = 1.880999e+03	FES = 13100	Diversity = 6.1603e+00
Generation	140:	Best = 1.712088e+03	FES = 14100	Diversity = 6.0532e+00
Generation	150:	Best = 1.469765e+03	FES = 15100	Diversity = 6.2208e+00
Generation	160:	Best = 1.293469e+03	FES = 16100	Diversity = 6.3390e+00
Generation	170:	Best = 1.192351e+03	FES = 17100	Diversity = 5.4885e+00
Generation	180:	Best = 1.088405e+03	FES = 18100	Diversity = 4.5903e+00
Generation	190:	Best = 1.048561e+03	FES = 19100	Diversity = 3.9837e+00
Generation	200:	Best = 1.027619e+03	FES = 20100	Diversity = 3.3115e+00
Generation	210:	Best = 9.771510e+02	FES = 21100	Diversity = 2.6659e+00
Generation	220:	Best = 9.482717e+02	FES = 22100	Diversity = 2.1340e+00
Generation	230:	Best = 9.354652e+02	FES = 23100	Diversity = 1.5698e+00
Generation	240:	Best = 9.332561e+02	FES = 24100	Diversity = 1.4108e+00
Generation	250:	Best = 9.258320e+02	FES = 25100	Diversity = 1.2994e+00
Generation	260:	Best = 9.180929e+02	FES = 26100	Diversity = 1.1556e+00
Generation	270:	Best = 9.114298e+02	FES = 27100	Diversity = 1.0293e+00
Generation	280:	Best = 9.058359e+02	FES = 28100	Diversity = 8.0487e-01
Generation	290:	Best = 9.040225e+02	FES = 29100	Diversity = 6.1615e-01
Generation	300:	Best = 9.025391e+02	FES = 30100	Diversity = 4.9272e-01
Generation	310:	Best = 9.018052e+02	FES = 31100	Diversity = 3.8263e-01
Generation	320:	Best = 9.016250e+02	FES = 32100	Diversity = 2.4719e-01
Generation	330:	Best = 9.014571e+02	FES = 33100	Diversity = 1.6377e-01
Generation	340:	Best = 9.013404e+02	FES = 34100	Diversity = 7.4190e-02
Generation	350:	Best = 9.011566e+02	FES = 35100	Diversity = 6.5113e-02
Generation	360:	Best = 9.010847e+02	FES = 36100	Diversity = 4.0986e-02
Generation	370:	Best = 9.010559e+02	FES = 37100	Diversity = 2.4017e-02
Generation	380:	Best = 9.010314e+02	FES = 38100	Diversity = 1.7372e-02
Generation	390:	Best = 9.010173e+02	FES = 39100	Diversity = 1.4832e-02
Generation	400:	Best = 9.010043e+02	FES = 40100	Diversity = 1.1493e-02
Generation	410:	Best = 9.009990e+02	FES = 41100	Diversity = 8.0884e-03
Generation	420:	Best = 9.009924e+02	FES = 42100	Diversity = 7.0957e-03
Generation	430:	Best = 9.009871e+02	FES = 43100	Diversity = 6.2219e-03
Generation	440:	Best = 9.009840e+02	FES = 44100	Diversity = 4.6313e-03

Generation	450:	Best = 9.009816e+02	FES = 45100	Diversity = 3.7416e-03
Generation	460:	Best = 9.009790e+02	FES = 46100	Diversity = 3.4285e-03
Generation	470:	Best = 9.009773e+02	FES = 47100	Diversity = 2.7721e-03
Generation	480:	Best = 9.009760e+02	FES = 48100	Diversity = 2.2371e-03
Generation	490:	Best = 9.009751e+02	FES = 49100	Diversity = 1.9900e-03
Generation	500:	Best = 9.009744e+02	FES = 50100	Diversity = 1.5157e-03
Generation	510:	Best = 9.009737e+02	FES = 51100	Diversity = 1.1681e-03
Generation	520:	Best = 9.009731e+02	FES = 52100	Diversity = 1.1283e-03
Generation	530:	Best = 9.009726e+02	FES = 53100	Diversity = 9.3399e-04
Generation	540:	Best = 9.009722e+02	FES = 54100	Diversity = 9.9148e-04
Generation	550:	Best = 9.009718e+02	FES = 55100	Diversity = 8.5276e-04
Generation	560:	Best = 9.009716e+02	FES = 56100	Diversity = 7.3682e-04
Generation	570:	Best = 9.009713e+02	FES = 57100	Diversity = 7.0855e-04
Generation	580:	Best = 9.009711e+02	FES = 58100	Diversity = 5.1761e-04
Generation	590:	Best = 9.009710e+02	FES = 59100	Diversity = 4.1864e-04
Generation	600:	Best = 9.009710e+02	FES = 60100	Diversity = 3.6112e-04
Generation	610:	Best = 9.009709e+02	FES = 61100	Diversity = 3.0076e-04
Generation	620:	Best = 9.009708e+02	FES = 62100	Diversity = 2.7483e-04
Generation	630:	Best = 9.009708e+02	FES = 63100	Diversity = 2.3541e-04
Generation	640:	Best = 9.009707e+02	FES = 64100	Diversity = 1.7471e-04
Generation	650:	Best = 9.009707e+02	FES = 65100	Diversity = 2.1011e-04
Generation	660:	Best = 9.009707e+02	FES = 66100	Diversity = 1.5368e-04
Generation	670:	Best = 9.009707e+02	FES = 67100	Diversity = 1.3782e-04
Generation	680:	Best = 9.009707e+02	FES = 68100	Diversity = 1.6160e-04
Generation	690:	Best = 9.009706e+02	FES = 69100	Diversity = 1.1151e-04
Generation	700:	Best = 9.009706e+02	FES = 70100	Diversity = 8.1385e-05
Generation	710:	Best = 9.009706e+02	FES = 71100	Diversity = 7.6358e-05
Generation	720:	Best = 9.009706e+02	FES = 72100	Diversity = 7.2871e-05
Generation	730:	Best = 9.009706e+02	FES = 73100	Diversity = 6.2429e-05
Generation	740:	Best = 9.009706e+02	FES = 74100	Diversity = 6.8475e-05
Generation	750:	Best = 9.009706e+02	FES = 75100	Diversity = 6.9204e-05
Generation	760:	Best = 9.009706e+02	FES = 76100	Diversity = 7.8089e-05
Generation	770:	Best = 9.009706e+02	FES = 77100	Diversity = 7.0535e-05
Generation	780:	Best = 9.009706e+02	FES = 78100	Diversity = 6.5380e-05
Generation	790:	Best = 9.009706e+02	FES = 79100	Diversity = 5.4304e-05
Generation	800:	Best = 9.009706e+02	FES = 80100	Diversity = 5.8122e-05
Generation	810:	Best = 9.009706e+02	FES = 81100	Diversity = 3.8570e-05
Generation	820:	Best = 9.009706e+02	FES = 82100	Diversity = 3.8697e-05
Generation	830:	Best = 9.009706e+02	FES = 83100	Diversity = 2.8906e-05
Generation	840:	Best = 9.009706e+02	FES = 84100	Diversity = 2.3557e-05
Generation	850:	Best = 9.009706e+02	FES = 85100	Diversity = 2.4785e-05
Generation	860:	Best = 9.009706e+02	FES = 86100	Diversity = 1.9499e-05
Generation	870:	Best = 9.009706e+02	FES = 87100	Diversity = 1.1912e-05
Generation	880:	Best = 9.009706e+02	FES = 88100	Diversity = 1.0027e-05
Generation	890:	Best = 9.009706e+02	FES = 89100	Diversity = 9.3902e-06
Generation	900:	Best = 9.009706e+02	FES = 90100	Diversity = 6.1639e-06
Generation	910:	Best = 9.009706e+02	FES = 91100	Diversity = 3.8685e-06
Generation	920:	Best = 9.009706e+02	FES = 92100	Diversity = 4.1415e-06
Generation	930:	Best = 9.009706e+02	FES = 93100	Diversity = 2.1918e-06
Generation	940:	Best = 9.009706e+02	FES = 94100	Diversity = 2.4134e-06
Generation	950:	Best = 9.009706e+02	FES = 95100	Diversity = 2.2371e-06
Generation	960:	Best = 9.009706e+02	FES = 96100	Diversity = 1.5973e-06
Generation	970:	Best = 9.009706e+02	FES = 97100	Diversity = 9.3623e-07
Generation	980:	Best = 9.009706e+02	FES = 98100	Diversity = 1.0319e-06

Generation 990:	Best = 9.009706e+02	FES = 99100	Diversity = 8.6745e-07
Generation 1000:	Best = 9.009706e+02	FES = 100100	Diversity = 6.0029e-07
Generation 1010:	Best = 9.009706e+02	FES = 101100	Diversity = 6.9659e-07
Generation 1020:	Best = 9.009706e+02	FES = 102100	Diversity = 5.5474e-07
Generation 1030:	Best = 9.009706e+02	FES = 103100	Diversity = 4.8182e-07
Generation 1040:	Best = 9.009706e+02	FES = 104100	Diversity = 2.3044e-07
Generation 1050:	Best = 9.009706e+02	FES = 105100	Diversity = 2.1049e-07
Generation 1060:	Best = 9.009706e+02	FES = 106100	Diversity = 1.7043e-07
Generation 1070:	Best = 9.009706e+02	FES = 107100	Diversity = 1.6231e-07
Generation 1080:	Best = 9.009706e+02	FES = 108100	Diversity = 1.5128e-07
Generation 1090:	Best = 9.009706e+02	FES = 109100	Diversity = 1.0739e-07
Generation 1100:	Best = 9.009706e+02	FES = 110100	Diversity = 8.4840e-08
Generation 1110:	Best = 9.009706e+02	FES = 111100	Diversity = 9.5659e-08
Generation 1120:	Best = 9.009706e+02	FES = 112100	Diversity = 5.9899e-08
Generation 1130:	Best = 9.009706e+02	FES = 113100	Diversity = 4.6139e-08
Generation 1140:	Best = 9.009706e+02	FES = 114100	Diversity = 2.8739e-08
Generation 1150:	Best = 9.009706e+02	FES = 115100	Diversity = 2.4738e-08
Generation 1160:	Best = 9.009706e+02	FES = 116100	Diversity = 1.7297e-08
Generation 1170:	Best = 9.009706e+02	FES = 117100	Diversity = 1.1748e-08
Generation 1180:	Best = 9.009706e+02	FES = 118100	Diversity = 2.1338e-09
Generation 1190:	Best = 9.009706e+02	FES = 119100	Diversity = 9.1048e-12
Generation 1200:	Best = 9.009706e+02	FES = 120100	Diversity = 7.0162e-14
Generation 1210:	Best = 9.009706e+02	FES = 121100	Diversity = 4.0876e-14
Generation 1220:	Best = 9.009706e+02	FES = 122100	Diversity = 4.1042e-14
Generation 1230:	Best = 9.009706e+02	FES = 123100	Diversity = 4.1071e-14
Generation 1240:	Best = 9.009706e+02	FES = 124100	Diversity = 4.1071e-14
Generation 1250:	Best = 9.009706e+02	FES = 125100	Diversity = 4.1071e-14
Generation 1260:	Best = 9.009706e+02	FES = 126100	Diversity = 4.1071e-14
Generation 1270:	Best = 9.009706e+02	FES = 127100	Diversity = 4.1071e-14
Generation 1280:	Best = 9.009706e+02	FES = 128100	Diversity = 4.1071e-14
Generation 1290:	Best = 9.009706e+02	FES = 129100	Diversity = 4.1071e-14
Generation 1300:	Best = 9.009706e+02	FES = 130100	Diversity = 4.1071e-14
Generation 1310:	Best = 9.009706e+02	FES = 131100	Diversity = 4.1071e-14
Generation 1320:	Best = 9.009706e+02	FES = 132100	Diversity = 4.1071e-14
Generation 1330:	Best = 9.009706e+02	FES = 133100	Diversity = 4.1071e-14
Generation 1340:	Best = 9.009706e+02	FES = 134100	Diversity = 4.1071e-14
Generation 1350:	Best = 9.009706e+02	FES = 135100	Diversity = 4.1071e-14
Generation 1360:	Best = 9.009706e+02	FES = 136100	Diversity = 4.1071e-14
Generation 1370:	Best = 9.009706e+02	FES = 137100	Diversity = 4.1071e-14
Generation 1380:	Best = 9.009706e+02	FES = 138100	Diversity = 4.1071e-14
Generation 1390:	Best = 9.009706e+02	FES = 139100	Diversity = 4.1071e-14
Generation 1400:	Best = 9.009706e+02	FES = 140100	Diversity = 4.1071e-14
Generation 1410:	Best = 9.009706e+02	FES = 141100	Diversity = 4.1071e-14
Generation 1420:	Best = 9.009706e+02	FES = 142100	Diversity = 4.1071e-14
Generation 1430:	Best = 9.009706e+02	FES = 143100	Diversity = 4.1071e-14
Generation 1440:	Best = 9.009706e+02	FES = 144100	Diversity = 4.1071e-14
Generation 1450:	Best = 9.009706e+02	FES = 145100	Diversity = 4.1071e-14
Generation 1460:	Best = 9.009706e+02	FES = 146100	Diversity = 4.1071e-14
Generation 1470:	Best = 9.009706e+02	FES = 147100	Diversity = 4.1071e-14
Generation 1480:	Best = 9.009706e+02	FES = 148100	Diversity = 4.1071e-14
Generation 1490:	Best = 9.009706e+02	FES = 149100	Diversity = 4.1071e-14
Generation 1500:	Best = 9.009706e+02	FES = 150100	Diversity = 4.1071e-14
Generation 1510:	Best = 9.009706e+02	FES = 151100	Diversity = 4.1071e-14
Generation 1520:	Best = 9.009706e+02	FES = 152100	Diversity = 4.1071e-14

[illegible]

[illegible]

Generation 2610:	Best = 9.009706e+02	FES = 261100	Diversity = 4.1071e-14
Generation 2620:	Best = 9.009706e+02	FES = 262100	Diversity = 4.1071e-14
Generation 2630:	Best = 9.009706e+02	FES = 263100	Diversity = 4.1071e-14
Generation 2640:	Best = 9.009706e+02	FES = 264100	Diversity = 4.1071e-14
Generation 2650:	Best = 9.009706e+02	FES = 265100	Diversity = 4.1071e-14
Generation 2660:	Best = 9.009706e+02	FES = 266100	Diversity = 4.1071e-14
Generation 2670:	Best = 9.009706e+02	FES = 267100	Diversity = 4.1071e-14
Generation 2680:	Best = 9.009706e+02	FES = 268100	Diversity = 4.1071e-14
Generation 2690:	Best = 9.009706e+02	FES = 269100	Diversity = 4.1071e-14
Generation 2700:	Best = 9.009706e+02	FES = 270100	Diversity = 4.1071e-14
Generation 2710:	Best = 9.009706e+02	FES = 271100	Diversity = 4.1071e-14
Generation 2720:	Best = 9.009706e+02	FES = 272100	Diversity = 4.1071e-14
Generation 2730:	Best = 9.009706e+02	FES = 273100	Diversity = 4.1071e-14
Generation 2740:	Best = 9.009706e+02	FES = 274100	Diversity = 4.1071e-14
Generation 2750:	Best = 9.009706e+02	FES = 275100	Diversity = 4.1071e-14
Generation 2760:	Best = 9.009706e+02	FES = 276100	Diversity = 4.1071e-14
Generation 2770:	Best = 9.009706e+02	FES = 277100	Diversity = 4.1071e-14
Generation 2780:	Best = 9.009706e+02	FES = 278100	Diversity = 4.1071e-14
Generation 2790:	Best = 9.009706e+02	FES = 279100	Diversity = 4.1071e-14
Generation 2800:	Best = 9.009706e+02	FES = 280100	Diversity = 4.1071e-14
Generation 2810:	Best = 9.009706e+02	FES = 281100	Diversity = 4.1071e-14
Generation 2820:	Best = 9.009706e+02	FES = 282100	Diversity = 4.1071e-14
Generation 2830:	Best = 9.009706e+02	FES = 283100	Diversity = 4.1071e-14
Generation 2840:	Best = 9.009706e+02	FES = 284100	Diversity = 4.1071e-14
Generation 2850:	Best = 9.009706e+02	FES = 285100	Diversity = 4.1071e-14
Generation 2860:	Best = 9.009706e+02	FES = 286100	Diversity = 4.1071e-14
Generation 2870:	Best = 9.009706e+02	FES = 287100	Diversity = 4.1071e-14
Generation 2880:	Best = 9.009706e+02	FES = 288100	Diversity = 4.1071e-14
Generation 2890:	Best = 9.009706e+02	FES = 289100	Diversity = 4.1071e-14
Generation 2900:	Best = 9.009706e+02	FES = 290100	Diversity = 4.1071e-14
Generation 2910:	Best = 9.009706e+02	FES = 291100	Diversity = 4.1071e-14
Generation 2920:	Best = 9.009706e+02	FES = 292100	Diversity = 4.1071e-14
Generation 2930:	Best = 9.009706e+02	FES = 293100	Diversity = 4.1071e-14
Generation 2940:	Best = 9.009706e+02	FES = 294100	Diversity = 4.1071e-14
Generation 2950:	Best = 9.009706e+02	FES = 295100	Diversity = 4.1071e-14
Generation 2960:	Best = 9.009706e+02	FES = 296100	Diversity = 4.1071e-14
Generation 2970:	Best = 9.009706e+02	FES = 297100	Diversity = 4.1071e-14
Generation 2980:	Best = 9.009706e+02	FES = 298100	Diversity = 4.1071e-14
Generation 2990:	Best = 9.009706e+02	FES = 299100	Diversity = 4.1071e-14

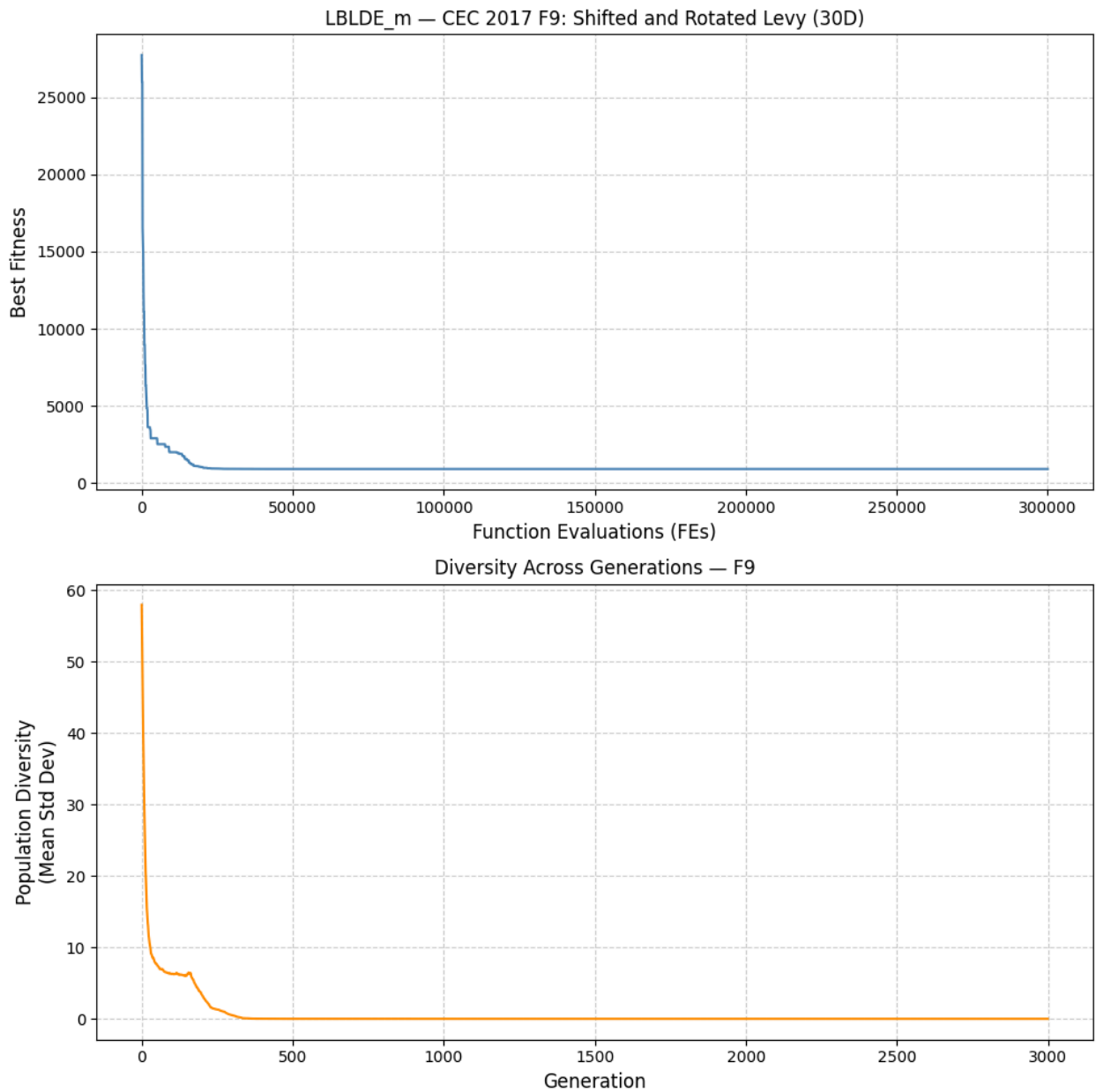
Optimization completed!

Final Best Fitness = 9.009706e+02

Total Function Evaluations = 300000

```
=====
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Results:
  Best Fitness           : 9.009706e+02
  Global Optimum         : 9.000000e+02
  Error (|best-optimum|) : 9.705759e-01
  Best Solution (5 dim)  : [-22.14625612   1.01249479  21.71307528  52.24615496
-16.94418268]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F9.png

```
In [36]: # =====  
# F10 — Shifted and Rotated Schwefel  
# =====  
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and Rotated Sch
```

=====

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LBLDE_m on CEC 2017 F10 – Single Run

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Function : Shifted and Rotated Schwefel's (30D)

Bounds : [-100, 100]

Global optimum : 1000.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 9.881110e+03 | Diversity = 5.7976e+01

Generation 10: Best = 9.577537e+03 FES = 1100 Diversity = 4.9919e+01

Generation 20: Best = 9.519649e+03 FES = 2100 Diversity = 4.7100e+01

Generation 30: Best = 9.219749e+03 FES = 3100 Diversity = 4.7038e+01

Generation 40: Best = 9.174364e+03 FES = 4100 Diversity = 4.7857e+01

Generation 50: Best = 8.889282e+03 FES = 5100 Diversity = 4.7447e+01

Generation 60: Best = 8.305010e+03 FES = 6100 Diversity = 4.7617e+01

Generation 70: Best = 8.202421e+03 FES = 7100 Diversity = 4.7382e+01

Generation 80: Best = 8.202421e+03 FES = 8100 Diversity = 4.7465e+01

Generation 90: Best = 8.046014e+03 FES = 9100 Diversity = 4.7197e+01

Generation 100: Best = 7.963080e+03 FES = 10100 Diversity = 4.7311e+01

Generation 110: Best = 7.867222e+03 FES = 11100 Diversity = 4.6962e+01

Generation 120: Best = 7.867222e+03 FES = 12100 Diversity = 4.6877e+01

Generation 130: Best = 7.798892e+03 FES = 13100 Diversity = 4.6913e+01

Generation 140: Best = 7.487243e+03 FES = 14100 Diversity = 4.6824e+01

Generation 150: Best = 7.487243e+03 FES = 15100 Diversity = 4.6843e+01

Generation 160: Best = 7.487243e+03 FES = 16100 Diversity = 4.6883e+01

Generation 170: Best = 7.487243e+03 FES = 17100 Diversity = 4.6360e+01

Generation 180: Best = 7.141196e+03 FES = 18100 Diversity = 4.5961e+01

Generation 190: Best = 7.141196e+03 FES = 19100 Diversity = 4.5945e+01

Generation 200: Best = 7.141196e+03 FES = 20100 Diversity = 4.5666e+01

Generation 210: Best = 7.028640e+03 FES = 21100 Diversity = 4.5311e+01

Generation 220: Best = 7.028640e+03 FES = 22100 Diversity = 4.5311e+01

Generation 230: Best = 7.028640e+03 FES = 23100 Diversity = 4.5604e+01

Generation 240: Best = 7.028640e+03 FES = 24100 Diversity = 4.5591e+01

Generation 250: Best = 7.028640e+03 FES = 25100 Diversity = 4.5514e+01

Generation 260: Best = 6.698617e+03 FES = 26100 Diversity = 4.5463e+01

Generation 270: Best = 6.698617e+03 FES = 27100 Diversity = 4.5278e+01

Generation 280: Best = 6.698617e+03 FES = 28100 Diversity = 4.5399e+01

Generation 290: Best = 6.698617e+03 FES = 29100 Diversity = 4.5214e+01

Generation 300: Best = 6.698617e+03 FES = 30100 Diversity = 4.5252e+01

Generation 310: Best = 6.698617e+03 FES = 31100 Diversity = 4.5119e+01

Generation 320: Best = 6.698617e+03 FES = 32100 Diversity = 4.4492e+01

Generation 330: Best = 6.698617e+03 FES = 33100 Diversity = 4.4576e+01

Generation 340: Best = 6.698617e+03 FES = 34100 Diversity = 4.4574e+01

Generation 350: Best = 6.698617e+03 FES = 35100 Diversity = 4.4639e+01

Generation 360: Best = 6.698617e+03 FES = 36100 Diversity = 4.4680e+01

Generation 370: Best = 6.698617e+03 FES = 37100 Diversity = 4.4653e+01

Generation 380: Best = 6.698617e+03 FES = 38100 Diversity = 4.4596e+01

Generation 390: Best = 6.698617e+03 FES = 39100 Diversity = 4.4595e+01

Generation 400: Best = 6.698617e+03 FES = 40100 Diversity = 4.4291e+01

Generation 410: Best = 6.578252e+03 FES = 41100 Diversity = 4.4168e+01

Generation 420: Best = 6.578252e+03 FES = 42100 Diversity = 4.4259e+01

Generation 430: Best = 6.578252e+03 FES = 43100 Diversity = 4.4254e+01

Generation 440: Best = 6.545294e+03 FES = 44100 Diversity = 4.4324e+01

Generation	450:	Best = 6.545294e+03	FES = 45100	Diversity = 4.4358e+01
Generation	460:	Best = 6.545294e+03	FES = 46100	Diversity = 4.4599e+01
Generation	470:	Best = 6.335542e+03	FES = 47100	Diversity = 4.4604e+01
Generation	480:	Best = 6.335542e+03	FES = 48100	Diversity = 4.4578e+01
Generation	490:	Best = 6.335542e+03	FES = 49100	Diversity = 4.4488e+01
Generation	500:	Best = 6.279692e+03	FES = 50100	Diversity = 4.4493e+01
Generation	510:	Best = 6.279692e+03	FES = 51100	Diversity = 4.4493e+01
Generation	520:	Best = 6.279692e+03	FES = 52100	Diversity = 4.4443e+01
Generation	530:	Best = 6.279692e+03	FES = 53100	Diversity = 4.4462e+01
Generation	540:	Best = 5.875771e+03	FES = 54100	Diversity = 4.4438e+01
Generation	550:	Best = 5.875771e+03	FES = 55100	Diversity = 4.4409e+01
Generation	560:	Best = 5.875771e+03	FES = 56100	Diversity = 4.4467e+01
Generation	570:	Best = 5.875771e+03	FES = 57100	Diversity = 4.4503e+01
Generation	580:	Best = 5.875771e+03	FES = 58100	Diversity = 4.4504e+01
Generation	590:	Best = 5.875771e+03	FES = 59100	Diversity = 4.4511e+01
Generation	600:	Best = 5.785122e+03	FES = 60100	Diversity = 4.4503e+01
Generation	610:	Best = 5.785122e+03	FES = 61100	Diversity = 4.4558e+01
Generation	620:	Best = 5.785122e+03	FES = 62100	Diversity = 4.4535e+01
Generation	630:	Best = 5.785122e+03	FES = 63100	Diversity = 4.4749e+01
Generation	640:	Best = 5.785122e+03	FES = 64100	Diversity = 4.4669e+01
Generation	650:	Best = 5.785122e+03	FES = 65100	Diversity = 4.4672e+01
Generation	660:	Best = 5.785122e+03	FES = 66100	Diversity = 4.4662e+01
Generation	670:	Best = 5.785122e+03	FES = 67100	Diversity = 4.4668e+01
Generation	680:	Best = 5.785122e+03	FES = 68100	Diversity = 4.4566e+01
Generation	690:	Best = 5.785122e+03	FES = 69100	Diversity = 4.4596e+01
Generation	700:	Best = 5.785122e+03	FES = 70100	Diversity = 4.4607e+01
Generation	710:	Best = 5.785122e+03	FES = 71100	Diversity = 4.4610e+01
Generation	720:	Best = 5.785122e+03	FES = 72100	Diversity = 4.4612e+01
Generation	730:	Best = 5.785122e+03	FES = 73100	Diversity = 4.4606e+01
Generation	740:	Best = 5.785122e+03	FES = 74100	Diversity = 4.4565e+01
Generation	750:	Best = 5.785122e+03	FES = 75100	Diversity = 4.4523e+01
Generation	760:	Best = 5.785122e+03	FES = 76100	Diversity = 4.4478e+01
Generation	770:	Best = 5.785122e+03	FES = 77100	Diversity = 4.4493e+01
Generation	780:	Best = 5.785122e+03	FES = 78100	Diversity = 4.4537e+01
Generation	790:	Best = 5.785122e+03	FES = 79100	Diversity = 4.4598e+01
Generation	800:	Best = 5.785122e+03	FES = 80100	Diversity = 4.4605e+01
Generation	810:	Best = 5.785122e+03	FES = 81100	Diversity = 4.4927e+01
Generation	820:	Best = 5.785122e+03	FES = 82100	Diversity = 4.4921e+01
Generation	830:	Best = 5.785122e+03	FES = 83100	Diversity = 4.4902e+01
Generation	840:	Best = 5.785122e+03	FES = 84100	Diversity = 4.4883e+01
Generation	850:	Best = 5.785122e+03	FES = 85100	Diversity = 4.4962e+01
Generation	860:	Best = 5.785122e+03	FES = 86100	Diversity = 4.4964e+01
Generation	870:	Best = 5.785122e+03	FES = 87100	Diversity = 4.4919e+01
Generation	880:	Best = 5.578383e+03	FES = 88100	Diversity = 4.4923e+01
Generation	890:	Best = 5.343708e+03	FES = 89100	Diversity = 4.4922e+01
Generation	900:	Best = 5.343708e+03	FES = 90100	Diversity = 4.4922e+01
Generation	910:	Best = 5.343708e+03	FES = 91100	Diversity = 4.4989e+01
Generation	920:	Best = 5.343708e+03	FES = 92100	Diversity = 4.5005e+01
Generation	930:	Best = 5.343708e+03	FES = 93100	Diversity = 4.4777e+01
Generation	940:	Best = 5.343708e+03	FES = 94100	Diversity = 4.4804e+01
Generation	950:	Best = 5.343708e+03	FES = 95100	Diversity = 4.4799e+01
Generation	960:	Best = 5.343708e+03	FES = 96100	Diversity = 4.4802e+01
Generation	970:	Best = 5.343708e+03	FES = 97100	Diversity = 4.4786e+01
Generation	980:	Best = 5.343708e+03	FES = 98100	Diversity = 4.4776e+01

Generation 990:	Best = 5.291352e+03	FES = 99100	Diversity = 4.4782e+01
Generation 1000:	Best = 5.291352e+03	FES = 100100	Diversity = 4.4772e+01
Generation 1010:	Best = 5.291352e+03	FES = 101100	Diversity = 4.4703e+01
Generation 1020:	Best = 5.291352e+03	FES = 102100	Diversity = 4.4679e+01
Generation 1030:	Best = 5.291352e+03	FES = 103100	Diversity = 4.4671e+01
Generation 1040:	Best = 5.291352e+03	FES = 104100	Diversity = 4.4645e+01
Generation 1050:	Best = 5.291352e+03	FES = 105100	Diversity = 4.4612e+01
Generation 1060:	Best = 5.291352e+03	FES = 106100	Diversity = 4.4587e+01
Generation 1070:	Best = 5.291352e+03	FES = 107100	Diversity = 4.4559e+01
Generation 1080:	Best = 5.291352e+03	FES = 108100	Diversity = 4.4537e+01
Generation 1090:	Best = 5.291352e+03	FES = 109100	Diversity = 4.4531e+01
Generation 1100:	Best = 5.291352e+03	FES = 110100	Diversity = 4.4505e+01
Generation 1110:	Best = 5.291352e+03	FES = 111100	Diversity = 4.4503e+01
Generation 1120:	Best = 5.291352e+03	FES = 112100	Diversity = 4.4604e+01
Generation 1130:	Best = 5.245025e+03	FES = 113100	Diversity = 4.4465e+01
Generation 1140:	Best = 5.245025e+03	FES = 114100	Diversity = 4.4467e+01
Generation 1150:	Best = 5.245025e+03	FES = 115100	Diversity = 4.4484e+01
Generation 1160:	Best = 5.245025e+03	FES = 116100	Diversity = 4.4483e+01
Generation 1170:	Best = 5.245025e+03	FES = 117100	Diversity = 4.4611e+01
Generation 1180:	Best = 5.245025e+03	FES = 118100	Diversity = 4.4611e+01
Generation 1190:	Best = 5.245025e+03	FES = 119100	Diversity = 4.4604e+01
Generation 1200:	Best = 5.245025e+03	FES = 120100	Diversity = 4.4653e+01
Generation 1210:	Best = 5.245025e+03	FES = 121100	Diversity = 4.4628e+01
Generation 1220:	Best = 5.090909e+03	FES = 122100	Diversity = 4.4637e+01
Generation 1230:	Best = 5.090909e+03	FES = 123100	Diversity = 4.4637e+01
Generation 1240:	Best = 5.090909e+03	FES = 124100	Diversity = 4.4612e+01
Generation 1250:	Best = 5.090909e+03	FES = 125100	Diversity = 4.4609e+01
Generation 1260:	Best = 5.090909e+03	FES = 126100	Diversity = 4.4602e+01
Generation 1270:	Best = 5.090909e+03	FES = 127100	Diversity = 4.4582e+01
Generation 1280:	Best = 5.090909e+03	FES = 128100	Diversity = 4.4587e+01
Generation 1290:	Best = 5.090909e+03	FES = 129100	Diversity = 4.4564e+01
Generation 1300:	Best = 5.090909e+03	FES = 130100	Diversity = 4.4563e+01
Generation 1310:	Best = 5.090783e+03	FES = 131100	Diversity = 4.4521e+01
Generation 1320:	Best = 5.090783e+03	FES = 132100	Diversity = 4.4528e+01
Generation 1330:	Best = 5.090783e+03	FES = 133100	Diversity = 4.4519e+01
Generation 1340:	Best = 5.090783e+03	FES = 134100	Diversity = 4.4329e+01
Generation 1350:	Best = 5.090783e+03	FES = 135100	Diversity = 4.4425e+01
Generation 1360:	Best = 5.090783e+03	FES = 136100	Diversity = 4.4426e+01
Generation 1370:	Best = 5.086695e+03	FES = 137100	Diversity = 4.4392e+01
Generation 1380:	Best = 5.052315e+03	FES = 138100	Diversity = 4.4405e+01
Generation 1390:	Best = 5.052315e+03	FES = 139100	Diversity = 4.4333e+01
Generation 1400:	Best = 5.052315e+03	FES = 140100	Diversity = 4.4281e+01
Generation 1410:	Best = 5.052315e+03	FES = 141100	Diversity = 4.4287e+01
Generation 1420:	Best = 5.052315e+03	FES = 142100	Diversity = 4.4278e+01
Generation 1430:	Best = 5.052315e+03	FES = 143100	Diversity = 4.4275e+01
Generation 1440:	Best = 5.052315e+03	FES = 144100	Diversity = 4.4265e+01
Generation 1450:	Best = 5.008715e+03	FES = 145100	Diversity = 4.4031e+01
Generation 1460:	Best = 5.008715e+03	FES = 146100	Diversity = 4.4026e+01
Generation 1470:	Best = 4.985312e+03	FES = 147100	Diversity = 4.3975e+01
Generation 1480:	Best = 4.980614e+03	FES = 148100	Diversity = 4.3970e+01
Generation 1490:	Best = 4.980614e+03	FES = 149100	Diversity = 4.3969e+01
Generation 1500:	Best = 4.980614e+03	FES = 150100	Diversity = 4.3954e+01
Generation 1510:	Best = 4.980614e+03	FES = 151100	Diversity = 4.3936e+01
Generation 1520:	Best = 4.980614e+03	FES = 152100	Diversity = 4.3686e+01

Generation 1530:	Best = 4.980614e+03	FES = 153100	Diversity = 4.3699e+01
Generation 1540:	Best = 4.980614e+03	FES = 154100	Diversity = 4.3698e+01
Generation 1550:	Best = 4.980614e+03	FES = 155100	Diversity = 4.3658e+01
Generation 1560:	Best = 4.980614e+03	FES = 156100	Diversity = 4.3654e+01
Generation 1570:	Best = 4.979869e+03	FES = 157100	Diversity = 4.3652e+01
Generation 1580:	Best = 4.979869e+03	FES = 158100	Diversity = 4.3650e+01
Generation 1590:	Best = 4.979869e+03	FES = 159100	Diversity = 4.3649e+01
Generation 1600:	Best = 4.979869e+03	FES = 160100	Diversity = 4.3644e+01
Generation 1610:	Best = 4.979869e+03	FES = 161100	Diversity = 4.3545e+01
Generation 1620:	Best = 4.979869e+03	FES = 162100	Diversity = 4.3546e+01
Generation 1630:	Best = 4.979869e+03	FES = 163100	Diversity = 4.3551e+01
Generation 1640:	Best = 4.979869e+03	FES = 164100	Diversity = 4.3552e+01
Generation 1650:	Best = 4.979869e+03	FES = 165100	Diversity = 4.3552e+01
Generation 1660:	Best = 4.950429e+03	FES = 166100	Diversity = 4.3575e+01
Generation 1670:	Best = 4.950429e+03	FES = 167100	Diversity = 4.3570e+01
Generation 1680:	Best = 4.950429e+03	FES = 168100	Diversity = 4.3591e+01
Generation 1690:	Best = 4.950429e+03	FES = 169100	Diversity = 4.3589e+01
Generation 1700:	Best = 4.950429e+03	FES = 170100	Diversity = 4.3586e+01
Generation 1710:	Best = 4.950429e+03	FES = 171100	Diversity = 4.3596e+01
Generation 1720:	Best = 4.950429e+03	FES = 172100	Diversity = 4.3578e+01
Generation 1730:	Best = 4.950429e+03	FES = 173100	Diversity = 4.3576e+01
Generation 1740:	Best = 4.950429e+03	FES = 174100	Diversity = 4.3455e+01
Generation 1750:	Best = 4.944772e+03	FES = 175100	Diversity = 4.3453e+01
Generation 1760:	Best = 4.944772e+03	FES = 176100	Diversity = 4.3443e+01
Generation 1770:	Best = 4.944772e+03	FES = 177100	Diversity = 4.3435e+01
Generation 1780:	Best = 4.944772e+03	FES = 178100	Diversity = 4.3429e+01
Generation 1790:	Best = 4.944772e+03	FES = 179100	Diversity = 4.3430e+01
Generation 1800:	Best = 4.944772e+03	FES = 180100	Diversity = 4.3430e+01
Generation 1810:	Best = 4.944772e+03	FES = 181100	Diversity = 4.3203e+01
Generation 1820:	Best = 4.944772e+03	FES = 182100	Diversity = 4.3204e+01
Generation 1830:	Best = 4.935403e+03	FES = 183100	Diversity = 4.3198e+01
Generation 1840:	Best = 4.935403e+03	FES = 184100	Diversity = 4.3199e+01
Generation 1850:	Best = 4.935403e+03	FES = 185100	Diversity = 4.3174e+01
Generation 1860:	Best = 4.935403e+03	FES = 186100	Diversity = 4.3175e+01
Generation 1870:	Best = 4.935403e+03	FES = 187100	Diversity = 4.3149e+01
Generation 1880:	Best = 4.935403e+03	FES = 188100	Diversity = 4.3148e+01
Generation 1890:	Best = 4.935403e+03	FES = 189100	Diversity = 4.3260e+01
Generation 1900:	Best = 4.935352e+03	FES = 190100	Diversity = 4.3260e+01
Generation 1910:	Best = 4.935352e+03	FES = 191100	Diversity = 4.3261e+01
Generation 1920:	Best = 4.935352e+03	FES = 192100	Diversity = 4.3252e+01
Generation 1930:	Best = 4.935352e+03	FES = 193100	Diversity = 4.3252e+01
Generation 1940:	Best = 4.935352e+03	FES = 194100	Diversity = 4.3252e+01
Generation 1950:	Best = 4.935352e+03	FES = 195100	Diversity = 4.3252e+01
Generation 1960:	Best = 4.935352e+03	FES = 196100	Diversity = 4.3258e+01
Generation 1970:	Best = 4.935352e+03	FES = 197100	Diversity = 4.3250e+01
Generation 1980:	Best = 4.935352e+03	FES = 198100	Diversity = 4.3251e+01
Generation 1990:	Best = 4.935352e+03	FES = 199100	Diversity = 4.3251e+01
Generation 2000:	Best = 4.935352e+03	FES = 200100	Diversity = 4.3251e+01
Generation 2010:	Best = 4.935352e+03	FES = 201100	Diversity = 4.3251e+01
Generation 2020:	Best = 4.935352e+03	FES = 202100	Diversity = 4.3178e+01
Generation 2030:	Best = 4.935352e+03	FES = 203100	Diversity = 4.3174e+01
Generation 2040:	Best = 4.935352e+03	FES = 204100	Diversity = 4.3174e+01
Generation 2050:	Best = 4.935352e+03	FES = 205100	Diversity = 4.3174e+01
Generation 2060:	Best = 4.935352e+03	FES = 206100	Diversity = 4.3167e+01

Generation 2070:	Best = 4.935352e+03	FES = 207100	Diversity = 4.3253e+01
Generation 2080:	Best = 4.935352e+03	FES = 208100	Diversity = 4.3223e+01
Generation 2090:	Best = 4.935352e+03	FES = 209100	Diversity = 4.3213e+01
Generation 2100:	Best = 4.935352e+03	FES = 210100	Diversity = 4.3211e+01
Generation 2110:	Best = 4.935352e+03	FES = 211100	Diversity = 4.3213e+01
Generation 2120:	Best = 4.935352e+03	FES = 212100	Diversity = 4.3215e+01
Generation 2130:	Best = 4.727597e+03	FES = 213100	Diversity = 4.3204e+01
Generation 2140:	Best = 4.727597e+03	FES = 214100	Diversity = 4.3194e+01
Generation 2150:	Best = 4.727597e+03	FES = 215100	Diversity = 4.3172e+01
Generation 2160:	Best = 4.727597e+03	FES = 216100	Diversity = 4.3172e+01
Generation 2170:	Best = 4.688096e+03	FES = 217100	Diversity = 4.3156e+01
Generation 2180:	Best = 4.688096e+03	FES = 218100	Diversity = 4.3134e+01
Generation 2190:	Best = 4.688096e+03	FES = 219100	Diversity = 4.3133e+01
Generation 2200:	Best = 4.688096e+03	FES = 220100	Diversity = 4.3134e+01
Generation 2210:	Best = 4.688096e+03	FES = 221100	Diversity = 4.3133e+01
Generation 2220:	Best = 4.688096e+03	FES = 222100	Diversity = 4.3132e+01
Generation 2230:	Best = 4.688096e+03	FES = 223100	Diversity = 4.3132e+01
Generation 2240:	Best = 4.688096e+03	FES = 224100	Diversity = 4.3131e+01
Generation 2250:	Best = 4.688096e+03	FES = 225100	Diversity = 4.3131e+01
Generation 2260:	Best = 4.688096e+03	FES = 226100	Diversity = 4.3133e+01
Generation 2270:	Best = 4.688096e+03	FES = 227100	Diversity = 4.3132e+01
Generation 2280:	Best = 4.688096e+03	FES = 228100	Diversity = 4.3129e+01
Generation 2290:	Best = 4.688096e+03	FES = 229100	Diversity = 4.3113e+01
Generation 2300:	Best = 4.688096e+03	FES = 230100	Diversity = 4.3105e+01
Generation 2310:	Best = 4.688096e+03	FES = 231100	Diversity = 4.3102e+01
Generation 2320:	Best = 4.688096e+03	FES = 232100	Diversity = 4.3095e+01
Generation 2330:	Best = 4.602375e+03	FES = 233100	Diversity = 4.3096e+01
Generation 2340:	Best = 4.602375e+03	FES = 234100	Diversity = 4.3091e+01
Generation 2350:	Best = 4.602375e+03	FES = 235100	Diversity = 4.3091e+01
Generation 2360:	Best = 4.602375e+03	FES = 236100	Diversity = 4.3088e+01
Generation 2370:	Best = 4.602375e+03	FES = 237100	Diversity = 4.3087e+01
Generation 2380:	Best = 4.602375e+03	FES = 238100	Diversity = 4.3079e+01
Generation 2390:	Best = 4.602375e+03	FES = 239100	Diversity = 4.3074e+01
Generation 2400:	Best = 4.602375e+03	FES = 240100	Diversity = 4.3073e+01
Generation 2410:	Best = 4.602375e+03	FES = 241100	Diversity = 4.3060e+01
Generation 2420:	Best = 4.602375e+03	FES = 242100	Diversity = 4.3056e+01
Generation 2430:	Best = 4.602375e+03	FES = 243100	Diversity = 4.3052e+01
Generation 2440:	Best = 4.602375e+03	FES = 244100	Diversity = 4.3052e+01
Generation 2450:	Best = 4.602375e+03	FES = 245100	Diversity = 4.3056e+01
Generation 2460:	Best = 4.602375e+03	FES = 246100	Diversity = 4.3056e+01
Generation 2470:	Best = 4.595022e+03	FES = 247100	Diversity = 4.3047e+01
Generation 2480:	Best = 4.595022e+03	FES = 248100	Diversity = 4.3016e+01
Generation 2490:	Best = 4.595022e+03	FES = 249100	Diversity = 4.3015e+01
Generation 2500:	Best = 4.595022e+03	FES = 250100	Diversity = 4.2913e+01
Generation 2510:	Best = 4.595022e+03	FES = 251100	Diversity = 4.2913e+01
Generation 2520:	Best = 4.595022e+03	FES = 252100	Diversity = 4.2916e+01
Generation 2530:	Best = 4.595022e+03	FES = 253100	Diversity = 4.2926e+01
Generation 2540:	Best = 4.595022e+03	FES = 254100	Diversity = 4.2923e+01
Generation 2550:	Best = 4.595022e+03	FES = 255100	Diversity = 4.2923e+01
Generation 2560:	Best = 4.595022e+03	FES = 256100	Diversity = 4.2925e+01
Generation 2570:	Best = 4.595022e+03	FES = 257100	Diversity = 4.2926e+01
Generation 2580:	Best = 4.595022e+03	FES = 258100	Diversity = 4.2921e+01
Generation 2590:	Best = 4.595022e+03	FES = 259100	Diversity = 4.2936e+01
Generation 2600:	Best = 4.595022e+03	FES = 260100	Diversity = 4.2946e+01

Generation 2610:	Best = 4.595022e+03	FES = 261100	Diversity = 4.2943e+01
Generation 2620:	Best = 4.595022e+03	FES = 262100	Diversity = 4.2943e+01
Generation 2630:	Best = 4.595022e+03	FES = 263100	Diversity = 4.2945e+01
Generation 2640:	Best = 4.595022e+03	FES = 264100	Diversity = 4.2944e+01
Generation 2650:	Best = 4.595022e+03	FES = 265100	Diversity = 4.2943e+01
Generation 2660:	Best = 4.595022e+03	FES = 266100	Diversity = 4.2942e+01
Generation 2670:	Best = 4.595022e+03	FES = 267100	Diversity = 4.2949e+01
Generation 2680:	Best = 4.595022e+03	FES = 268100	Diversity = 4.2949e+01
Generation 2690:	Best = 4.595022e+03	FES = 269100	Diversity = 4.2949e+01
Generation 2700:	Best = 4.595022e+03	FES = 270100	Diversity = 4.2947e+01
Generation 2710:	Best = 4.595022e+03	FES = 271100	Diversity = 4.2952e+01
Generation 2720:	Best = 4.595022e+03	FES = 272100	Diversity = 4.2952e+01
Generation 2730:	Best = 4.595022e+03	FES = 273100	Diversity = 4.2953e+01
Generation 2740:	Best = 4.595022e+03	FES = 274100	Diversity = 4.2953e+01
Generation 2750:	Best = 4.595022e+03	FES = 275100	Diversity = 4.2904e+01
Generation 2760:	Best = 4.595022e+03	FES = 276100	Diversity = 4.2904e+01
Generation 2770:	Best = 4.595022e+03	FES = 277100	Diversity = 4.2899e+01
Generation 2780:	Best = 4.595022e+03	FES = 278100	Diversity = 4.2922e+01
Generation 2790:	Best = 4.595022e+03	FES = 279100	Diversity = 4.2919e+01
Generation 2800:	Best = 4.595022e+03	FES = 280100	Diversity = 4.2920e+01
Generation 2810:	Best = 4.595022e+03	FES = 281100	Diversity = 4.2919e+01
Generation 2820:	Best = 4.595022e+03	FES = 282100	Diversity = 4.2918e+01
Generation 2830:	Best = 4.595022e+03	FES = 283100	Diversity = 4.2919e+01
Generation 2840:	Best = 4.595022e+03	FES = 284100	Diversity = 4.2939e+01
Generation 2850:	Best = 4.595022e+03	FES = 285100	Diversity = 4.2939e+01
Generation 2860:	Best = 4.595022e+03	FES = 286100	Diversity = 4.2939e+01
Generation 2870:	Best = 4.595022e+03	FES = 287100	Diversity = 4.2939e+01
Generation 2880:	Best = 4.595022e+03	FES = 288100	Diversity = 4.2937e+01
Generation 2890:	Best = 4.595022e+03	FES = 289100	Diversity = 4.2917e+01
Generation 2900:	Best = 4.595022e+03	FES = 290100	Diversity = 4.2916e+01
Generation 2910:	Best = 4.595022e+03	FES = 291100	Diversity = 4.2935e+01
Generation 2920:	Best = 4.595022e+03	FES = 292100	Diversity = 4.2935e+01
Generation 2930:	Best = 4.595022e+03	FES = 293100	Diversity = 4.2935e+01
Generation 2940:	Best = 4.595022e+03	FES = 294100	Diversity = 4.2933e+01
Generation 2950:	Best = 4.595022e+03	FES = 295100	Diversity = 4.2929e+01
Generation 2960:	Best = 4.595022e+03	FES = 296100	Diversity = 4.2932e+01
Generation 2970:	Best = 4.595022e+03	FES = 297100	Diversity = 4.2932e+01
Generation 2980:	Best = 4.595022e+03	FES = 298100	Diversity = 4.2924e+01
Generation 2990:	Best = 4.595022e+03	FES = 299100	Diversity = 4.2924e+01

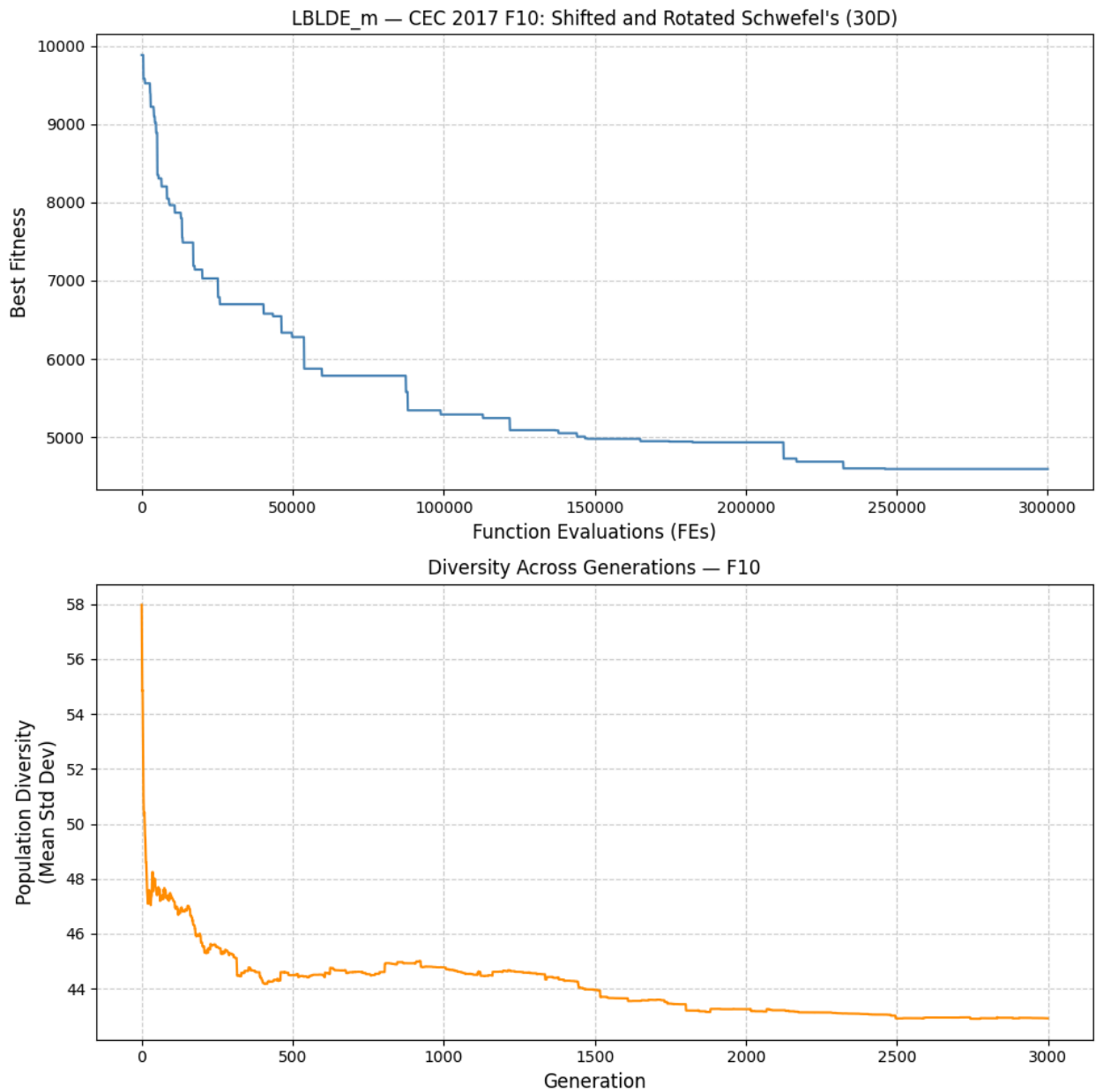
Optimization completed!

Final Best Fitness = 4.595022e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 4.595022e+03
  Global Optimum         : 1.000000e+03
  Error (|best-optimum|) : 3.595022e+03
  Best Solution (5 dim)  : [-8.25741543 25.47937889 54.97533444 38.11382008 7
8.48198089]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F10.png

6.3 Hybrid Functions (F11-F20)

```
In [37]: # =====  
# F11 – Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)  
# =====  
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid: Zakharov; Roser
```

=====

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LBLDE_m on CEC 2017 F11 – Single Run

=====

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Function : Hybrid: Zakharov; Rosenbrock; Rastrigin (30D)

Bounds : [-100, 100]

Global optimum : 1100.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 1.525880e+04 | Diversity = 5.7976e+01

Generation	10:	Best = 1.050780e+04	FES = 1100	Diversity = 3.6945e+01
Generation	20:	Best = 3.893727e+03	FES = 2100	Diversity = 2.1010e+01
Generation	30:	Best = 2.646429e+03	FES = 3100	Diversity = 1.3498e+01
Generation	40:	Best = 2.015285e+03	FES = 4100	Diversity = 8.6887e+00
Generation	50:	Best = 1.843732e+03	FES = 5100	Diversity = 5.9187e+00
Generation	60:	Best = 1.736010e+03	FES = 6100	Diversity = 4.2865e+00
Generation	70:	Best = 1.619711e+03	FES = 7100	Diversity = 3.6280e+00
Generation	80:	Best = 1.562480e+03	FES = 8100	Diversity = 2.7052e+00
Generation	90:	Best = 1.528429e+03	FES = 9100	Diversity = 1.1519e+00
Generation	100:	Best = 1.504257e+03	FES = 10100	Diversity = 6.0853e-01
Generation	110:	Best = 1.486992e+03	FES = 11100	Diversity = 6.0798e-01
Generation	120:	Best = 1.467496e+03	FES = 12100	Diversity = 6.8070e-01
Generation	130:	Best = 1.449455e+03	FES = 13100	Diversity = 8.0975e-01
Generation	140:	Best = 1.430135e+03	FES = 14100	Diversity = 6.7382e-01
Generation	150:	Best = 1.415992e+03	FES = 15100	Diversity = 6.1803e-01
Generation	160:	Best = 1.403218e+03	FES = 16100	Diversity = 7.2939e-01
Generation	170:	Best = 1.394849e+03	FES = 17100	Diversity = 6.8483e-01
Generation	180:	Best = 1.386856e+03	FES = 18100	Diversity = 4.9384e-01
Generation	190:	Best = 1.382312e+03	FES = 19100	Diversity = 4.1603e-01
Generation	200:	Best = 1.380578e+03	FES = 20100	Diversity = 2.6150e-01
Generation	210:	Best = 1.379329e+03	FES = 21100	Diversity = 2.3359e-01
Generation	220:	Best = 1.378547e+03	FES = 22100	Diversity = 1.4488e-01
Generation	230:	Best = 1.377827e+03	FES = 23100	Diversity = 1.3521e-01
Generation	240:	Best = 1.377352e+03	FES = 24100	Diversity = 1.1753e-01
Generation	250:	Best = 1.376920e+03	FES = 25100	Diversity = 1.2651e-01
Generation	260:	Best = 1.376466e+03	FES = 26100	Diversity = 1.2575e-01
Generation	270:	Best = 1.375977e+03	FES = 27100	Diversity = 6.5250e-02
Generation	280:	Best = 1.375626e+03	FES = 28100	Diversity = 6.3651e-02
Generation	290:	Best = 1.375306e+03	FES = 29100	Diversity = 7.9779e-02
Generation	300:	Best = 1.375116e+03	FES = 30100	Diversity = 4.0031e-02
Generation	310:	Best = 1.375033e+03	FES = 31100	Diversity = 2.8616e-02
Generation	320:	Best = 1.374975e+03	FES = 32100	Diversity = 2.7717e-02
Generation	330:	Best = 1.374921e+03	FES = 33100	Diversity = 2.6497e-02
Generation	340:	Best = 1.374878e+03	FES = 34100	Diversity = 3.1530e-02
Generation	350:	Best = 1.374851e+03	FES = 35100	Diversity = 2.2856e-02
Generation	360:	Best = 1.374826e+03	FES = 36100	Diversity = 2.0340e-02
Generation	370:	Best = 1.374803e+03	FES = 37100	Diversity = 2.3065e-02
Generation	380:	Best = 1.374768e+03	FES = 38100	Diversity = 2.0279e-02
Generation	390:	Best = 1.374740e+03	FES = 39100	Diversity = 2.1901e-02
Generation	400:	Best = 1.374717e+03	FES = 40100	Diversity = 1.6934e-02
Generation	410:	Best = 1.374706e+03	FES = 41100	Diversity = 1.3811e-02
Generation	420:	Best = 1.374696e+03	FES = 42100	Diversity = 8.9684e-03
Generation	430:	Best = 1.374688e+03	FES = 43100	Diversity = 8.0764e-03
Generation	440:	Best = 1.374683e+03	FES = 44100	Diversity = 5.5868e-03

Generation	450:	Best = 1.374681e+03	FES = 45100	Diversity = 4.6166e-03
Generation	460:	Best = 1.374678e+03	FES = 46100	Diversity = 4.7688e-03
Generation	470:	Best = 1.374677e+03	FES = 47100	Diversity = 2.8590e-03
Generation	480:	Best = 1.374676e+03	FES = 48100	Diversity = 1.3281e-03
Generation	490:	Best = 1.374675e+03	FES = 49100	Diversity = 1.0263e-03
Generation	500:	Best = 1.374675e+03	FES = 50100	Diversity = 1.5752e-03
Generation	510:	Best = 1.374675e+03	FES = 51100	Diversity = 1.2909e-03
Generation	520:	Best = 1.374674e+03	FES = 52100	Diversity = 1.0479e-03
Generation	530:	Best = 1.374674e+03	FES = 53100	Diversity = 1.2395e-03
Generation	540:	Best = 1.374674e+03	FES = 54100	Diversity = 5.8168e-04
Generation	550:	Best = 1.374674e+03	FES = 55100	Diversity = 5.2935e-04
Generation	560:	Best = 1.374674e+03	FES = 56100	Diversity = 3.7266e-04
Generation	570:	Best = 1.374674e+03	FES = 57100	Diversity = 3.7035e-04
Generation	580:	Best = 1.374674e+03	FES = 58100	Diversity = 2.0740e-04
Generation	590:	Best = 1.374674e+03	FES = 59100	Diversity = 1.3146e-04
Generation	600:	Best = 1.374674e+03	FES = 60100	Diversity = 1.8095e-04
Generation	610:	Best = 1.374674e+03	FES = 61100	Diversity = 1.7015e-04
Generation	620:	Best = 1.374674e+03	FES = 62100	Diversity = 1.1057e-04
Generation	630:	Best = 1.374674e+03	FES = 63100	Diversity = 5.8452e-05
Generation	640:	Best = 1.374674e+03	FES = 64100	Diversity = 4.9376e-05
Generation	650:	Best = 1.374674e+03	FES = 65100	Diversity = 3.4897e-05
Generation	660:	Best = 1.374674e+03	FES = 66100	Diversity = 2.3521e-05
Generation	670:	Best = 1.374674e+03	FES = 67100	Diversity = 1.7427e-05
Generation	680:	Best = 1.374674e+03	FES = 68100	Diversity = 1.0794e-05
Generation	690:	Best = 1.374674e+03	FES = 69100	Diversity = 4.4691e-06
Generation	700:	Best = 1.374674e+03	FES = 70100	Diversity = 4.9172e-06
Generation	710:	Best = 1.374674e+03	FES = 71100	Diversity = 3.5276e-06
Generation	720:	Best = 1.374674e+03	FES = 72100	Diversity = 2.5915e-06
Generation	730:	Best = 1.374674e+03	FES = 73100	Diversity = 1.5497e-06
Generation	740:	Best = 1.374674e+03	FES = 74100	Diversity = 1.1946e-06
Generation	750:	Best = 1.374674e+03	FES = 75100	Diversity = 6.3353e-07
Generation	760:	Best = 1.374674e+03	FES = 76100	Diversity = 4.5534e-07
Generation	770:	Best = 1.374674e+03	FES = 77100	Diversity = 3.0597e-07
Generation	780:	Best = 1.374674e+03	FES = 78100	Diversity = 1.5374e-07
Generation	790:	Best = 1.374674e+03	FES = 79100	Diversity = 1.7710e-07
Generation	800:	Best = 1.374674e+03	FES = 80100	Diversity = 8.3816e-08
Generation	810:	Best = 1.374674e+03	FES = 81100	Diversity = 6.5268e-08
Generation	820:	Best = 1.374674e+03	FES = 82100	Diversity = 2.4379e-08
Generation	830:	Best = 1.374674e+03	FES = 83100	Diversity = 1.5247e-08
Generation	840:	Best = 1.374674e+03	FES = 84100	Diversity = 1.2211e-08
Generation	850:	Best = 1.374674e+03	FES = 85100	Diversity = 1.8980e-09
Generation	860:	Best = 1.374674e+03	FES = 86100	Diversity = 3.2459e-10
Generation	870:	Best = 1.374674e+03	FES = 87100	Diversity = 1.9912e-11
Generation	880:	Best = 1.374674e+03	FES = 88100	Diversity = 7.0075e-14
Generation	890:	Best = 1.374674e+03	FES = 89100	Diversity = 2.8420e-14
Generation	900:	Best = 1.374674e+03	FES = 90100	Diversity = 2.8422e-14
Generation	910:	Best = 1.374674e+03	FES = 91100	Diversity = 2.8422e-14
Generation	920:	Best = 1.374674e+03	FES = 92100	Diversity = 2.8422e-14
Generation	930:	Best = 1.374674e+03	FES = 93100	Diversity = 2.8422e-14
Generation	940:	Best = 1.374674e+03	FES = 94100	Diversity = 2.8422e-14
Generation	950:	Best = 1.374674e+03	FES = 95100	Diversity = 2.8422e-14
Generation	960:	Best = 1.374674e+03	FES = 96100	Diversity = 2.8422e-14
Generation	970:	Best = 1.374674e+03	FES = 97100	Diversity = 2.8422e-14
Generation	980:	Best = 1.374674e+03	FES = 98100	Diversity = 2.8422e-14

[illegible]

[illegible]

[illegible]

Generation 2610:	Best = 1.374674e+03	FES = 261100	Diversity = 2.8422e-14
Generation 2620:	Best = 1.374674e+03	FES = 262100	Diversity = 2.8422e-14
Generation 2630:	Best = 1.374674e+03	FES = 263100	Diversity = 2.8422e-14
Generation 2640:	Best = 1.374674e+03	FES = 264100	Diversity = 2.8422e-14
Generation 2650:	Best = 1.374674e+03	FES = 265100	Diversity = 2.8422e-14
Generation 2660:	Best = 1.374674e+03	FES = 266100	Diversity = 2.8422e-14
Generation 2670:	Best = 1.374674e+03	FES = 267100	Diversity = 2.8422e-14
Generation 2680:	Best = 1.374674e+03	FES = 268100	Diversity = 2.8422e-14
Generation 2690:	Best = 1.374674e+03	FES = 269100	Diversity = 2.8422e-14
Generation 2700:	Best = 1.374674e+03	FES = 270100	Diversity = 2.8422e-14
Generation 2710:	Best = 1.374674e+03	FES = 271100	Diversity = 2.8422e-14
Generation 2720:	Best = 1.374674e+03	FES = 272100	Diversity = 2.8422e-14
Generation 2730:	Best = 1.374674e+03	FES = 273100	Diversity = 2.8422e-14
Generation 2740:	Best = 1.374674e+03	FES = 274100	Diversity = 2.8422e-14
Generation 2750:	Best = 1.374674e+03	FES = 275100	Diversity = 2.8422e-14
Generation 2760:	Best = 1.374674e+03	FES = 276100	Diversity = 2.8422e-14
Generation 2770:	Best = 1.374674e+03	FES = 277100	Diversity = 2.8422e-14
Generation 2780:	Best = 1.374674e+03	FES = 278100	Diversity = 2.8422e-14
Generation 2790:	Best = 1.374674e+03	FES = 279100	Diversity = 2.8422e-14
Generation 2800:	Best = 1.374674e+03	FES = 280100	Diversity = 2.8422e-14
Generation 2810:	Best = 1.374674e+03	FES = 281100	Diversity = 2.8422e-14
Generation 2820:	Best = 1.374674e+03	FES = 282100	Diversity = 2.8422e-14
Generation 2830:	Best = 1.374674e+03	FES = 283100	Diversity = 2.8422e-14
Generation 2840:	Best = 1.374674e+03	FES = 284100	Diversity = 2.8422e-14
Generation 2850:	Best = 1.374674e+03	FES = 285100	Diversity = 2.8422e-14
Generation 2860:	Best = 1.374674e+03	FES = 286100	Diversity = 2.8422e-14
Generation 2870:	Best = 1.374674e+03	FES = 287100	Diversity = 2.8422e-14
Generation 2880:	Best = 1.374674e+03	FES = 288100	Diversity = 2.8422e-14
Generation 2890:	Best = 1.374674e+03	FES = 289100	Diversity = 2.8422e-14
Generation 2900:	Best = 1.374674e+03	FES = 290100	Diversity = 2.8422e-14
Generation 2910:	Best = 1.374674e+03	FES = 291100	Diversity = 2.8422e-14
Generation 2920:	Best = 1.374674e+03	FES = 292100	Diversity = 2.8422e-14
Generation 2930:	Best = 1.374674e+03	FES = 293100	Diversity = 2.8422e-14
Generation 2940:	Best = 1.374674e+03	FES = 294100	Diversity = 2.8422e-14
Generation 2950:	Best = 1.374674e+03	FES = 295100	Diversity = 2.8422e-14
Generation 2960:	Best = 1.374674e+03	FES = 296100	Diversity = 2.8422e-14
Generation 2970:	Best = 1.374674e+03	FES = 297100	Diversity = 2.8422e-14
Generation 2980:	Best = 1.374674e+03	FES = 298100	Diversity = 2.8422e-14
Generation 2990:	Best = 1.374674e+03	FES = 299100	Diversity = 2.8422e-14

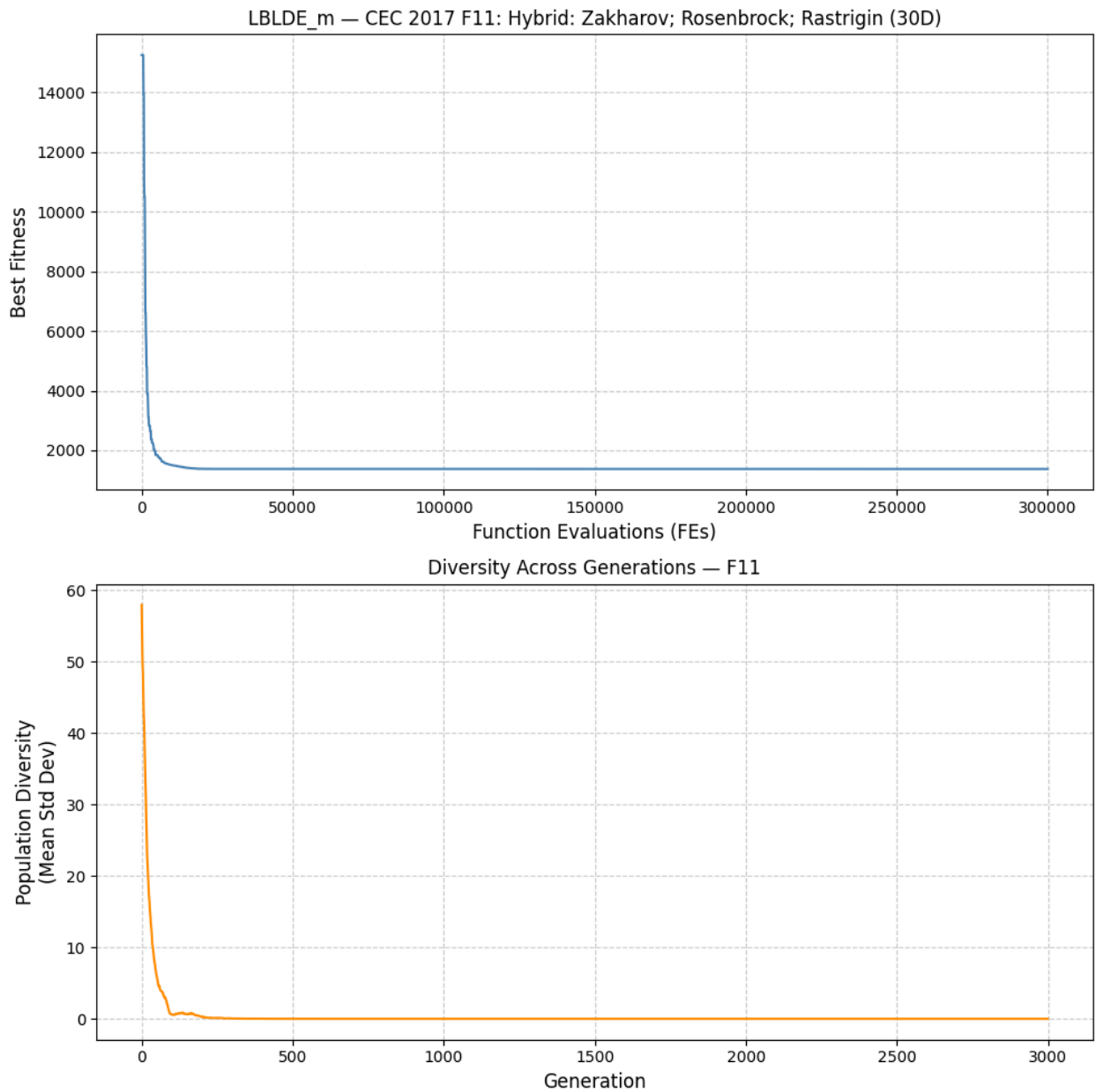
Optimization completed!

Final Best Fitness = 1.374674e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 1.374674e+03
  Global Optimum         : 1.100000e+03
  Error (|best-optimum|) : 2.746740e+02
  Best Solution (5 dim)  : [-15.79350584 -25.24779017  33.23213363  37.43008022
12.85781988]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F11.png

```
In [38]: # =====  
# F12 – Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)  
# =====  
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid: Elliptic; Schwe
```

=====

=

LBLDE_m on CEC 2017 F12 – Single Run

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Function : Hybrid: Elliptic; Schwefel; Bent Cigar (30D)
Bounds : [-100, 100]
Global optimum : 1200.0
Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 1.806776e+11 | Diversity = 5.7976e+01

Generation	10:	Best = 2.735263e+10	FES = 1100	Diversity = 2.3727e+01
Generation	20:	Best = 7.247495e+09	FES = 2100	Diversity = 1.0909e+01
Generation	30:	Best = 5.637403e+09	FES = 3100	Diversity = 6.1081e+00
Generation	40:	Best = 4.419309e+09	FES = 4100	Diversity = 4.0211e+00
Generation	50:	Best = 3.764370e+09	FES = 5100	Diversity = 3.0354e+00
Generation	60:	Best = 3.528186e+09	FES = 6100	Diversity = 2.6700e+00
Generation	70:	Best = 3.321066e+09	FES = 7100	Diversity = 2.7642e+00
Generation	80:	Best = 3.201017e+09	FES = 8100	Diversity = 2.6395e+00
Generation	90:	Best = 3.064013e+09	FES = 9100	Diversity = 3.2952e+00
Generation	100:	Best = 2.983512e+09	FES = 10100	Diversity = 3.2127e+00
Generation	110:	Best = 2.914421e+09	FES = 11100	Diversity = 3.7073e+00
Generation	120:	Best = 2.882315e+09	FES = 12100	Diversity = 4.0597e+00
Generation	130:	Best = 2.854978e+09	FES = 13100	Diversity = 3.8166e+00
Generation	140:	Best = 2.828234e+09	FES = 14100	Diversity = 4.1100e+00
Generation	150:	Best = 2.805733e+09	FES = 15100	Diversity = 4.6300e+00
Generation	160:	Best = 2.791502e+09	FES = 16100	Diversity = 4.1480e+00
Generation	170:	Best = 2.770955e+09	FES = 17100	Diversity = 5.2284e+00
Generation	180:	Best = 2.747485e+09	FES = 18100	Diversity = 6.3888e+00
Generation	190:	Best = 2.735975e+09	FES = 19100	Diversity = 4.7335e+00
Generation	200:	Best = 2.727385e+09	FES = 20100	Diversity = 4.5208e+00
Generation	210:	Best = 2.719940e+09	FES = 21100	Diversity = 5.2467e+00
Generation	220:	Best = 2.712940e+09	FES = 22100	Diversity = 5.2847e+00
Generation	230:	Best = 2.707758e+09	FES = 23100	Diversity = 5.1772e+00
Generation	240:	Best = 2.704705e+09	FES = 24100	Diversity = 5.0117e+00
Generation	250:	Best = 2.702149e+09	FES = 25100	Diversity = 5.0484e+00
Generation	260:	Best = 2.700675e+09	FES = 26100	Diversity = 4.6960e+00
Generation	270:	Best = 2.699048e+09	FES = 27100	Diversity = 4.3471e+00
Generation	280:	Best = 2.697281e+09	FES = 28100	Diversity = 5.6040e+00
Generation	290:	Best = 2.695617e+09	FES = 29100	Diversity = 5.9398e+00
Generation	300:	Best = 2.693965e+09	FES = 30100	Diversity = 6.3846e+00
Generation	310:	Best = 2.692331e+09	FES = 31100	Diversity = 6.0417e+00
Generation	320:	Best = 2.690698e+09	FES = 32100	Diversity = 5.8131e+00
Generation	330:	Best = 2.689068e+09	FES = 33100	Diversity = 6.3102e+00
Generation	340:	Best = 2.687523e+09	FES = 34100	Diversity = 6.6633e+00
Generation	350:	Best = 2.685822e+09	FES = 35100	Diversity = 6.3040e+00
Generation	360:	Best = 2.683280e+09	FES = 36100	Diversity = 6.1455e+00
Generation	370:	Best = 2.682213e+09	FES = 37100	Diversity = 5.8648e+00
Generation	380:	Best = 2.681587e+09	FES = 38100	Diversity = 5.0404e+00
Generation	390:	Best = 2.680933e+09	FES = 39100	Diversity = 4.0705e+00
Generation	400:	Best = 2.680485e+09	FES = 40100	Diversity = 4.2573e+00
Generation	410:	Best = 2.680016e+09	FES = 41100	Diversity = 3.1211e+00
Generation	420:	Best = 2.679662e+09	FES = 42100	Diversity = 4.1929e+00
Generation	430:	Best = 2.679430e+09	FES = 43100	Diversity = 4.7475e+00
Generation	440:	Best = 2.679152e+09	FES = 44100	Diversity = 3.3609e+00

Generation	450:	Best = 2.678895e+09	FES = 45100	Diversity = 3.8238e+00
Generation	460:	Best = 2.678602e+09	FES = 46100	Diversity = 3.3198e+00
Generation	470:	Best = 2.678344e+09	FES = 47100	Diversity = 2.7279e+00
Generation	480:	Best = 2.678129e+09	FES = 48100	Diversity = 3.1485e+00
Generation	490:	Best = 2.677946e+09	FES = 49100	Diversity = 2.9474e+00
Generation	500:	Best = 2.677858e+09	FES = 50100	Diversity = 2.5306e+00
Generation	510:	Best = 2.677782e+09	FES = 51100	Diversity = 2.3090e+00
Generation	520:	Best = 2.677694e+09	FES = 52100	Diversity = 2.5371e+00
Generation	530:	Best = 2.677615e+09	FES = 53100	Diversity = 2.4514e+00
Generation	540:	Best = 2.677570e+09	FES = 54100	Diversity = 1.9084e+00
Generation	550:	Best = 2.677515e+09	FES = 55100	Diversity = 1.4116e+00
Generation	560:	Best = 2.677477e+09	FES = 56100	Diversity = 1.6935e+00
Generation	570:	Best = 2.677412e+09	FES = 57100	Diversity = 2.1050e+00
Generation	580:	Best = 2.677370e+09	FES = 58100	Diversity = 2.0258e+00
Generation	590:	Best = 2.677330e+09	FES = 59100	Diversity = 2.1016e+00
Generation	600:	Best = 2.677301e+09	FES = 60100	Diversity = 1.7482e+00
Generation	610:	Best = 2.677268e+09	FES = 61100	Diversity = 1.3945e+00
Generation	620:	Best = 2.677243e+09	FES = 62100	Diversity = 1.2431e+00
Generation	630:	Best = 2.677220e+09	FES = 63100	Diversity = 1.3640e+00
Generation	640:	Best = 2.677199e+09	FES = 64100	Diversity = 9.6413e-01
Generation	650:	Best = 2.677190e+09	FES = 65100	Diversity = 8.9709e-01
Generation	660:	Best = 2.677180e+09	FES = 66100	Diversity = 7.4959e-01
Generation	670:	Best = 2.677175e+09	FES = 67100	Diversity = 3.3156e-01
Generation	680:	Best = 2.677172e+09	FES = 68100	Diversity = 3.3224e-01
Generation	690:	Best = 2.677168e+09	FES = 69100	Diversity = 3.0719e-01
Generation	700:	Best = 2.677167e+09	FES = 70100	Diversity = 1.2217e-01
Generation	710:	Best = 2.677166e+09	FES = 71100	Diversity = 1.0256e-01
Generation	720:	Best = 2.677166e+09	FES = 72100	Diversity = 7.7010e-02
Generation	730:	Best = 2.677165e+09	FES = 73100	Diversity = 9.5204e-02
Generation	740:	Best = 2.677165e+09	FES = 74100	Diversity = 7.2405e-02
Generation	750:	Best = 2.677164e+09	FES = 75100	Diversity = 5.7494e-02
Generation	760:	Best = 2.677164e+09	FES = 76100	Diversity = 3.0190e-02
Generation	770:	Best = 2.677164e+09	FES = 77100	Diversity = 1.5680e-02
Generation	780:	Best = 2.677164e+09	FES = 78100	Diversity = 1.3084e-02
Generation	790:	Best = 2.677164e+09	FES = 79100	Diversity = 8.2381e-03
Generation	800:	Best = 2.677164e+09	FES = 80100	Diversity = 4.6421e-03
Generation	810:	Best = 2.677164e+09	FES = 81100	Diversity = 5.4344e-03
Generation	820:	Best = 2.677164e+09	FES = 82100	Diversity = 2.1631e-03
Generation	830:	Best = 2.677164e+09	FES = 83100	Diversity = 1.8089e-03
Generation	840:	Best = 2.677164e+09	FES = 84100	Diversity = 1.5046e-03
Generation	850:	Best = 2.677164e+09	FES = 85100	Diversity = 6.8007e-04
Generation	860:	Best = 2.677164e+09	FES = 86100	Diversity = 6.3462e-04
Generation	870:	Best = 2.677164e+09	FES = 87100	Diversity = 3.4347e-04
Generation	880:	Best = 2.677164e+09	FES = 88100	Diversity = 1.9424e-04
Generation	890:	Best = 2.677164e+09	FES = 89100	Diversity = 1.5941e-04
Generation	900:	Best = 2.677164e+09	FES = 90100	Diversity = 1.3563e-04
Generation	910:	Best = 2.677164e+09	FES = 91100	Diversity = 7.2194e-05
Generation	920:	Best = 2.677164e+09	FES = 92100	Diversity = 5.6556e-05
Generation	930:	Best = 2.677164e+09	FES = 93100	Diversity = 4.3232e-05
Generation	940:	Best = 2.677164e+09	FES = 94100	Diversity = 2.6121e-05
Generation	950:	Best = 2.677164e+09	FES = 95100	Diversity = 1.9974e-05
Generation	960:	Best = 2.677164e+09	FES = 96100	Diversity = 9.3136e-06
Generation	970:	Best = 2.677164e+09	FES = 97100	Diversity = 8.2875e-06
Generation	980:	Best = 2.677164e+09	FES = 98100	Diversity = 4.8912e-06

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Generation 2610:	Best = 2.677164e+09	FES = 261100	Diversity = 5.1703e-14
Generation 2620:	Best = 2.677164e+09	FES = 262100	Diversity = 5.1703e-14
Generation 2630:	Best = 2.677164e+09	FES = 263100	Diversity = 5.1703e-14
Generation 2640:	Best = 2.677164e+09	FES = 264100	Diversity = 5.1703e-14
Generation 2650:	Best = 2.677164e+09	FES = 265100	Diversity = 5.1703e-14
Generation 2660:	Best = 2.677164e+09	FES = 266100	Diversity = 5.1703e-14
Generation 2670:	Best = 2.677164e+09	FES = 267100	Diversity = 5.1703e-14
Generation 2680:	Best = 2.677164e+09	FES = 268100	Diversity = 5.1703e-14
Generation 2690:	Best = 2.677164e+09	FES = 269100	Diversity = 5.1703e-14
Generation 2700:	Best = 2.677164e+09	FES = 270100	Diversity = 5.1703e-14
Generation 2710:	Best = 2.677164e+09	FES = 271100	Diversity = 5.1703e-14
Generation 2720:	Best = 2.677164e+09	FES = 272100	Diversity = 5.1703e-14
Generation 2730:	Best = 2.677164e+09	FES = 273100	Diversity = 5.1703e-14
Generation 2740:	Best = 2.677164e+09	FES = 274100	Diversity = 5.1703e-14
Generation 2750:	Best = 2.677164e+09	FES = 275100	Diversity = 5.1703e-14
Generation 2760:	Best = 2.677164e+09	FES = 276100	Diversity = 5.1703e-14
Generation 2770:	Best = 2.677164e+09	FES = 277100	Diversity = 5.1703e-14
Generation 2780:	Best = 2.677164e+09	FES = 278100	Diversity = 5.1703e-14
Generation 2790:	Best = 2.677164e+09	FES = 279100	Diversity = 5.1703e-14
Generation 2800:	Best = 2.677164e+09	FES = 280100	Diversity = 5.1703e-14
Generation 2810:	Best = 2.677164e+09	FES = 281100	Diversity = 5.1703e-14
Generation 2820:	Best = 2.677164e+09	FES = 282100	Diversity = 5.1703e-14
Generation 2830:	Best = 2.677164e+09	FES = 283100	Diversity = 5.1703e-14
Generation 2840:	Best = 2.677164e+09	FES = 284100	Diversity = 5.1703e-14
Generation 2850:	Best = 2.677164e+09	FES = 285100	Diversity = 5.1703e-14
Generation 2860:	Best = 2.677164e+09	FES = 286100	Diversity = 5.1703e-14
Generation 2870:	Best = 2.677164e+09	FES = 287100	Diversity = 5.1703e-14
Generation 2880:	Best = 2.677164e+09	FES = 288100	Diversity = 5.1703e-14
Generation 2890:	Best = 2.677164e+09	FES = 289100	Diversity = 5.1703e-14
Generation 2900:	Best = 2.677164e+09	FES = 290100	Diversity = 5.1703e-14
Generation 2910:	Best = 2.677164e+09	FES = 291100	Diversity = 5.1703e-14
Generation 2920:	Best = 2.677164e+09	FES = 292100	Diversity = 5.1703e-14
Generation 2930:	Best = 2.677164e+09	FES = 293100	Diversity = 5.1703e-14
Generation 2940:	Best = 2.677164e+09	FES = 294100	Diversity = 5.1703e-14
Generation 2950:	Best = 2.677164e+09	FES = 295100	Diversity = 5.1703e-14
Generation 2960:	Best = 2.677164e+09	FES = 296100	Diversity = 5.1703e-14
Generation 2970:	Best = 2.677164e+09	FES = 297100	Diversity = 5.1703e-14
Generation 2980:	Best = 2.677164e+09	FES = 298100	Diversity = 5.1703e-14
Generation 2990:	Best = 2.677164e+09	FES = 299100	Diversity = 5.1703e-14

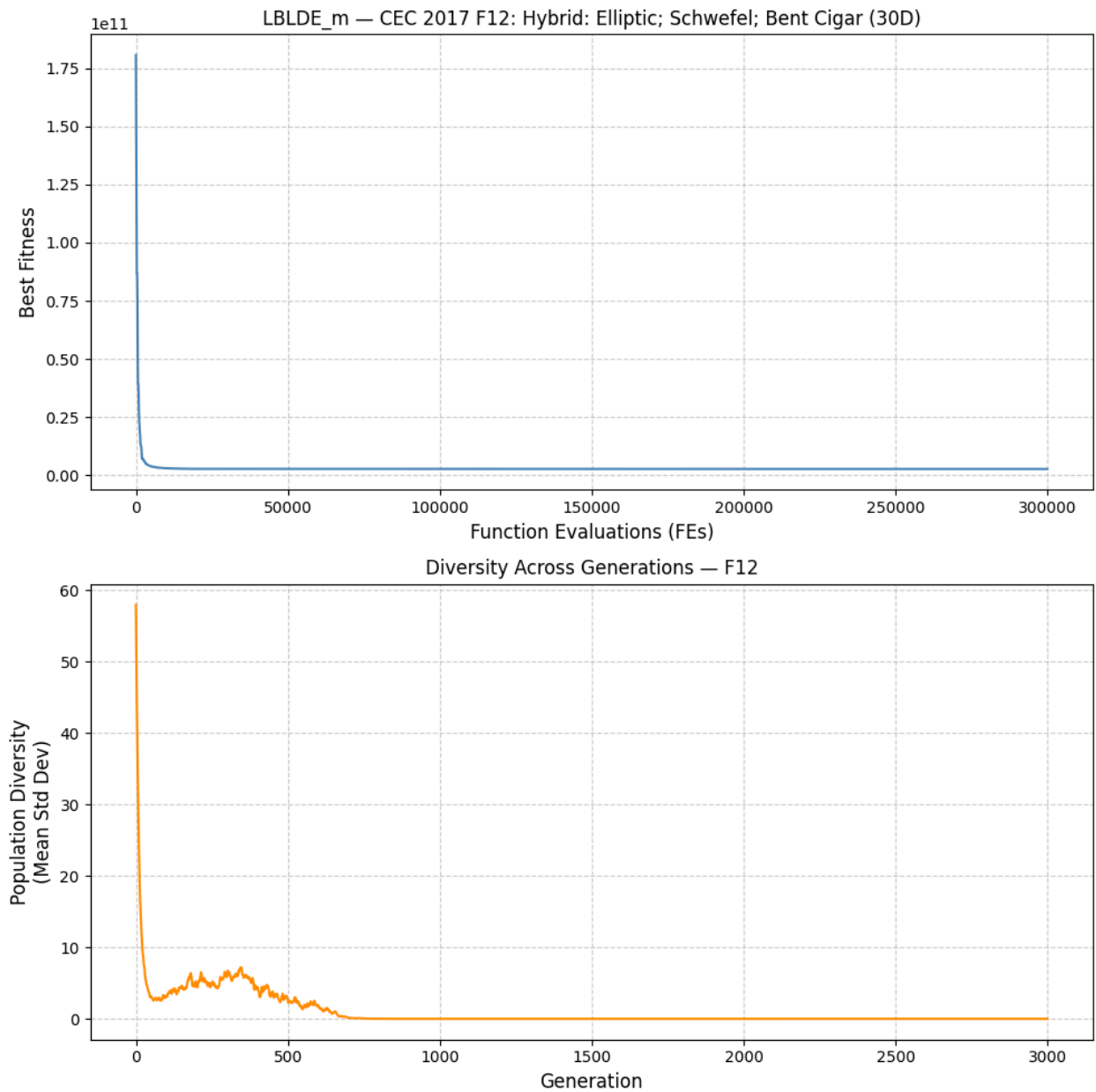
Optimization completed!

Final Best Fitness = 2.677164e+09

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness      : 2.677164e+09
  Global Optimum    : 1.200000e+03
  Error (|best-optimum|) : 2.677163e+09
  Best Solution (5 dim) : [ 54.80024184 -95.88915747 -65.24990348 25.67187405
-96.0082264 ]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F12.png

```
In [39]: # =====  
# F13 – Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)  
# =====  
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent Cigar; Ros
```

=====

=

LBLDE_m on CEC 2017 F13 – Single Run

=====

=

Function : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (30D)

Bounds : [-100, 100]

Global optimum : 1300.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 1.700763e+11 | Diversity = 5.7976e+01

Generation	10:	Best = 1.112298e+10	FES = 1100	Diversity = 2.5925e+01
Generation	20:	Best = 3.481860e+08	FES = 2100	Diversity = 1.2376e+01
Generation	30:	Best = 3.810995e+07	FES = 3100	Diversity = 6.4196e+00
Generation	40:	Best = 3.468346e+06	FES = 4100	Diversity = 4.2043e+00
Generation	50:	Best = 3.318512e+05	FES = 5100	Diversity = 3.2250e+00
Generation	60:	Best = 9.761217e+04	FES = 6100	Diversity = 2.2097e+00
Generation	70:	Best = 3.547642e+04	FES = 7100	Diversity = 1.8226e+00
Generation	80:	Best = 2.985761e+04	FES = 8100	Diversity = 1.1932e+00
Generation	90:	Best = 2.869527e+04	FES = 9100	Diversity = 6.1921e-01
Generation	100:	Best = 2.831316e+04	FES = 10100	Diversity = 5.6561e-01
Generation	110:	Best = 2.786120e+04	FES = 11100	Diversity = 6.4309e-01
Generation	120:	Best = 2.745112e+04	FES = 12100	Diversity = 7.5825e-01
Generation	130:	Best = 2.710610e+04	FES = 13100	Diversity = 8.0657e-01
Generation	140:	Best = 2.683231e+04	FES = 14100	Diversity = 8.2191e-01
Generation	150:	Best = 2.660965e+04	FES = 15100	Diversity = 6.5411e-01
Generation	160:	Best = 2.643306e+04	FES = 16100	Diversity = 6.8879e-01
Generation	170:	Best = 2.619282e+04	FES = 17100	Diversity = 8.4479e-01
Generation	180:	Best = 2.579188e+04	FES = 18100	Diversity = 8.6743e-01
Generation	190:	Best = 2.556294e+04	FES = 19100	Diversity = 9.1747e-01
Generation	200:	Best = 2.516217e+04	FES = 20100	Diversity = 1.0070e+00
Generation	210:	Best = 2.475327e+04	FES = 21100	Diversity = 9.8959e-01
Generation	220:	Best = 2.429497e+04	FES = 22100	Diversity = 1.1242e+00
Generation	230:	Best = 2.372937e+04	FES = 23100	Diversity = 1.4375e+00
Generation	240:	Best = 2.332571e+04	FES = 24100	Diversity = 1.5404e+00
Generation	250:	Best = 2.277408e+04	FES = 25100	Diversity = 1.4250e+00
Generation	260:	Best = 2.243555e+04	FES = 26100	Diversity = 1.2130e+00
Generation	270:	Best = 2.214711e+04	FES = 27100	Diversity = 1.3018e+00
Generation	280:	Best = 2.196788e+04	FES = 28100	Diversity = 1.3656e+00
Generation	290:	Best = 2.174865e+04	FES = 29100	Diversity = 1.2814e+00
Generation	300:	Best = 2.160309e+04	FES = 30100	Diversity = 1.2312e+00
Generation	310:	Best = 2.144425e+04	FES = 31100	Diversity = 1.0712e+00
Generation	320:	Best = 2.128831e+04	FES = 32100	Diversity = 1.1884e+00
Generation	330:	Best = 2.118947e+04	FES = 33100	Diversity = 1.0200e+00
Generation	340:	Best = 2.104528e+04	FES = 34100	Diversity = 9.8210e-01
Generation	350:	Best = 2.089017e+04	FES = 35100	Diversity = 1.3012e+00
Generation	360:	Best = 2.076945e+04	FES = 36100	Diversity = 1.2645e+00
Generation	370:	Best = 2.056041e+04	FES = 37100	Diversity = 1.4300e+00
Generation	380:	Best = 2.035490e+04	FES = 38100	Diversity = 1.2901e+00
Generation	390:	Best = 2.022430e+04	FES = 39100	Diversity = 1.1016e+00
Generation	400:	Best = 2.007331e+04	FES = 40100	Diversity = 1.0529e+00
Generation	410:	Best = 1.992895e+04	FES = 41100	Diversity = 1.1869e+00
Generation	420:	Best = 1.983655e+04	FES = 42100	Diversity = 8.8919e-01
Generation	430:	Best = 1.982105e+04	FES = 43100	Diversity = 8.5819e-01
Generation	440:	Best = 1.974873e+04	FES = 44100	Diversity = 8.5890e-01

Generation	450:	Best = 1.967439e+04	FES = 45100	Diversity = 7.0223e-01
Generation	460:	Best = 1.963603e+04	FES = 46100	Diversity = 5.0272e-01
Generation	470:	Best = 1.962772e+04	FES = 47100	Diversity = 3.1791e-02
Generation	480:	Best = 1.962396e+04	FES = 48100	Diversity = 1.8862e-02
Generation	490:	Best = 1.962291e+04	FES = 49100	Diversity = 7.2669e-03
Generation	500:	Best = 1.962237e+04	FES = 50100	Diversity = 5.3048e-03
Generation	510:	Best = 1.962217e+04	FES = 51100	Diversity = 2.9775e-03
Generation	520:	Best = 1.962200e+04	FES = 52100	Diversity = 3.0417e-03
Generation	530:	Best = 1.962192e+04	FES = 53100	Diversity = 1.8825e-03
Generation	540:	Best = 1.962187e+04	FES = 54100	Diversity = 1.0218e-03
Generation	550:	Best = 1.962184e+04	FES = 55100	Diversity = 6.6072e-04
Generation	560:	Best = 1.962183e+04	FES = 56100	Diversity = 6.4041e-04
Generation	570:	Best = 1.962183e+04	FES = 57100	Diversity = 4.9219e-04
Generation	580:	Best = 1.962182e+04	FES = 58100	Diversity = 2.7241e-04
Generation	590:	Best = 1.962182e+04	FES = 59100	Diversity = 2.4443e-04
Generation	600:	Best = 1.962182e+04	FES = 60100	Diversity = 1.5848e-04
Generation	610:	Best = 1.962182e+04	FES = 61100	Diversity = 1.0645e-04
Generation	620:	Best = 1.962182e+04	FES = 62100	Diversity = 7.1315e-05
Generation	630:	Best = 1.962182e+04	FES = 63100	Diversity = 8.3764e-05
Generation	640:	Best = 1.962182e+04	FES = 64100	Diversity = 5.7264e-05
Generation	650:	Best = 1.962182e+04	FES = 65100	Diversity = 3.6295e-05
Generation	660:	Best = 1.962182e+04	FES = 66100	Diversity = 2.8513e-05
Generation	670:	Best = 1.962182e+04	FES = 67100	Diversity = 2.0259e-05
Generation	680:	Best = 1.962182e+04	FES = 68100	Diversity = 1.2454e-05
Generation	690:	Best = 1.962182e+04	FES = 69100	Diversity = 8.9005e-06
Generation	700:	Best = 1.962182e+04	FES = 70100	Diversity = 6.7613e-06
Generation	710:	Best = 1.962182e+04	FES = 71100	Diversity = 6.4007e-06
Generation	720:	Best = 1.962182e+04	FES = 72100	Diversity = 4.6317e-06
Generation	730:	Best = 1.962182e+04	FES = 73100	Diversity = 3.2695e-06
Generation	740:	Best = 1.962182e+04	FES = 74100	Diversity = 2.2893e-06
Generation	750:	Best = 1.962182e+04	FES = 75100	Diversity = 1.4771e-06
Generation	760:	Best = 1.962182e+04	FES = 76100	Diversity = 6.7581e-07
Generation	770:	Best = 1.962182e+04	FES = 77100	Diversity = 5.0746e-07
Generation	780:	Best = 1.962182e+04	FES = 78100	Diversity = 4.5701e-07
Generation	790:	Best = 1.962182e+04	FES = 79100	Diversity = 4.3337e-07
Generation	800:	Best = 1.962182e+04	FES = 80100	Diversity = 2.6993e-07
Generation	810:	Best = 1.962182e+04	FES = 81100	Diversity = 1.9404e-07
Generation	820:	Best = 1.962182e+04	FES = 82100	Diversity = 1.4912e-07
Generation	830:	Best = 1.962182e+04	FES = 83100	Diversity = 1.0819e-07
Generation	840:	Best = 1.962182e+04	FES = 84100	Diversity = 7.1133e-08
Generation	850:	Best = 1.962182e+04	FES = 85100	Diversity = 8.1505e-08
Generation	860:	Best = 1.962182e+04	FES = 86100	Diversity = 7.6391e-08
Generation	870:	Best = 1.962182e+04	FES = 87100	Diversity = 3.7143e-08
Generation	880:	Best = 1.962182e+04	FES = 88100	Diversity = 2.3751e-08
Generation	890:	Best = 1.962182e+04	FES = 89100	Diversity = 1.3816e-08
Generation	900:	Best = 1.962182e+04	FES = 90100	Diversity = 6.4918e-09
Generation	910:	Best = 1.962182e+04	FES = 91100	Diversity = 1.0062e-08
Generation	920:	Best = 1.962182e+04	FES = 92100	Diversity = 9.0438e-09
Generation	930:	Best = 1.962182e+04	FES = 93100	Diversity = 8.7762e-09
Generation	940:	Best = 1.962182e+04	FES = 94100	Diversity = 8.0175e-09
Generation	950:	Best = 1.962182e+04	FES = 95100	Diversity = 7.2751e-09
Generation	960:	Best = 1.962182e+04	FES = 96100	Diversity = 7.3468e-09
Generation	970:	Best = 1.962182e+04	FES = 97100	Diversity = 6.6288e-09
Generation	980:	Best = 1.962182e+04	FES = 98100	Diversity = 4.3645e-09

[illegible]

[illegible]

[illegible]

Generation 2610:	Best = 1.962182e+04	FES = 261100	Diversity = 4.9523e-14
Generation 2620:	Best = 1.962182e+04	FES = 262100	Diversity = 4.9523e-14
Generation 2630:	Best = 1.962182e+04	FES = 263100	Diversity = 4.9523e-14
Generation 2640:	Best = 1.962182e+04	FES = 264100	Diversity = 4.9523e-14
Generation 2650:	Best = 1.962182e+04	FES = 265100	Diversity = 4.9523e-14
Generation 2660:	Best = 1.962182e+04	FES = 266100	Diversity = 4.9523e-14
Generation 2670:	Best = 1.962182e+04	FES = 267100	Diversity = 4.9523e-14
Generation 2680:	Best = 1.962182e+04	FES = 268100	Diversity = 4.9523e-14
Generation 2690:	Best = 1.962182e+04	FES = 269100	Diversity = 4.9523e-14
Generation 2700:	Best = 1.962182e+04	FES = 270100	Diversity = 4.9523e-14
Generation 2710:	Best = 1.962182e+04	FES = 271100	Diversity = 4.9523e-14
Generation 2720:	Best = 1.962182e+04	FES = 272100	Diversity = 4.9523e-14
Generation 2730:	Best = 1.962182e+04	FES = 273100	Diversity = 4.9523e-14
Generation 2740:	Best = 1.962182e+04	FES = 274100	Diversity = 4.9523e-14
Generation 2750:	Best = 1.962182e+04	FES = 275100	Diversity = 4.9523e-14
Generation 2760:	Best = 1.962182e+04	FES = 276100	Diversity = 4.9523e-14
Generation 2770:	Best = 1.962182e+04	FES = 277100	Diversity = 4.9523e-14
Generation 2780:	Best = 1.962182e+04	FES = 278100	Diversity = 4.9523e-14
Generation 2790:	Best = 1.962182e+04	FES = 279100	Diversity = 4.9523e-14
Generation 2800:	Best = 1.962182e+04	FES = 280100	Diversity = 4.9523e-14
Generation 2810:	Best = 1.962182e+04	FES = 281100	Diversity = 4.9523e-14
Generation 2820:	Best = 1.962182e+04	FES = 282100	Diversity = 4.9523e-14
Generation 2830:	Best = 1.962182e+04	FES = 283100	Diversity = 4.9523e-14
Generation 2840:	Best = 1.962182e+04	FES = 284100	Diversity = 4.9523e-14
Generation 2850:	Best = 1.962182e+04	FES = 285100	Diversity = 4.9523e-14
Generation 2860:	Best = 1.962182e+04	FES = 286100	Diversity = 4.9523e-14
Generation 2870:	Best = 1.962182e+04	FES = 287100	Diversity = 4.9523e-14
Generation 2880:	Best = 1.962182e+04	FES = 288100	Diversity = 4.9523e-14
Generation 2890:	Best = 1.962182e+04	FES = 289100	Diversity = 4.9523e-14
Generation 2900:	Best = 1.962182e+04	FES = 290100	Diversity = 4.9523e-14
Generation 2910:	Best = 1.962182e+04	FES = 291100	Diversity = 4.9523e-14
Generation 2920:	Best = 1.962182e+04	FES = 292100	Diversity = 4.9523e-14
Generation 2930:	Best = 1.962182e+04	FES = 293100	Diversity = 4.9523e-14
Generation 2940:	Best = 1.962182e+04	FES = 294100	Diversity = 4.9523e-14
Generation 2950:	Best = 1.962182e+04	FES = 295100	Diversity = 4.9523e-14
Generation 2960:	Best = 1.962182e+04	FES = 296100	Diversity = 4.9523e-14
Generation 2970:	Best = 1.962182e+04	FES = 297100	Diversity = 4.9523e-14
Generation 2980:	Best = 1.962182e+04	FES = 298100	Diversity = 4.9523e-14
Generation 2990:	Best = 1.962182e+04	FES = 299100	Diversity = 4.9523e-14

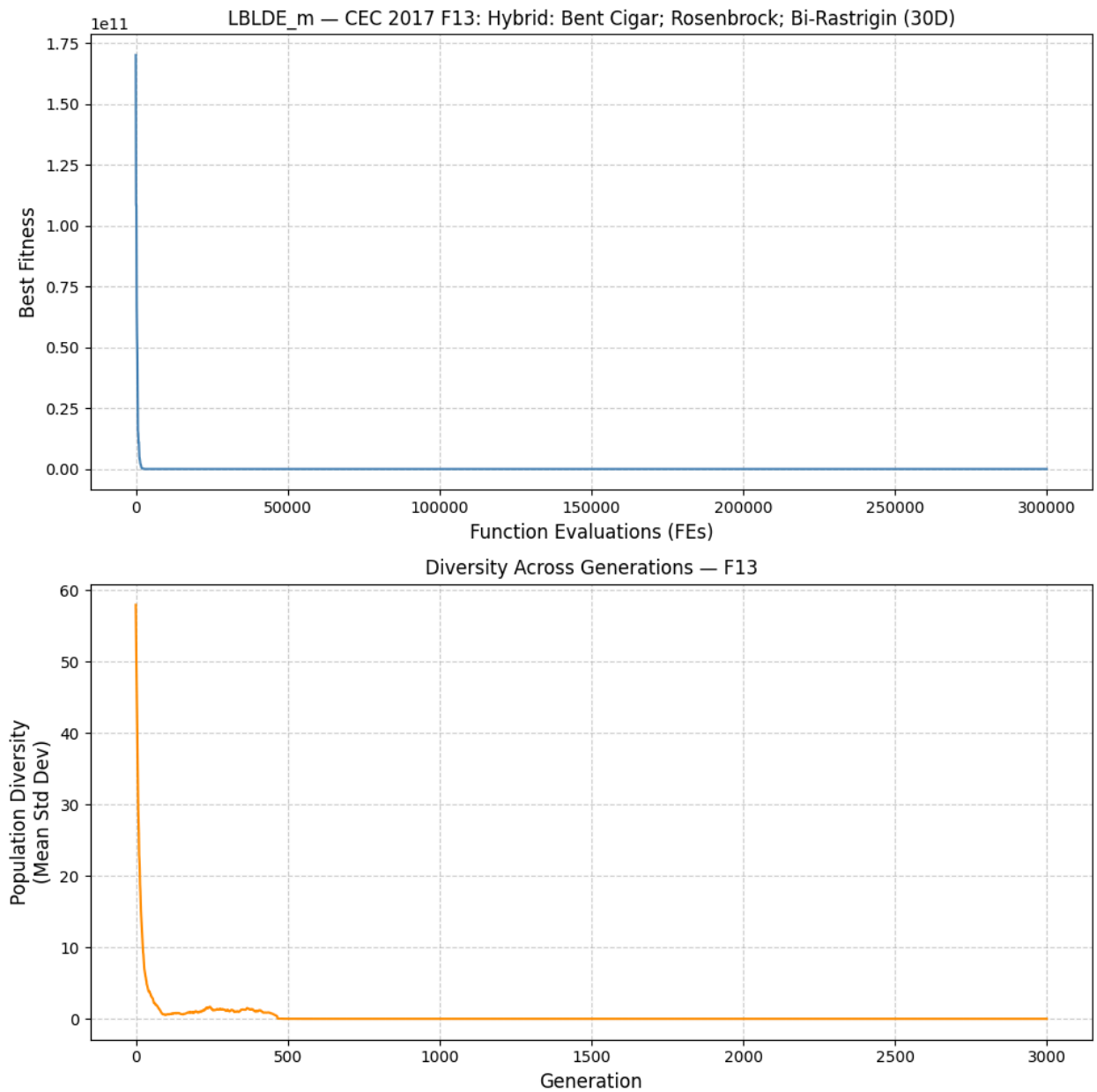
Optimization completed!

Final Best Fitness = 1.962182e+04

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness      : 1.962182e+04
  Global Optimum    : 1.300000e+03
  Error (|best-optimum|) : 1.832182e+04
  Best Solution (5 dim) : [-16.53055022 -32.66459549 -3.15768391 18.88315358
-4.94298271]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F13.png

```
In [40]: # =====  
# F14 – Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)  
# =====  
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid: Elliptic; Ackle
```

=====

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LBLDE_m on CEC 2017 F14 – Single Run

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Function : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (30D)

Bounds : [-100, 100]

Global optimum : 1400.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.706702e+07 | Diversity = 5.7976e+01

Generation	10:	Best = 1.139212e+06	FES = 1100	Diversity = 3.8220e+01
Generation	20:	Best = 1.338199e+05	FES = 2100	Diversity = 2.4963e+01
Generation	30:	Best = 2.393042e+04	FES = 3100	Diversity = 2.3472e+01
Generation	40:	Best = 9.764619e+03	FES = 4100	Diversity = 2.1381e+01
Generation	50:	Best = 6.455497e+03	FES = 5100	Diversity = 2.2223e+01
Generation	60:	Best = 3.519286e+03	FES = 6100	Diversity = 2.3925e+01
Generation	70:	Best = 2.850837e+03	FES = 7100	Diversity = 2.6254e+01
Generation	80:	Best = 2.354363e+03	FES = 8100	Diversity = 2.7834e+01
Generation	90:	Best = 2.239145e+03	FES = 9100	Diversity = 2.5503e+01
Generation	100:	Best = 2.025286e+03	FES = 10100	Diversity = 2.3216e+01
Generation	110:	Best = 1.934169e+03	FES = 11100	Diversity = 2.1190e+01
Generation	120:	Best = 1.881321e+03	FES = 12100	Diversity = 2.0069e+01
Generation	130:	Best = 1.815944e+03	FES = 13100	Diversity = 1.8452e+01
Generation	140:	Best = 1.815944e+03	FES = 14100	Diversity = 1.8147e+01
Generation	150:	Best = 1.793849e+03	FES = 15100	Diversity = 1.8973e+01
Generation	160:	Best = 1.776007e+03	FES = 16100	Diversity = 1.8683e+01
Generation	170:	Best = 1.747319e+03	FES = 17100	Diversity = 1.8112e+01
Generation	180:	Best = 1.717557e+03	FES = 18100	Diversity = 1.7971e+01
Generation	190:	Best = 1.705398e+03	FES = 19100	Diversity = 1.7503e+01
Generation	200:	Best = 1.631089e+03	FES = 20100	Diversity = 1.7027e+01
Generation	210:	Best = 1.631089e+03	FES = 21100	Diversity = 1.6522e+01
Generation	220:	Best = 1.627688e+03	FES = 22100	Diversity = 1.5974e+01
Generation	230:	Best = 1.603663e+03	FES = 23100	Diversity = 1.5409e+01
Generation	240:	Best = 1.596367e+03	FES = 24100	Diversity = 1.5396e+01
Generation	250:	Best = 1.554843e+03	FES = 25100	Diversity = 1.5124e+01
Generation	260:	Best = 1.546610e+03	FES = 26100	Diversity = 1.4596e+01
Generation	270:	Best = 1.546610e+03	FES = 27100	Diversity = 1.4022e+01
Generation	280:	Best = 1.546610e+03	FES = 28100	Diversity = 1.4179e+01
Generation	290:	Best = 1.546294e+03	FES = 29100	Diversity = 1.3732e+01
Generation	300:	Best = 1.533439e+03	FES = 30100	Diversity = 1.3332e+01
Generation	310:	Best = 1.533439e+03	FES = 31100	Diversity = 1.3069e+01
Generation	320:	Best = 1.533439e+03	FES = 32100	Diversity = 1.3115e+01
Generation	330:	Best = 1.530185e+03	FES = 33100	Diversity = 1.3024e+01
Generation	340:	Best = 1.530185e+03	FES = 34100	Diversity = 1.2888e+01
Generation	350:	Best = 1.530180e+03	FES = 35100	Diversity = 1.2632e+01
Generation	360:	Best = 1.528964e+03	FES = 36100	Diversity = 1.2413e+01
Generation	370:	Best = 1.527192e+03	FES = 37100	Diversity = 1.2261e+01
Generation	380:	Best = 1.526341e+03	FES = 38100	Diversity = 1.1874e+01
Generation	390:	Best = 1.520259e+03	FES = 39100	Diversity = 1.1823e+01
Generation	400:	Best = 1.514812e+03	FES = 40100	Diversity = 1.1729e+01
Generation	410:	Best = 1.512228e+03	FES = 41100	Diversity = 1.1615e+01
Generation	420:	Best = 1.512228e+03	FES = 42100	Diversity = 1.1576e+01
Generation	430:	Best = 1.512109e+03	FES = 43100	Diversity = 1.1102e+01
Generation	440:	Best = 1.507493e+03	FES = 44100	Diversity = 1.0922e+01

Generation	450:	Best = 1.507493e+03	FES = 45100	Diversity = 1.0836e+01
Generation	460:	Best = 1.505889e+03	FES = 46100	Diversity = 1.0799e+01
Generation	470:	Best = 1.502375e+03	FES = 47100	Diversity = 1.0735e+01
Generation	480:	Best = 1.502283e+03	FES = 48100	Diversity = 1.0650e+01
Generation	490:	Best = 1.501419e+03	FES = 49100	Diversity = 1.0649e+01
Generation	500:	Best = 1.501332e+03	FES = 50100	Diversity = 1.0626e+01
Generation	510:	Best = 1.486427e+03	FES = 51100	Diversity = 1.0528e+01
Generation	520:	Best = 1.486411e+03	FES = 52100	Diversity = 1.0499e+01
Generation	530:	Best = 1.486124e+03	FES = 53100	Diversity = 1.0274e+01
Generation	540:	Best = 1.484892e+03	FES = 54100	Diversity = 1.0175e+01
Generation	550:	Best = 1.484892e+03	FES = 55100	Diversity = 1.0116e+01
Generation	560:	Best = 1.484892e+03	FES = 56100	Diversity = 1.0020e+01
Generation	570:	Best = 1.484892e+03	FES = 57100	Diversity = 9.9893e+00
Generation	580:	Best = 1.484892e+03	FES = 58100	Diversity = 9.9214e+00
Generation	590:	Best = 1.483322e+03	FES = 59100	Diversity = 9.9283e+00
Generation	600:	Best = 1.482482e+03	FES = 60100	Diversity = 9.9112e+00
Generation	610:	Best = 1.476753e+03	FES = 61100	Diversity = 9.9039e+00
Generation	620:	Best = 1.475714e+03	FES = 62100	Diversity = 9.8431e+00
Generation	630:	Best = 1.475714e+03	FES = 63100	Diversity = 9.8794e+00
Generation	640:	Best = 1.475714e+03	FES = 64100	Diversity = 9.8193e+00
Generation	650:	Best = 1.475714e+03	FES = 65100	Diversity = 9.8245e+00
Generation	660:	Best = 1.475680e+03	FES = 66100	Diversity = 9.7680e+00
Generation	670:	Best = 1.475680e+03	FES = 67100	Diversity = 9.7164e+00
Generation	680:	Best = 1.475680e+03	FES = 68100	Diversity = 9.7256e+00
Generation	690:	Best = 1.475680e+03	FES = 69100	Diversity = 9.7216e+00
Generation	700:	Best = 1.475467e+03	FES = 70100	Diversity = 9.7135e+00
Generation	710:	Best = 1.475467e+03	FES = 71100	Diversity = 9.6664e+00
Generation	720:	Best = 1.475467e+03	FES = 72100	Diversity = 9.6285e+00
Generation	730:	Best = 1.475467e+03	FES = 73100	Diversity = 9.6493e+00
Generation	740:	Best = 1.471189e+03	FES = 74100	Diversity = 9.5898e+00
Generation	750:	Best = 1.471189e+03	FES = 75100	Diversity = 9.5728e+00
Generation	760:	Best = 1.471189e+03	FES = 76100	Diversity = 9.5035e+00
Generation	770:	Best = 1.471189e+03	FES = 77100	Diversity = 9.4569e+00
Generation	780:	Best = 1.471189e+03	FES = 78100	Diversity = 9.4256e+00
Generation	790:	Best = 1.471175e+03	FES = 79100	Diversity = 9.4057e+00
Generation	800:	Best = 1.471175e+03	FES = 80100	Diversity = 9.4027e+00
Generation	810:	Best = 1.471175e+03	FES = 81100	Diversity = 9.3797e+00
Generation	820:	Best = 1.471175e+03	FES = 82100	Diversity = 9.2538e+00
Generation	830:	Best = 1.466156e+03	FES = 83100	Diversity = 9.2120e+00
Generation	840:	Best = 1.466156e+03	FES = 84100	Diversity = 9.2544e+00
Generation	850:	Best = 1.466156e+03	FES = 85100	Diversity = 9.2167e+00
Generation	860:	Best = 1.466156e+03	FES = 86100	Diversity = 9.2072e+00
Generation	870:	Best = 1.466156e+03	FES = 87100	Diversity = 9.1932e+00
Generation	880:	Best = 1.464721e+03	FES = 88100	Diversity = 9.1655e+00
Generation	890:	Best = 1.464721e+03	FES = 89100	Diversity = 9.1364e+00
Generation	900:	Best = 1.463736e+03	FES = 90100	Diversity = 9.1309e+00
Generation	910:	Best = 1.463736e+03	FES = 91100	Diversity = 9.0855e+00
Generation	920:	Best = 1.463736e+03	FES = 92100	Diversity = 9.0689e+00
Generation	930:	Best = 1.463591e+03	FES = 93100	Diversity = 9.0777e+00
Generation	940:	Best = 1.463591e+03	FES = 94100	Diversity = 9.0977e+00
Generation	950:	Best = 1.463591e+03	FES = 95100	Diversity = 9.0641e+00
Generation	960:	Best = 1.463591e+03	FES = 96100	Diversity = 9.0827e+00
Generation	970:	Best = 1.463591e+03	FES = 97100	Diversity = 9.0675e+00
Generation	980:	Best = 1.463457e+03	FES = 98100	Diversity = 9.0559e+00

Generation 990:	Best = 1.463457e+03	FES = 99100	Diversity = 9.0512e+00
Generation 1000:	Best = 1.463457e+03	FES = 100100	Diversity = 9.0572e+00
Generation 1010:	Best = 1.462534e+03	FES = 101100	Diversity = 9.0549e+00
Generation 1020:	Best = 1.462534e+03	FES = 102100	Diversity = 9.0548e+00
Generation 1030:	Best = 1.459981e+03	FES = 103100	Diversity = 9.1084e+00
Generation 1040:	Best = 1.459981e+03	FES = 104100	Diversity = 9.1283e+00
Generation 1050:	Best = 1.458287e+03	FES = 105100	Diversity = 9.0572e+00
Generation 1060:	Best = 1.457973e+03	FES = 106100	Diversity = 9.0504e+00
Generation 1070:	Best = 1.457973e+03	FES = 107100	Diversity = 9.0149e+00
Generation 1080:	Best = 1.457973e+03	FES = 108100	Diversity = 9.0290e+00
Generation 1090:	Best = 1.457973e+03	FES = 109100	Diversity = 9.0234e+00
Generation 1100:	Best = 1.457973e+03	FES = 110100	Diversity = 8.9790e+00
Generation 1110:	Best = 1.453052e+03	FES = 111100	Diversity = 8.9637e+00
Generation 1120:	Best = 1.452849e+03	FES = 112100	Diversity = 8.9643e+00
Generation 1130:	Best = 1.452849e+03	FES = 113100	Diversity = 8.9291e+00
Generation 1140:	Best = 1.452849e+03	FES = 114100	Diversity = 8.9165e+00
Generation 1150:	Best = 1.452849e+03	FES = 115100	Diversity = 8.8886e+00
Generation 1160:	Best = 1.452849e+03	FES = 116100	Diversity = 8.9143e+00
Generation 1170:	Best = 1.452697e+03	FES = 117100	Diversity = 8.9119e+00
Generation 1180:	Best = 1.452697e+03	FES = 118100	Diversity = 8.8035e+00
Generation 1190:	Best = 1.452451e+03	FES = 119100	Diversity = 8.8022e+00
Generation 1200:	Best = 1.452451e+03	FES = 120100	Diversity = 8.7861e+00
Generation 1210:	Best = 1.452451e+03	FES = 121100	Diversity = 8.7822e+00
Generation 1220:	Best = 1.452451e+03	FES = 122100	Diversity = 8.7760e+00
Generation 1230:	Best = 1.452451e+03	FES = 123100	Diversity = 8.7832e+00
Generation 1240:	Best = 1.452451e+03	FES = 124100	Diversity = 8.6928e+00
Generation 1250:	Best = 1.452451e+03	FES = 125100	Diversity = 8.5090e+00
Generation 1260:	Best = 1.452451e+03	FES = 126100	Diversity = 8.4927e+00
Generation 1270:	Best = 1.452451e+03	FES = 127100	Diversity = 8.4479e+00
Generation 1280:	Best = 1.452451e+03	FES = 128100	Diversity = 8.4081e+00
Generation 1290:	Best = 1.452451e+03	FES = 129100	Diversity = 8.3747e+00
Generation 1300:	Best = 1.451730e+03	FES = 130100	Diversity = 8.3512e+00
Generation 1310:	Best = 1.451730e+03	FES = 131100	Diversity = 8.3289e+00
Generation 1320:	Best = 1.451419e+03	FES = 132100	Diversity = 8.2649e+00
Generation 1330:	Best = 1.451419e+03	FES = 133100	Diversity = 8.2558e+00
Generation 1340:	Best = 1.451419e+03	FES = 134100	Diversity = 8.2658e+00
Generation 1350:	Best = 1.451419e+03	FES = 135100	Diversity = 8.2581e+00
Generation 1360:	Best = 1.451419e+03	FES = 136100	Diversity = 8.2503e+00
Generation 1370:	Best = 1.451419e+03	FES = 137100	Diversity = 8.2276e+00
Generation 1380:	Best = 1.451097e+03	FES = 138100	Diversity = 8.2094e+00
Generation 1390:	Best = 1.450568e+03	FES = 139100	Diversity = 8.2180e+00
Generation 1400:	Best = 1.450568e+03	FES = 140100	Diversity = 8.2326e+00
Generation 1410:	Best = 1.450568e+03	FES = 141100	Diversity = 8.2351e+00
Generation 1420:	Best = 1.450568e+03	FES = 142100	Diversity = 8.2166e+00
Generation 1430:	Best = 1.450071e+03	FES = 143100	Diversity = 8.2137e+00
Generation 1440:	Best = 1.450071e+03	FES = 144100	Diversity = 8.1720e+00
Generation 1450:	Best = 1.450071e+03	FES = 145100	Diversity = 8.1821e+00
Generation 1460:	Best = 1.450071e+03	FES = 146100	Diversity = 8.1676e+00
Generation 1470:	Best = 1.450071e+03	FES = 147100	Diversity = 8.1585e+00
Generation 1480:	Best = 1.450071e+03	FES = 148100	Diversity = 8.1502e+00
Generation 1490:	Best = 1.450071e+03	FES = 149100	Diversity = 8.1511e+00
Generation 1500:	Best = 1.450071e+03	FES = 150100	Diversity = 8.1488e+00
Generation 1510:	Best = 1.450071e+03	FES = 151100	Diversity = 8.1115e+00
Generation 1520:	Best = 1.450071e+03	FES = 152100	Diversity = 8.0834e+00

Generation 1530:	Best = 1.450071e+03	FES = 153100	Diversity = 8.0363e+00
Generation 1540:	Best = 1.450071e+03	FES = 154100	Diversity = 8.0353e+00
Generation 1550:	Best = 1.450071e+03	FES = 155100	Diversity = 7.9754e+00
Generation 1560:	Best = 1.450071e+03	FES = 156100	Diversity = 7.9792e+00
Generation 1570:	Best = 1.450071e+03	FES = 157100	Diversity = 7.9749e+00
Generation 1580:	Best = 1.449991e+03	FES = 158100	Diversity = 7.9662e+00
Generation 1590:	Best = 1.449991e+03	FES = 159100	Diversity = 7.9670e+00
Generation 1600:	Best = 1.449991e+03	FES = 160100	Diversity = 7.9660e+00
Generation 1610:	Best = 1.449991e+03	FES = 161100	Diversity = 7.9520e+00
Generation 1620:	Best = 1.449991e+03	FES = 162100	Diversity = 7.9491e+00
Generation 1630:	Best = 1.449991e+03	FES = 163100	Diversity = 7.9457e+00
Generation 1640:	Best = 1.449991e+03	FES = 164100	Diversity = 7.9465e+00
Generation 1650:	Best = 1.449991e+03	FES = 165100	Diversity = 7.9394e+00
Generation 1660:	Best = 1.449991e+03	FES = 166100	Diversity = 7.9316e+00
Generation 1670:	Best = 1.449991e+03	FES = 167100	Diversity = 7.9334e+00
Generation 1680:	Best = 1.449991e+03	FES = 168100	Diversity = 7.9412e+00
Generation 1690:	Best = 1.449991e+03	FES = 169100	Diversity = 7.9374e+00
Generation 1700:	Best = 1.449991e+03	FES = 170100	Diversity = 7.9374e+00
Generation 1710:	Best = 1.449991e+03	FES = 171100	Diversity = 7.9297e+00
Generation 1720:	Best = 1.449991e+03	FES = 172100	Diversity = 7.9168e+00
Generation 1730:	Best = 1.449902e+03	FES = 173100	Diversity = 7.9135e+00
Generation 1740:	Best = 1.449902e+03	FES = 174100	Diversity = 7.8595e+00
Generation 1750:	Best = 1.449902e+03	FES = 175100	Diversity = 7.9157e+00
Generation 1760:	Best = 1.449902e+03	FES = 176100	Diversity = 7.8832e+00
Generation 1770:	Best = 1.449902e+03	FES = 177100	Diversity = 7.8880e+00
Generation 1780:	Best = 1.449902e+03	FES = 178100	Diversity = 7.8543e+00
Generation 1790:	Best = 1.449902e+03	FES = 179100	Diversity = 7.8506e+00
Generation 1800:	Best = 1.449902e+03	FES = 180100	Diversity = 7.8299e+00
Generation 1810:	Best = 1.449902e+03	FES = 181100	Diversity = 7.8315e+00
Generation 1820:	Best = 1.449902e+03	FES = 182100	Diversity = 7.8191e+00
Generation 1830:	Best = 1.449902e+03	FES = 183100	Diversity = 7.8179e+00
Generation 1840:	Best = 1.449583e+03	FES = 184100	Diversity = 7.8289e+00
Generation 1850:	Best = 1.449495e+03	FES = 185100	Diversity = 7.8452e+00
Generation 1860:	Best = 1.449377e+03	FES = 186100	Diversity = 7.8505e+00
Generation 1870:	Best = 1.449295e+03	FES = 187100	Diversity = 7.8417e+00
Generation 1880:	Best = 1.449295e+03	FES = 188100	Diversity = 7.8414e+00
Generation 1890:	Best = 1.449295e+03	FES = 189100	Diversity = 7.8504e+00
Generation 1900:	Best = 1.449295e+03	FES = 190100	Diversity = 7.8396e+00
Generation 1910:	Best = 1.449295e+03	FES = 191100	Diversity = 7.8173e+00
Generation 1920:	Best = 1.449295e+03	FES = 192100	Diversity = 7.8173e+00
Generation 1930:	Best = 1.449295e+03	FES = 193100	Diversity = 7.7741e+00
Generation 1940:	Best = 1.449295e+03	FES = 194100	Diversity = 7.7546e+00
Generation 1950:	Best = 1.449295e+03	FES = 195100	Diversity = 7.7514e+00
Generation 1960:	Best = 1.449295e+03	FES = 196100	Diversity = 7.7496e+00
Generation 1970:	Best = 1.449286e+03	FES = 197100	Diversity = 7.7390e+00
Generation 1980:	Best = 1.449286e+03	FES = 198100	Diversity = 7.7328e+00
Generation 1990:	Best = 1.449075e+03	FES = 199100	Diversity = 7.7328e+00
Generation 2000:	Best = 1.449075e+03	FES = 200100	Diversity = 7.7042e+00
Generation 2010:	Best = 1.449075e+03	FES = 201100	Diversity = 7.7138e+00
Generation 2020:	Best = 1.449075e+03	FES = 202100	Diversity = 7.7123e+00
Generation 2030:	Best = 1.449075e+03	FES = 203100	Diversity = 7.6943e+00
Generation 2040:	Best = 1.449075e+03	FES = 204100	Diversity = 7.6996e+00
Generation 2050:	Best = 1.449075e+03	FES = 205100	Diversity = 7.6989e+00
Generation 2060:	Best = 1.449075e+03	FES = 206100	Diversity = 7.6971e+00

Generation 2070:	Best = 1.449075e+03	FES = 207100	Diversity = 7.6994e+00
Generation 2080:	Best = 1.449075e+03	FES = 208100	Diversity = 7.6929e+00
Generation 2090:	Best = 1.449075e+03	FES = 209100	Diversity = 7.6725e+00
Generation 2100:	Best = 1.449075e+03	FES = 210100	Diversity = 7.7095e+00
Generation 2110:	Best = 1.449075e+03	FES = 211100	Diversity = 7.7060e+00
Generation 2120:	Best = 1.449075e+03	FES = 212100	Diversity = 7.6966e+00
Generation 2130:	Best = 1.449075e+03	FES = 213100	Diversity = 7.6816e+00
Generation 2140:	Best = 1.449075e+03	FES = 214100	Diversity = 7.6659e+00
Generation 2150:	Best = 1.449064e+03	FES = 215100	Diversity = 7.6606e+00
Generation 2160:	Best = 1.449064e+03	FES = 216100	Diversity = 7.4796e+00
Generation 2170:	Best = 1.449064e+03	FES = 217100	Diversity = 7.4773e+00
Generation 2180:	Best = 1.449064e+03	FES = 218100	Diversity = 7.4736e+00
Generation 2190:	Best = 1.449064e+03	FES = 219100	Diversity = 7.4772e+00
Generation 2200:	Best = 1.449064e+03	FES = 220100	Diversity = 7.4745e+00
Generation 2210:	Best = 1.449064e+03	FES = 221100	Diversity = 7.4725e+00
Generation 2220:	Best = 1.449064e+03	FES = 222100	Diversity = 7.4692e+00
Generation 2230:	Best = 1.449064e+03	FES = 223100	Diversity = 7.4014e+00
Generation 2240:	Best = 1.449064e+03	FES = 224100	Diversity = 7.3251e+00
Generation 2250:	Best = 1.449064e+03	FES = 225100	Diversity = 7.3235e+00
Generation 2260:	Best = 1.449064e+03	FES = 226100	Diversity = 7.3313e+00
Generation 2270:	Best = 1.449064e+03	FES = 227100	Diversity = 7.3123e+00
Generation 2280:	Best = 1.449064e+03	FES = 228100	Diversity = 7.2898e+00
Generation 2290:	Best = 1.449064e+03	FES = 229100	Diversity = 7.2730e+00
Generation 2300:	Best = 1.449064e+03	FES = 230100	Diversity = 7.2717e+00
Generation 2310:	Best = 1.449064e+03	FES = 231100	Diversity = 7.2721e+00
Generation 2320:	Best = 1.449064e+03	FES = 232100	Diversity = 7.2539e+00
Generation 2330:	Best = 1.449064e+03	FES = 233100	Diversity = 7.2700e+00
Generation 2340:	Best = 1.449056e+03	FES = 234100	Diversity = 7.2643e+00
Generation 2350:	Best = 1.449056e+03	FES = 235100	Diversity = 7.2648e+00
Generation 2360:	Best = 1.449056e+03	FES = 236100	Diversity = 7.2711e+00
Generation 2370:	Best = 1.449056e+03	FES = 237100	Diversity = 7.2621e+00
Generation 2380:	Best = 1.449056e+03	FES = 238100	Diversity = 7.2542e+00
Generation 2390:	Best = 1.449056e+03	FES = 239100	Diversity = 7.2536e+00
Generation 2400:	Best = 1.449056e+03	FES = 240100	Diversity = 7.2528e+00
Generation 2410:	Best = 1.448880e+03	FES = 241100	Diversity = 7.2532e+00
Generation 2420:	Best = 1.448880e+03	FES = 242100	Diversity = 7.2535e+00
Generation 2430:	Best = 1.448880e+03	FES = 243100	Diversity = 7.2416e+00
Generation 2440:	Best = 1.448880e+03	FES = 244100	Diversity = 7.2423e+00
Generation 2450:	Best = 1.448880e+03	FES = 245100	Diversity = 7.2403e+00
Generation 2460:	Best = 1.448880e+03	FES = 246100	Diversity = 7.2161e+00
Generation 2470:	Best = 1.448880e+03	FES = 247100	Diversity = 7.1984e+00
Generation 2480:	Best = 1.448880e+03	FES = 248100	Diversity = 7.2078e+00
Generation 2490:	Best = 1.448880e+03	FES = 249100	Diversity = 7.2029e+00
Generation 2500:	Best = 1.448880e+03	FES = 250100	Diversity = 7.1960e+00
Generation 2510:	Best = 1.448148e+03	FES = 251100	Diversity = 7.1929e+00
Generation 2520:	Best = 1.448148e+03	FES = 252100	Diversity = 7.2044e+00
Generation 2530:	Best = 1.448148e+03	FES = 253100	Diversity = 7.2032e+00
Generation 2540:	Best = 1.448148e+03	FES = 254100	Diversity = 7.2001e+00
Generation 2550:	Best = 1.448148e+03	FES = 255100	Diversity = 7.1975e+00
Generation 2560:	Best = 1.448148e+03	FES = 256100	Diversity = 7.1771e+00
Generation 2570:	Best = 1.448148e+03	FES = 257100	Diversity = 7.1944e+00
Generation 2580:	Best = 1.448148e+03	FES = 258100	Diversity = 7.1786e+00
Generation 2590:	Best = 1.448148e+03	FES = 259100	Diversity = 7.1727e+00
Generation 2600:	Best = 1.448148e+03	FES = 260100	Diversity = 7.1830e+00

Generation 2610:	Best = 1.448148e+03	FES = 261100	Diversity = 7.1831e+00
Generation 2620:	Best = 1.448148e+03	FES = 262100	Diversity = 7.1927e+00
Generation 2630:	Best = 1.447717e+03	FES = 263100	Diversity = 7.1916e+00
Generation 2640:	Best = 1.447717e+03	FES = 264100	Diversity = 7.1900e+00
Generation 2650:	Best = 1.447717e+03	FES = 265100	Diversity = 7.1864e+00
Generation 2660:	Best = 1.447717e+03	FES = 266100	Diversity = 7.1849e+00
Generation 2670:	Best = 1.444756e+03	FES = 267100	Diversity = 7.2381e+00
Generation 2680:	Best = 1.444527e+03	FES = 268100	Diversity = 7.2403e+00
Generation 2690:	Best = 1.444527e+03	FES = 269100	Diversity = 7.2018e+00
Generation 2700:	Best = 1.444527e+03	FES = 270100	Diversity = 7.2001e+00
Generation 2710:	Best = 1.444527e+03	FES = 271100	Diversity = 7.1988e+00
Generation 2720:	Best = 1.444527e+03	FES = 272100	Diversity = 7.1439e+00
Generation 2730:	Best = 1.442976e+03	FES = 273100	Diversity = 7.1331e+00
Generation 2740:	Best = 1.442976e+03	FES = 274100	Diversity = 7.1324e+00
Generation 2750:	Best = 1.442976e+03	FES = 275100	Diversity = 7.1218e+00
Generation 2760:	Best = 1.442834e+03	FES = 276100	Diversity = 7.1207e+00
Generation 2770:	Best = 1.442834e+03	FES = 277100	Diversity = 7.1172e+00
Generation 2780:	Best = 1.442410e+03	FES = 278100	Diversity = 7.1160e+00
Generation 2790:	Best = 1.442410e+03	FES = 279100	Diversity = 7.1157e+00
Generation 2800:	Best = 1.442410e+03	FES = 280100	Diversity = 7.1107e+00
Generation 2810:	Best = 1.442410e+03	FES = 281100	Diversity = 7.0313e+00
Generation 2820:	Best = 1.442326e+03	FES = 282100	Diversity = 7.0290e+00
Generation 2830:	Best = 1.442326e+03	FES = 283100	Diversity = 7.0278e+00
Generation 2840:	Best = 1.442326e+03	FES = 284100	Diversity = 7.0293e+00
Generation 2850:	Best = 1.442326e+03	FES = 285100	Diversity = 7.0310e+00
Generation 2860:	Best = 1.442326e+03	FES = 286100	Diversity = 6.9902e+00
Generation 2870:	Best = 1.442326e+03	FES = 287100	Diversity = 6.9900e+00
Generation 2880:	Best = 1.442326e+03	FES = 288100	Diversity = 6.9901e+00
Generation 2890:	Best = 1.442326e+03	FES = 289100	Diversity = 7.0031e+00
Generation 2900:	Best = 1.442326e+03	FES = 290100	Diversity = 6.9907e+00
Generation 2910:	Best = 1.442326e+03	FES = 291100	Diversity = 6.9920e+00
Generation 2920:	Best = 1.442326e+03	FES = 292100	Diversity = 7.0042e+00
Generation 2930:	Best = 1.442326e+03	FES = 293100	Diversity = 7.0099e+00
Generation 2940:	Best = 1.442326e+03	FES = 294100	Diversity = 7.0096e+00
Generation 2950:	Best = 1.442326e+03	FES = 295100	Diversity = 7.0026e+00
Generation 2960:	Best = 1.442326e+03	FES = 296100	Diversity = 7.0286e+00
Generation 2970:	Best = 1.442326e+03	FES = 297100	Diversity = 7.0289e+00
Generation 2980:	Best = 1.441525e+03	FES = 298100	Diversity = 7.0100e+00
Generation 2990:	Best = 1.441525e+03	FES = 299100	Diversity = 7.0131e+00

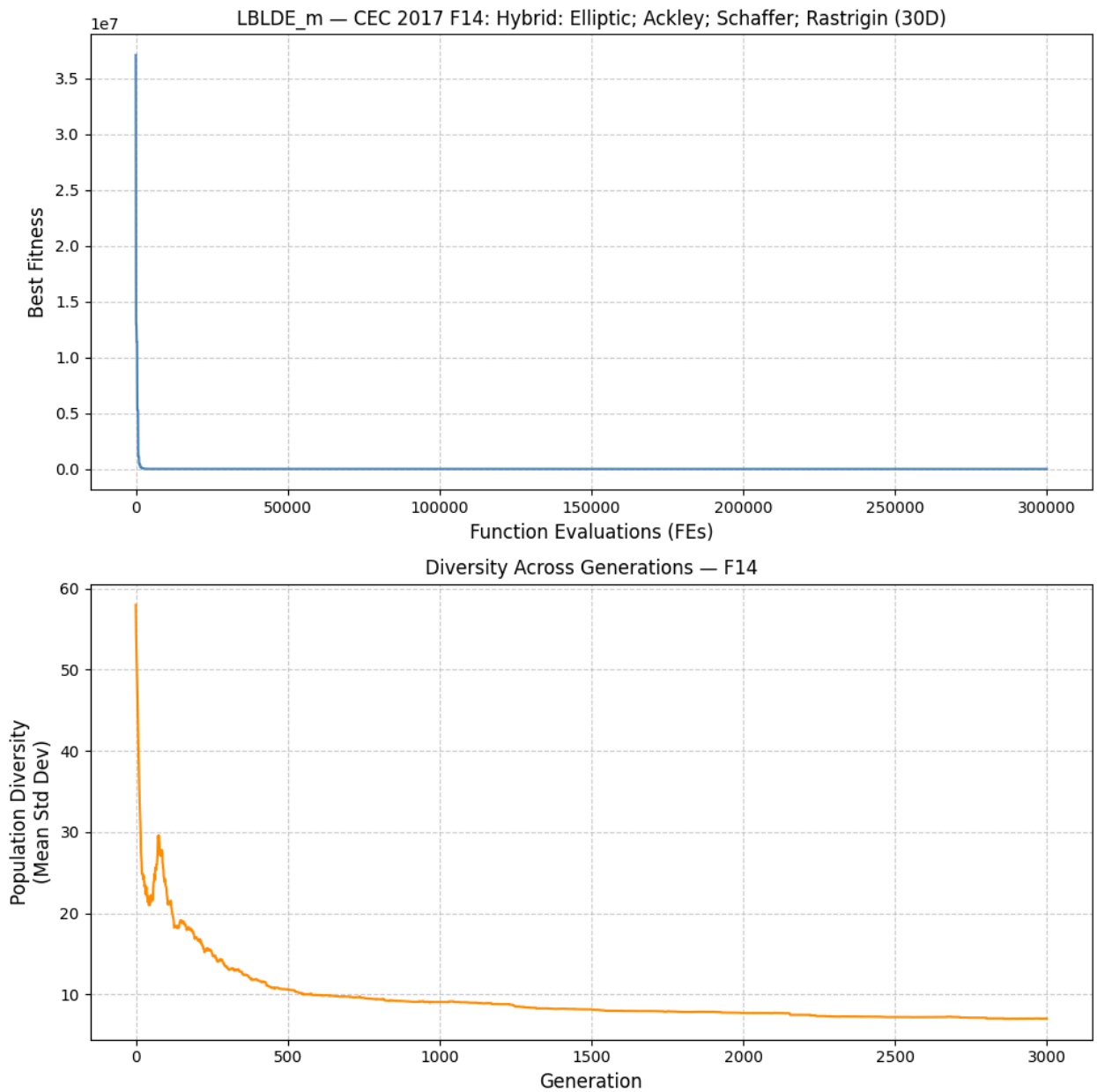
Optimization completed!

Final Best Fitness = 1.441525e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 1.441525e+03
  Global Optimum         : 1.400000e+03
  Error (|best-optimum|) : 4.152461e+01
  Best Solution (5 dim)  : [-36.57418265 -87.92474777 -27.90250679 -34.75192586
62.01869031]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F14.png

```
In [41]: # =====  
# F15 – Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)  
# =====  
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent Cigar; HGE
```

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LBLDE_m on CEC 2017 F15 – Single Run

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Function : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock (30D)

Bounds : [-100, 100]

Global optimum : 1500.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 4.037230e+10 | Diversity = 5.7976e+01

Generation	10:	Best = 4.973661e+08	FES = 1100	Diversity = 2.7877e+01
Generation	20:	Best = 1.577700e+07	FES = 2100	Diversity = 1.3945e+01
Generation	30:	Best = 1.532166e+06	FES = 3100	Diversity = 8.9186e+00
Generation	40:	Best = 1.064571e+05	FES = 4100	Diversity = 8.2968e+00
Generation	50:	Best = 3.004270e+04	FES = 5100	Diversity = 8.5958e+00
Generation	60:	Best = 1.193923e+04	FES = 6100	Diversity = 9.3781e+00
Generation	70:	Best = 9.484422e+03	FES = 7100	Diversity = 1.0027e+01
Generation	80:	Best = 7.360337e+03	FES = 8100	Diversity = 1.0134e+01
Generation	90:	Best = 6.783270e+03	FES = 9100	Diversity = 1.0347e+01
Generation	100:	Best = 5.254216e+03	FES = 10100	Diversity = 1.0366e+01
Generation	110:	Best = 3.700685e+03	FES = 11100	Diversity = 1.0944e+01
Generation	120:	Best = 2.963381e+03	FES = 12100	Diversity = 1.3053e+01
Generation	130:	Best = 2.819234e+03	FES = 13100	Diversity = 1.2849e+01
Generation	140:	Best = 2.561839e+03	FES = 14100	Diversity = 1.2047e+01
Generation	150:	Best = 2.328980e+03	FES = 15100	Diversity = 1.1253e+01
Generation	160:	Best = 2.134517e+03	FES = 16100	Diversity = 1.0267e+01
Generation	170:	Best = 2.012596e+03	FES = 17100	Diversity = 9.9332e+00
Generation	180:	Best = 1.889929e+03	FES = 18100	Diversity = 1.0062e+01
Generation	190:	Best = 1.843535e+03	FES = 19100	Diversity = 9.5791e+00
Generation	200:	Best = 1.791100e+03	FES = 20100	Diversity = 9.4382e+00
Generation	210:	Best = 1.771875e+03	FES = 21100	Diversity = 8.9743e+00
Generation	220:	Best = 1.757527e+03	FES = 22100	Diversity = 8.4938e+00
Generation	230:	Best = 1.718860e+03	FES = 23100	Diversity = 7.7083e+00
Generation	240:	Best = 1.704359e+03	FES = 24100	Diversity = 7.6083e+00
Generation	250:	Best = 1.686814e+03	FES = 25100	Diversity = 6.7353e+00
Generation	260:	Best = 1.680852e+03	FES = 26100	Diversity = 6.4907e+00
Generation	270:	Best = 1.664190e+03	FES = 27100	Diversity = 6.2437e+00
Generation	280:	Best = 1.641675e+03	FES = 28100	Diversity = 6.3135e+00
Generation	290:	Best = 1.639962e+03	FES = 29100	Diversity = 6.0020e+00
Generation	300:	Best = 1.624868e+03	FES = 30100	Diversity = 5.6658e+00
Generation	310:	Best = 1.619516e+03	FES = 31100	Diversity = 5.3201e+00
Generation	320:	Best = 1.613340e+03	FES = 32100	Diversity = 5.0197e+00
Generation	330:	Best = 1.610521e+03	FES = 33100	Diversity = 4.7274e+00
Generation	340:	Best = 1.610521e+03	FES = 34100	Diversity = 4.5863e+00
Generation	350:	Best = 1.610377e+03	FES = 35100	Diversity = 4.4950e+00
Generation	360:	Best = 1.606814e+03	FES = 36100	Diversity = 4.4271e+00
Generation	370:	Best = 1.605991e+03	FES = 37100	Diversity = 4.3276e+00
Generation	380:	Best = 1.604252e+03	FES = 38100	Diversity = 4.2147e+00
Generation	390:	Best = 1.601017e+03	FES = 39100	Diversity = 4.1774e+00
Generation	400:	Best = 1.587792e+03	FES = 40100	Diversity = 4.0531e+00
Generation	410:	Best = 1.587066e+03	FES = 41100	Diversity = 3.9553e+00
Generation	420:	Best = 1.584753e+03	FES = 42100	Diversity = 3.9331e+00
Generation	430:	Best = 1.584753e+03	FES = 43100	Diversity = 3.8676e+00
Generation	440:	Best = 1.584753e+03	FES = 44100	Diversity = 3.7698e+00

Generation	450:	Best = 1.575006e+03	FES = 45100	Diversity = 3.6411e+00
Generation	460:	Best = 1.575006e+03	FES = 46100	Diversity = 3.6036e+00
Generation	470:	Best = 1.572655e+03	FES = 47100	Diversity = 3.4981e+00
Generation	480:	Best = 1.572655e+03	FES = 48100	Diversity = 3.4500e+00
Generation	490:	Best = 1.571346e+03	FES = 49100	Diversity = 3.3629e+00
Generation	500:	Best = 1.571346e+03	FES = 50100	Diversity = 3.3316e+00
Generation	510:	Best = 1.570315e+03	FES = 51100	Diversity = 3.3524e+00
Generation	520:	Best = 1.570315e+03	FES = 52100	Diversity = 3.2884e+00
Generation	530:	Best = 1.570222e+03	FES = 53100	Diversity = 3.1537e+00
Generation	540:	Best = 1.570222e+03	FES = 54100	Diversity = 3.1220e+00
Generation	550:	Best = 1.570179e+03	FES = 55100	Diversity = 3.0784e+00
Generation	560:	Best = 1.569602e+03	FES = 56100	Diversity = 3.0403e+00
Generation	570:	Best = 1.569602e+03	FES = 57100	Diversity = 3.0097e+00
Generation	580:	Best = 1.569366e+03	FES = 58100	Diversity = 2.9476e+00
Generation	590:	Best = 1.568887e+03	FES = 59100	Diversity = 2.9424e+00
Generation	600:	Best = 1.566980e+03	FES = 60100	Diversity = 2.8635e+00
Generation	610:	Best = 1.563191e+03	FES = 61100	Diversity = 2.7900e+00
Generation	620:	Best = 1.561936e+03	FES = 62100	Diversity = 2.8124e+00
Generation	630:	Best = 1.561936e+03	FES = 63100	Diversity = 2.8074e+00
Generation	640:	Best = 1.559733e+03	FES = 64100	Diversity = 2.8084e+00
Generation	650:	Best = 1.559578e+03	FES = 65100	Diversity = 2.7901e+00
Generation	660:	Best = 1.559531e+03	FES = 66100	Diversity = 2.7412e+00
Generation	670:	Best = 1.558888e+03	FES = 67100	Diversity = 2.6814e+00
Generation	680:	Best = 1.558888e+03	FES = 68100	Diversity = 2.6478e+00
Generation	690:	Best = 1.558888e+03	FES = 69100	Diversity = 2.6158e+00
Generation	700:	Best = 1.557452e+03	FES = 70100	Diversity = 2.6164e+00
Generation	710:	Best = 1.557071e+03	FES = 71100	Diversity = 2.5438e+00
Generation	720:	Best = 1.557071e+03	FES = 72100	Diversity = 2.5292e+00
Generation	730:	Best = 1.557033e+03	FES = 73100	Diversity = 2.4905e+00
Generation	740:	Best = 1.557033e+03	FES = 74100	Diversity = 2.4843e+00
Generation	750:	Best = 1.557033e+03	FES = 75100	Diversity = 2.4716e+00
Generation	760:	Best = 1.557001e+03	FES = 76100	Diversity = 2.4216e+00
Generation	770:	Best = 1.557001e+03	FES = 77100	Diversity = 2.4314e+00
Generation	780:	Best = 1.556377e+03	FES = 78100	Diversity = 2.4242e+00
Generation	790:	Best = 1.555170e+03	FES = 79100	Diversity = 2.4025e+00
Generation	800:	Best = 1.554943e+03	FES = 80100	Diversity = 2.4092e+00
Generation	810:	Best = 1.554707e+03	FES = 81100	Diversity = 2.3774e+00
Generation	820:	Best = 1.554707e+03	FES = 82100	Diversity = 2.3765e+00
Generation	830:	Best = 1.554197e+03	FES = 83100	Diversity = 2.3333e+00
Generation	840:	Best = 1.554197e+03	FES = 84100	Diversity = 2.2659e+00
Generation	850:	Best = 1.553344e+03	FES = 85100	Diversity = 2.2586e+00
Generation	860:	Best = 1.553321e+03	FES = 86100	Diversity = 2.2234e+00
Generation	870:	Best = 1.553321e+03	FES = 87100	Diversity = 2.1928e+00
Generation	880:	Best = 1.553114e+03	FES = 88100	Diversity = 2.1956e+00
Generation	890:	Best = 1.552516e+03	FES = 89100	Diversity = 2.1625e+00
Generation	900:	Best = 1.552201e+03	FES = 90100	Diversity = 2.1604e+00
Generation	910:	Best = 1.552171e+03	FES = 91100	Diversity = 2.1296e+00
Generation	920:	Best = 1.551475e+03	FES = 92100	Diversity = 2.1154e+00
Generation	930:	Best = 1.551120e+03	FES = 93100	Diversity = 2.1109e+00
Generation	940:	Best = 1.550983e+03	FES = 94100	Diversity = 2.0902e+00
Generation	950:	Best = 1.550760e+03	FES = 95100	Diversity = 2.0880e+00
Generation	960:	Best = 1.550262e+03	FES = 96100	Diversity = 2.0529e+00
Generation	970:	Best = 1.550154e+03	FES = 97100	Diversity = 2.0161e+00
Generation	980:	Best = 1.550154e+03	FES = 98100	Diversity = 2.0248e+00

Generation 990:	Best = 1.550146e+03	FES = 99100	Diversity = 2.0278e+00
Generation 1000:	Best = 1.550146e+03	FES = 100100	Diversity = 2.0121e+00
Generation 1010:	Best = 1.550135e+03	FES = 101100	Diversity = 2.0012e+00
Generation 1020:	Best = 1.550134e+03	FES = 102100	Diversity = 1.9876e+00
Generation 1030:	Best = 1.550121e+03	FES = 103100	Diversity = 1.9710e+00
Generation 1040:	Best = 1.550115e+03	FES = 104100	Diversity = 1.9694e+00
Generation 1050:	Best = 1.550015e+03	FES = 105100	Diversity = 1.9298e+00
Generation 1060:	Best = 1.550015e+03	FES = 106100	Diversity = 1.9379e+00
Generation 1070:	Best = 1.550015e+03	FES = 107100	Diversity = 1.9350e+00
Generation 1080:	Best = 1.550006e+03	FES = 108100	Diversity = 1.9303e+00
Generation 1090:	Best = 1.549986e+03	FES = 109100	Diversity = 1.9216e+00
Generation 1100:	Best = 1.549984e+03	FES = 110100	Diversity = 1.9181e+00
Generation 1110:	Best = 1.549923e+03	FES = 111100	Diversity = 1.9119e+00
Generation 1120:	Best = 1.549915e+03	FES = 112100	Diversity = 1.8801e+00
Generation 1130:	Best = 1.549915e+03	FES = 113100	Diversity = 1.8694e+00
Generation 1140:	Best = 1.549905e+03	FES = 114100	Diversity = 1.8715e+00
Generation 1150:	Best = 1.549754e+03	FES = 115100	Diversity = 1.8596e+00
Generation 1160:	Best = 1.548912e+03	FES = 116100	Diversity = 1.8604e+00
Generation 1170:	Best = 1.548905e+03	FES = 117100	Diversity = 1.8368e+00
Generation 1180:	Best = 1.548905e+03	FES = 118100	Diversity = 1.8415e+00
Generation 1190:	Best = 1.548905e+03	FES = 119100	Diversity = 1.8308e+00
Generation 1200:	Best = 1.548905e+03	FES = 120100	Diversity = 1.8274e+00
Generation 1210:	Best = 1.548872e+03	FES = 121100	Diversity = 1.7953e+00
Generation 1220:	Best = 1.548148e+03	FES = 122100	Diversity = 1.7848e+00
Generation 1230:	Best = 1.548135e+03	FES = 123100	Diversity = 1.7590e+00
Generation 1240:	Best = 1.546985e+03	FES = 124100	Diversity = 1.7125e+00
Generation 1250:	Best = 1.546116e+03	FES = 125100	Diversity = 1.6649e+00
Generation 1260:	Best = 1.546116e+03	FES = 126100	Diversity = 1.6520e+00
Generation 1270:	Best = 1.546055e+03	FES = 127100	Diversity = 1.6482e+00
Generation 1280:	Best = 1.545982e+03	FES = 128100	Diversity = 1.6396e+00
Generation 1290:	Best = 1.545896e+03	FES = 129100	Diversity = 1.6199e+00
Generation 1300:	Best = 1.545896e+03	FES = 130100	Diversity = 1.5952e+00
Generation 1310:	Best = 1.545896e+03	FES = 131100	Diversity = 1.5865e+00
Generation 1320:	Best = 1.545891e+03	FES = 132100	Diversity = 1.5854e+00
Generation 1330:	Best = 1.545886e+03	FES = 133100	Diversity = 1.5777e+00
Generation 1340:	Best = 1.545886e+03	FES = 134100	Diversity = 1.5722e+00
Generation 1350:	Best = 1.545883e+03	FES = 135100	Diversity = 1.5620e+00
Generation 1360:	Best = 1.545883e+03	FES = 136100	Diversity = 1.5523e+00
Generation 1370:	Best = 1.545870e+03	FES = 137100	Diversity = 1.5520e+00
Generation 1380:	Best = 1.545862e+03	FES = 138100	Diversity = 1.5508e+00
Generation 1390:	Best = 1.545862e+03	FES = 139100	Diversity = 1.5506e+00
Generation 1400:	Best = 1.545830e+03	FES = 140100	Diversity = 1.5521e+00
Generation 1410:	Best = 1.545829e+03	FES = 141100	Diversity = 1.5422e+00
Generation 1420:	Best = 1.545827e+03	FES = 142100	Diversity = 1.5229e+00
Generation 1430:	Best = 1.545821e+03	FES = 143100	Diversity = 1.5053e+00
Generation 1440:	Best = 1.545819e+03	FES = 144100	Diversity = 1.5086e+00
Generation 1450:	Best = 1.545794e+03	FES = 145100	Diversity = 1.5079e+00
Generation 1460:	Best = 1.545244e+03	FES = 146100	Diversity = 1.4934e+00
Generation 1470:	Best = 1.545242e+03	FES = 147100	Diversity = 1.4683e+00
Generation 1480:	Best = 1.545242e+03	FES = 148100	Diversity = 1.4686e+00
Generation 1490:	Best = 1.544903e+03	FES = 149100	Diversity = 1.4486e+00
Generation 1500:	Best = 1.544884e+03	FES = 150100	Diversity = 1.4477e+00
Generation 1510:	Best = 1.544884e+03	FES = 151100	Diversity = 1.4283e+00
Generation 1520:	Best = 1.544808e+03	FES = 152100	Diversity = 1.4195e+00

Generation 1530:	Best = 1.544800e+03	FES = 153100	Diversity = 1.4201e+00
Generation 1540:	Best = 1.544799e+03	FES = 154100	Diversity = 1.4060e+00
Generation 1550:	Best = 1.544774e+03	FES = 155100	Diversity = 1.4056e+00
Generation 1560:	Best = 1.544770e+03	FES = 156100	Diversity = 1.4037e+00
Generation 1570:	Best = 1.544770e+03	FES = 157100	Diversity = 1.3994e+00
Generation 1580:	Best = 1.544770e+03	FES = 158100	Diversity = 1.3960e+00
Generation 1590:	Best = 1.544698e+03	FES = 159100	Diversity = 1.3726e+00
Generation 1600:	Best = 1.544698e+03	FES = 160100	Diversity = 1.3728e+00
Generation 1610:	Best = 1.544698e+03	FES = 161100	Diversity = 1.3722e+00
Generation 1620:	Best = 1.544691e+03	FES = 162100	Diversity = 1.3657e+00
Generation 1630:	Best = 1.544655e+03	FES = 163100	Diversity = 1.3650e+00
Generation 1640:	Best = 1.544655e+03	FES = 164100	Diversity = 1.3661e+00
Generation 1650:	Best = 1.544649e+03	FES = 165100	Diversity = 1.3631e+00
Generation 1660:	Best = 1.544649e+03	FES = 166100	Diversity = 1.3553e+00
Generation 1670:	Best = 1.544648e+03	FES = 167100	Diversity = 1.3594e+00
Generation 1680:	Best = 1.544648e+03	FES = 168100	Diversity = 1.3601e+00
Generation 1690:	Best = 1.544648e+03	FES = 169100	Diversity = 1.3309e+00
Generation 1700:	Best = 1.544647e+03	FES = 170100	Diversity = 1.3301e+00
Generation 1710:	Best = 1.544622e+03	FES = 171100	Diversity = 1.3290e+00
Generation 1720:	Best = 1.544622e+03	FES = 172100	Diversity = 1.3310e+00
Generation 1730:	Best = 1.544622e+03	FES = 173100	Diversity = 1.3356e+00
Generation 1740:	Best = 1.544622e+03	FES = 174100	Diversity = 1.3359e+00
Generation 1750:	Best = 1.544621e+03	FES = 175100	Diversity = 1.3350e+00
Generation 1760:	Best = 1.544621e+03	FES = 176100	Diversity = 1.3374e+00
Generation 1770:	Best = 1.544619e+03	FES = 177100	Diversity = 1.3190e+00
Generation 1780:	Best = 1.544517e+03	FES = 178100	Diversity = 1.3114e+00
Generation 1790:	Best = 1.544517e+03	FES = 179100	Diversity = 1.3096e+00
Generation 1800:	Best = 1.544517e+03	FES = 180100	Diversity = 1.3049e+00
Generation 1810:	Best = 1.544517e+03	FES = 181100	Diversity = 1.3089e+00
Generation 1820:	Best = 1.544514e+03	FES = 182100	Diversity = 1.3065e+00
Generation 1830:	Best = 1.544508e+03	FES = 183100	Diversity = 1.2978e+00
Generation 1840:	Best = 1.544507e+03	FES = 184100	Diversity = 1.2969e+00
Generation 1850:	Best = 1.544504e+03	FES = 185100	Diversity = 1.2973e+00
Generation 1860:	Best = 1.544316e+03	FES = 186100	Diversity = 1.2851e+00
Generation 1870:	Best = 1.544316e+03	FES = 187100	Diversity = 1.2852e+00
Generation 1880:	Best = 1.544316e+03	FES = 188100	Diversity = 1.2847e+00
Generation 1890:	Best = 1.544297e+03	FES = 189100	Diversity = 1.2729e+00
Generation 1900:	Best = 1.544294e+03	FES = 190100	Diversity = 1.2695e+00
Generation 1910:	Best = 1.544294e+03	FES = 191100	Diversity = 1.2694e+00
Generation 1920:	Best = 1.544294e+03	FES = 192100	Diversity = 1.2644e+00
Generation 1930:	Best = 1.544294e+03	FES = 193100	Diversity = 1.2705e+00
Generation 1940:	Best = 1.544293e+03	FES = 194100	Diversity = 1.2745e+00
Generation 1950:	Best = 1.544293e+03	FES = 195100	Diversity = 1.2744e+00
Generation 1960:	Best = 1.544289e+03	FES = 196100	Diversity = 1.2744e+00
Generation 1970:	Best = 1.544289e+03	FES = 197100	Diversity = 1.2637e+00
Generation 1980:	Best = 1.544289e+03	FES = 198100	Diversity = 1.2646e+00
Generation 1990:	Best = 1.544289e+03	FES = 199100	Diversity = 1.2592e+00
Generation 2000:	Best = 1.544288e+03	FES = 200100	Diversity = 1.2594e+00
Generation 2010:	Best = 1.544288e+03	FES = 201100	Diversity = 1.2518e+00
Generation 2020:	Best = 1.544284e+03	FES = 202100	Diversity = 1.2529e+00
Generation 2030:	Best = 1.544282e+03	FES = 203100	Diversity = 1.2448e+00
Generation 2040:	Best = 1.544282e+03	FES = 204100	Diversity = 1.2446e+00
Generation 2050:	Best = 1.544282e+03	FES = 205100	Diversity = 1.2432e+00
Generation 2060:	Best = 1.544282e+03	FES = 206100	Diversity = 1.2430e+00

Generation 2070:	Best = 1.544199e+03	FES = 207100	Diversity = 1.2420e+00
Generation 2080:	Best = 1.544199e+03	FES = 208100	Diversity = 1.2414e+00
Generation 2090:	Best = 1.544199e+03	FES = 209100	Diversity = 1.2314e+00
Generation 2100:	Best = 1.544199e+03	FES = 210100	Diversity = 1.2214e+00
Generation 2110:	Best = 1.544193e+03	FES = 211100	Diversity = 1.2234e+00
Generation 2120:	Best = 1.544193e+03	FES = 212100	Diversity = 1.2236e+00
Generation 2130:	Best = 1.544193e+03	FES = 213100	Diversity = 1.2233e+00
Generation 2140:	Best = 1.544193e+03	FES = 214100	Diversity = 1.2227e+00
Generation 2150:	Best = 1.543739e+03	FES = 215100	Diversity = 1.2227e+00
Generation 2160:	Best = 1.543739e+03	FES = 216100	Diversity = 1.2226e+00
Generation 2170:	Best = 1.543727e+03	FES = 217100	Diversity = 1.2222e+00
Generation 2180:	Best = 1.543727e+03	FES = 218100	Diversity = 1.2303e+00
Generation 2190:	Best = 1.543727e+03	FES = 219100	Diversity = 1.2249e+00
Generation 2200:	Best = 1.543726e+03	FES = 220100	Diversity = 1.2213e+00
Generation 2210:	Best = 1.543726e+03	FES = 221100	Diversity = 1.2187e+00
Generation 2220:	Best = 1.543726e+03	FES = 222100	Diversity = 1.2187e+00
Generation 2230:	Best = 1.543597e+03	FES = 223100	Diversity = 1.2187e+00
Generation 2240:	Best = 1.543597e+03	FES = 224100	Diversity = 1.2188e+00
Generation 2250:	Best = 1.543597e+03	FES = 225100	Diversity = 1.2125e+00
Generation 2260:	Best = 1.543597e+03	FES = 226100	Diversity = 1.2127e+00
Generation 2270:	Best = 1.543597e+03	FES = 227100	Diversity = 1.2140e+00
Generation 2280:	Best = 1.543596e+03	FES = 228100	Diversity = 1.2137e+00
Generation 2290:	Best = 1.543591e+03	FES = 229100	Diversity = 1.2034e+00
Generation 2300:	Best = 1.543591e+03	FES = 230100	Diversity = 1.1931e+00
Generation 2310:	Best = 1.543591e+03	FES = 231100	Diversity = 1.1935e+00
Generation 2320:	Best = 1.543589e+03	FES = 232100	Diversity = 1.1931e+00
Generation 2330:	Best = 1.543589e+03	FES = 233100	Diversity = 1.1941e+00
Generation 2340:	Best = 1.543589e+03	FES = 234100	Diversity = 1.1941e+00
Generation 2350:	Best = 1.543589e+03	FES = 235100	Diversity = 1.1948e+00
Generation 2360:	Best = 1.543589e+03	FES = 236100	Diversity = 1.1947e+00
Generation 2370:	Best = 1.543589e+03	FES = 237100	Diversity = 1.2087e+00
Generation 2380:	Best = 1.543588e+03	FES = 238100	Diversity = 1.2088e+00
Generation 2390:	Best = 1.543588e+03	FES = 239100	Diversity = 1.2089e+00
Generation 2400:	Best = 1.543586e+03	FES = 240100	Diversity = 1.2089e+00
Generation 2410:	Best = 1.543568e+03	FES = 241100	Diversity = 1.2087e+00
Generation 2420:	Best = 1.543568e+03	FES = 242100	Diversity = 1.2087e+00
Generation 2430:	Best = 1.543568e+03	FES = 243100	Diversity = 1.2088e+00
Generation 2440:	Best = 1.543530e+03	FES = 244100	Diversity = 1.2087e+00
Generation 2450:	Best = 1.543530e+03	FES = 245100	Diversity = 1.2011e+00
Generation 2460:	Best = 1.543529e+03	FES = 246100	Diversity = 1.2011e+00
Generation 2470:	Best = 1.543524e+03	FES = 247100	Diversity = 1.2015e+00
Generation 2480:	Best = 1.543524e+03	FES = 248100	Diversity = 1.2022e+00
Generation 2490:	Best = 1.543524e+03	FES = 249100	Diversity = 1.2046e+00
Generation 2500:	Best = 1.543523e+03	FES = 250100	Diversity = 1.1910e+00
Generation 2510:	Best = 1.543523e+03	FES = 251100	Diversity = 1.1674e+00
Generation 2520:	Best = 1.543523e+03	FES = 252100	Diversity = 1.1663e+00
Generation 2530:	Best = 1.543522e+03	FES = 253100	Diversity = 1.1675e+00
Generation 2540:	Best = 1.543522e+03	FES = 254100	Diversity = 1.1688e+00
Generation 2550:	Best = 1.543522e+03	FES = 255100	Diversity = 1.1582e+00
Generation 2560:	Best = 1.543522e+03	FES = 256100	Diversity = 1.1575e+00
Generation 2570:	Best = 1.543522e+03	FES = 257100	Diversity = 1.1528e+00
Generation 2580:	Best = 1.543522e+03	FES = 258100	Diversity = 1.1528e+00
Generation 2590:	Best = 1.543521e+03	FES = 259100	Diversity = 1.1425e+00
Generation 2600:	Best = 1.543521e+03	FES = 260100	Diversity = 1.1353e+00

Generation 2610:	Best = 1.543521e+03	FES = 261100	Diversity = 1.1343e+00
Generation 2620:	Best = 1.543521e+03	FES = 262100	Diversity = 1.1329e+00
Generation 2630:	Best = 1.543521e+03	FES = 263100	Diversity = 1.1372e+00
Generation 2640:	Best = 1.543521e+03	FES = 264100	Diversity = 1.1361e+00
Generation 2650:	Best = 1.543521e+03	FES = 265100	Diversity = 1.1112e+00
Generation 2660:	Best = 1.543521e+03	FES = 266100	Diversity = 1.1178e+00
Generation 2670:	Best = 1.543521e+03	FES = 267100	Diversity = 1.1184e+00
Generation 2680:	Best = 1.543521e+03	FES = 268100	Diversity = 1.1206e+00
Generation 2690:	Best = 1.543521e+03	FES = 269100	Diversity = 1.1203e+00
Generation 2700:	Best = 1.543521e+03	FES = 270100	Diversity = 1.1192e+00
Generation 2710:	Best = 1.543521e+03	FES = 271100	Diversity = 1.1147e+00
Generation 2720:	Best = 1.543513e+03	FES = 272100	Diversity = 1.1164e+00
Generation 2730:	Best = 1.543494e+03	FES = 273100	Diversity = 1.1103e+00
Generation 2740:	Best = 1.543478e+03	FES = 274100	Diversity = 1.1101e+00
Generation 2750:	Best = 1.543478e+03	FES = 275100	Diversity = 1.1101e+00
Generation 2760:	Best = 1.543478e+03	FES = 276100	Diversity = 1.1096e+00
Generation 2770:	Best = 1.543476e+03	FES = 277100	Diversity = 1.1093e+00
Generation 2780:	Best = 1.543475e+03	FES = 278100	Diversity = 1.1095e+00
Generation 2790:	Best = 1.543475e+03	FES = 279100	Diversity = 1.1123e+00
Generation 2800:	Best = 1.543472e+03	FES = 280100	Diversity = 1.1165e+00
Generation 2810:	Best = 1.543472e+03	FES = 281100	Diversity = 1.1152e+00
Generation 2820:	Best = 1.543472e+03	FES = 282100	Diversity = 1.1154e+00
Generation 2830:	Best = 1.543472e+03	FES = 283100	Diversity = 1.1154e+00
Generation 2840:	Best = 1.543472e+03	FES = 284100	Diversity = 1.1146e+00
Generation 2850:	Best = 1.543472e+03	FES = 285100	Diversity = 1.1146e+00
Generation 2860:	Best = 1.543456e+03	FES = 286100	Diversity = 1.1144e+00
Generation 2870:	Best = 1.543455e+03	FES = 287100	Diversity = 1.1102e+00
Generation 2880:	Best = 1.543455e+03	FES = 288100	Diversity = 1.1101e+00
Generation 2890:	Best = 1.543455e+03	FES = 289100	Diversity = 1.1171e+00
Generation 2900:	Best = 1.543455e+03	FES = 290100	Diversity = 1.1187e+00
Generation 2910:	Best = 1.543455e+03	FES = 291100	Diversity = 1.1156e+00
Generation 2920:	Best = 1.543455e+03	FES = 292100	Diversity = 1.1153e+00
Generation 2930:	Best = 1.543455e+03	FES = 293100	Diversity = 1.1154e+00
Generation 2940:	Best = 1.543455e+03	FES = 294100	Diversity = 1.1151e+00
Generation 2950:	Best = 1.543191e+03	FES = 295100	Diversity = 1.1153e+00
Generation 2960:	Best = 1.543183e+03	FES = 296100	Diversity = 1.1154e+00
Generation 2970:	Best = 1.543183e+03	FES = 297100	Diversity = 1.1107e+00
Generation 2980:	Best = 1.543183e+03	FES = 298100	Diversity = 1.1106e+00
Generation 2990:	Best = 1.543183e+03	FES = 299100	Diversity = 1.1031e+00

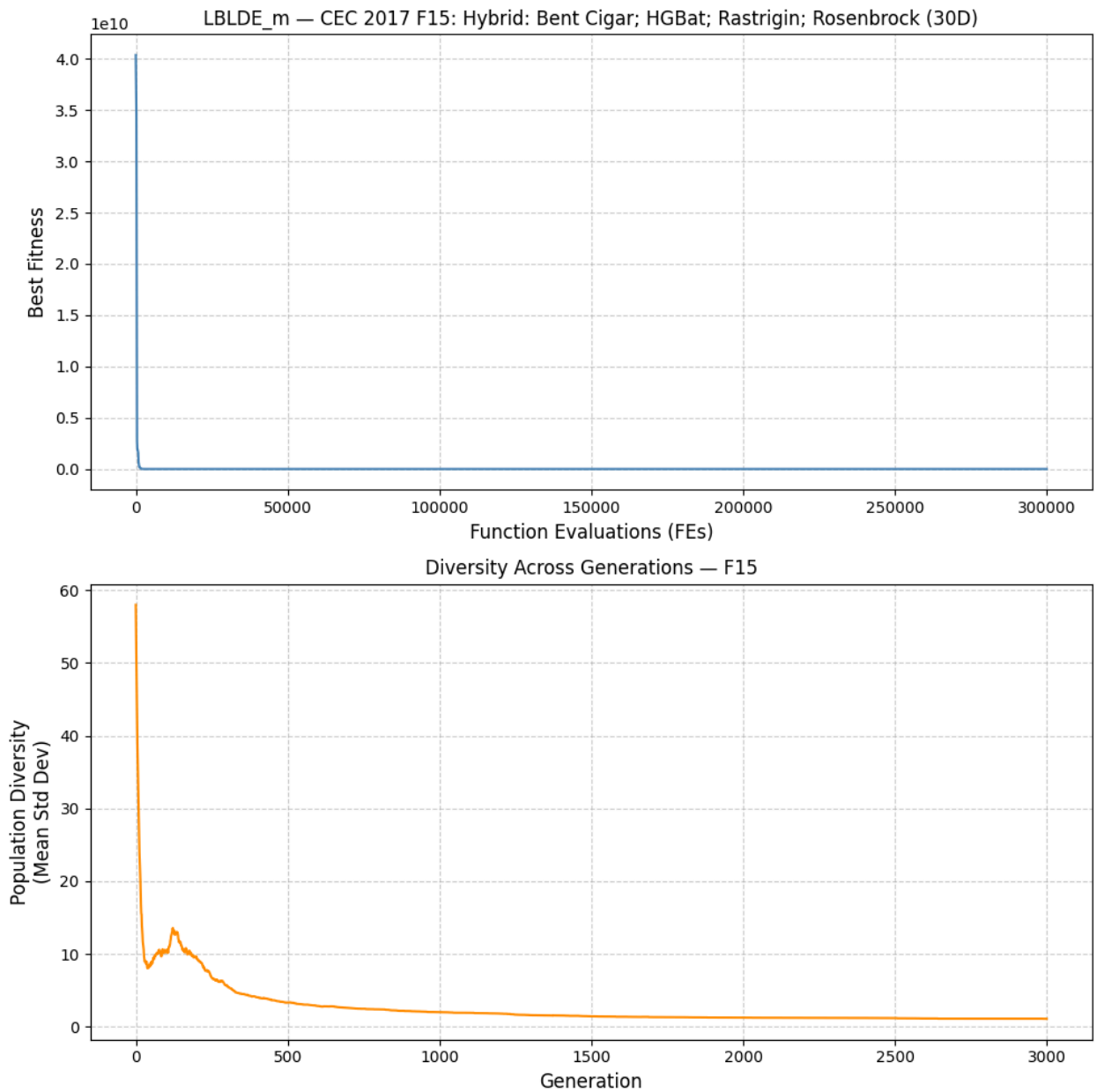
Optimization completed!

Final Best Fitness = 1.543183e+03

Total Function Evaluations = 300000

```
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Results:
  Best Fitness           : 1.543183e+03
  Global Optimum         : 1.500000e+03
  Error (|best-optimum|) : 4.318287e+01
  Best Solution (5 dim)  : [ 5.02838289 25.62184777 45.10938633 -66.61122842
41.19662324]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F15.png

```
In [42]: # =====  
# F16 – Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)  
# =====  
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid: Schaffer; HGBat
```

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LBLDE_m on CEC 2017 F16 – Single Run

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Function : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (30D)

Bounds : [-100, 100]

Global optimum : 1600.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 8.664946e+03 | Diversity = 5.7976e+01

Generation	10:	Best = 4.402257e+03	FES = 1100	Diversity = 3.0379e+01
Generation	20:	Best = 4.191102e+03	FES = 2100	Diversity = 2.3835e+01
Generation	30:	Best = 3.928740e+03	FES = 3100	Diversity = 2.3614e+01
Generation	40:	Best = 3.887490e+03	FES = 4100	Diversity = 2.3136e+01
Generation	50:	Best = 3.777875e+03	FES = 5100	Diversity = 2.2845e+01
Generation	60:	Best = 2.992188e+03	FES = 6100	Diversity = 2.2522e+01
Generation	70:	Best = 2.992188e+03	FES = 7100	Diversity = 2.1963e+01
Generation	80:	Best = 2.992050e+03	FES = 8100	Diversity = 2.1463e+01
Generation	90:	Best = 2.992050e+03	FES = 9100	Diversity = 2.1080e+01
Generation	100:	Best = 2.992050e+03	FES = 10100	Diversity = 2.0767e+01
Generation	110:	Best = 2.985086e+03	FES = 11100	Diversity = 2.0720e+01
Generation	120:	Best = 2.985086e+03	FES = 12100	Diversity = 2.0106e+01
Generation	130:	Best = 2.985086e+03	FES = 13100	Diversity = 2.0129e+01
Generation	140:	Best = 2.985064e+03	FES = 14100	Diversity = 1.9520e+01
Generation	150:	Best = 2.985064e+03	FES = 15100	Diversity = 1.9045e+01
Generation	160:	Best = 2.977493e+03	FES = 16100	Diversity = 1.8643e+01
Generation	170:	Best = 2.972846e+03	FES = 17100	Diversity = 1.8350e+01
Generation	180:	Best = 2.972534e+03	FES = 18100	Diversity = 1.8113e+01
Generation	190:	Best = 2.857712e+03	FES = 19100	Diversity = 1.7780e+01
Generation	200:	Best = 2.856713e+03	FES = 20100	Diversity = 1.7495e+01
Generation	210:	Best = 2.840831e+03	FES = 21100	Diversity = 1.7039e+01
Generation	220:	Best = 2.834761e+03	FES = 22100	Diversity = 1.7015e+01
Generation	230:	Best = 2.834719e+03	FES = 23100	Diversity = 1.6940e+01
Generation	240:	Best = 2.727953e+03	FES = 24100	Diversity = 1.6851e+01
Generation	250:	Best = 2.727934e+03	FES = 25100	Diversity = 1.6485e+01
Generation	260:	Best = 2.727117e+03	FES = 26100	Diversity = 1.6217e+01
Generation	270:	Best = 2.727108e+03	FES = 27100	Diversity = 1.5888e+01
Generation	280:	Best = 2.726870e+03	FES = 28100	Diversity = 1.5875e+01
Generation	290:	Best = 2.726870e+03	FES = 29100	Diversity = 1.5606e+01
Generation	300:	Best = 2.725500e+03	FES = 30100	Diversity = 1.5602e+01
Generation	310:	Best = 2.725170e+03	FES = 31100	Diversity = 1.5655e+01
Generation	320:	Best = 2.724931e+03	FES = 32100	Diversity = 1.5574e+01
Generation	330:	Best = 2.724931e+03	FES = 33100	Diversity = 1.5739e+01
Generation	340:	Best = 2.724930e+03	FES = 34100	Diversity = 1.5549e+01
Generation	350:	Best = 2.724930e+03	FES = 35100	Diversity = 1.5286e+01
Generation	360:	Best = 2.660039e+03	FES = 36100	Diversity = 1.5217e+01
Generation	370:	Best = 2.660039e+03	FES = 37100	Diversity = 1.4975e+01
Generation	380:	Best = 2.660039e+03	FES = 38100	Diversity = 1.4996e+01
Generation	390:	Best = 2.653631e+03	FES = 39100	Diversity = 1.4823e+01
Generation	400:	Best = 2.651016e+03	FES = 40100	Diversity = 1.4824e+01
Generation	410:	Best = 2.650381e+03	FES = 41100	Diversity = 1.4684e+01
Generation	420:	Best = 2.649070e+03	FES = 42100	Diversity = 1.4626e+01
Generation	430:	Best = 2.648303e+03	FES = 43100	Diversity = 1.4506e+01
Generation	440:	Best = 2.631799e+03	FES = 44100	Diversity = 1.4482e+01

Generation	450:	Best = 2.631748e+03	FES = 45100	Diversity = 1.4224e+01
Generation	460:	Best = 2.631747e+03	FES = 46100	Diversity = 1.4170e+01
Generation	470:	Best = 2.631704e+03	FES = 47100	Diversity = 1.4170e+01
Generation	480:	Best = 2.631704e+03	FES = 48100	Diversity = 1.4139e+01
Generation	490:	Best = 2.631650e+03	FES = 49100	Diversity = 1.4125e+01
Generation	500:	Best = 2.628932e+03	FES = 50100	Diversity = 1.4011e+01
Generation	510:	Best = 2.628932e+03	FES = 51100	Diversity = 1.3894e+01
Generation	520:	Best = 2.628433e+03	FES = 52100	Diversity = 1.3895e+01
Generation	530:	Best = 2.628433e+03	FES = 53100	Diversity = 1.3743e+01
Generation	540:	Best = 2.628433e+03	FES = 54100	Diversity = 1.3658e+01
Generation	550:	Best = 2.628247e+03	FES = 55100	Diversity = 1.3630e+01
Generation	560:	Best = 2.628216e+03	FES = 56100	Diversity = 1.3524e+01
Generation	570:	Best = 2.628181e+03	FES = 57100	Diversity = 1.3488e+01
Generation	580:	Best = 2.628065e+03	FES = 58100	Diversity = 1.3562e+01
Generation	590:	Best = 2.627996e+03	FES = 59100	Diversity = 1.3416e+01
Generation	600:	Best = 2.627968e+03	FES = 60100	Diversity = 1.3369e+01
Generation	610:	Best = 2.627870e+03	FES = 61100	Diversity = 1.3303e+01
Generation	620:	Best = 2.627870e+03	FES = 62100	Diversity = 1.3247e+01
Generation	630:	Best = 2.627870e+03	FES = 63100	Diversity = 1.3228e+01
Generation	640:	Best = 2.627870e+03	FES = 64100	Diversity = 1.3205e+01
Generation	650:	Best = 2.627867e+03	FES = 65100	Diversity = 1.3215e+01
Generation	660:	Best = 2.627650e+03	FES = 66100	Diversity = 1.3254e+01
Generation	670:	Best = 2.622964e+03	FES = 67100	Diversity = 1.3242e+01
Generation	680:	Best = 2.622705e+03	FES = 68100	Diversity = 1.3005e+01
Generation	690:	Best = 2.622705e+03	FES = 69100	Diversity = 1.2980e+01
Generation	700:	Best = 2.622705e+03	FES = 70100	Diversity = 1.2959e+01
Generation	710:	Best = 2.622656e+03	FES = 71100	Diversity = 1.2931e+01
Generation	720:	Best = 2.621784e+03	FES = 72100	Diversity = 1.2884e+01
Generation	730:	Best = 2.621784e+03	FES = 73100	Diversity = 1.2924e+01
Generation	740:	Best = 2.621755e+03	FES = 74100	Diversity = 1.2914e+01
Generation	750:	Best = 2.621753e+03	FES = 75100	Diversity = 1.2835e+01
Generation	760:	Best = 2.621750e+03	FES = 76100	Diversity = 1.2812e+01
Generation	770:	Best = 2.621750e+03	FES = 77100	Diversity = 1.2800e+01
Generation	780:	Best = 2.621697e+03	FES = 78100	Diversity = 1.2794e+01
Generation	790:	Best = 2.611007e+03	FES = 79100	Diversity = 1.2787e+01
Generation	800:	Best = 2.610860e+03	FES = 80100	Diversity = 1.2855e+01
Generation	810:	Best = 2.610860e+03	FES = 81100	Diversity = 1.2839e+01
Generation	820:	Best = 2.610766e+03	FES = 82100	Diversity = 1.2772e+01
Generation	830:	Best = 2.610763e+03	FES = 83100	Diversity = 1.2757e+01
Generation	840:	Best = 2.610736e+03	FES = 84100	Diversity = 1.2742e+01
Generation	850:	Best = 2.610736e+03	FES = 85100	Diversity = 1.2732e+01
Generation	860:	Best = 2.610703e+03	FES = 86100	Diversity = 1.2738e+01
Generation	870:	Best = 2.610703e+03	FES = 87100	Diversity = 1.2738e+01
Generation	880:	Best = 2.610703e+03	FES = 88100	Diversity = 1.2732e+01
Generation	890:	Best = 2.610686e+03	FES = 89100	Diversity = 1.2725e+01
Generation	900:	Best = 2.584784e+03	FES = 90100	Diversity = 1.2717e+01
Generation	910:	Best = 2.584783e+03	FES = 91100	Diversity = 1.2609e+01
Generation	920:	Best = 2.584783e+03	FES = 92100	Diversity = 1.2605e+01
Generation	930:	Best = 2.584783e+03	FES = 93100	Diversity = 1.2549e+01
Generation	940:	Best = 2.584774e+03	FES = 94100	Diversity = 1.2536e+01
Generation	950:	Best = 2.584772e+03	FES = 95100	Diversity = 1.2522e+01
Generation	960:	Best = 2.584772e+03	FES = 96100	Diversity = 1.2527e+01
Generation	970:	Best = 2.584768e+03	FES = 97100	Diversity = 1.2519e+01
Generation	980:	Best = 2.584758e+03	FES = 98100	Diversity = 1.2589e+01

Generation 990:	Best = 2.584755e+03	FES = 99100	Diversity = 1.2529e+01
Generation 1000:	Best = 2.518584e+03	FES = 100100	Diversity = 1.2522e+01
Generation 1010:	Best = 2.518577e+03	FES = 101100	Diversity = 1.2515e+01
Generation 1020:	Best = 2.518576e+03	FES = 102100	Diversity = 1.2384e+01
Generation 1030:	Best = 2.518576e+03	FES = 103100	Diversity = 1.2375e+01
Generation 1040:	Best = 2.518576e+03	FES = 104100	Diversity = 1.2306e+01
Generation 1050:	Best = 2.518576e+03	FES = 105100	Diversity = 1.2296e+01
Generation 1060:	Best = 2.518575e+03	FES = 106100	Diversity = 1.2297e+01
Generation 1070:	Best = 2.518575e+03	FES = 107100	Diversity = 1.2263e+01
Generation 1080:	Best = 2.518575e+03	FES = 108100	Diversity = 1.2250e+01
Generation 1090:	Best = 2.518571e+03	FES = 109100	Diversity = 1.2242e+01
Generation 1100:	Best = 2.518571e+03	FES = 110100	Diversity = 1.2239e+01
Generation 1110:	Best = 2.518571e+03	FES = 111100	Diversity = 1.2235e+01
Generation 1120:	Best = 2.518571e+03	FES = 112100	Diversity = 1.2238e+01
Generation 1130:	Best = 2.518571e+03	FES = 113100	Diversity = 1.2217e+01
Generation 1140:	Best = 2.518571e+03	FES = 114100	Diversity = 1.2137e+01
Generation 1150:	Best = 2.518117e+03	FES = 115100	Diversity = 1.2133e+01
Generation 1160:	Best = 2.517665e+03	FES = 116100	Diversity = 1.2133e+01
Generation 1170:	Best = 2.517665e+03	FES = 117100	Diversity = 1.2124e+01
Generation 1180:	Best = 2.517665e+03	FES = 118100	Diversity = 1.2114e+01
Generation 1190:	Best = 2.517665e+03	FES = 119100	Diversity = 1.2114e+01
Generation 1200:	Best = 2.517665e+03	FES = 120100	Diversity = 1.2111e+01
Generation 1210:	Best = 2.517665e+03	FES = 121100	Diversity = 1.2109e+01
Generation 1220:	Best = 2.516459e+03	FES = 122100	Diversity = 1.2102e+01
Generation 1230:	Best = 2.516099e+03	FES = 123100	Diversity = 1.2102e+01
Generation 1240:	Best = 2.516099e+03	FES = 124100	Diversity = 1.2098e+01
Generation 1250:	Best = 2.516099e+03	FES = 125100	Diversity = 1.2093e+01
Generation 1260:	Best = 2.516095e+03	FES = 126100	Diversity = 1.2089e+01
Generation 1270:	Best = 2.515956e+03	FES = 127100	Diversity = 1.2090e+01
Generation 1280:	Best = 2.515939e+03	FES = 128100	Diversity = 1.2096e+01
Generation 1290:	Best = 2.475218e+03	FES = 129100	Diversity = 1.2095e+01
Generation 1300:	Best = 2.454840e+03	FES = 130100	Diversity = 1.2095e+01
Generation 1310:	Best = 2.454840e+03	FES = 131100	Diversity = 1.2088e+01
Generation 1320:	Best = 2.454840e+03	FES = 132100	Diversity = 1.2085e+01
Generation 1330:	Best = 2.441709e+03	FES = 133100	Diversity = 1.2150e+01
Generation 1340:	Best = 2.441430e+03	FES = 134100	Diversity = 1.2149e+01
Generation 1350:	Best = 2.440600e+03	FES = 135100	Diversity = 1.2146e+01
Generation 1360:	Best = 2.440600e+03	FES = 136100	Diversity = 1.2146e+01
Generation 1370:	Best = 2.440416e+03	FES = 137100	Diversity = 1.2143e+01
Generation 1380:	Best = 2.440412e+03	FES = 138100	Diversity = 1.2141e+01
Generation 1390:	Best = 2.440145e+03	FES = 139100	Diversity = 1.2140e+01
Generation 1400:	Best = 2.439815e+03	FES = 140100	Diversity = 1.2161e+01
Generation 1410:	Best = 2.439815e+03	FES = 141100	Diversity = 1.2162e+01
Generation 1420:	Best = 2.438024e+03	FES = 142100	Diversity = 1.2157e+01
Generation 1430:	Best = 2.437864e+03	FES = 143100	Diversity = 1.2154e+01
Generation 1440:	Best = 2.437856e+03	FES = 144100	Diversity = 1.2154e+01
Generation 1450:	Best = 2.437778e+03	FES = 145100	Diversity = 1.2151e+01
Generation 1460:	Best = 2.437778e+03	FES = 146100	Diversity = 1.2149e+01
Generation 1470:	Best = 2.437778e+03	FES = 147100	Diversity = 1.2138e+01
Generation 1480:	Best = 2.437717e+03	FES = 148100	Diversity = 1.2087e+01
Generation 1490:	Best = 2.437707e+03	FES = 149100	Diversity = 1.2082e+01
Generation 1500:	Best = 2.437707e+03	FES = 150100	Diversity = 1.2080e+01
Generation 1510:	Best = 2.437707e+03	FES = 151100	Diversity = 1.2075e+01
Generation 1520:	Best = 2.437662e+03	FES = 152100	Diversity = 1.2071e+01

Generation 1530:	Best = 2.437662e+03	FES = 153100	Diversity = 1.2069e+01
Generation 1540:	Best = 2.437662e+03	FES = 154100	Diversity = 1.2067e+01
Generation 1550:	Best = 2.437662e+03	FES = 155100	Diversity = 1.2061e+01
Generation 1560:	Best = 2.437660e+03	FES = 156100	Diversity = 1.2058e+01
Generation 1570:	Best = 2.437660e+03	FES = 157100	Diversity = 1.2064e+01
Generation 1580:	Best = 2.437653e+03	FES = 158100	Diversity = 1.2064e+01
Generation 1590:	Best = 2.437635e+03	FES = 159100	Diversity = 1.2064e+01
Generation 1600:	Best = 2.437367e+03	FES = 160100	Diversity = 1.2064e+01
Generation 1610:	Best = 2.437274e+03	FES = 161100	Diversity = 1.2067e+01
Generation 1620:	Best = 2.437264e+03	FES = 162100	Diversity = 1.2067e+01
Generation 1630:	Best = 2.437264e+03	FES = 163100	Diversity = 1.2091e+01
Generation 1640:	Best = 2.379535e+03	FES = 164100	Diversity = 1.2070e+01
Generation 1650:	Best = 2.379535e+03	FES = 165100	Diversity = 1.2070e+01
Generation 1660:	Best = 2.353624e+03	FES = 166100	Diversity = 1.2067e+01
Generation 1670:	Best = 2.353586e+03	FES = 167100	Diversity = 1.2032e+01
Generation 1680:	Best = 2.353586e+03	FES = 168100	Diversity = 1.1974e+01
Generation 1690:	Best = 2.353586e+03	FES = 169100	Diversity = 1.1974e+01
Generation 1700:	Best = 2.353586e+03	FES = 170100	Diversity = 1.1973e+01
Generation 1710:	Best = 2.353586e+03	FES = 171100	Diversity = 1.1969e+01
Generation 1720:	Best = 2.353577e+03	FES = 172100	Diversity = 1.1970e+01
Generation 1730:	Best = 2.353577e+03	FES = 173100	Diversity = 1.1970e+01
Generation 1740:	Best = 2.353575e+03	FES = 174100	Diversity = 1.1970e+01
Generation 1750:	Best = 2.353575e+03	FES = 175100	Diversity = 1.1968e+01
Generation 1760:	Best = 2.353575e+03	FES = 176100	Diversity = 1.2012e+01
Generation 1770:	Best = 2.353575e+03	FES = 177100	Diversity = 1.2012e+01
Generation 1780:	Best = 2.353575e+03	FES = 178100	Diversity = 1.1937e+01
Generation 1790:	Best = 2.353574e+03	FES = 179100	Diversity = 1.1936e+01
Generation 1800:	Best = 2.353572e+03	FES = 180100	Diversity = 1.1932e+01
Generation 1810:	Best = 2.353548e+03	FES = 181100	Diversity = 1.1710e+01
Generation 1820:	Best = 2.353548e+03	FES = 182100	Diversity = 1.1707e+01
Generation 1830:	Best = 2.353547e+03	FES = 183100	Diversity = 1.1709e+01
Generation 1840:	Best = 2.353546e+03	FES = 184100	Diversity = 1.1732e+01
Generation 1850:	Best = 2.353546e+03	FES = 185100	Diversity = 1.1730e+01
Generation 1860:	Best = 2.353541e+03	FES = 186100	Diversity = 1.1730e+01
Generation 1870:	Best = 2.353541e+03	FES = 187100	Diversity = 1.1730e+01
Generation 1880:	Best = 2.353541e+03	FES = 188100	Diversity = 1.1725e+01
Generation 1890:	Best = 2.353539e+03	FES = 189100	Diversity = 1.1725e+01
Generation 1900:	Best = 2.353539e+03	FES = 190100	Diversity = 1.1721e+01
Generation 1910:	Best = 2.353539e+03	FES = 191100	Diversity = 1.1717e+01
Generation 1920:	Best = 2.353539e+03	FES = 192100	Diversity = 1.1718e+01
Generation 1930:	Best = 2.353534e+03	FES = 193100	Diversity = 1.1717e+01
Generation 1940:	Best = 2.353534e+03	FES = 194100	Diversity = 1.1670e+01
Generation 1950:	Best = 2.353534e+03	FES = 195100	Diversity = 1.1671e+01
Generation 1960:	Best = 2.353534e+03	FES = 196100	Diversity = 1.1669e+01
Generation 1970:	Best = 2.353505e+03	FES = 197100	Diversity = 1.1667e+01
Generation 1980:	Best = 2.353485e+03	FES = 198100	Diversity = 1.1732e+01
Generation 1990:	Best = 2.353485e+03	FES = 199100	Diversity = 1.1732e+01
Generation 2000:	Best = 2.353481e+03	FES = 200100	Diversity = 1.1718e+01
Generation 2010:	Best = 2.353480e+03	FES = 201100	Diversity = 1.1556e+01
Generation 2020:	Best = 2.353480e+03	FES = 202100	Diversity = 1.1556e+01
Generation 2030:	Best = 2.353480e+03	FES = 203100	Diversity = 1.1557e+01
Generation 2040:	Best = 2.353480e+03	FES = 204100	Diversity = 1.1556e+01
Generation 2050:	Best = 2.353480e+03	FES = 205100	Diversity = 1.1557e+01
Generation 2060:	Best = 2.353480e+03	FES = 206100	Diversity = 1.1557e+01

[illegible]

Generation 2610:	Best = 2.251600e+03	FES = 261100	Diversity = 1.1465e+01
Generation 2620:	Best = 2.251600e+03	FES = 262100	Diversity = 1.1465e+01
Generation 2630:	Best = 2.251600e+03	FES = 263100	Diversity = 1.1464e+01
Generation 2640:	Best = 2.251600e+03	FES = 264100	Diversity = 1.1464e+01
Generation 2650:	Best = 2.251600e+03	FES = 265100	Diversity = 1.1465e+01
Generation 2660:	Best = 2.251600e+03	FES = 266100	Diversity = 1.1466e+01
Generation 2670:	Best = 2.251600e+03	FES = 267100	Diversity = 1.1466e+01
Generation 2680:	Best = 2.251600e+03	FES = 268100	Diversity = 1.1462e+01
Generation 2690:	Best = 2.251600e+03	FES = 269100	Diversity = 1.1462e+01
Generation 2700:	Best = 2.251600e+03	FES = 270100	Diversity = 1.1463e+01
Generation 2710:	Best = 2.251600e+03	FES = 271100	Diversity = 1.1464e+01
Generation 2720:	Best = 2.251600e+03	FES = 272100	Diversity = 1.1208e+01
Generation 2730:	Best = 2.251600e+03	FES = 273100	Diversity = 1.1208e+01
Generation 2740:	Best = 2.251600e+03	FES = 274100	Diversity = 1.1207e+01
Generation 2750:	Best = 2.251600e+03	FES = 275100	Diversity = 1.1208e+01
Generation 2760:	Best = 2.251600e+03	FES = 276100	Diversity = 1.1208e+01
Generation 2770:	Best = 2.251600e+03	FES = 277100	Diversity = 1.1209e+01
Generation 2780:	Best = 2.251600e+03	FES = 278100	Diversity = 1.1183e+01
Generation 2790:	Best = 2.251600e+03	FES = 279100	Diversity = 1.1182e+01
Generation 2800:	Best = 2.251600e+03	FES = 280100	Diversity = 1.1182e+01
Generation 2810:	Best = 2.251600e+03	FES = 281100	Diversity = 1.1184e+01
Generation 2820:	Best = 2.251600e+03	FES = 282100	Diversity = 1.1184e+01
Generation 2830:	Best = 2.251600e+03	FES = 283100	Diversity = 1.1183e+01
Generation 2840:	Best = 2.251600e+03	FES = 284100	Diversity = 1.1183e+01
Generation 2850:	Best = 2.251600e+03	FES = 285100	Diversity = 1.1205e+01
Generation 2860:	Best = 2.251600e+03	FES = 286100	Diversity = 1.1203e+01
Generation 2870:	Best = 2.251600e+03	FES = 287100	Diversity = 1.1201e+01
Generation 2880:	Best = 2.251600e+03	FES = 288100	Diversity = 1.1202e+01
Generation 2890:	Best = 2.251600e+03	FES = 289100	Diversity = 1.1202e+01
Generation 2900:	Best = 2.251600e+03	FES = 290100	Diversity = 1.1203e+01
Generation 2910:	Best = 2.251600e+03	FES = 291100	Diversity = 1.1202e+01
Generation 2920:	Best = 2.251600e+03	FES = 292100	Diversity = 1.1203e+01
Generation 2930:	Best = 2.251600e+03	FES = 293100	Diversity = 1.1203e+01
Generation 2940:	Best = 2.251600e+03	FES = 294100	Diversity = 1.1203e+01
Generation 2950:	Best = 2.251600e+03	FES = 295100	Diversity = 1.1165e+01
Generation 2960:	Best = 2.251600e+03	FES = 296100	Diversity = 1.1165e+01
Generation 2970:	Best = 2.251600e+03	FES = 297100	Diversity = 1.1165e+01
Generation 2980:	Best = 2.251600e+03	FES = 298100	Diversity = 1.1165e+01
Generation 2990:	Best = 2.251600e+03	FES = 299100	Diversity = 1.1164e+01

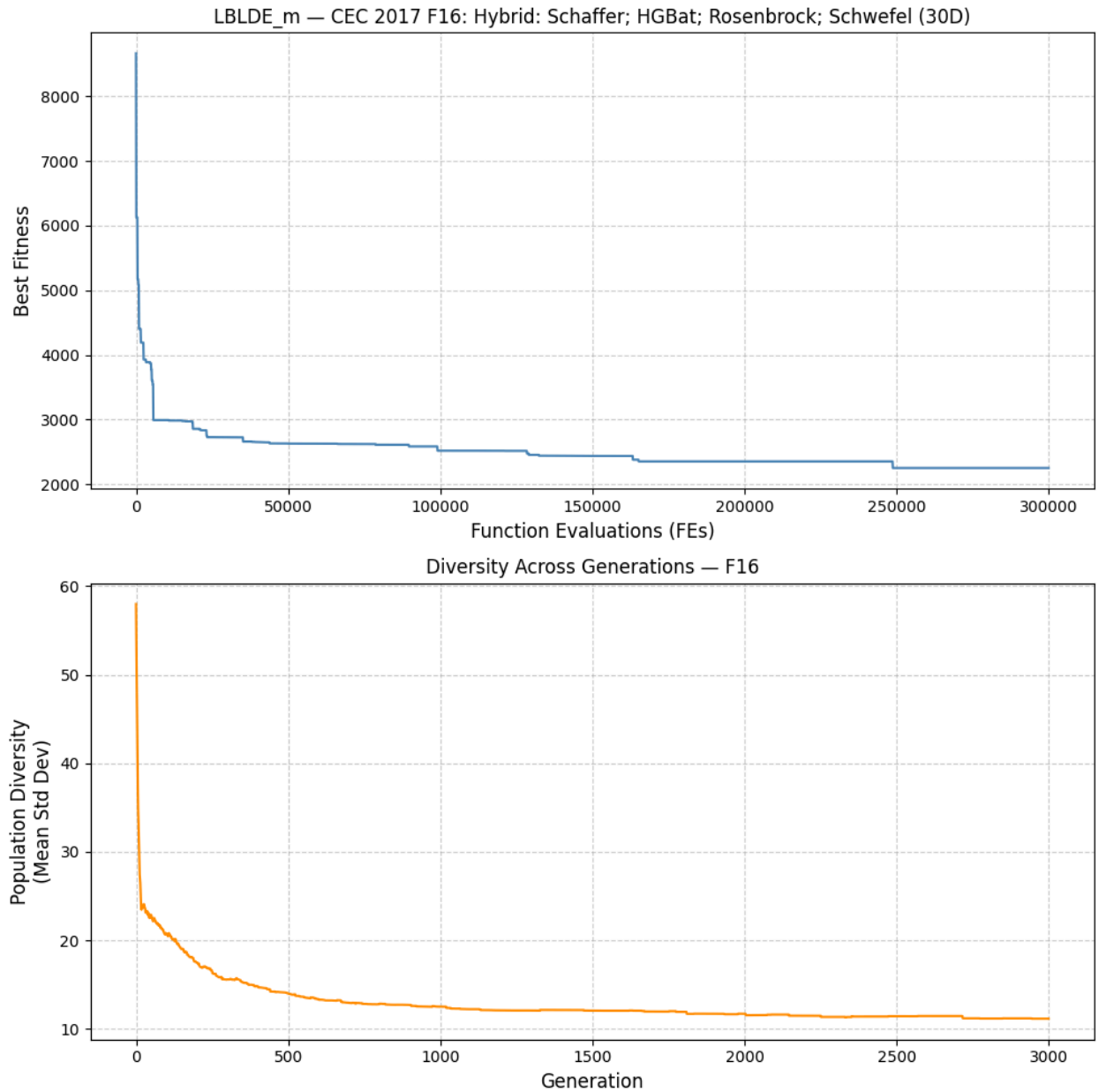
Optimization completed!

Final Best Fitness = 2.251600e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 2.251600e+03
  Global Optimum         : 1.600000e+03
  Error (|best-optimum|) : 6.515999e+02
  Best Solution (5 dim)  : [ 38.02952911 -8.20433856  58.66615523 -26.29713422
4.90720184]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F16.png

```
In [43]: # =====  
# F17 – Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Ra  
# =====  
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid: Katsuura; Ackle
```

=====

=

LBLDE_m on CEC 2017 F17 – Single Run

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Function : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin (30D)

Bounds : [-100, 100]

Global optimum : 1700.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 4.634765e+03 | Diversity = 5.7976e+01

Generation 10: Best = 3.003171e+03 FES = 1100 Diversity = 2.9345e+01

Generation 20: Best = 2.711482e+03 FES = 2100 Diversity = 2.6012e+01

Generation 30: Best = 2.494339e+03 FES = 3100 Diversity = 2.4373e+01

Generation 40: Best = 2.494339e+03 FES = 4100 Diversity = 2.3564e+01

Generation 50: Best = 2.494339e+03 FES = 5100 Diversity = 2.2707e+01

Generation 60: Best = 2.431759e+03 FES = 6100 Diversity = 2.2179e+01

Generation 70: Best = 2.400005e+03 FES = 7100 Diversity = 2.1934e+01

Generation 80: Best = 2.162585e+03 FES = 8100 Diversity = 2.1626e+01

Generation 90: Best = 2.150831e+03 FES = 9100 Diversity = 2.1553e+01

Generation 100: Best = 2.150831e+03 FES = 10100 Diversity = 2.1141e+01

Generation 110: Best = 2.124891e+03 FES = 11100 Diversity = 2.0538e+01

Generation 120: Best = 2.115147e+03 FES = 12100 Diversity = 2.0102e+01

Generation 130: Best = 2.112031e+03 FES = 13100 Diversity = 2.0015e+01

Generation 140: Best = 2.093245e+03 FES = 14100 Diversity = 1.9734e+01

Generation 150: Best = 2.093245e+03 FES = 15100 Diversity = 1.9260e+01

Generation 160: Best = 2.084836e+03 FES = 16100 Diversity = 1.9172e+01

Generation 170: Best = 2.030850e+03 FES = 17100 Diversity = 1.8978e+01

Generation 180: Best = 2.013910e+03 FES = 18100 Diversity = 1.8451e+01

Generation 190: Best = 1.985646e+03 FES = 19100 Diversity = 1.7758e+01

Generation 200: Best = 1.952908e+03 FES = 20100 Diversity = 1.7574e+01

Generation 210: Best = 1.952105e+03 FES = 21100 Diversity = 1.7310e+01

Generation 220: Best = 1.948522e+03 FES = 22100 Diversity = 1.7067e+01

Generation 230: Best = 1.948057e+03 FES = 23100 Diversity = 1.6867e+01

Generation 240: Best = 1.947591e+03 FES = 24100 Diversity = 1.6801e+01

Generation 250: Best = 1.947591e+03 FES = 25100 Diversity = 1.6821e+01

Generation 260: Best = 1.947591e+03 FES = 26100 Diversity = 1.6770e+01

Generation 270: Best = 1.942040e+03 FES = 27100 Diversity = 1.6519e+01

Generation 280: Best = 1.942040e+03 FES = 28100 Diversity = 1.6488e+01

Generation 290: Best = 1.938314e+03 FES = 29100 Diversity = 1.6330e+01

Generation 300: Best = 1.932086e+03 FES = 30100 Diversity = 1.6206e+01

Generation 310: Best = 1.932086e+03 FES = 31100 Diversity = 1.6043e+01

Generation 320: Best = 1.932086e+03 FES = 32100 Diversity = 1.5913e+01

Generation 330: Best = 1.931983e+03 FES = 33100 Diversity = 1.5814e+01

Generation 340: Best = 1.931983e+03 FES = 34100 Diversity = 1.5715e+01

Generation 350: Best = 1.931983e+03 FES = 35100 Diversity = 1.5733e+01

Generation 360: Best = 1.931983e+03 FES = 36100 Diversity = 1.5648e+01

Generation 370: Best = 1.916558e+03 FES = 37100 Diversity = 1.5616e+01

Generation 380: Best = 1.916558e+03 FES = 38100 Diversity = 1.5417e+01

Generation 390: Best = 1.916558e+03 FES = 39100 Diversity = 1.5087e+01

Generation 400: Best = 1.916558e+03 FES = 40100 Diversity = 1.4819e+01

Generation 410: Best = 1.903495e+03 FES = 41100 Diversity = 1.4691e+01

Generation 420: Best = 1.903495e+03 FES = 42100 Diversity = 1.4734e+01

Generation 430: Best = 1.903495e+03 FES = 43100 Diversity = 1.4257e+01

Generation	440:	Best = 1.903495e+03	FES = 44100	Diversity = 1.4126e+01
Generation	450:	Best = 1.902478e+03	FES = 45100	Diversity = 1.3884e+01
Generation	460:	Best = 1.896015e+03	FES = 46100	Diversity = 1.3559e+01
Generation	470:	Best = 1.890552e+03	FES = 47100	Diversity = 1.3546e+01
Generation	480:	Best = 1.890379e+03	FES = 48100	Diversity = 1.3662e+01
Generation	490:	Best = 1.890379e+03	FES = 49100	Diversity = 1.3699e+01
Generation	500:	Best = 1.890379e+03	FES = 50100	Diversity = 1.3592e+01
Generation	510:	Best = 1.879431e+03	FES = 51100	Diversity = 1.3453e+01
Generation	520:	Best = 1.857875e+03	FES = 52100	Diversity = 1.3409e+01
Generation	530:	Best = 1.853618e+03	FES = 53100	Diversity = 1.3276e+01
Generation	540:	Best = 1.844639e+03	FES = 54100	Diversity = 1.3080e+01
Generation	550:	Best = 1.837406e+03	FES = 55100	Diversity = 1.2973e+01
Generation	560:	Best = 1.830682e+03	FES = 56100	Diversity = 1.2969e+01
Generation	570:	Best = 1.825815e+03	FES = 57100	Diversity = 1.2935e+01
Generation	580:	Best = 1.823942e+03	FES = 58100	Diversity = 1.2911e+01
Generation	590:	Best = 1.823275e+03	FES = 59100	Diversity = 1.2930e+01
Generation	600:	Best = 1.822169e+03	FES = 60100	Diversity = 1.2778e+01
Generation	610:	Best = 1.821542e+03	FES = 61100	Diversity = 1.2652e+01
Generation	620:	Best = 1.819851e+03	FES = 62100	Diversity = 1.2507e+01
Generation	630:	Best = 1.818777e+03	FES = 63100	Diversity = 1.2486e+01
Generation	640:	Best = 1.818777e+03	FES = 64100	Diversity = 1.2467e+01
Generation	650:	Best = 1.818262e+03	FES = 65100	Diversity = 1.2298e+01
Generation	660:	Best = 1.816678e+03	FES = 66100	Diversity = 1.2214e+01
Generation	670:	Best = 1.814619e+03	FES = 67100	Diversity = 1.2181e+01
Generation	680:	Best = 1.814045e+03	FES = 68100	Diversity = 1.2205e+01
Generation	690:	Best = 1.813644e+03	FES = 69100	Diversity = 1.2086e+01
Generation	700:	Best = 1.813128e+03	FES = 70100	Diversity = 1.2036e+01
Generation	710:	Best = 1.813051e+03	FES = 71100	Diversity = 1.1923e+01
Generation	720:	Best = 1.813051e+03	FES = 72100	Diversity = 1.1772e+01
Generation	730:	Best = 1.812916e+03	FES = 73100	Diversity = 1.1592e+01
Generation	740:	Best = 1.812488e+03	FES = 74100	Diversity = 1.1607e+01
Generation	750:	Best = 1.812468e+03	FES = 75100	Diversity = 1.1237e+01
Generation	760:	Best = 1.812456e+03	FES = 76100	Diversity = 1.1218e+01
Generation	770:	Best = 1.811846e+03	FES = 77100	Diversity = 1.1199e+01
Generation	780:	Best = 1.811791e+03	FES = 78100	Diversity = 1.1061e+01
Generation	790:	Best = 1.811791e+03	FES = 79100	Diversity = 1.0825e+01
Generation	800:	Best = 1.811791e+03	FES = 80100	Diversity = 1.0793e+01
Generation	810:	Best = 1.811791e+03	FES = 81100	Diversity = 1.0598e+01
Generation	820:	Best = 1.811663e+03	FES = 82100	Diversity = 1.0493e+01
Generation	830:	Best = 1.811655e+03	FES = 83100	Diversity = 1.0353e+01
Generation	840:	Best = 1.811655e+03	FES = 84100	Diversity = 1.0119e+01
Generation	850:	Best = 1.811655e+03	FES = 85100	Diversity = 1.0052e+01
Generation	860:	Best = 1.811655e+03	FES = 86100	Diversity = 1.0002e+01
Generation	870:	Best = 1.811546e+03	FES = 87100	Diversity = 9.9618e+00
Generation	880:	Best = 1.811540e+03	FES = 88100	Diversity = 9.9463e+00
Generation	890:	Best = 1.811525e+03	FES = 89100	Diversity = 9.9309e+00
Generation	900:	Best = 1.811411e+03	FES = 90100	Diversity = 9.8872e+00
Generation	910:	Best = 1.811053e+03	FES = 91100	Diversity = 9.8432e+00
Generation	920:	Best = 1.803330e+03	FES = 92100	Diversity = 9.6867e+00
Generation	930:	Best = 1.782910e+03	FES = 93100	Diversity = 9.6608e+00
Generation	940:	Best = 1.782910e+03	FES = 94100	Diversity = 9.3614e+00
Generation	950:	Best = 1.779064e+03	FES = 95100	Diversity = 9.3229e+00
Generation	960:	Best = 1.778271e+03	FES = 96100	Diversity = 9.1290e+00
Generation	970:	Best = 1.777767e+03	FES = 97100	Diversity = 8.9835e+00

Generation 980:	Best = 1.777525e+03	FES = 98100	Diversity = 8.9074e+00
Generation 990:	Best = 1.777340e+03	FES = 99100	Diversity = 8.8009e+00
Generation 1000:	Best = 1.777209e+03	FES = 100100	Diversity = 8.6138e+00
Generation 1010:	Best = 1.777209e+03	FES = 101100	Diversity = 8.6134e+00
Generation 1020:	Best = 1.777170e+03	FES = 102100	Diversity = 8.5311e+00
Generation 1030:	Best = 1.777121e+03	FES = 103100	Diversity = 8.5073e+00
Generation 1040:	Best = 1.775335e+03	FES = 104100	Diversity = 8.2972e+00
Generation 1050:	Best = 1.772897e+03	FES = 105100	Diversity = 8.0330e+00
Generation 1060:	Best = 1.771323e+03	FES = 106100	Diversity = 8.0257e+00
Generation 1070:	Best = 1.770805e+03	FES = 107100	Diversity = 7.9341e+00
Generation 1080:	Best = 1.770491e+03	FES = 108100	Diversity = 7.8529e+00
Generation 1090:	Best = 1.767058e+03	FES = 109100	Diversity = 7.8211e+00
Generation 1100:	Best = 1.766989e+03	FES = 110100	Diversity = 7.8101e+00
Generation 1110:	Best = 1.766107e+03	FES = 111100	Diversity = 7.7608e+00
Generation 1120:	Best = 1.765223e+03	FES = 112100	Diversity = 7.6654e+00
Generation 1130:	Best = 1.765098e+03	FES = 113100	Diversity = 7.6490e+00
Generation 1140:	Best = 1.764955e+03	FES = 114100	Diversity = 7.6456e+00
Generation 1150:	Best = 1.764907e+03	FES = 115100	Diversity = 7.5692e+00
Generation 1160:	Best = 1.764899e+03	FES = 116100	Diversity = 7.2981e+00
Generation 1170:	Best = 1.764899e+03	FES = 117100	Diversity = 7.3041e+00
Generation 1180:	Best = 1.764815e+03	FES = 118100	Diversity = 7.0141e+00
Generation 1190:	Best = 1.764815e+03	FES = 119100	Diversity = 7.0368e+00
Generation 1200:	Best = 1.764798e+03	FES = 120100	Diversity = 6.8423e+00
Generation 1210:	Best = 1.764796e+03	FES = 121100	Diversity = 6.8003e+00
Generation 1220:	Best = 1.764768e+03	FES = 122100	Diversity = 6.6058e+00
Generation 1230:	Best = 1.764766e+03	FES = 123100	Diversity = 6.4584e+00
Generation 1240:	Best = 1.764764e+03	FES = 124100	Diversity = 5.9767e+00
Generation 1250:	Best = 1.764740e+03	FES = 125100	Diversity = 6.0724e+00
Generation 1260:	Best = 1.764740e+03	FES = 126100	Diversity = 5.8522e+00
Generation 1270:	Best = 1.764740e+03	FES = 127100	Diversity = 5.7496e+00
Generation 1280:	Best = 1.764730e+03	FES = 128100	Diversity = 5.6257e+00
Generation 1290:	Best = 1.764684e+03	FES = 129100	Diversity = 5.6197e+00
Generation 1300:	Best = 1.764566e+03	FES = 130100	Diversity = 5.3891e+00
Generation 1310:	Best = 1.764502e+03	FES = 131100	Diversity = 5.2926e+00
Generation 1320:	Best = 1.763759e+03	FES = 132100	Diversity = 5.2373e+00
Generation 1330:	Best = 1.763122e+03	FES = 133100	Diversity = 4.9573e+00
Generation 1340:	Best = 1.763047e+03	FES = 134100	Diversity = 4.6954e+00
Generation 1350:	Best = 1.762976e+03	FES = 135100	Diversity = 4.6473e+00
Generation 1360:	Best = 1.762964e+03	FES = 136100	Diversity = 4.1788e+00
Generation 1370:	Best = 1.762861e+03	FES = 137100	Diversity = 3.8112e+00
Generation 1380:	Best = 1.762693e+03	FES = 138100	Diversity = 3.5303e+00
Generation 1390:	Best = 1.761876e+03	FES = 139100	Diversity = 3.4303e+00
Generation 1400:	Best = 1.761842e+03	FES = 140100	Diversity = 3.2531e+00
Generation 1410:	Best = 1.761731e+03	FES = 141100	Diversity = 2.9851e+00
Generation 1420:	Best = 1.761726e+03	FES = 142100	Diversity = 2.9647e+00
Generation 1430:	Best = 1.761627e+03	FES = 143100	Diversity = 2.8550e+00
Generation 1440:	Best = 1.761565e+03	FES = 144100	Diversity = 2.7526e+00
Generation 1450:	Best = 1.761555e+03	FES = 145100	Diversity = 2.7201e+00
Generation 1460:	Best = 1.761549e+03	FES = 146100	Diversity = 2.4824e+00
Generation 1470:	Best = 1.761548e+03	FES = 147100	Diversity = 2.4711e+00
Generation 1480:	Best = 1.761140e+03	FES = 148100	Diversity = 2.4159e+00
Generation 1490:	Best = 1.761009e+03	FES = 149100	Diversity = 2.3884e+00
Generation 1500:	Best = 1.760997e+03	FES = 150100	Diversity = 1.7685e+00
Generation 1510:	Best = 1.760977e+03	FES = 151100	Diversity = 1.6662e+00

Generation 1520:	Best = 1.760689e+03	FES = 152100	Diversity = 1.3606e+00
Generation 1530:	Best = 1.759786e+03	FES = 153100	Diversity = 1.1384e+00
Generation 1540:	Best = 1.758921e+03	FES = 154100	Diversity = 1.0439e+00
Generation 1550:	Best = 1.757986e+03	FES = 155100	Diversity = 1.0433e+00
Generation 1560:	Best = 1.757624e+03	FES = 156100	Diversity = 8.5568e-01
Generation 1570:	Best = 1.757612e+03	FES = 157100	Diversity = 9.1924e-01
Generation 1580:	Best = 1.757516e+03	FES = 158100	Diversity = 1.0438e+00
Generation 1590:	Best = 1.757413e+03	FES = 159100	Diversity = 9.9051e-01
Generation 1600:	Best = 1.757381e+03	FES = 160100	Diversity = 1.0341e+00
Generation 1610:	Best = 1.757274e+03	FES = 161100	Diversity = 1.0562e+00
Generation 1620:	Best = 1.757253e+03	FES = 162100	Diversity = 1.0913e+00
Generation 1630:	Best = 1.757215e+03	FES = 163100	Diversity = 1.1717e+00
Generation 1640:	Best = 1.757134e+03	FES = 164100	Diversity = 1.1902e+00
Generation 1650:	Best = 1.757114e+03	FES = 165100	Diversity = 1.2778e+00
Generation 1660:	Best = 1.757094e+03	FES = 166100	Diversity = 1.3436e+00
Generation 1670:	Best = 1.757077e+03	FES = 167100	Diversity = 1.3705e+00
Generation 1680:	Best = 1.757071e+03	FES = 168100	Diversity = 1.3245e+00
Generation 1690:	Best = 1.757049e+03	FES = 169100	Diversity = 1.3599e+00
Generation 1700:	Best = 1.757028e+03	FES = 170100	Diversity = 1.3339e+00
Generation 1710:	Best = 1.756958e+03	FES = 171100	Diversity = 1.3402e+00
Generation 1720:	Best = 1.756955e+03	FES = 172100	Diversity = 1.3927e+00
Generation 1730:	Best = 1.756954e+03	FES = 173100	Diversity = 1.3914e+00
Generation 1740:	Best = 1.756953e+03	FES = 174100	Diversity = 1.4204e+00
Generation 1750:	Best = 1.756953e+03	FES = 175100	Diversity = 1.4252e+00
Generation 1760:	Best = 1.756952e+03	FES = 176100	Diversity = 1.5289e+00
Generation 1770:	Best = 1.756952e+03	FES = 177100	Diversity = 1.4747e+00
Generation 1780:	Best = 1.756952e+03	FES = 178100	Diversity = 1.4177e+00
Generation 1790:	Best = 1.756945e+03	FES = 179100	Diversity = 1.3861e+00
Generation 1800:	Best = 1.756944e+03	FES = 180100	Diversity = 1.3823e+00
Generation 1810:	Best = 1.756944e+03	FES = 181100	Diversity = 1.3828e+00
Generation 1820:	Best = 1.756924e+03	FES = 182100	Diversity = 1.4354e+00
Generation 1830:	Best = 1.756924e+03	FES = 183100	Diversity = 1.4337e+00
Generation 1840:	Best = 1.756903e+03	FES = 184100	Diversity = 1.4962e+00
Generation 1850:	Best = 1.756895e+03	FES = 185100	Diversity = 1.4766e+00
Generation 1860:	Best = 1.756893e+03	FES = 186100	Diversity = 1.4802e+00
Generation 1870:	Best = 1.756892e+03	FES = 187100	Diversity = 1.4842e+00
Generation 1880:	Best = 1.756888e+03	FES = 188100	Diversity = 1.5795e+00
Generation 1890:	Best = 1.756888e+03	FES = 189100	Diversity = 1.5978e+00
Generation 1900:	Best = 1.756888e+03	FES = 190100	Diversity = 1.6056e+00
Generation 1910:	Best = 1.756888e+03	FES = 191100	Diversity = 1.7038e+00
Generation 1920:	Best = 1.756888e+03	FES = 192100	Diversity = 1.7038e+00
Generation 1930:	Best = 1.756888e+03	FES = 193100	Diversity = 1.7192e+00
Generation 1940:	Best = 1.756865e+03	FES = 194100	Diversity = 1.7627e+00
Generation 1950:	Best = 1.756852e+03	FES = 195100	Diversity = 1.7986e+00
Generation 1960:	Best = 1.756852e+03	FES = 196100	Diversity = 1.7985e+00
Generation 1970:	Best = 1.756850e+03	FES = 197100	Diversity = 1.7984e+00
Generation 1980:	Best = 1.756849e+03	FES = 198100	Diversity = 1.8052e+00
Generation 1990:	Best = 1.756849e+03	FES = 199100	Diversity = 1.8504e+00
Generation 2000:	Best = 1.756848e+03	FES = 200100	Diversity = 1.8509e+00
Generation 2010:	Best = 1.756848e+03	FES = 201100	Diversity = 1.8551e+00
Generation 2020:	Best = 1.756824e+03	FES = 202100	Diversity = 1.8389e+00
Generation 2030:	Best = 1.756821e+03	FES = 203100	Diversity = 1.8516e+00
Generation 2040:	Best = 1.756819e+03	FES = 204100	Diversity = 1.8904e+00
Generation 2050:	Best = 1.756819e+03	FES = 205100	Diversity = 1.8572e+00

Generation 2060:	Best = 1.756818e+03	FES = 206100	Diversity = 1.8802e+00
Generation 2070:	Best = 1.756818e+03	FES = 207100	Diversity = 1.8464e+00
Generation 2080:	Best = 1.756818e+03	FES = 208100	Diversity = 1.8199e+00
Generation 2090:	Best = 1.756818e+03	FES = 209100	Diversity = 1.8575e+00
Generation 2100:	Best = 1.756818e+03	FES = 210100	Diversity = 1.8484e+00
Generation 2110:	Best = 1.756817e+03	FES = 211100	Diversity = 1.8483e+00
Generation 2120:	Best = 1.756817e+03	FES = 212100	Diversity = 1.8483e+00
Generation 2130:	Best = 1.756817e+03	FES = 213100	Diversity = 1.8483e+00
Generation 2140:	Best = 1.756817e+03	FES = 214100	Diversity = 1.8409e+00
Generation 2150:	Best = 1.756817e+03	FES = 215100	Diversity = 1.8409e+00
Generation 2160:	Best = 1.756817e+03	FES = 216100	Diversity = 1.8181e+00
Generation 2170:	Best = 1.756817e+03	FES = 217100	Diversity = 1.8796e+00
Generation 2180:	Best = 1.756817e+03	FES = 218100	Diversity = 1.8796e+00
Generation 2190:	Best = 1.756774e+03	FES = 219100	Diversity = 1.8626e+00
Generation 2200:	Best = 1.756774e+03	FES = 220100	Diversity = 1.8919e+00
Generation 2210:	Best = 1.756774e+03	FES = 221100	Diversity = 1.8864e+00
Generation 2220:	Best = 1.756774e+03	FES = 222100	Diversity = 1.8887e+00
Generation 2230:	Best = 1.756774e+03	FES = 223100	Diversity = 1.8862e+00
Generation 2240:	Best = 1.756774e+03	FES = 224100	Diversity = 1.8862e+00
Generation 2250:	Best = 1.756774e+03	FES = 225100	Diversity = 1.8861e+00
Generation 2260:	Best = 1.756774e+03	FES = 226100	Diversity = 1.8861e+00
Generation 2270:	Best = 1.756774e+03	FES = 227100	Diversity = 1.8861e+00
Generation 2280:	Best = 1.756774e+03	FES = 228100	Diversity = 1.9180e+00
Generation 2290:	Best = 1.756774e+03	FES = 229100	Diversity = 1.9324e+00
Generation 2300:	Best = 1.756774e+03	FES = 230100	Diversity = 1.9324e+00
Generation 2310:	Best = 1.756774e+03	FES = 231100	Diversity = 1.9030e+00
Generation 2320:	Best = 1.756774e+03	FES = 232100	Diversity = 1.8811e+00
Generation 2330:	Best = 1.756774e+03	FES = 233100	Diversity = 1.8802e+00
Generation 2340:	Best = 1.756774e+03	FES = 234100	Diversity = 1.8802e+00
Generation 2350:	Best = 1.756774e+03	FES = 235100	Diversity = 1.9019e+00
Generation 2360:	Best = 1.756774e+03	FES = 236100	Diversity = 1.9019e+00
Generation 2370:	Best = 1.756774e+03	FES = 237100	Diversity = 1.9254e+00
Generation 2380:	Best = 1.756774e+03	FES = 238100	Diversity = 1.9074e+00
Generation 2390:	Best = 1.756774e+03	FES = 239100	Diversity = 1.9074e+00
Generation 2400:	Best = 1.756774e+03	FES = 240100	Diversity = 1.9387e+00
Generation 2410:	Best = 1.756774e+03	FES = 241100	Diversity = 1.9387e+00
Generation 2420:	Best = 1.756774e+03	FES = 242100	Diversity = 1.9387e+00
Generation 2430:	Best = 1.756774e+03	FES = 243100	Diversity = 1.9387e+00
Generation 2440:	Best = 1.756774e+03	FES = 244100	Diversity = 1.9077e+00
Generation 2450:	Best = 1.756774e+03	FES = 245100	Diversity = 1.9127e+00
Generation 2460:	Best = 1.756774e+03	FES = 246100	Diversity = 1.9127e+00
Generation 2470:	Best = 1.756774e+03	FES = 247100	Diversity = 1.9123e+00
Generation 2480:	Best = 1.756774e+03	FES = 248100	Diversity = 1.9694e+00
Generation 2490:	Best = 1.756774e+03	FES = 249100	Diversity = 1.9694e+00
Generation 2500:	Best = 1.756774e+03	FES = 250100	Diversity = 1.9618e+00
Generation 2510:	Best = 1.756774e+03	FES = 251100	Diversity = 1.9618e+00
Generation 2520:	Best = 1.756774e+03	FES = 252100	Diversity = 1.9614e+00
Generation 2530:	Best = 1.756774e+03	FES = 253100	Diversity = 1.9614e+00
Generation 2540:	Best = 1.756774e+03	FES = 254100	Diversity = 1.9663e+00
Generation 2550:	Best = 1.756774e+03	FES = 255100	Diversity = 1.9663e+00
Generation 2560:	Best = 1.756774e+03	FES = 256100	Diversity = 1.9662e+00
Generation 2570:	Best = 1.756774e+03	FES = 257100	Diversity = 1.9662e+00
Generation 2580:	Best = 1.756774e+03	FES = 258100	Diversity = 1.9662e+00
Generation 2590:	Best = 1.756774e+03	FES = 259100	Diversity = 1.9662e+00

Generation 2600:	Best = 1.756774e+03	FES = 260100	Diversity = 1.9717e+00
Generation 2610:	Best = 1.756774e+03	FES = 261100	Diversity = 1.9676e+00
Generation 2620:	Best = 1.756774e+03	FES = 262100	Diversity = 1.8878e+00
Generation 2630:	Best = 1.756774e+03	FES = 263100	Diversity = 1.8821e+00
Generation 2640:	Best = 1.756774e+03	FES = 264100	Diversity = 1.8883e+00
Generation 2650:	Best = 1.756774e+03	FES = 265100	Diversity = 1.8883e+00
Generation 2660:	Best = 1.756774e+03	FES = 266100	Diversity = 1.8883e+00
Generation 2670:	Best = 1.756774e+03	FES = 267100	Diversity = 1.8883e+00
Generation 2680:	Best = 1.756774e+03	FES = 268100	Diversity = 1.9037e+00
Generation 2690:	Best = 1.756774e+03	FES = 269100	Diversity = 1.9037e+00
Generation 2700:	Best = 1.756774e+03	FES = 270100	Diversity = 1.9037e+00
Generation 2710:	Best = 1.756774e+03	FES = 271100	Diversity = 1.9102e+00
Generation 2720:	Best = 1.756774e+03	FES = 272100	Diversity = 1.9102e+00
Generation 2730:	Best = 1.756774e+03	FES = 273100	Diversity = 1.9102e+00
Generation 2740:	Best = 1.756774e+03	FES = 274100	Diversity = 1.9102e+00
Generation 2750:	Best = 1.756774e+03	FES = 275100	Diversity = 1.9101e+00
Generation 2760:	Best = 1.756774e+03	FES = 276100	Diversity = 1.9101e+00
Generation 2770:	Best = 1.756774e+03	FES = 277100	Diversity = 1.9052e+00
Generation 2780:	Best = 1.756774e+03	FES = 278100	Diversity = 1.8463e+00
Generation 2790:	Best = 1.756774e+03	FES = 279100	Diversity = 1.8405e+00
Generation 2800:	Best = 1.756774e+03	FES = 280100	Diversity = 1.8405e+00
Generation 2810:	Best = 1.756774e+03	FES = 281100	Diversity = 1.8093e+00
Generation 2820:	Best = 1.756774e+03	FES = 282100	Diversity = 1.8093e+00
Generation 2830:	Best = 1.756774e+03	FES = 283100	Diversity = 1.8142e+00
Generation 2840:	Best = 1.756774e+03	FES = 284100	Diversity = 1.8142e+00
Generation 2850:	Best = 1.756774e+03	FES = 285100	Diversity = 1.8142e+00
Generation 2860:	Best = 1.756774e+03	FES = 286100	Diversity = 1.8350e+00
Generation 2870:	Best = 1.756774e+03	FES = 287100	Diversity = 1.8350e+00
Generation 2880:	Best = 1.756774e+03	FES = 288100	Diversity = 1.8298e+00
Generation 2890:	Best = 1.756774e+03	FES = 289100	Diversity = 1.8298e+00
Generation 2900:	Best = 1.756774e+03	FES = 290100	Diversity = 1.8297e+00
Generation 2910:	Best = 1.756774e+03	FES = 291100	Diversity = 1.8297e+00
Generation 2920:	Best = 1.756774e+03	FES = 292100	Diversity = 1.8297e+00
Generation 2930:	Best = 1.756774e+03	FES = 293100	Diversity = 1.8297e+00
Generation 2940:	Best = 1.756774e+03	FES = 294100	Diversity = 1.8609e+00
Generation 2950:	Best = 1.756774e+03	FES = 295100	Diversity = 1.8609e+00
Generation 2960:	Best = 1.756774e+03	FES = 296100	Diversity = 1.8205e+00
Generation 2970:	Best = 1.756774e+03	FES = 297100	Diversity = 1.7987e+00
Generation 2980:	Best = 1.756774e+03	FES = 298100	Diversity = 1.7988e+00
Generation 2990:	Best = 1.756774e+03	FES = 299100	Diversity = 1.7936e+00

Optimization completed!

Final Best Fitness = 1.756774e+03

Total Function Evaluations = 300000

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Results:

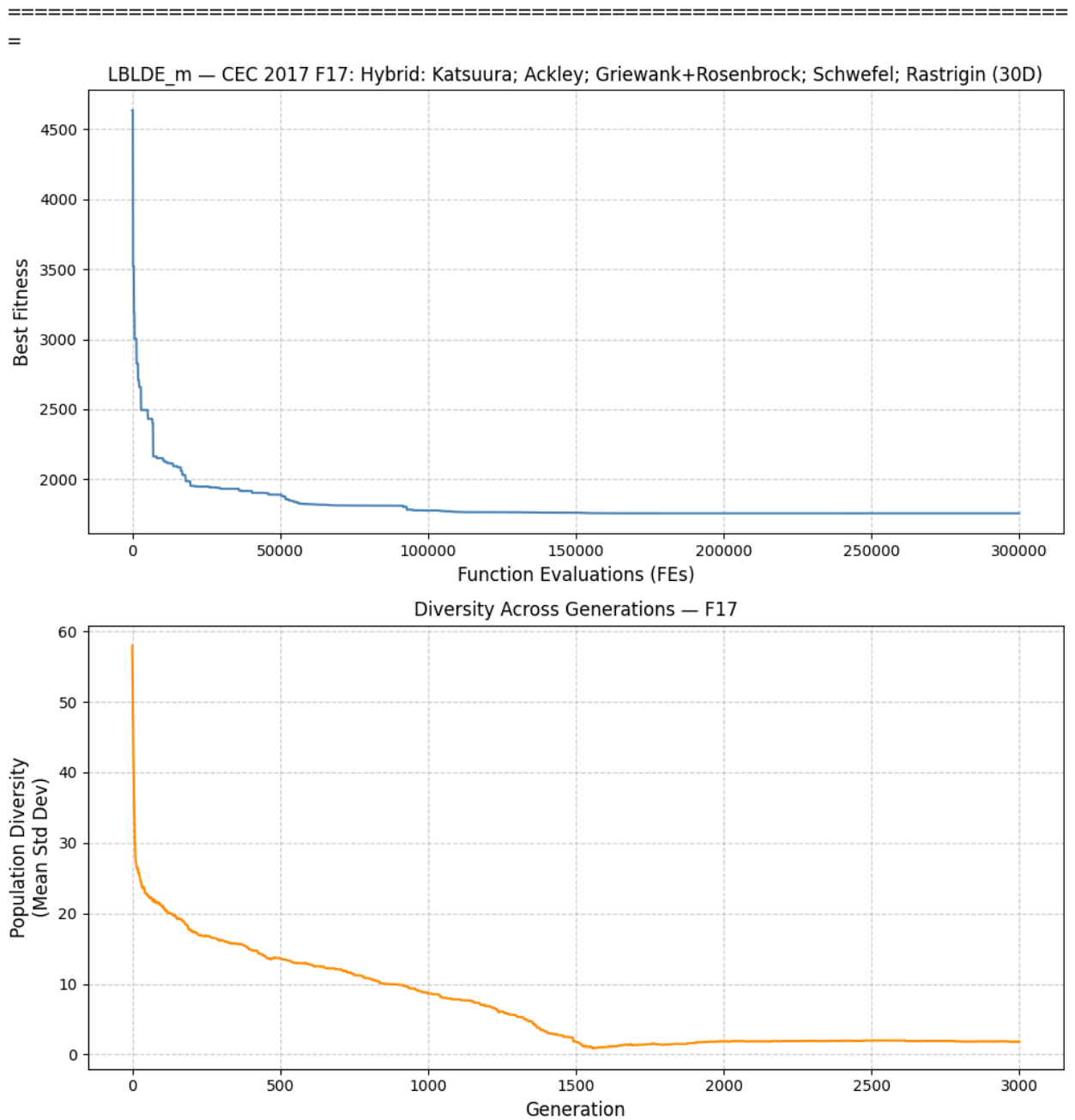
Best Fitness : 1.756774e+03

Global Optimum : 1.700000e+03

Error (|best-optimum|) : 5.677415e+01

Best Solution (5 dim) : [-22.65724923 3.5317961 -2.46839632 -17.1516615 -21.65808313]

D^(0) (init. diversity): 5.7976e+01



Graph saved as convergence_F17.png

```
In [44]: # =====
# F18 — Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# =====
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid: Elliptic; Ackle
```

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LBLDE_m on CEC 2017 F18 – Single Run

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Function : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus (30D)
Bounds : [-100, 100]
Global optimum : 1800.0
Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 3.522341e+08 | Diversity = 5.7976e+01

Generation	10:	Best = 4.824881e+06	FES = 1100	Diversity = 3.7070e+01
Generation	20:	Best = 4.350086e+06	FES = 2100	Diversity = 2.7299e+01
Generation	30:	Best = 1.064912e+06	FES = 3100	Diversity = 2.3708e+01
Generation	40:	Best = 5.749111e+05	FES = 4100	Diversity = 2.2536e+01
Generation	50:	Best = 4.716240e+05	FES = 5100	Diversity = 2.3237e+01
Generation	60:	Best = 3.768937e+05	FES = 6100	Diversity = 2.1323e+01
Generation	70:	Best = 3.768937e+05	FES = 7100	Diversity = 1.7536e+01
Generation	80:	Best = 3.305038e+05	FES = 8100	Diversity = 1.5924e+01
Generation	90:	Best = 3.290584e+05	FES = 9100	Diversity = 1.5565e+01
Generation	100:	Best = 3.051087e+05	FES = 10100	Diversity = 1.6511e+01
Generation	110:	Best = 2.925217e+05	FES = 11100	Diversity = 1.5868e+01
Generation	120:	Best = 2.844801e+05	FES = 12100	Diversity = 1.5540e+01
Generation	130:	Best = 2.757726e+05	FES = 13100	Diversity = 1.5248e+01
Generation	140:	Best = 2.757726e+05	FES = 14100	Diversity = 1.5692e+01
Generation	150:	Best = 2.733273e+05	FES = 15100	Diversity = 1.6109e+01
Generation	160:	Best = 2.677375e+05	FES = 16100	Diversity = 1.5145e+01
Generation	170:	Best = 2.644317e+05	FES = 17100	Diversity = 1.4618e+01
Generation	180:	Best = 2.630474e+05	FES = 18100	Diversity = 1.3271e+01
Generation	190:	Best = 2.612716e+05	FES = 19100	Diversity = 1.3961e+01
Generation	200:	Best = 2.115546e+05	FES = 20100	Diversity = 1.5541e+01
Generation	210:	Best = 2.049227e+05	FES = 21100	Diversity = 1.9979e+01
Generation	220:	Best = 1.962806e+05	FES = 22100	Diversity = 2.2309e+01
Generation	230:	Best = 1.946691e+05	FES = 23100	Diversity = 2.1588e+01
Generation	240:	Best = 1.871133e+05	FES = 24100	Diversity = 2.3847e+01
Generation	250:	Best = 1.819863e+05	FES = 25100	Diversity = 2.2595e+01
Generation	260:	Best = 1.819660e+05	FES = 26100	Diversity = 2.2345e+01
Generation	270:	Best = 1.798697e+05	FES = 27100	Diversity = 2.0517e+01
Generation	280:	Best = 1.789149e+05	FES = 28100	Diversity = 2.1071e+01
Generation	290:	Best = 1.771018e+05	FES = 29100	Diversity = 1.8827e+01
Generation	300:	Best = 1.756649e+05	FES = 30100	Diversity = 1.9802e+01
Generation	310:	Best = 1.748153e+05	FES = 31100	Diversity = 1.9704e+01
Generation	320:	Best = 1.739981e+05	FES = 32100	Diversity = 1.8584e+01
Generation	330:	Best = 1.723898e+05	FES = 33100	Diversity = 1.8737e+01
Generation	340:	Best = 1.719950e+05	FES = 34100	Diversity = 1.6907e+01
Generation	350:	Best = 1.708239e+05	FES = 35100	Diversity = 1.7137e+01
Generation	360:	Best = 1.702786e+05	FES = 36100	Diversity = 1.5708e+01
Generation	370:	Best = 1.637556e+05	FES = 37100	Diversity = 1.5641e+01
Generation	380:	Best = 1.544299e+05	FES = 38100	Diversity = 1.6951e+01
Generation	390:	Best = 1.458742e+05	FES = 39100	Diversity = 1.8394e+01
Generation	400:	Best = 1.409366e+05	FES = 40100	Diversity = 1.9953e+01
Generation	410:	Best = 1.387628e+05	FES = 41100	Diversity = 2.1002e+01
Generation	420:	Best = 1.375028e+05	FES = 42100	Diversity = 1.9679e+01
Generation	430:	Best = 1.352803e+05	FES = 43100	Diversity = 1.8600e+01
Generation	440:	Best = 1.343079e+05	FES = 44100	Diversity = 1.9120e+01

Generation	450:	Best = 1.325700e+05	FES = 45100	Diversity = 1.9607e+01
Generation	460:	Best = 1.316767e+05	FES = 46100	Diversity = 1.8964e+01
Generation	470:	Best = 1.305621e+05	FES = 47100	Diversity = 1.8383e+01
Generation	480:	Best = 1.300560e+05	FES = 48100	Diversity = 1.8951e+01
Generation	490:	Best = 1.294935e+05	FES = 49100	Diversity = 1.7404e+01
Generation	500:	Best = 1.286249e+05	FES = 50100	Diversity = 1.7561e+01
Generation	510:	Best = 1.284456e+05	FES = 51100	Diversity = 1.7138e+01
Generation	520:	Best = 1.269292e+05	FES = 52100	Diversity = 1.6672e+01
Generation	530:	Best = 1.131864e+05	FES = 53100	Diversity = 1.8517e+01
Generation	540:	Best = 1.035156e+05	FES = 54100	Diversity = 1.9875e+01
Generation	550:	Best = 9.740546e+04	FES = 55100	Diversity = 2.2118e+01
Generation	560:	Best = 7.771555e+04	FES = 56100	Diversity = 2.1934e+01
Generation	570:	Best = 7.148971e+04	FES = 57100	Diversity = 2.5185e+01
Generation	580:	Best = 6.791733e+04	FES = 58100	Diversity = 2.2124e+01
Generation	590:	Best = 6.496737e+04	FES = 59100	Diversity = 1.9316e+01
Generation	600:	Best = 5.821042e+04	FES = 60100	Diversity = 2.0796e+01
Generation	610:	Best = 5.493898e+04	FES = 61100	Diversity = 1.8249e+01
Generation	620:	Best = 5.352987e+04	FES = 62100	Diversity = 1.7034e+01
Generation	630:	Best = 4.821846e+04	FES = 63100	Diversity = 1.6402e+01
Generation	640:	Best = 4.482765e+04	FES = 64100	Diversity = 1.7379e+01
Generation	650:	Best = 4.246054e+04	FES = 65100	Diversity = 1.7940e+01
Generation	660:	Best = 4.050013e+04	FES = 66100	Diversity = 1.8663e+01
Generation	670:	Best = 3.896746e+04	FES = 67100	Diversity = 1.7168e+01
Generation	680:	Best = 3.740049e+04	FES = 68100	Diversity = 1.6021e+01
Generation	690:	Best = 3.714338e+04	FES = 69100	Diversity = 1.6683e+01
Generation	700:	Best = 3.654838e+04	FES = 70100	Diversity = 1.5816e+01
Generation	710:	Best = 3.604576e+04	FES = 71100	Diversity = 1.5086e+01
Generation	720:	Best = 3.570641e+04	FES = 72100	Diversity = 1.4428e+01
Generation	730:	Best = 3.532651e+04	FES = 73100	Diversity = 1.4281e+01
Generation	740:	Best = 3.497280e+04	FES = 74100	Diversity = 1.3478e+01
Generation	750:	Best = 3.424000e+04	FES = 75100	Diversity = 1.2549e+01
Generation	760:	Best = 3.407045e+04	FES = 76100	Diversity = 1.3546e+01
Generation	770:	Best = 3.393725e+04	FES = 77100	Diversity = 1.1174e+01
Generation	780:	Best = 3.385642e+04	FES = 78100	Diversity = 9.6908e+00
Generation	790:	Best = 3.377112e+04	FES = 79100	Diversity = 9.9646e+00
Generation	800:	Best = 3.354055e+04	FES = 80100	Diversity = 1.0188e+01
Generation	810:	Best = 3.227235e+04	FES = 81100	Diversity = 1.2057e+01
Generation	820:	Best = 3.141235e+04	FES = 82100	Diversity = 1.4501e+01
Generation	830:	Best = 3.102095e+04	FES = 83100	Diversity = 1.3107e+01
Generation	840:	Best = 3.065822e+04	FES = 84100	Diversity = 1.5529e+01
Generation	850:	Best = 3.021820e+04	FES = 85100	Diversity = 1.4945e+01
Generation	860:	Best = 2.992719e+04	FES = 86100	Diversity = 1.3117e+01
Generation	870:	Best = 2.921615e+04	FES = 87100	Diversity = 1.4002e+01
Generation	880:	Best = 2.885896e+04	FES = 88100	Diversity = 1.4220e+01
Generation	890:	Best = 2.866246e+04	FES = 89100	Diversity = 1.5171e+01
Generation	900:	Best = 2.803430e+04	FES = 90100	Diversity = 1.5480e+01
Generation	910:	Best = 2.781084e+04	FES = 91100	Diversity = 1.4505e+01
Generation	920:	Best = 2.722392e+04	FES = 92100	Diversity = 1.3793e+01
Generation	930:	Best = 2.722392e+04	FES = 93100	Diversity = 1.5110e+01
Generation	940:	Best = 2.696429e+04	FES = 94100	Diversity = 1.4069e+01
Generation	950:	Best = 2.434635e+04	FES = 95100	Diversity = 1.4260e+01
Generation	960:	Best = 2.337166e+04	FES = 96100	Diversity = 1.3901e+01
Generation	970:	Best = 2.283789e+04	FES = 97100	Diversity = 1.2086e+01
Generation	980:	Best = 2.240730e+04	FES = 98100	Diversity = 1.2321e+01

Generation 990:	Best = 2.170348e+04	FES = 99100	Diversity = 1.1922e+01
Generation 1000:	Best = 2.155399e+04	FES = 100100	Diversity = 1.1953e+01
Generation 1010:	Best = 2.122145e+04	FES = 101100	Diversity = 1.1764e+01
Generation 1020:	Best = 2.048914e+04	FES = 102100	Diversity = 1.0934e+01
Generation 1030:	Best = 2.029023e+04	FES = 103100	Diversity = 1.2164e+01
Generation 1040:	Best = 2.004913e+04	FES = 104100	Diversity = 1.1802e+01
Generation 1050:	Best = 1.985172e+04	FES = 105100	Diversity = 1.0990e+01
Generation 1060:	Best = 1.959369e+04	FES = 106100	Diversity = 1.1039e+01
Generation 1070:	Best = 1.936306e+04	FES = 107100	Diversity = 1.1427e+01
Generation 1080:	Best = 1.916615e+04	FES = 108100	Diversity = 1.0794e+01
Generation 1090:	Best = 1.911741e+04	FES = 109100	Diversity = 9.7782e+00
Generation 1100:	Best = 1.906528e+04	FES = 110100	Diversity = 9.0172e+00
Generation 1110:	Best = 1.893968e+04	FES = 111100	Diversity = 8.3483e+00
Generation 1120:	Best = 1.880978e+04	FES = 112100	Diversity = 8.2783e+00
Generation 1130:	Best = 1.874612e+04	FES = 113100	Diversity = 7.8099e+00
Generation 1140:	Best = 1.870911e+04	FES = 114100	Diversity = 7.0844e+00
Generation 1150:	Best = 1.860694e+04	FES = 115100	Diversity = 6.4713e+00
Generation 1160:	Best = 1.854578e+04	FES = 116100	Diversity = 6.0246e+00
Generation 1170:	Best = 1.845413e+04	FES = 117100	Diversity = 6.1546e+00
Generation 1180:	Best = 1.840630e+04	FES = 118100	Diversity = 7.2875e+00
Generation 1190:	Best = 1.831696e+04	FES = 119100	Diversity = 7.6382e+00
Generation 1200:	Best = 1.831626e+04	FES = 120100	Diversity = 6.7696e+00
Generation 1210:	Best = 1.826226e+04	FES = 121100	Diversity = 6.8805e+00
Generation 1220:	Best = 1.820897e+04	FES = 122100	Diversity = 6.5082e+00
Generation 1230:	Best = 1.816420e+04	FES = 123100	Diversity = 7.6452e+00
Generation 1240:	Best = 1.805545e+04	FES = 124100	Diversity = 8.1876e+00
Generation 1250:	Best = 1.797918e+04	FES = 125100	Diversity = 9.1411e+00
Generation 1260:	Best = 1.789687e+04	FES = 126100	Diversity = 9.1200e+00
Generation 1270:	Best = 1.786847e+04	FES = 127100	Diversity = 8.0231e+00
Generation 1280:	Best = 1.779946e+04	FES = 128100	Diversity = 8.5593e+00
Generation 1290:	Best = 1.765871e+04	FES = 129100	Diversity = 1.0645e+01
Generation 1300:	Best = 1.750227e+04	FES = 130100	Diversity = 1.2029e+01
Generation 1310:	Best = 1.746135e+04	FES = 131100	Diversity = 1.2905e+01
Generation 1320:	Best = 1.739421e+04	FES = 132100	Diversity = 1.1949e+01
Generation 1330:	Best = 1.734793e+04	FES = 133100	Diversity = 1.1491e+01
Generation 1340:	Best = 1.726574e+04	FES = 134100	Diversity = 1.1165e+01
Generation 1350:	Best = 1.695914e+04	FES = 135100	Diversity = 1.1160e+01
Generation 1360:	Best = 1.686720e+04	FES = 136100	Diversity = 1.2417e+01
Generation 1370:	Best = 1.673631e+04	FES = 137100	Diversity = 1.2532e+01
Generation 1380:	Best = 1.665261e+04	FES = 138100	Diversity = 1.2585e+01
Generation 1390:	Best = 1.652477e+04	FES = 139100	Diversity = 1.2784e+01
Generation 1400:	Best = 1.630832e+04	FES = 140100	Diversity = 1.4925e+01
Generation 1410:	Best = 1.609980e+04	FES = 141100	Diversity = 1.7954e+01
Generation 1420:	Best = 1.604783e+04	FES = 142100	Diversity = 1.9101e+01
Generation 1430:	Best = 1.589213e+04	FES = 143100	Diversity = 1.7717e+01
Generation 1440:	Best = 1.582644e+04	FES = 144100	Diversity = 1.5449e+01
Generation 1450:	Best = 1.494257e+04	FES = 145100	Diversity = 1.6975e+01
Generation 1460:	Best = 1.483869e+04	FES = 146100	Diversity = 1.6000e+01
Generation 1470:	Best = 1.443872e+04	FES = 147100	Diversity = 1.5604e+01
Generation 1480:	Best = 1.443084e+04	FES = 148100	Diversity = 1.5895e+01
Generation 1490:	Best = 1.430999e+04	FES = 149100	Diversity = 1.3767e+01
Generation 1500:	Best = 1.418005e+04	FES = 150100	Diversity = 1.2265e+01
Generation 1510:	Best = 1.408803e+04	FES = 151100	Diversity = 1.2529e+01
Generation 1520:	Best = 1.389601e+04	FES = 152100	Diversity = 1.1325e+01

Generation 1530:	Best = 1.387220e+04	FES = 153100	Diversity = 1.1749e+01
Generation 1540:	Best = 1.363656e+04	FES = 154100	Diversity = 1.0469e+01
Generation 1550:	Best = 1.363656e+04	FES = 155100	Diversity = 1.1332e+01
Generation 1560:	Best = 1.360774e+04	FES = 156100	Diversity = 1.1652e+01
Generation 1570:	Best = 1.349089e+04	FES = 157100	Diversity = 1.0685e+01
Generation 1580:	Best = 1.344079e+04	FES = 158100	Diversity = 9.6549e+00
Generation 1590:	Best = 1.334572e+04	FES = 159100	Diversity = 9.6628e+00
Generation 1600:	Best = 1.328750e+04	FES = 160100	Diversity = 1.0030e+01
Generation 1610:	Best = 1.322762e+04	FES = 161100	Diversity = 1.0099e+01
Generation 1620:	Best = 1.310082e+04	FES = 162100	Diversity = 1.1232e+01
Generation 1630:	Best = 1.304553e+04	FES = 163100	Diversity = 1.0490e+01
Generation 1640:	Best = 1.297259e+04	FES = 164100	Diversity = 9.8637e+00
Generation 1650:	Best = 1.277200e+04	FES = 165100	Diversity = 1.0144e+01
Generation 1660:	Best = 1.271796e+04	FES = 166100	Diversity = 1.0572e+01
Generation 1670:	Best = 1.267557e+04	FES = 167100	Diversity = 1.0063e+01
Generation 1680:	Best = 1.261731e+04	FES = 168100	Diversity = 9.7801e+00
Generation 1690:	Best = 1.252098e+04	FES = 169100	Diversity = 9.6068e+00
Generation 1700:	Best = 1.248171e+04	FES = 170100	Diversity = 8.3442e+00
Generation 1710:	Best = 1.240882e+04	FES = 171100	Diversity = 7.7499e+00
Generation 1720:	Best = 1.238732e+04	FES = 172100	Diversity = 7.3255e+00
Generation 1730:	Best = 1.235165e+04	FES = 173100	Diversity = 7.8102e+00
Generation 1740:	Best = 1.228175e+04	FES = 174100	Diversity = 7.7997e+00
Generation 1750:	Best = 1.219175e+04	FES = 175100	Diversity = 8.4630e+00
Generation 1760:	Best = 1.216669e+04	FES = 176100	Diversity = 8.7650e+00
Generation 1770:	Best = 1.211787e+04	FES = 177100	Diversity = 8.2827e+00
Generation 1780:	Best = 1.197717e+04	FES = 178100	Diversity = 9.3966e+00
Generation 1790:	Best = 1.194549e+04	FES = 179100	Diversity = 9.4796e+00
Generation 1800:	Best = 1.191928e+04	FES = 180100	Diversity = 9.6555e+00
Generation 1810:	Best = 1.178967e+04	FES = 181100	Diversity = 1.1157e+01
Generation 1820:	Best = 1.172573e+04	FES = 182100	Diversity = 1.1463e+01
Generation 1830:	Best = 1.170121e+04	FES = 183100	Diversity = 1.1121e+01
Generation 1840:	Best = 1.164802e+04	FES = 184100	Diversity = 1.1002e+01
Generation 1850:	Best = 1.162623e+04	FES = 185100	Diversity = 1.0461e+01
Generation 1860:	Best = 1.145658e+04	FES = 186100	Diversity = 1.1185e+01
Generation 1870:	Best = 1.145658e+04	FES = 187100	Diversity = 1.1891e+01
Generation 1880:	Best = 1.141544e+04	FES = 188100	Diversity = 1.1758e+01
Generation 1890:	Best = 1.134600e+04	FES = 189100	Diversity = 1.2066e+01
Generation 1900:	Best = 1.124222e+04	FES = 190100	Diversity = 1.3000e+01
Generation 1910:	Best = 1.122124e+04	FES = 191100	Diversity = 1.1716e+01
Generation 1920:	Best = 1.120159e+04	FES = 192100	Diversity = 1.0476e+01
Generation 1930:	Best = 1.113827e+04	FES = 193100	Diversity = 1.0341e+01
Generation 1940:	Best = 1.112770e+04	FES = 194100	Diversity = 9.7032e+00
Generation 1950:	Best = 1.108238e+04	FES = 195100	Diversity = 9.6120e+00
Generation 1960:	Best = 1.105751e+04	FES = 196100	Diversity = 9.1284e+00
Generation 1970:	Best = 1.102445e+04	FES = 197100	Diversity = 8.2956e+00
Generation 1980:	Best = 1.097864e+04	FES = 198100	Diversity = 8.1104e+00
Generation 1990:	Best = 1.094829e+04	FES = 199100	Diversity = 7.5611e+00
Generation 2000:	Best = 1.090235e+04	FES = 200100	Diversity = 7.2075e+00
Generation 2010:	Best = 1.083294e+04	FES = 201100	Diversity = 7.0802e+00
Generation 2020:	Best = 1.081260e+04	FES = 202100	Diversity = 8.0521e+00
Generation 2030:	Best = 1.075323e+04	FES = 203100	Diversity = 7.4741e+00
Generation 2040:	Best = 1.068576e+04	FES = 204100	Diversity = 7.7090e+00
Generation 2050:	Best = 1.037941e+04	FES = 205100	Diversity = 7.9605e+00
Generation 2060:	Best = 1.028558e+04	FES = 206100	Diversity = 9.9220e+00

Generation 2070:	Best = 1.018018e+04	FES = 207100	Diversity = 9.6881e+00
Generation 2080:	Best = 1.012017e+04	FES = 208100	Diversity = 9.6593e+00
Generation 2090:	Best = 1.005885e+04	FES = 209100	Diversity = 9.7923e+00
Generation 2100:	Best = 9.969868e+03	FES = 210100	Diversity = 9.4502e+00
Generation 2110:	Best = 9.926914e+03	FES = 211100	Diversity = 8.9644e+00
Generation 2120:	Best = 9.860934e+03	FES = 212100	Diversity = 8.7917e+00
Generation 2130:	Best = 9.795855e+03	FES = 213100	Diversity = 8.4622e+00
Generation 2140:	Best = 9.747274e+03	FES = 214100	Diversity = 8.4960e+00
Generation 2150:	Best = 9.693628e+03	FES = 215100	Diversity = 8.4030e+00
Generation 2160:	Best = 9.615714e+03	FES = 216100	Diversity = 8.6905e+00
Generation 2170:	Best = 9.550667e+03	FES = 217100	Diversity = 8.8799e+00
Generation 2180:	Best = 9.472658e+03	FES = 218100	Diversity = 8.5340e+00
Generation 2190:	Best = 9.468033e+03	FES = 219100	Diversity = 8.5384e+00
Generation 2200:	Best = 9.410434e+03	FES = 220100	Diversity = 7.8377e+00
Generation 2210:	Best = 9.389876e+03	FES = 221100	Diversity = 7.6208e+00
Generation 2220:	Best = 9.351885e+03	FES = 222100	Diversity = 6.9379e+00
Generation 2230:	Best = 9.344404e+03	FES = 223100	Diversity = 6.8584e+00
Generation 2240:	Best = 9.261836e+03	FES = 224100	Diversity = 7.2192e+00
Generation 2250:	Best = 9.212521e+03	FES = 225100	Diversity = 8.1558e+00
Generation 2260:	Best = 9.165498e+03	FES = 226100	Diversity = 8.4351e+00
Generation 2270:	Best = 9.093960e+03	FES = 227100	Diversity = 7.9183e+00
Generation 2280:	Best = 9.074233e+03	FES = 228100	Diversity = 8.3613e+00
Generation 2290:	Best = 8.997976e+03	FES = 229100	Diversity = 9.0275e+00
Generation 2300:	Best = 8.938399e+03	FES = 230100	Diversity = 9.8980e+00
Generation 2310:	Best = 8.912843e+03	FES = 231100	Diversity = 9.7315e+00
Generation 2320:	Best = 8.846354e+03	FES = 232100	Diversity = 9.7331e+00
Generation 2330:	Best = 8.821541e+03	FES = 233100	Diversity = 1.0392e+01
Generation 2340:	Best = 8.766536e+03	FES = 234100	Diversity = 1.1325e+01
Generation 2350:	Best = 8.678012e+03	FES = 235100	Diversity = 1.1812e+01
Generation 2360:	Best = 8.637838e+03	FES = 236100	Diversity = 1.1010e+01
Generation 2370:	Best = 8.594606e+03	FES = 237100	Diversity = 1.0595e+01
Generation 2380:	Best = 8.556582e+03	FES = 238100	Diversity = 1.1268e+01
Generation 2390:	Best = 8.471332e+03	FES = 239100	Diversity = 1.0444e+01
Generation 2400:	Best = 8.425988e+03	FES = 240100	Diversity = 9.8079e+00
Generation 2410:	Best = 8.376687e+03	FES = 241100	Diversity = 9.3229e+00
Generation 2420:	Best = 8.311538e+03	FES = 242100	Diversity = 9.3422e+00
Generation 2430:	Best = 8.273684e+03	FES = 243100	Diversity = 9.3912e+00
Generation 2440:	Best = 8.233368e+03	FES = 244100	Diversity = 1.0438e+01
Generation 2450:	Best = 8.144683e+03	FES = 245100	Diversity = 1.0198e+01
Generation 2460:	Best = 8.095726e+03	FES = 246100	Diversity = 1.1637e+01
Generation 2470:	Best = 8.063414e+03	FES = 247100	Diversity = 9.9973e+00
Generation 2480:	Best = 8.003837e+03	FES = 248100	Diversity = 9.6114e+00
Generation 2490:	Best = 7.986374e+03	FES = 249100	Diversity = 9.8718e+00
Generation 2500:	Best = 7.928512e+03	FES = 250100	Diversity = 1.0033e+01
Generation 2510:	Best = 7.817138e+03	FES = 251100	Diversity = 9.9182e+00
Generation 2520:	Best = 7.807779e+03	FES = 252100	Diversity = 1.0338e+01
Generation 2530:	Best = 7.748489e+03	FES = 253100	Diversity = 1.0370e+01
Generation 2540:	Best = 7.698486e+03	FES = 254100	Diversity = 1.0773e+01
Generation 2550:	Best = 7.624406e+03	FES = 255100	Diversity = 1.1351e+01
Generation 2560:	Best = 7.564445e+03	FES = 256100	Diversity = 1.0679e+01
Generation 2570:	Best = 7.495446e+03	FES = 257100	Diversity = 1.1342e+01
Generation 2580:	Best = 7.444575e+03	FES = 258100	Diversity = 1.1275e+01
Generation 2590:	Best = 7.396051e+03	FES = 259100	Diversity = 1.1235e+01
Generation 2600:	Best = 7.377465e+03	FES = 260100	Diversity = 1.1273e+01

Generation 2610:	Best = 7.308812e+03	FES = 261100	Diversity = 1.1881e+01
Generation 2620:	Best = 7.203691e+03	FES = 262100	Diversity = 1.2096e+01
Generation 2630:	Best = 7.189266e+03	FES = 263100	Diversity = 1.2392e+01
Generation 2640:	Best = 7.143105e+03	FES = 264100	Diversity = 1.1405e+01
Generation 2650:	Best = 7.104341e+03	FES = 265100	Diversity = 1.0970e+01
Generation 2660:	Best = 7.060615e+03	FES = 266100	Diversity = 9.7030e+00
Generation 2670:	Best = 7.010031e+03	FES = 267100	Diversity = 1.0391e+01
Generation 2680:	Best = 6.965519e+03	FES = 268100	Diversity = 1.0413e+01
Generation 2690:	Best = 6.941319e+03	FES = 269100	Diversity = 1.0888e+01
Generation 2700:	Best = 6.902268e+03	FES = 270100	Diversity = 1.0476e+01
Generation 2710:	Best = 6.861366e+03	FES = 271100	Diversity = 9.9261e+00
Generation 2720:	Best = 6.828467e+03	FES = 272100	Diversity = 8.8799e+00
Generation 2730:	Best = 6.787950e+03	FES = 273100	Diversity = 8.9885e+00
Generation 2740:	Best = 6.689583e+03	FES = 274100	Diversity = 9.8613e+00
Generation 2750:	Best = 6.621387e+03	FES = 275100	Diversity = 9.5510e+00
Generation 2760:	Best = 6.559374e+03	FES = 276100	Diversity = 9.4221e+00
Generation 2770:	Best = 6.516943e+03	FES = 277100	Diversity = 9.5877e+00
Generation 2780:	Best = 6.476745e+03	FES = 278100	Diversity = 9.9774e+00
Generation 2790:	Best = 6.431029e+03	FES = 279100	Diversity = 9.8136e+00
Generation 2800:	Best = 6.361906e+03	FES = 280100	Diversity = 9.8609e+00
Generation 2810:	Best = 6.327853e+03	FES = 281100	Diversity = 9.5036e+00
Generation 2820:	Best = 6.253841e+03	FES = 282100	Diversity = 9.7496e+00
Generation 2830:	Best = 6.226289e+03	FES = 283100	Diversity = 9.5308e+00
Generation 2840:	Best = 6.203014e+03	FES = 284100	Diversity = 9.0694e+00
Generation 2850:	Best = 6.164004e+03	FES = 285100	Diversity = 9.0742e+00
Generation 2860:	Best = 6.107613e+03	FES = 286100	Diversity = 9.0251e+00
Generation 2870:	Best = 6.086720e+03	FES = 287100	Diversity = 9.1105e+00
Generation 2880:	Best = 6.048082e+03	FES = 288100	Diversity = 8.4747e+00
Generation 2890:	Best = 5.996906e+03	FES = 289100	Diversity = 9.1419e+00
Generation 2900:	Best = 5.902705e+03	FES = 290100	Diversity = 1.0282e+01
Generation 2910:	Best = 5.835112e+03	FES = 291100	Diversity = 1.0150e+01
Generation 2920:	Best = 5.799620e+03	FES = 292100	Diversity = 9.8850e+00
Generation 2930:	Best = 5.735931e+03	FES = 293100	Diversity = 9.5741e+00
Generation 2940:	Best = 5.712067e+03	FES = 294100	Diversity = 9.0062e+00
Generation 2950:	Best = 5.675125e+03	FES = 295100	Diversity = 8.9050e+00
Generation 2960:	Best = 5.643185e+03	FES = 296100	Diversity = 8.9644e+00
Generation 2970:	Best = 5.602136e+03	FES = 297100	Diversity = 8.9486e+00
Generation 2980:	Best = 5.547107e+03	FES = 298100	Diversity = 9.7353e+00
Generation 2990:	Best = 5.516976e+03	FES = 299100	Diversity = 1.0529e+01

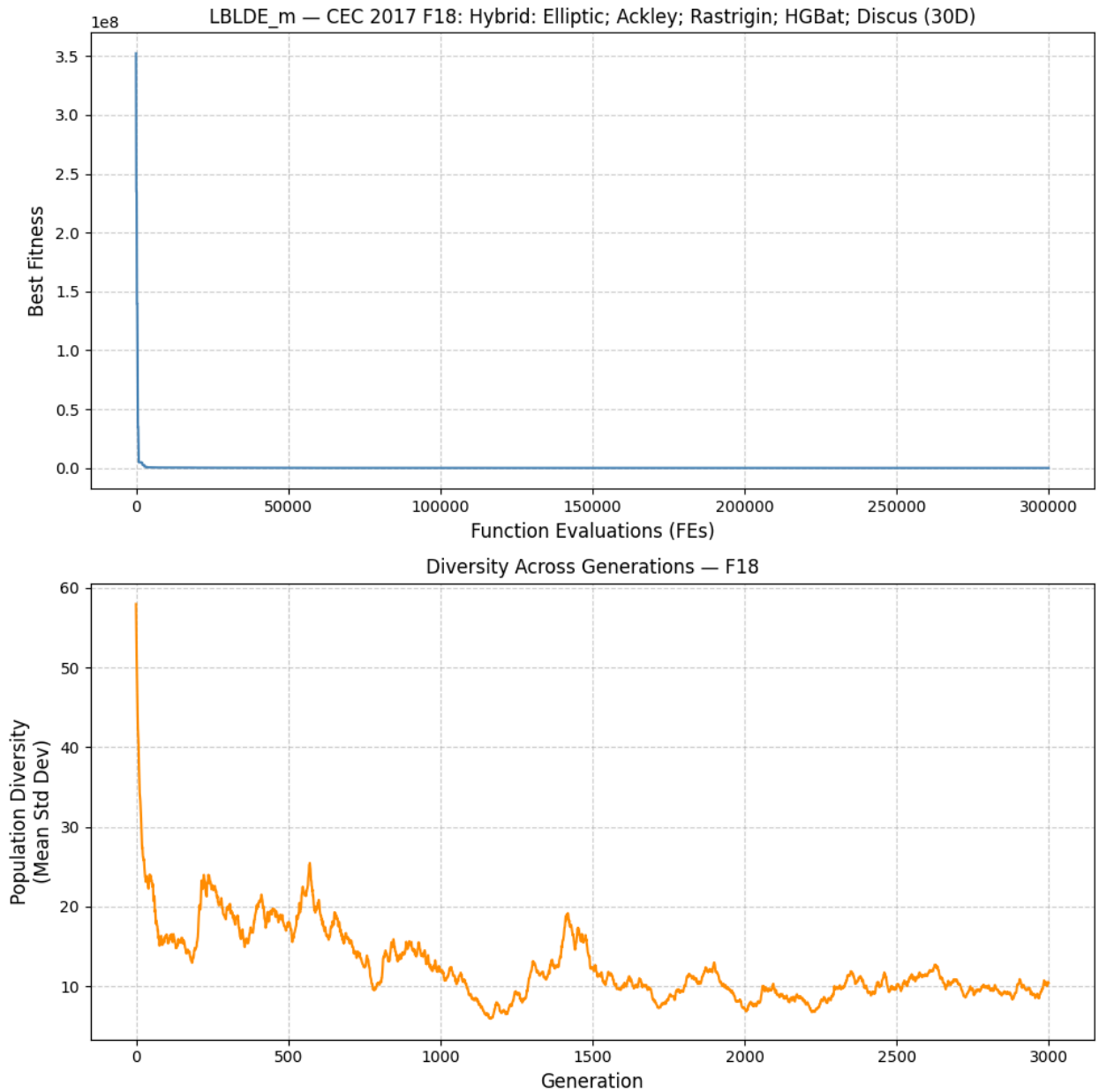
Optimization completed!

Final Best Fitness = 5.479425e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 5.479425e+03
  Global Optimum         : 1.800000e+03
  Error (|best-optimum|) : 3.679425e+03
  Best Solution (5 dim)  : [ -5.08398006  61.16125813  34.74254472 -84.13952506
46.39458768]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F18.png

```
In [45]: # =====  
# F19 – Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierst  
# =====  
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent Cigar; Ras
```

=====

=

LBLDE_m on CEC 2017 F19 – Single Run

=====

=

Function : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstras
s; Schaffer (30D)

Bounds : [-100, 100]

Global optimum : 1900.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 7.567444e+10 | Diversity = 5.7976e+01

Generation 10: Best = 1.537828e+09 FES = 1100 Diversity = 2.7094e+01

Generation 20: Best = 3.379348e+07 FES = 2100 Diversity = 1.6196e+01

Generation 30: Best = 3.043828e+06 FES = 3100 Diversity = 1.0147e+01

Generation 40: Best = 9.486817e+05 FES = 4100 Diversity = 6.8011e+00

Generation 50: Best = 5.399948e+05 FES = 5100 Diversity = 4.7169e+00

Generation 60: Best = 3.107463e+05 FES = 6100 Diversity = 4.4856e+00

Generation 70: Best = 1.860535e+05 FES = 7100 Diversity = 4.3068e+00

Generation 80: Best = 1.293818e+05 FES = 8100 Diversity = 4.3358e+00

Generation 90: Best = 8.933489e+04 FES = 9100 Diversity = 4.7953e+00

Generation 100: Best = 4.956655e+04 FES = 10100 Diversity = 6.0382e+00

Generation 110: Best = 3.451081e+04 FES = 11100 Diversity = 6.9431e+00

Generation 120: Best = 2.980083e+04 FES = 12100 Diversity = 7.0855e+00

Generation 130: Best = 2.544230e+04 FES = 13100 Diversity = 7.6947e+00

Generation 140: Best = 2.046131e+04 FES = 14100 Diversity = 9.4630e+00

Generation 150: Best = 1.962534e+04 FES = 15100 Diversity = 9.6476e+00

Generation 160: Best = 1.686219e+04 FES = 16100 Diversity = 9.8587e+00

Generation 170: Best = 1.524511e+04 FES = 17100 Diversity = 1.1316e+01

Generation 180: Best = 1.491947e+04 FES = 18100 Diversity = 1.1010e+01

Generation 190: Best = 1.443128e+04 FES = 19100 Diversity = 1.0946e+01

Generation 200: Best = 1.407405e+04 FES = 20100 Diversity = 1.2549e+01

Generation 210: Best = 1.351551e+04 FES = 21100 Diversity = 1.4552e+01

Generation 220: Best = 1.324223e+04 FES = 22100 Diversity = 1.4083e+01

Generation 230: Best = 1.291149e+04 FES = 23100 Diversity = 1.4145e+01

Generation 240: Best = 1.263924e+04 FES = 24100 Diversity = 1.5192e+01

Generation 250: Best = 1.254764e+04 FES = 25100 Diversity = 1.5804e+01

Generation 260: Best = 1.236670e+04 FES = 26100 Diversity = 1.4636e+01

Generation 270: Best = 1.221464e+04 FES = 27100 Diversity = 1.4117e+01

Generation 280: Best = 1.206788e+04 FES = 28100 Diversity = 1.6073e+01

Generation 290: Best = 1.200733e+04 FES = 29100 Diversity = 1.4828e+01

Generation 300: Best = 1.191070e+04 FES = 30100 Diversity = 1.4632e+01

Generation 310: Best = 1.182107e+04 FES = 31100 Diversity = 1.5596e+01

Generation 320: Best = 1.172229e+04 FES = 32100 Diversity = 1.7265e+01

Generation 330: Best = 1.155739e+04 FES = 33100 Diversity = 1.6889e+01

Generation 340: Best = 1.154271e+04 FES = 34100 Diversity = 1.6390e+01

Generation 350: Best = 1.148468e+04 FES = 35100 Diversity = 1.6121e+01

Generation 360: Best = 1.145317e+04 FES = 36100 Diversity = 1.6830e+01

Generation 370: Best = 1.138436e+04 FES = 37100 Diversity = 1.5753e+01

Generation 380: Best = 1.133439e+04 FES = 38100 Diversity = 1.5320e+01

Generation 390: Best = 1.128572e+04 FES = 39100 Diversity = 1.4704e+01

Generation 400: Best = 1.126635e+04 FES = 40100 Diversity = 1.3966e+01

Generation 410: Best = 1.119532e+04 FES = 41100 Diversity = 1.4214e+01

Generation 420: Best = 1.115907e+04 FES = 42100 Diversity = 1.4075e+01

Generation 430: Best = 1.109710e+04 FES = 43100 Diversity = 1.4320e+01

Generation	440:	Best = 1.107124e+04	FES = 44100	Diversity = 1.4862e+01
Generation	450:	Best = 1.097604e+04	FES = 45100	Diversity = 1.3668e+01
Generation	460:	Best = 1.096837e+04	FES = 46100	Diversity = 1.5136e+01
Generation	470:	Best = 1.090461e+04	FES = 47100	Diversity = 1.4563e+01
Generation	480:	Best = 1.076737e+04	FES = 48100	Diversity = 1.5013e+01
Generation	490:	Best = 1.068403e+04	FES = 49100	Diversity = 1.4396e+01
Generation	500:	Best = 1.061436e+04	FES = 50100	Diversity = 1.6222e+01
Generation	510:	Best = 1.055817e+04	FES = 51100	Diversity = 1.6127e+01
Generation	520:	Best = 1.051564e+04	FES = 52100	Diversity = 1.4668e+01
Generation	530:	Best = 1.044262e+04	FES = 53100	Diversity = 1.5251e+01
Generation	540:	Best = 1.040159e+04	FES = 54100	Diversity = 1.5779e+01
Generation	550:	Best = 1.030337e+04	FES = 55100	Diversity = 1.6007e+01
Generation	560:	Best = 1.018599e+04	FES = 56100	Diversity = 1.8040e+01
Generation	570:	Best = 1.012926e+04	FES = 57100	Diversity = 1.6655e+01
Generation	580:	Best = 1.001939e+04	FES = 58100	Diversity = 1.5356e+01
Generation	590:	Best = 9.983820e+03	FES = 59100	Diversity = 1.5184e+01
Generation	600:	Best = 9.938648e+03	FES = 60100	Diversity = 1.4752e+01
Generation	610:	Best = 9.913231e+03	FES = 61100	Diversity = 1.5117e+01
Generation	620:	Best = 9.880973e+03	FES = 62100	Diversity = 1.4566e+01
Generation	630:	Best = 9.828344e+03	FES = 63100	Diversity = 1.5101e+01
Generation	640:	Best = 9.776529e+03	FES = 64100	Diversity = 1.5701e+01
Generation	650:	Best = 9.742120e+03	FES = 65100	Diversity = 1.4863e+01
Generation	660:	Best = 9.656448e+03	FES = 66100	Diversity = 1.2067e+01
Generation	670:	Best = 9.633581e+03	FES = 67100	Diversity = 1.3669e+01
Generation	680:	Best = 9.563821e+03	FES = 68100	Diversity = 1.3815e+01
Generation	690:	Best = 9.545271e+03	FES = 69100	Diversity = 1.4185e+01
Generation	700:	Best = 9.515658e+03	FES = 70100	Diversity = 1.2701e+01
Generation	710:	Best = 9.491485e+03	FES = 71100	Diversity = 1.2588e+01
Generation	720:	Best = 9.468898e+03	FES = 72100	Diversity = 1.1714e+01
Generation	730:	Best = 9.410725e+03	FES = 73100	Diversity = 1.2331e+01
Generation	740:	Best = 9.402864e+03	FES = 74100	Diversity = 1.2515e+01
Generation	750:	Best = 9.359855e+03	FES = 75100	Diversity = 1.2018e+01
Generation	760:	Best = 9.359855e+03	FES = 76100	Diversity = 1.1544e+01
Generation	770:	Best = 9.344564e+03	FES = 77100	Diversity = 1.2453e+01
Generation	780:	Best = 9.334015e+03	FES = 78100	Diversity = 1.3733e+01
Generation	790:	Best = 9.308205e+03	FES = 79100	Diversity = 1.4632e+01
Generation	800:	Best = 9.269205e+03	FES = 80100	Diversity = 1.5169e+01
Generation	810:	Best = 9.261986e+03	FES = 81100	Diversity = 1.4646e+01
Generation	820:	Best = 9.219351e+03	FES = 82100	Diversity = 1.4049e+01
Generation	830:	Best = 9.165275e+03	FES = 83100	Diversity = 1.4499e+01
Generation	840:	Best = 9.157292e+03	FES = 84100	Diversity = 1.6030e+01
Generation	850:	Best = 9.119976e+03	FES = 85100	Diversity = 1.6972e+01
Generation	860:	Best = 9.084378e+03	FES = 86100	Diversity = 1.7043e+01
Generation	870:	Best = 9.060742e+03	FES = 87100	Diversity = 1.6859e+01
Generation	880:	Best = 9.045167e+03	FES = 88100	Diversity = 1.6487e+01
Generation	890:	Best = 9.004237e+03	FES = 89100	Diversity = 1.5443e+01
Generation	900:	Best = 8.985052e+03	FES = 90100	Diversity = 1.3881e+01
Generation	910:	Best = 8.920928e+03	FES = 91100	Diversity = 1.3947e+01
Generation	920:	Best = 8.905112e+03	FES = 92100	Diversity = 1.4535e+01
Generation	930:	Best = 8.863189e+03	FES = 93100	Diversity = 1.4446e+01
Generation	940:	Best = 8.846899e+03	FES = 94100	Diversity = 1.3914e+01
Generation	950:	Best = 8.765176e+03	FES = 95100	Diversity = 1.3910e+01
Generation	960:	Best = 8.765176e+03	FES = 96100	Diversity = 1.3837e+01
Generation	970:	Best = 8.743926e+03	FES = 97100	Diversity = 1.4325e+01

Generation 980:	Best = 8.664023e+03	FES = 98100	Diversity = 1.4602e+01
Generation 990:	Best = 8.649553e+03	FES = 99100	Diversity = 1.3854e+01
Generation 1000:	Best = 8.540758e+03	FES = 100100	Diversity = 1.5578e+01
Generation 1010:	Best = 8.453744e+03	FES = 101100	Diversity = 1.6444e+01
Generation 1020:	Best = 8.414419e+03	FES = 102100	Diversity = 1.4146e+01
Generation 1030:	Best = 8.386401e+03	FES = 103100	Diversity = 1.3389e+01
Generation 1040:	Best = 8.359973e+03	FES = 104100	Diversity = 1.3993e+01
Generation 1050:	Best = 8.319606e+03	FES = 105100	Diversity = 1.4161e+01
Generation 1060:	Best = 8.302354e+03	FES = 106100	Diversity = 1.4280e+01
Generation 1070:	Best = 8.287274e+03	FES = 107100	Diversity = 1.3367e+01
Generation 1080:	Best = 8.252756e+03	FES = 108100	Diversity = 1.2713e+01
Generation 1090:	Best = 8.207362e+03	FES = 109100	Diversity = 1.3461e+01
Generation 1100:	Best = 8.188250e+03	FES = 110100	Diversity = 1.3129e+01
Generation 1110:	Best = 8.175381e+03	FES = 111100	Diversity = 1.2576e+01
Generation 1120:	Best = 8.143073e+03	FES = 112100	Diversity = 1.3889e+01
Generation 1130:	Best = 8.133785e+03	FES = 113100	Diversity = 1.4185e+01
Generation 1140:	Best = 8.105370e+03	FES = 114100	Diversity = 1.3464e+01
Generation 1150:	Best = 8.099731e+03	FES = 115100	Diversity = 1.2776e+01
Generation 1160:	Best = 8.050509e+03	FES = 116100	Diversity = 1.1877e+01
Generation 1170:	Best = 8.050506e+03	FES = 117100	Diversity = 1.1160e+01
Generation 1180:	Best = 8.050506e+03	FES = 118100	Diversity = 1.0806e+01
Generation 1190:	Best = 8.041994e+03	FES = 119100	Diversity = 1.0557e+01
Generation 1200:	Best = 8.036562e+03	FES = 120100	Diversity = 1.0188e+01
Generation 1210:	Best = 8.008809e+03	FES = 121100	Diversity = 9.6519e+00
Generation 1220:	Best = 7.965110e+03	FES = 122100	Diversity = 1.1086e+01
Generation 1230:	Best = 7.957639e+03	FES = 123100	Diversity = 1.1820e+01
Generation 1240:	Best = 7.923241e+03	FES = 124100	Diversity = 1.3449e+01
Generation 1250:	Best = 7.853683e+03	FES = 125100	Diversity = 1.3929e+01
Generation 1260:	Best = 7.802185e+03	FES = 126100	Diversity = 1.4971e+01
Generation 1270:	Best = 7.756013e+03	FES = 127100	Diversity = 1.5955e+01
Generation 1280:	Best = 7.662404e+03	FES = 128100	Diversity = 1.6214e+01
Generation 1290:	Best = 7.642111e+03	FES = 129100	Diversity = 1.6173e+01
Generation 1300:	Best = 7.532030e+03	FES = 130100	Diversity = 1.5568e+01
Generation 1310:	Best = 7.532030e+03	FES = 131100	Diversity = 1.7350e+01
Generation 1320:	Best = 7.482658e+03	FES = 132100	Diversity = 1.8192e+01
Generation 1330:	Best = 7.412400e+03	FES = 133100	Diversity = 1.8974e+01
Generation 1340:	Best = 7.357265e+03	FES = 134100	Diversity = 1.8970e+01
Generation 1350:	Best = 7.344564e+03	FES = 135100	Diversity = 1.7818e+01
Generation 1360:	Best = 7.329205e+03	FES = 136100	Diversity = 1.7347e+01
Generation 1370:	Best = 7.293208e+03	FES = 137100	Diversity = 1.5654e+01
Generation 1380:	Best = 7.238967e+03	FES = 138100	Diversity = 1.6035e+01
Generation 1390:	Best = 7.226268e+03	FES = 139100	Diversity = 1.5488e+01
Generation 1400:	Best = 7.189176e+03	FES = 140100	Diversity = 1.4750e+01
Generation 1410:	Best = 7.169270e+03	FES = 141100	Diversity = 1.4099e+01
Generation 1420:	Best = 7.158069e+03	FES = 142100	Diversity = 1.4236e+01
Generation 1430:	Best = 7.130638e+03	FES = 143100	Diversity = 1.3209e+01
Generation 1440:	Best = 7.096566e+03	FES = 144100	Diversity = 1.4109e+01
Generation 1450:	Best = 7.029626e+03	FES = 145100	Diversity = 1.6711e+01
Generation 1460:	Best = 6.984342e+03	FES = 146100	Diversity = 1.8866e+01
Generation 1470:	Best = 6.949182e+03	FES = 147100	Diversity = 1.8203e+01
Generation 1480:	Best = 6.913546e+03	FES = 148100	Diversity = 1.7308e+01
Generation 1490:	Best = 6.893944e+03	FES = 149100	Diversity = 1.7261e+01
Generation 1500:	Best = 6.880031e+03	FES = 150100	Diversity = 1.6789e+01
Generation 1510:	Best = 6.848271e+03	FES = 151100	Diversity = 1.6436e+01

Generation 1520:	Best = 6.817398e+03	FES = 152100	Diversity = 1.6786e+01
Generation 1530:	Best = 6.811590e+03	FES = 153100	Diversity = 1.7684e+01
Generation 1540:	Best = 6.796350e+03	FES = 154100	Diversity = 1.8414e+01
Generation 1550:	Best = 6.768041e+03	FES = 155100	Diversity = 1.9161e+01
Generation 1560:	Best = 6.758749e+03	FES = 156100	Diversity = 1.8577e+01
Generation 1570:	Best = 6.742429e+03	FES = 157100	Diversity = 1.7378e+01
Generation 1580:	Best = 6.709436e+03	FES = 158100	Diversity = 1.6423e+01
Generation 1590:	Best = 6.675532e+03	FES = 159100	Diversity = 1.5949e+01
Generation 1600:	Best = 6.669268e+03	FES = 160100	Diversity = 1.5345e+01
Generation 1610:	Best = 6.660942e+03	FES = 161100	Diversity = 1.4762e+01
Generation 1620:	Best = 6.636134e+03	FES = 162100	Diversity = 1.5622e+01
Generation 1630:	Best = 6.601605e+03	FES = 163100	Diversity = 1.5168e+01
Generation 1640:	Best = 6.549045e+03	FES = 164100	Diversity = 1.4134e+01
Generation 1650:	Best = 6.492105e+03	FES = 165100	Diversity = 1.2925e+01
Generation 1660:	Best = 6.447854e+03	FES = 166100	Diversity = 1.2492e+01
Generation 1670:	Best = 6.433685e+03	FES = 167100	Diversity = 1.2185e+01
Generation 1680:	Best = 6.395160e+03	FES = 168100	Diversity = 1.2545e+01
Generation 1690:	Best = 6.347021e+03	FES = 169100	Diversity = 1.2304e+01
Generation 1700:	Best = 6.320849e+03	FES = 170100	Diversity = 1.3032e+01
Generation 1710:	Best = 6.286786e+03	FES = 171100	Diversity = 1.2999e+01
Generation 1720:	Best = 6.236499e+03	FES = 172100	Diversity = 1.2769e+01
Generation 1730:	Best = 6.211045e+03	FES = 173100	Diversity = 1.2936e+01
Generation 1740:	Best = 6.171328e+03	FES = 174100	Diversity = 1.2906e+01
Generation 1750:	Best = 6.133783e+03	FES = 175100	Diversity = 1.3124e+01
Generation 1760:	Best = 6.092484e+03	FES = 176100	Diversity = 1.1830e+01
Generation 1770:	Best = 6.039554e+03	FES = 177100	Diversity = 1.0809e+01
Generation 1780:	Best = 5.996385e+03	FES = 178100	Diversity = 1.0682e+01
Generation 1790:	Best = 5.965385e+03	FES = 179100	Diversity = 1.1789e+01
Generation 1800:	Best = 5.912429e+03	FES = 180100	Diversity = 1.2284e+01
Generation 1810:	Best = 5.897829e+03	FES = 181100	Diversity = 1.2645e+01
Generation 1820:	Best = 5.757300e+03	FES = 182100	Diversity = 1.2721e+01
Generation 1830:	Best = 5.666266e+03	FES = 183100	Diversity = 1.3216e+01
Generation 1840:	Best = 5.621602e+03	FES = 184100	Diversity = 1.3371e+01
Generation 1850:	Best = 5.473703e+03	FES = 185100	Diversity = 1.4734e+01
Generation 1860:	Best = 5.376674e+03	FES = 186100	Diversity = 1.5492e+01
Generation 1870:	Best = 5.346557e+03	FES = 187100	Diversity = 1.4966e+01
Generation 1880:	Best = 5.273394e+03	FES = 188100	Diversity = 1.3424e+01
Generation 1890:	Best = 5.237256e+03	FES = 189100	Diversity = 1.3245e+01
Generation 1900:	Best = 5.139247e+03	FES = 190100	Diversity = 1.2811e+01
Generation 1910:	Best = 5.129352e+03	FES = 191100	Diversity = 1.3276e+01
Generation 1920:	Best = 5.077263e+03	FES = 192100	Diversity = 1.3190e+01
Generation 1930:	Best = 5.004504e+03	FES = 193100	Diversity = 1.3417e+01
Generation 1940:	Best = 4.928875e+03	FES = 194100	Diversity = 1.3754e+01
Generation 1950:	Best = 4.872060e+03	FES = 195100	Diversity = 1.4214e+01
Generation 1960:	Best = 4.823008e+03	FES = 196100	Diversity = 1.5145e+01
Generation 1970:	Best = 4.807249e+03	FES = 197100	Diversity = 1.3748e+01
Generation 1980:	Best = 4.727933e+03	FES = 198100	Diversity = 1.3166e+01
Generation 1990:	Best = 4.674893e+03	FES = 199100	Diversity = 1.3008e+01
Generation 2000:	Best = 4.629436e+03	FES = 200100	Diversity = 1.2365e+01
Generation 2010:	Best = 4.595552e+03	FES = 201100	Diversity = 1.2291e+01
Generation 2020:	Best = 4.537434e+03	FES = 202100	Diversity = 1.1996e+01
Generation 2030:	Best = 4.524988e+03	FES = 203100	Diversity = 1.1846e+01
Generation 2040:	Best = 4.486915e+03	FES = 204100	Diversity = 1.0706e+01
Generation 2050:	Best = 4.464399e+03	FES = 205100	Diversity = 1.1669e+01

Generation 2060:	Best = 4.412137e+03	FES = 206100	Diversity = 1.2220e+01
Generation 2070:	Best = 4.397852e+03	FES = 207100	Diversity = 1.1316e+01
Generation 2080:	Best = 4.346260e+03	FES = 208100	Diversity = 1.1112e+01
Generation 2090:	Best = 4.310603e+03	FES = 209100	Diversity = 1.1949e+01
Generation 2100:	Best = 4.269361e+03	FES = 210100	Diversity = 1.1725e+01
Generation 2110:	Best = 4.257959e+03	FES = 211100	Diversity = 1.2386e+01
Generation 2120:	Best = 4.244604e+03	FES = 212100	Diversity = 1.2219e+01
Generation 2130:	Best = 4.221639e+03	FES = 213100	Diversity = 1.2813e+01
Generation 2140:	Best = 4.208947e+03	FES = 214100	Diversity = 1.3956e+01
Generation 2150:	Best = 4.181412e+03	FES = 215100	Diversity = 1.3569e+01
Generation 2160:	Best = 4.152274e+03	FES = 216100	Diversity = 1.3234e+01
Generation 2170:	Best = 4.126122e+03	FES = 217100	Diversity = 1.2591e+01
Generation 2180:	Best = 4.107543e+03	FES = 218100	Diversity = 1.1633e+01
Generation 2190:	Best = 4.095605e+03	FES = 219100	Diversity = 1.1532e+01
Generation 2200:	Best = 4.065762e+03	FES = 220100	Diversity = 1.1132e+01
Generation 2210:	Best = 4.051258e+03	FES = 221100	Diversity = 1.1677e+01
Generation 2220:	Best = 3.996218e+03	FES = 222100	Diversity = 1.2546e+01
Generation 2230:	Best = 3.943996e+03	FES = 223100	Diversity = 1.2772e+01
Generation 2240:	Best = 3.907830e+03	FES = 224100	Diversity = 1.4597e+01
Generation 2250:	Best = 3.894764e+03	FES = 225100	Diversity = 1.4296e+01
Generation 2260:	Best = 3.865621e+03	FES = 226100	Diversity = 1.4723e+01
Generation 2270:	Best = 3.850417e+03	FES = 227100	Diversity = 1.4583e+01
Generation 2280:	Best = 3.842340e+03	FES = 228100	Diversity = 1.3679e+01
Generation 2290:	Best = 3.812377e+03	FES = 229100	Diversity = 1.3317e+01
Generation 2300:	Best = 3.786999e+03	FES = 230100	Diversity = 1.3379e+01
Generation 2310:	Best = 3.754388e+03	FES = 231100	Diversity = 1.3713e+01
Generation 2320:	Best = 3.725682e+03	FES = 232100	Diversity = 1.3937e+01
Generation 2330:	Best = 3.716569e+03	FES = 233100	Diversity = 1.3687e+01
Generation 2340:	Best = 3.677603e+03	FES = 234100	Diversity = 1.3859e+01
Generation 2350:	Best = 3.672678e+03	FES = 235100	Diversity = 1.3528e+01
Generation 2360:	Best = 3.622643e+03	FES = 236100	Diversity = 1.2572e+01
Generation 2370:	Best = 3.600720e+03	FES = 237100	Diversity = 1.3067e+01
Generation 2380:	Best = 3.577988e+03	FES = 238100	Diversity = 1.3109e+01
Generation 2390:	Best = 3.555003e+03	FES = 239100	Diversity = 1.4295e+01
Generation 2400:	Best = 3.541996e+03	FES = 240100	Diversity = 1.2860e+01
Generation 2410:	Best = 3.496272e+03	FES = 241100	Diversity = 1.2833e+01
Generation 2420:	Best = 3.458117e+03	FES = 242100	Diversity = 1.3416e+01
Generation 2430:	Best = 3.408845e+03	FES = 243100	Diversity = 1.3314e+01
Generation 2440:	Best = 3.391534e+03	FES = 244100	Diversity = 1.2071e+01
Generation 2450:	Best = 3.378861e+03	FES = 245100	Diversity = 1.1313e+01
Generation 2460:	Best = 3.363892e+03	FES = 246100	Diversity = 1.1125e+01
Generation 2470:	Best = 3.345318e+03	FES = 247100	Diversity = 1.1483e+01
Generation 2480:	Best = 3.325086e+03	FES = 248100	Diversity = 1.0990e+01
Generation 2490:	Best = 3.314564e+03	FES = 249100	Diversity = 1.0736e+01
Generation 2500:	Best = 3.289720e+03	FES = 250100	Diversity = 1.0842e+01
Generation 2510:	Best = 3.237120e+03	FES = 251100	Diversity = 1.2655e+01
Generation 2520:	Best = 3.184104e+03	FES = 252100	Diversity = 1.4100e+01
Generation 2530:	Best = 3.148883e+03	FES = 253100	Diversity = 1.2884e+01
Generation 2540:	Best = 3.127265e+03	FES = 254100	Diversity = 1.1931e+01
Generation 2550:	Best = 3.097871e+03	FES = 255100	Diversity = 1.2196e+01
Generation 2560:	Best = 3.083735e+03	FES = 256100	Diversity = 1.3296e+01
Generation 2570:	Best = 3.056912e+03	FES = 257100	Diversity = 1.3592e+01
Generation 2580:	Best = 3.029017e+03	FES = 258100	Diversity = 1.3455e+01
Generation 2590:	Best = 3.009798e+03	FES = 259100	Diversity = 1.4328e+01

Generation 2600:	Best = 2.978522e+03	FES = 260100	Diversity = 1.4209e+01
Generation 2610:	Best = 2.936745e+03	FES = 261100	Diversity = 1.3868e+01
Generation 2620:	Best = 2.911465e+03	FES = 262100	Diversity = 1.4254e+01
Generation 2630:	Best = 2.892859e+03	FES = 263100	Diversity = 1.3957e+01
Generation 2640:	Best = 2.846088e+03	FES = 264100	Diversity = 1.2620e+01
Generation 2650:	Best = 2.829007e+03	FES = 265100	Diversity = 1.2953e+01
Generation 2660:	Best = 2.800451e+03	FES = 266100	Diversity = 1.3445e+01
Generation 2670:	Best = 2.758053e+03	FES = 267100	Diversity = 1.3188e+01
Generation 2680:	Best = 2.715149e+03	FES = 268100	Diversity = 1.2521e+01
Generation 2690:	Best = 2.676219e+03	FES = 269100	Diversity = 1.2099e+01
Generation 2700:	Best = 2.655170e+03	FES = 270100	Diversity = 1.2387e+01
Generation 2710:	Best = 2.622885e+03	FES = 271100	Diversity = 1.3131e+01
Generation 2720:	Best = 2.591854e+03	FES = 272100	Diversity = 1.2572e+01
Generation 2730:	Best = 2.574499e+03	FES = 273100	Diversity = 1.2703e+01
Generation 2740:	Best = 2.565939e+03	FES = 274100	Diversity = 1.2765e+01
Generation 2750:	Best = 2.544289e+03	FES = 275100	Diversity = 1.3746e+01
Generation 2760:	Best = 2.470574e+03	FES = 276100	Diversity = 1.3639e+01
Generation 2770:	Best = 2.440307e+03	FES = 277100	Diversity = 1.3165e+01
Generation 2780:	Best = 2.410542e+03	FES = 278100	Diversity = 1.4457e+01
Generation 2790:	Best = 2.386090e+03	FES = 279100	Diversity = 1.3928e+01
Generation 2800:	Best = 2.362591e+03	FES = 280100	Diversity = 1.3232e+01
Generation 2810:	Best = 2.348809e+03	FES = 281100	Diversity = 1.3132e+01
Generation 2820:	Best = 2.316493e+03	FES = 282100	Diversity = 1.2961e+01
Generation 2830:	Best = 2.299979e+03	FES = 283100	Diversity = 1.3367e+01
Generation 2840:	Best = 2.277776e+03	FES = 284100	Diversity = 1.4287e+01
Generation 2850:	Best = 2.254530e+03	FES = 285100	Diversity = 1.5203e+01
Generation 2860:	Best = 2.227504e+03	FES = 286100	Diversity = 1.5796e+01
Generation 2870:	Best = 2.201033e+03	FES = 287100	Diversity = 1.6595e+01
Generation 2880:	Best = 2.180837e+03	FES = 288100	Diversity = 1.6560e+01
Generation 2890:	Best = 2.164916e+03	FES = 289100	Diversity = 1.6882e+01
Generation 2900:	Best = 2.113990e+03	FES = 290100	Diversity = 1.6459e+01
Generation 2910:	Best = 2.088808e+03	FES = 291100	Diversity = 1.4683e+01
Generation 2920:	Best = 2.073718e+03	FES = 292100	Diversity = 1.3869e+01
Generation 2930:	Best = 2.032363e+03	FES = 293100	Diversity = 1.3510e+01
Generation 2940:	Best = 2.021541e+03	FES = 294100	Diversity = 1.2559e+01
Generation 2950:	Best = 2.005599e+03	FES = 295100	Diversity = 1.1660e+01
Generation 2960:	Best = 1.996321e+03	FES = 296100	Diversity = 1.0853e+01
Generation 2970:	Best = 1.987526e+03	FES = 297100	Diversity = 1.0862e+01
Generation 2980:	Best = 1.980682e+03	FES = 298100	Diversity = 1.0936e+01
Generation 2990:	Best = 1.979372e+03	FES = 299100	Diversity = 1.0649e+01

Optimization completed!

Final Best Fitness = 1.962082e+03

Total Function Evaluations = 300000

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Results:

Best Fitness : 1.962082e+03

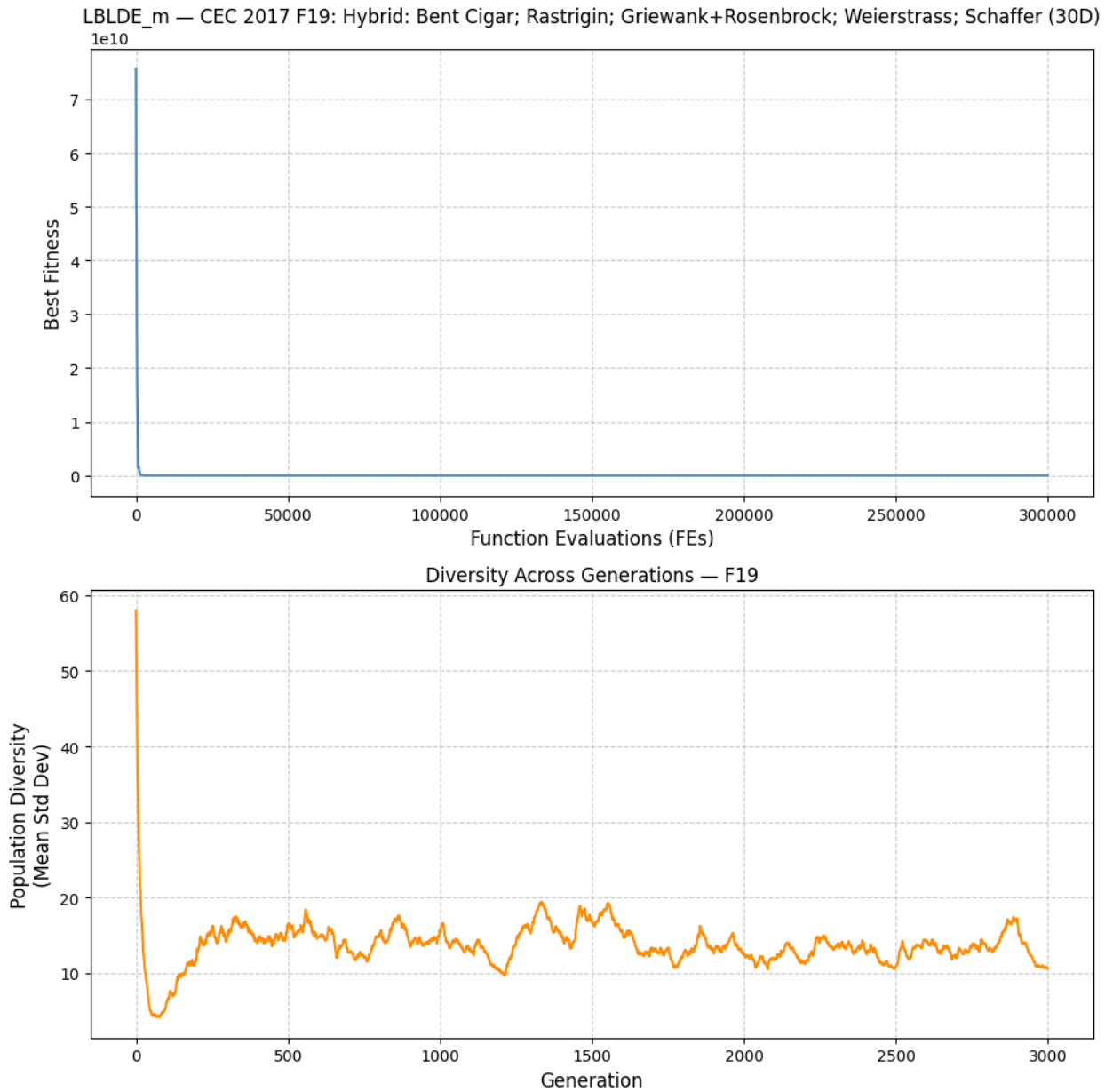
Global Optimum : 1.900000e+03

Error (|best-optimum|) : 6.208153e+01

Best Solution (5 dim) : [97.84736937 -86.99505671 -18.21852085 -100.
-23.12617195]

D^(0) (init. diversity): 5.7976e+01

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Graph saved as convergence_F19.png

```
In [46]: # =====  
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; S  
# =====  
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid: HappyCat; Katsu
```

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LBLDE_m on CEC 2017 F20 – Single Run

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Function : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer (30D)

Bounds : [-100, 100]

Global optimum : 2000.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.694893e+03 | Diversity = 5.7976e+01

Generation 10: Best = 2.988369e+03 FES = 1100 Diversity = 4.5696e+01

Generation 20: Best = 2.918161e+03 FES = 2100 Diversity = 4.1508e+01

Generation 30: Best = 2.811994e+03 FES = 3100 Diversity = 3.6795e+01

Generation 40: Best = 2.806079e+03 FES = 4100 Diversity = 3.4715e+01

Generation 50: Best = 2.667375e+03 FES = 5100 Diversity = 3.2726e+01

Generation 60: Best = 2.590846e+03 FES = 6100 Diversity = 3.1068e+01

Generation 70: Best = 2.565125e+03 FES = 7100 Diversity = 2.9849e+01

Generation 80: Best = 2.539528e+03 FES = 8100 Diversity = 2.9456e+01

Generation 90: Best = 2.539528e+03 FES = 9100 Diversity = 2.9133e+01

Generation 100: Best = 2.539528e+03 FES = 10100 Diversity = 2.8363e+01

Generation 110: Best = 2.514074e+03 FES = 11100 Diversity = 2.8024e+01

Generation 120: Best = 2.506676e+03 FES = 12100 Diversity = 2.6977e+01

Generation 130: Best = 2.496717e+03 FES = 13100 Diversity = 2.6724e+01

Generation 140: Best = 2.496186e+03 FES = 14100 Diversity = 2.6055e+01

Generation 150: Best = 2.489253e+03 FES = 15100 Diversity = 2.5892e+01

Generation 160: Best = 2.473931e+03 FES = 16100 Diversity = 2.4934e+01

Generation 170: Best = 2.470068e+03 FES = 17100 Diversity = 2.4148e+01

Generation 180: Best = 2.442492e+03 FES = 18100 Diversity = 2.4043e+01

Generation 190: Best = 2.426126e+03 FES = 19100 Diversity = 2.3749e+01

Generation 200: Best = 2.425022e+03 FES = 20100 Diversity = 2.3566e+01

Generation 210: Best = 2.425022e+03 FES = 21100 Diversity = 2.3526e+01

Generation 220: Best = 2.418519e+03 FES = 22100 Diversity = 2.3550e+01

Generation 230: Best = 2.407895e+03 FES = 23100 Diversity = 2.2774e+01

Generation 240: Best = 2.382547e+03 FES = 24100 Diversity = 2.2100e+01

Generation 250: Best = 2.372915e+03 FES = 25100 Diversity = 2.1865e+01

Generation 260: Best = 2.362778e+03 FES = 26100 Diversity = 2.1672e+01

Generation 270: Best = 2.362778e+03 FES = 27100 Diversity = 2.1301e+01

Generation 280: Best = 2.354905e+03 FES = 28100 Diversity = 2.1020e+01

Generation 290: Best = 2.348093e+03 FES = 29100 Diversity = 2.0409e+01

Generation 300: Best = 2.317801e+03 FES = 30100 Diversity = 2.0170e+01

Generation 310: Best = 2.293509e+03 FES = 31100 Diversity = 1.9891e+01

Generation 320: Best = 2.288183e+03 FES = 32100 Diversity = 1.9643e+01

Generation 330: Best = 2.287230e+03 FES = 33100 Diversity = 1.9073e+01

Generation 340: Best = 2.287230e+03 FES = 34100 Diversity = 1.8776e+01

Generation 350: Best = 2.287230e+03 FES = 35100 Diversity = 1.8701e+01

Generation 360: Best = 2.287230e+03 FES = 36100 Diversity = 1.8753e+01

Generation 370: Best = 2.284751e+03 FES = 37100 Diversity = 1.8526e+01

Generation 380: Best = 2.274067e+03 FES = 38100 Diversity = 1.8189e+01

Generation 390: Best = 2.274067e+03 FES = 39100 Diversity = 1.8008e+01

Generation 400: Best = 2.274067e+03 FES = 40100 Diversity = 1.7828e+01

Generation 410: Best = 2.268918e+03 FES = 41100 Diversity = 1.7773e+01

Generation 420: Best = 2.268918e+03 FES = 42100 Diversity = 1.7679e+01

Generation 430: Best = 2.268713e+03 FES = 43100 Diversity = 1.7700e+01

Generation	440:	Best = 2.268713e+03	FES = 44100	Diversity = 1.7365e+01
Generation	450:	Best = 2.268713e+03	FES = 45100	Diversity = 1.7316e+01
Generation	460:	Best = 2.268713e+03	FES = 46100	Diversity = 1.7147e+01
Generation	470:	Best = 2.268713e+03	FES = 47100	Diversity = 1.7065e+01
Generation	480:	Best = 2.268713e+03	FES = 48100	Diversity = 1.6819e+01
Generation	490:	Best = 2.268713e+03	FES = 49100	Diversity = 1.6685e+01
Generation	500:	Best = 2.268713e+03	FES = 50100	Diversity = 1.6352e+01
Generation	510:	Best = 2.268563e+03	FES = 51100	Diversity = 1.6258e+01
Generation	520:	Best = 2.268378e+03	FES = 52100	Diversity = 1.6060e+01
Generation	530:	Best = 2.268378e+03	FES = 53100	Diversity = 1.5888e+01
Generation	540:	Best = 2.267199e+03	FES = 54100	Diversity = 1.5667e+01
Generation	550:	Best = 2.239410e+03	FES = 55100	Diversity = 1.5738e+01
Generation	560:	Best = 2.239410e+03	FES = 56100	Diversity = 1.5636e+01
Generation	570:	Best = 2.239096e+03	FES = 57100	Diversity = 1.5657e+01
Generation	580:	Best = 2.238225e+03	FES = 58100	Diversity = 1.5702e+01
Generation	590:	Best = 2.238225e+03	FES = 59100	Diversity = 1.5607e+01
Generation	600:	Best = 2.233971e+03	FES = 60100	Diversity = 1.5457e+01
Generation	610:	Best = 2.232329e+03	FES = 61100	Diversity = 1.5441e+01
Generation	620:	Best = 2.232044e+03	FES = 62100	Diversity = 1.5373e+01
Generation	630:	Best = 2.229933e+03	FES = 63100	Diversity = 1.5422e+01
Generation	640:	Best = 2.229233e+03	FES = 64100	Diversity = 1.5377e+01
Generation	650:	Best = 2.229233e+03	FES = 65100	Diversity = 1.5367e+01
Generation	660:	Best = 2.228584e+03	FES = 66100	Diversity = 1.5339e+01
Generation	670:	Best = 2.228421e+03	FES = 67100	Diversity = 1.5289e+01
Generation	680:	Best = 2.228375e+03	FES = 68100	Diversity = 1.5189e+01
Generation	690:	Best = 2.228155e+03	FES = 69100	Diversity = 1.5166e+01
Generation	700:	Best = 2.227678e+03	FES = 70100	Diversity = 1.5124e+01
Generation	710:	Best = 2.227479e+03	FES = 71100	Diversity = 1.4567e+01
Generation	720:	Best = 2.227336e+03	FES = 72100	Diversity = 1.4587e+01
Generation	730:	Best = 2.222661e+03	FES = 73100	Diversity = 1.4647e+01
Generation	740:	Best = 2.222661e+03	FES = 74100	Diversity = 1.4622e+01
Generation	750:	Best = 2.222661e+03	FES = 75100	Diversity = 1.4633e+01
Generation	760:	Best = 2.222661e+03	FES = 76100	Diversity = 1.4398e+01
Generation	770:	Best = 2.222522e+03	FES = 77100	Diversity = 1.4187e+01
Generation	780:	Best = 2.222489e+03	FES = 78100	Diversity = 1.4143e+01
Generation	790:	Best = 2.22246e+03	FES = 79100	Diversity = 1.4056e+01
Generation	800:	Best = 2.222001e+03	FES = 80100	Diversity = 1.4003e+01
Generation	810:	Best = 2.221426e+03	FES = 81100	Diversity = 1.3751e+01
Generation	820:	Best = 2.221426e+03	FES = 82100	Diversity = 1.3854e+01
Generation	830:	Best = 2.221371e+03	FES = 83100	Diversity = 1.3505e+01
Generation	840:	Best = 2.221350e+03	FES = 84100	Diversity = 1.3421e+01
Generation	850:	Best = 2.221350e+03	FES = 85100	Diversity = 1.3315e+01
Generation	860:	Best = 2.207237e+03	FES = 86100	Diversity = 1.3291e+01
Generation	870:	Best = 2.207237e+03	FES = 87100	Diversity = 1.3314e+01
Generation	880:	Best = 2.207230e+03	FES = 88100	Diversity = 1.3193e+01
Generation	890:	Best = 2.207230e+03	FES = 89100	Diversity = 1.3034e+01
Generation	900:	Best = 2.207230e+03	FES = 90100	Diversity = 1.3034e+01
Generation	910:	Best = 2.207230e+03	FES = 91100	Diversity = 1.2878e+01
Generation	920:	Best = 2.207229e+03	FES = 92100	Diversity = 1.2683e+01
Generation	930:	Best = 2.207137e+03	FES = 93100	Diversity = 1.2688e+01
Generation	940:	Best = 2.206505e+03	FES = 94100	Diversity = 1.2559e+01
Generation	950:	Best = 2.206046e+03	FES = 95100	Diversity = 1.2427e+01
Generation	960:	Best = 2.203505e+03	FES = 96100	Diversity = 1.2225e+01
Generation	970:	Best = 2.201370e+03	FES = 97100	Diversity = 1.2168e+01

Generation 980:	Best = 2.201301e+03	FES = 98100	Diversity = 1.2297e+01
Generation 990:	Best = 2.197862e+03	FES = 99100	Diversity = 1.2285e+01
Generation 1000:	Best = 2.197862e+03	FES = 100100	Diversity = 1.2259e+01
Generation 1010:	Best = 2.195596e+03	FES = 101100	Diversity = 1.2167e+01
Generation 1020:	Best = 2.195374e+03	FES = 102100	Diversity = 1.2060e+01
Generation 1030:	Best = 2.195373e+03	FES = 103100	Diversity = 1.1996e+01
Generation 1040:	Best = 2.194807e+03	FES = 104100	Diversity = 1.2013e+01
Generation 1050:	Best = 2.194623e+03	FES = 105100	Diversity = 1.1991e+01
Generation 1060:	Best = 2.194238e+03	FES = 106100	Diversity = 1.1957e+01
Generation 1070:	Best = 2.194194e+03	FES = 107100	Diversity = 1.1920e+01
Generation 1080:	Best = 2.192026e+03	FES = 108100	Diversity = 1.1735e+01
Generation 1090:	Best = 2.191511e+03	FES = 109100	Diversity = 1.1600e+01
Generation 1100:	Best = 2.191069e+03	FES = 110100	Diversity = 1.1489e+01
Generation 1110:	Best = 2.191068e+03	FES = 111100	Diversity = 1.1276e+01
Generation 1120:	Best = 2.190862e+03	FES = 112100	Diversity = 1.1381e+01
Generation 1130:	Best = 2.189948e+03	FES = 113100	Diversity = 1.1033e+01
Generation 1140:	Best = 2.188598e+03	FES = 114100	Diversity = 1.0963e+01
Generation 1150:	Best = 2.186607e+03	FES = 115100	Diversity = 1.0921e+01
Generation 1160:	Best = 2.186607e+03	FES = 116100	Diversity = 1.0765e+01
Generation 1170:	Best = 2.185478e+03	FES = 117100	Diversity = 1.0739e+01
Generation 1180:	Best = 2.185478e+03	FES = 118100	Diversity = 1.0654e+01
Generation 1190:	Best = 2.185357e+03	FES = 119100	Diversity = 1.0657e+01
Generation 1200:	Best = 2.185100e+03	FES = 120100	Diversity = 1.0601e+01
Generation 1210:	Best = 2.183438e+03	FES = 121100	Diversity = 1.0134e+01
Generation 1220:	Best = 2.183129e+03	FES = 122100	Diversity = 9.9016e+00
Generation 1230:	Best = 2.183125e+03	FES = 123100	Diversity = 9.8617e+00
Generation 1240:	Best = 2.183117e+03	FES = 124100	Diversity = 9.7271e+00
Generation 1250:	Best = 2.183110e+03	FES = 125100	Diversity = 9.5687e+00
Generation 1260:	Best = 2.182134e+03	FES = 126100	Diversity = 8.9618e+00
Generation 1270:	Best = 2.182102e+03	FES = 127100	Diversity = 8.9179e+00
Generation 1280:	Best = 2.182097e+03	FES = 128100	Diversity = 8.8748e+00
Generation 1290:	Best = 2.182091e+03	FES = 129100	Diversity = 8.9142e+00
Generation 1300:	Best = 2.182090e+03	FES = 130100	Diversity = 8.8142e+00
Generation 1310:	Best = 2.182090e+03	FES = 131100	Diversity = 8.7999e+00
Generation 1320:	Best = 2.182088e+03	FES = 132100	Diversity = 8.7813e+00
Generation 1330:	Best = 2.182088e+03	FES = 133100	Diversity = 8.7471e+00
Generation 1340:	Best = 2.182088e+03	FES = 134100	Diversity = 8.7219e+00
Generation 1350:	Best = 2.182088e+03	FES = 135100	Diversity = 8.6806e+00
Generation 1360:	Best = 2.182088e+03	FES = 136100	Diversity = 8.5615e+00
Generation 1370:	Best = 2.182088e+03	FES = 137100	Diversity = 8.4359e+00
Generation 1380:	Best = 2.182088e+03	FES = 138100	Diversity = 8.3537e+00
Generation 1390:	Best = 2.182088e+03	FES = 139100	Diversity = 8.2817e+00
Generation 1400:	Best = 2.182088e+03	FES = 140100	Diversity = 8.1766e+00
Generation 1410:	Best = 2.182088e+03	FES = 141100	Diversity = 8.1173e+00
Generation 1420:	Best = 2.182088e+03	FES = 142100	Diversity = 8.0313e+00
Generation 1430:	Best = 2.177662e+03	FES = 143100	Diversity = 7.8201e+00
Generation 1440:	Best = 2.173951e+03	FES = 144100	Diversity = 7.6026e+00
Generation 1450:	Best = 2.173625e+03	FES = 145100	Diversity = 7.3490e+00
Generation 1460:	Best = 2.173522e+03	FES = 146100	Diversity = 7.1754e+00
Generation 1470:	Best = 2.173456e+03	FES = 147100	Diversity = 6.9971e+00
Generation 1480:	Best = 2.172634e+03	FES = 148100	Diversity = 6.8079e+00
Generation 1490:	Best = 2.171759e+03	FES = 149100	Diversity = 6.5401e+00
Generation 1500:	Best = 2.171756e+03	FES = 150100	Diversity = 6.4625e+00
Generation 1510:	Best = 2.169046e+03	FES = 151100	Diversity = 6.2180e+00

Generation 1520:	Best = 2.168763e+03	FES = 152100	Diversity = 5.7598e+00
Generation 1530:	Best = 2.168183e+03	FES = 153100	Diversity = 4.7325e+00
Generation 1540:	Best = 2.166472e+03	FES = 154100	Diversity = 4.0200e+00
Generation 1550:	Best = 2.163971e+03	FES = 155100	Diversity = 3.5348e+00
Generation 1560:	Best = 2.163344e+03	FES = 156100	Diversity = 2.5623e+00
Generation 1570:	Best = 2.163167e+03	FES = 157100	Diversity = 2.3614e+00
Generation 1580:	Best = 2.162906e+03	FES = 158100	Diversity = 2.0215e+00
Generation 1590:	Best = 2.162800e+03	FES = 159100	Diversity = 1.9862e+00
Generation 1600:	Best = 2.162680e+03	FES = 160100	Diversity = 1.9252e+00
Generation 1610:	Best = 2.162641e+03	FES = 161100	Diversity = 1.9619e+00
Generation 1620:	Best = 2.162603e+03	FES = 162100	Diversity = 1.9991e+00
Generation 1630:	Best = 2.162530e+03	FES = 163100	Diversity = 1.9027e+00
Generation 1640:	Best = 2.162523e+03	FES = 164100	Diversity = 1.8663e+00
Generation 1650:	Best = 2.162521e+03	FES = 165100	Diversity = 1.8726e+00
Generation 1660:	Best = 2.162520e+03	FES = 166100	Diversity = 1.8393e+00
Generation 1670:	Best = 2.162520e+03	FES = 167100	Diversity = 1.6522e+00
Generation 1680:	Best = 2.162520e+03	FES = 168100	Diversity = 1.6859e+00
Generation 1690:	Best = 2.162445e+03	FES = 169100	Diversity = 1.6466e+00
Generation 1700:	Best = 2.162032e+03	FES = 170100	Diversity = 1.6128e+00
Generation 1710:	Best = 2.161799e+03	FES = 171100	Diversity = 1.5788e+00
Generation 1720:	Best = 2.161776e+03	FES = 172100	Diversity = 1.7452e+00
Generation 1730:	Best = 2.161769e+03	FES = 173100	Diversity = 1.8397e+00
Generation 1740:	Best = 2.161768e+03	FES = 174100	Diversity = 1.8555e+00
Generation 1750:	Best = 2.161766e+03	FES = 175100	Diversity = 1.8875e+00
Generation 1760:	Best = 2.161766e+03	FES = 176100	Diversity = 1.8508e+00
Generation 1770:	Best = 2.161704e+03	FES = 177100	Diversity = 1.8907e+00
Generation 1780:	Best = 2.161649e+03	FES = 178100	Diversity = 1.8466e+00
Generation 1790:	Best = 2.161634e+03	FES = 179100	Diversity = 1.8007e+00
Generation 1800:	Best = 2.161627e+03	FES = 180100	Diversity = 1.8067e+00
Generation 1810:	Best = 2.161626e+03	FES = 181100	Diversity = 1.8882e+00
Generation 1820:	Best = 2.161626e+03	FES = 182100	Diversity = 1.8095e+00
Generation 1830:	Best = 2.161626e+03	FES = 183100	Diversity = 1.7265e+00
Generation 1840:	Best = 2.161626e+03	FES = 184100	Diversity = 1.7885e+00
Generation 1850:	Best = 2.161610e+03	FES = 185100	Diversity = 1.8373e+00
Generation 1860:	Best = 2.161596e+03	FES = 186100	Diversity = 1.8877e+00
Generation 1870:	Best = 2.161596e+03	FES = 187100	Diversity = 1.8791e+00
Generation 1880:	Best = 2.161595e+03	FES = 188100	Diversity = 1.8689e+00
Generation 1890:	Best = 2.161595e+03	FES = 189100	Diversity = 1.9258e+00
Generation 1900:	Best = 2.161595e+03	FES = 190100	Diversity = 1.9101e+00
Generation 1910:	Best = 2.161595e+03	FES = 191100	Diversity = 1.9063e+00
Generation 1920:	Best = 2.161595e+03	FES = 192100	Diversity = 1.9267e+00
Generation 1930:	Best = 2.161595e+03	FES = 193100	Diversity = 1.9440e+00
Generation 1940:	Best = 2.161594e+03	FES = 194100	Diversity = 1.9492e+00
Generation 1950:	Best = 2.161594e+03	FES = 195100	Diversity = 1.9493e+00
Generation 1960:	Best = 2.161594e+03	FES = 196100	Diversity = 1.9595e+00
Generation 1970:	Best = 2.161594e+03	FES = 197100	Diversity = 2.0076e+00
Generation 1980:	Best = 2.161569e+03	FES = 198100	Diversity = 1.9919e+00
Generation 1990:	Best = 2.161567e+03	FES = 199100	Diversity = 1.9919e+00
Generation 2000:	Best = 2.161567e+03	FES = 200100	Diversity = 2.0545e+00
Generation 2010:	Best = 2.161567e+03	FES = 201100	Diversity = 1.9691e+00
Generation 2020:	Best = 2.161567e+03	FES = 202100	Diversity = 1.9498e+00
Generation 2030:	Best = 2.161567e+03	FES = 203100	Diversity = 1.9856e+00
Generation 2040:	Best = 2.161567e+03	FES = 204100	Diversity = 1.9989e+00
Generation 2050:	Best = 2.161567e+03	FES = 205100	Diversity = 1.9731e+00

Generation 2060:	Best = 2.161567e+03	FES = 206100	Diversity = 1.9358e+00
Generation 2070:	Best = 2.161567e+03	FES = 207100	Diversity = 1.9922e+00
Generation 2080:	Best = 2.161567e+03	FES = 208100	Diversity = 1.9934e+00
Generation 2090:	Best = 2.161567e+03	FES = 209100	Diversity = 2.0210e+00
Generation 2100:	Best = 2.161567e+03	FES = 210100	Diversity = 2.0372e+00
Generation 2110:	Best = 2.161567e+03	FES = 211100	Diversity = 2.0257e+00
Generation 2120:	Best = 2.161567e+03	FES = 212100	Diversity = 2.0257e+00
Generation 2130:	Best = 2.161567e+03	FES = 213100	Diversity = 2.0020e+00
Generation 2140:	Best = 2.161567e+03	FES = 214100	Diversity = 2.0092e+00
Generation 2150:	Best = 2.161566e+03	FES = 215100	Diversity = 1.9992e+00
Generation 2160:	Best = 2.161566e+03	FES = 216100	Diversity = 2.0529e+00
Generation 2170:	Best = 2.161566e+03	FES = 217100	Diversity = 2.0901e+00
Generation 2180:	Best = 2.161566e+03	FES = 218100	Diversity = 2.0811e+00
Generation 2190:	Best = 2.161566e+03	FES = 219100	Diversity = 2.1346e+00
Generation 2200:	Best = 2.161566e+03	FES = 220100	Diversity = 2.1317e+00
Generation 2210:	Best = 2.161566e+03	FES = 221100	Diversity = 2.1472e+00
Generation 2220:	Best = 2.161566e+03	FES = 222100	Diversity = 2.1197e+00
Generation 2230:	Best = 2.161566e+03	FES = 223100	Diversity = 2.1359e+00
Generation 2240:	Best = 2.161566e+03	FES = 224100	Diversity = 2.1690e+00
Generation 2250:	Best = 2.161503e+03	FES = 225100	Diversity = 2.1715e+00
Generation 2260:	Best = 2.161501e+03	FES = 226100	Diversity = 2.1755e+00
Generation 2270:	Best = 2.161496e+03	FES = 227100	Diversity = 2.2065e+00
Generation 2280:	Best = 2.161495e+03	FES = 228100	Diversity = 2.2335e+00
Generation 2290:	Best = 2.161495e+03	FES = 229100	Diversity = 2.2306e+00
Generation 2300:	Best = 2.161495e+03	FES = 230100	Diversity = 2.2306e+00
Generation 2310:	Best = 2.161495e+03	FES = 231100	Diversity = 2.2435e+00
Generation 2320:	Best = 2.161495e+03	FES = 232100	Diversity = 2.2793e+00
Generation 2330:	Best = 2.161495e+03	FES = 233100	Diversity = 2.3138e+00
Generation 2340:	Best = 2.161495e+03	FES = 234100	Diversity = 2.2945e+00
Generation 2350:	Best = 2.161429e+03	FES = 235100	Diversity = 2.2913e+00
Generation 2360:	Best = 2.161380e+03	FES = 236100	Diversity = 2.1808e+00
Generation 2370:	Best = 2.161380e+03	FES = 237100	Diversity = 2.2010e+00
Generation 2380:	Best = 2.161371e+03	FES = 238100	Diversity = 2.1394e+00
Generation 2390:	Best = 2.161370e+03	FES = 239100	Diversity = 2.1644e+00
Generation 2400:	Best = 2.161366e+03	FES = 240100	Diversity = 2.0845e+00
Generation 2410:	Best = 2.161362e+03	FES = 241100	Diversity = 1.9908e+00
Generation 2420:	Best = 2.161360e+03	FES = 242100	Diversity = 1.9406e+00
Generation 2430:	Best = 2.161359e+03	FES = 243100	Diversity = 1.9800e+00
Generation 2440:	Best = 2.161358e+03	FES = 244100	Diversity = 1.9888e+00
Generation 2450:	Best = 2.161358e+03	FES = 245100	Diversity = 1.8838e+00
Generation 2460:	Best = 2.161357e+03	FES = 246100	Diversity = 1.8982e+00
Generation 2470:	Best = 2.161357e+03	FES = 247100	Diversity = 1.8943e+00
Generation 2480:	Best = 2.161357e+03	FES = 248100	Diversity = 1.9189e+00
Generation 2490:	Best = 2.161357e+03	FES = 249100	Diversity = 1.9110e+00
Generation 2500:	Best = 2.161328e+03	FES = 250100	Diversity = 1.9273e+00
Generation 2510:	Best = 2.161328e+03	FES = 251100	Diversity = 1.8086e+00
Generation 2520:	Best = 2.161328e+03	FES = 252100	Diversity = 1.8078e+00
Generation 2530:	Best = 2.161328e+03	FES = 253100	Diversity = 1.7987e+00
Generation 2540:	Best = 2.161328e+03	FES = 254100	Diversity = 1.8120e+00
Generation 2550:	Best = 2.161328e+03	FES = 255100	Diversity = 1.8071e+00
Generation 2560:	Best = 2.161328e+03	FES = 256100	Diversity = 1.8157e+00
Generation 2570:	Best = 2.161316e+03	FES = 257100	Diversity = 1.8526e+00
Generation 2580:	Best = 2.161314e+03	FES = 258100	Diversity = 1.9225e+00
Generation 2590:	Best = 2.161314e+03	FES = 259100	Diversity = 1.9880e+00

Generation 2600:	Best = 2.161309e+03	FES = 260100	Diversity = 1.8617e+00
Generation 2610:	Best = 2.161306e+03	FES = 261100	Diversity = 1.9109e+00
Generation 2620:	Best = 2.161304e+03	FES = 262100	Diversity = 1.9163e+00
Generation 2630:	Best = 2.161298e+03	FES = 263100	Diversity = 1.8550e+00
Generation 2640:	Best = 2.161297e+03	FES = 264100	Diversity = 1.9012e+00
Generation 2650:	Best = 2.161297e+03	FES = 265100	Diversity = 1.9446e+00
Generation 2660:	Best = 2.161297e+03	FES = 266100	Diversity = 1.9768e+00
Generation 2670:	Best = 2.161297e+03	FES = 267100	Diversity = 1.9778e+00
Generation 2680:	Best = 2.161297e+03	FES = 268100	Diversity = 1.9897e+00
Generation 2690:	Best = 2.161297e+03	FES = 269100	Diversity = 1.9544e+00
Generation 2700:	Best = 2.161294e+03	FES = 270100	Diversity = 1.9250e+00
Generation 2710:	Best = 2.161293e+03	FES = 271100	Diversity = 1.9214e+00
Generation 2720:	Best = 2.161290e+03	FES = 272100	Diversity = 1.9962e+00
Generation 2730:	Best = 2.161290e+03	FES = 273100	Diversity = 2.0014e+00
Generation 2740:	Best = 2.161290e+03	FES = 274100	Diversity = 2.1106e+00
Generation 2750:	Best = 2.161290e+03	FES = 275100	Diversity = 2.1233e+00
Generation 2760:	Best = 2.161290e+03	FES = 276100	Diversity = 2.1729e+00
Generation 2770:	Best = 2.161290e+03	FES = 277100	Diversity = 1.9460e+00
Generation 2780:	Best = 2.161290e+03	FES = 278100	Diversity = 1.9127e+00
Generation 2790:	Best = 2.161290e+03	FES = 279100	Diversity = 1.9294e+00
Generation 2800:	Best = 2.161290e+03	FES = 280100	Diversity = 1.9073e+00
Generation 2810:	Best = 2.161290e+03	FES = 281100	Diversity = 1.7969e+00
Generation 2820:	Best = 2.161290e+03	FES = 282100	Diversity = 1.7703e+00
Generation 2830:	Best = 2.161290e+03	FES = 283100	Diversity = 1.7295e+00
Generation 2840:	Best = 2.161290e+03	FES = 284100	Diversity = 1.6931e+00
Generation 2850:	Best = 2.161289e+03	FES = 285100	Diversity = 1.7207e+00
Generation 2860:	Best = 2.161289e+03	FES = 286100	Diversity = 1.8205e+00
Generation 2870:	Best = 2.161289e+03	FES = 287100	Diversity = 1.9433e+00
Generation 2880:	Best = 2.161287e+03	FES = 288100	Diversity = 1.9510e+00
Generation 2890:	Best = 2.161287e+03	FES = 289100	Diversity = 2.0120e+00
Generation 2900:	Best = 2.161287e+03	FES = 290100	Diversity = 1.9965e+00
Generation 2910:	Best = 2.161285e+03	FES = 291100	Diversity = 1.9289e+00
Generation 2920:	Best = 2.161285e+03	FES = 292100	Diversity = 1.9002e+00
Generation 2930:	Best = 2.161283e+03	FES = 293100	Diversity = 1.8687e+00
Generation 2940:	Best = 2.161283e+03	FES = 294100	Diversity = 1.8799e+00
Generation 2950:	Best = 2.161281e+03	FES = 295100	Diversity = 1.8162e+00
Generation 2960:	Best = 2.161277e+03	FES = 296100	Diversity = 1.8159e+00
Generation 2970:	Best = 2.161276e+03	FES = 297100	Diversity = 1.7797e+00
Generation 2980:	Best = 2.161276e+03	FES = 298100	Diversity = 1.7687e+00
Generation 2990:	Best = 2.161276e+03	FES = 299100	Diversity = 1.7672e+00

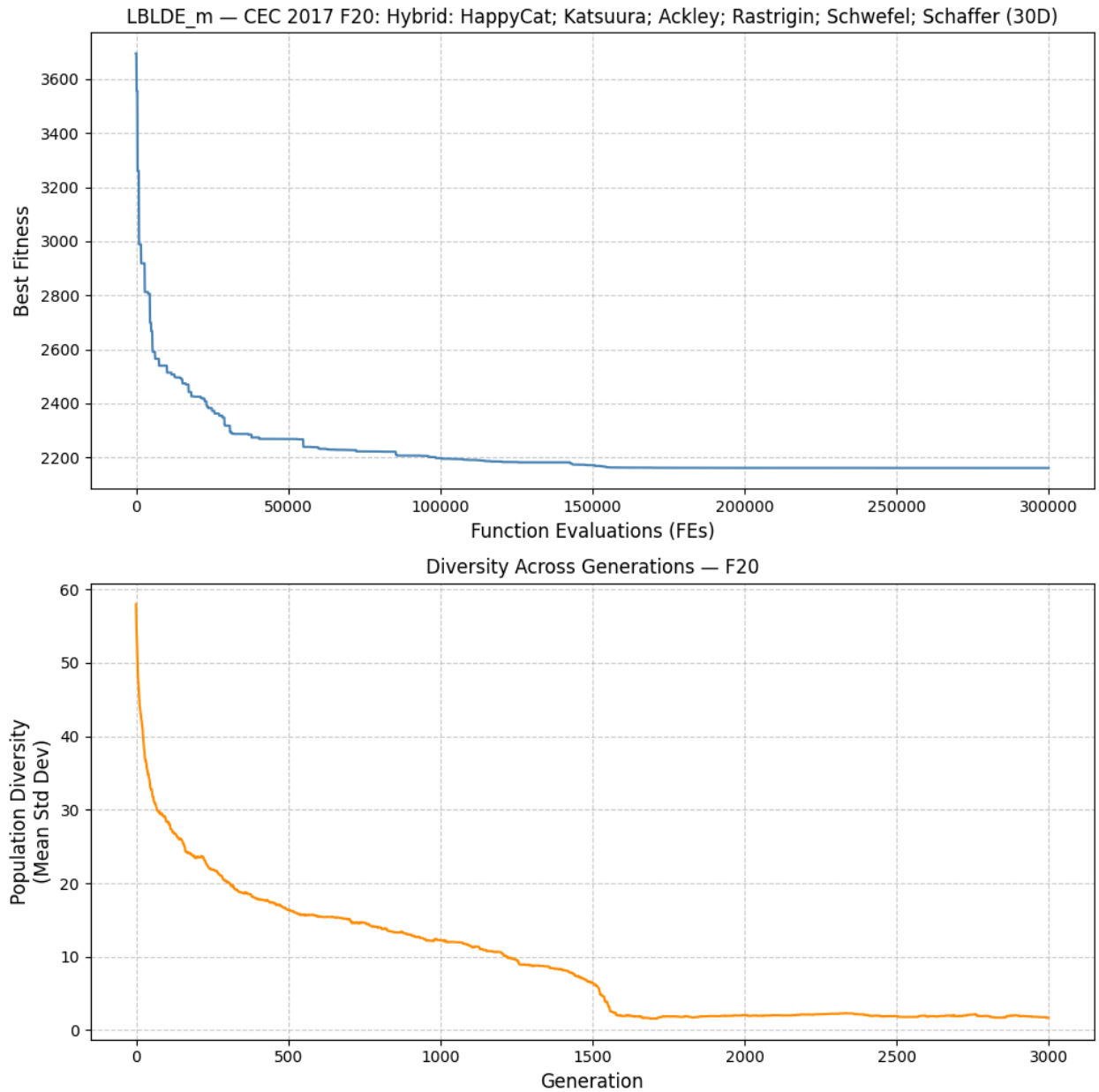
Optimization completed!

Final Best Fitness = 2.161276e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 2.161276e+03
  Global Optimum         : 2.000000e+03
  Error (|best-optimum|) : 1.612756e+02
  Best Solution (5 dim)  : [ 45.09615612 -36.68665957 -15.17633372  18.40799972
35.92435969]
  D^(0) (init. diversity): 5.7976e+01
```

II



Graph saved as convergence_F20.png

6.4 Composite Functions (F21-F30)

```
In [47]: # =====  
# F21 — Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)  
# =====  
run_and_plot('F21', make_wrapper(composition.f21), 2100.0, 'Composition: Rosen
```

=====

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LBLDE_m on CEC 2017 F21 – Single Run

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=

Function : Composition: Rosenbrock; Elliptic; Rastrigin (30D)

Bounds : [-100, 100]

Global optimum : 2100.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 2.946639e+03 | Diversity = 5.7976e+01

Generation	10:	Best = 2.641739e+03	FES = 1100	Diversity = 2.5062e+01
Generation	20:	Best = 2.576726e+03	FES = 2100	Diversity = 1.7605e+01
Generation	30:	Best = 2.541890e+03	FES = 3100	Diversity = 1.5757e+01
Generation	40:	Best = 2.534655e+03	FES = 4100	Diversity = 1.5027e+01
Generation	50:	Best = 2.515429e+03	FES = 5100	Diversity = 1.4356e+01
Generation	60:	Best = 2.509787e+03	FES = 6100	Diversity = 1.3601e+01
Generation	70:	Best = 2.489515e+03	FES = 7100	Diversity = 1.3527e+01
Generation	80:	Best = 2.485548e+03	FES = 8100	Diversity = 1.2991e+01
Generation	90:	Best = 2.475699e+03	FES = 9100	Diversity = 1.2694e+01
Generation	100:	Best = 2.475699e+03	FES = 10100	Diversity = 1.2749e+01
Generation	110:	Best = 2.475699e+03	FES = 11100	Diversity = 1.2595e+01
Generation	120:	Best = 2.461374e+03	FES = 12100	Diversity = 1.2682e+01
Generation	130:	Best = 2.457852e+03	FES = 13100	Diversity = 1.2498e+01
Generation	140:	Best = 2.450167e+03	FES = 14100	Diversity = 1.2287e+01
Generation	150:	Best = 2.450167e+03	FES = 15100	Diversity = 1.2171e+01
Generation	160:	Best = 2.450167e+03	FES = 16100	Diversity = 1.2076e+01
Generation	170:	Best = 2.444237e+03	FES = 17100	Diversity = 1.2032e+01
Generation	180:	Best = 2.444237e+03	FES = 18100	Diversity = 1.1940e+01
Generation	190:	Best = 2.442971e+03	FES = 19100	Diversity = 1.1911e+01
Generation	200:	Best = 2.442971e+03	FES = 20100	Diversity = 1.1840e+01
Generation	210:	Best = 2.442971e+03	FES = 21100	Diversity = 1.1702e+01
Generation	220:	Best = 2.440825e+03	FES = 22100	Diversity = 1.1686e+01
Generation	230:	Best = 2.440825e+03	FES = 23100	Diversity = 1.1684e+01
Generation	240:	Best = 2.440806e+03	FES = 24100	Diversity = 1.1627e+01
Generation	250:	Best = 2.437160e+03	FES = 25100	Diversity = 1.1598e+01
Generation	260:	Best = 2.437160e+03	FES = 26100	Diversity = 1.1583e+01
Generation	270:	Best = 2.431506e+03	FES = 27100	Diversity = 1.1590e+01
Generation	280:	Best = 2.429951e+03	FES = 28100	Diversity = 1.1496e+01
Generation	290:	Best = 2.421519e+03	FES = 29100	Diversity = 1.1463e+01
Generation	300:	Best = 2.421366e+03	FES = 30100	Diversity = 1.1336e+01
Generation	310:	Best = 2.421366e+03	FES = 31100	Diversity = 1.1294e+01
Generation	320:	Best = 2.421277e+03	FES = 32100	Diversity = 1.1258e+01
Generation	330:	Best = 2.421277e+03	FES = 33100	Diversity = 1.1291e+01
Generation	340:	Best = 2.421277e+03	FES = 34100	Diversity = 1.1266e+01
Generation	350:	Best = 2.419487e+03	FES = 35100	Diversity = 1.1295e+01
Generation	360:	Best = 2.419487e+03	FES = 36100	Diversity = 1.1151e+01
Generation	370:	Best = 2.419487e+03	FES = 37100	Diversity = 1.1083e+01
Generation	380:	Best = 2.419466e+03	FES = 38100	Diversity = 1.1073e+01
Generation	390:	Best = 2.419466e+03	FES = 39100	Diversity = 1.1070e+01
Generation	400:	Best = 2.414122e+03	FES = 40100	Diversity = 1.1056e+01
Generation	410:	Best = 2.414122e+03	FES = 41100	Diversity = 1.0982e+01
Generation	420:	Best = 2.401246e+03	FES = 42100	Diversity = 1.0968e+01
Generation	430:	Best = 2.400109e+03	FES = 43100	Diversity = 1.0927e+01
Generation	440:	Best = 2.399717e+03	FES = 44100	Diversity = 1.0756e+01

Generation	450:	Best = 2.399717e+03	FES = 45100	Diversity = 1.0774e+01
Generation	460:	Best = 2.399717e+03	FES = 46100	Diversity = 1.0769e+01
Generation	470:	Best = 2.399274e+03	FES = 47100	Diversity = 1.0778e+01
Generation	480:	Best = 2.399274e+03	FES = 48100	Diversity = 1.0761e+01
Generation	490:	Best = 2.397158e+03	FES = 49100	Diversity = 1.0761e+01
Generation	500:	Best = 2.397001e+03	FES = 50100	Diversity = 1.0638e+01
Generation	510:	Best = 2.397001e+03	FES = 51100	Diversity = 1.0664e+01
Generation	520:	Best = 2.397001e+03	FES = 52100	Diversity = 1.0651e+01
Generation	530:	Best = 2.397001e+03	FES = 53100	Diversity = 1.0601e+01
Generation	540:	Best = 2.397001e+03	FES = 54100	Diversity = 1.0601e+01
Generation	550:	Best = 2.397001e+03	FES = 55100	Diversity = 1.0555e+01
Generation	560:	Best = 2.397001e+03	FES = 56100	Diversity = 1.0524e+01
Generation	570:	Best = 2.396617e+03	FES = 57100	Diversity = 1.0522e+01
Generation	580:	Best = 2.396617e+03	FES = 58100	Diversity = 1.0528e+01
Generation	590:	Best = 2.384069e+03	FES = 59100	Diversity = 1.0481e+01
Generation	600:	Best = 2.384069e+03	FES = 60100	Diversity = 1.0523e+01
Generation	610:	Best = 2.384069e+03	FES = 61100	Diversity = 1.0573e+01
Generation	620:	Best = 2.383762e+03	FES = 62100	Diversity = 1.0564e+01
Generation	630:	Best = 2.383762e+03	FES = 63100	Diversity = 1.0606e+01
Generation	640:	Best = 2.383762e+03	FES = 64100	Diversity = 1.0605e+01
Generation	650:	Best = 2.383762e+03	FES = 65100	Diversity = 1.0598e+01
Generation	660:	Best = 2.383762e+03	FES = 66100	Diversity = 1.0551e+01
Generation	670:	Best = 2.381360e+03	FES = 67100	Diversity = 1.0544e+01
Generation	680:	Best = 2.381360e+03	FES = 68100	Diversity = 1.0505e+01
Generation	690:	Best = 2.381353e+03	FES = 69100	Diversity = 1.0495e+01
Generation	700:	Best = 2.381353e+03	FES = 70100	Diversity = 1.0484e+01
Generation	710:	Best = 2.381353e+03	FES = 71100	Diversity = 1.0483e+01
Generation	720:	Best = 2.381353e+03	FES = 72100	Diversity = 1.0449e+01
Generation	730:	Best = 2.381353e+03	FES = 73100	Diversity = 1.0440e+01
Generation	740:	Best = 2.379928e+03	FES = 74100	Diversity = 1.0439e+01
Generation	750:	Best = 2.379928e+03	FES = 75100	Diversity = 1.0433e+01
Generation	760:	Best = 2.379928e+03	FES = 76100	Diversity = 1.0432e+01
Generation	770:	Best = 2.379717e+03	FES = 77100	Diversity = 1.0378e+01
Generation	780:	Best = 2.379717e+03	FES = 78100	Diversity = 1.0377e+01
Generation	790:	Best = 2.379716e+03	FES = 79100	Diversity = 1.0462e+01
Generation	800:	Best = 2.379707e+03	FES = 80100	Diversity = 1.0463e+01
Generation	810:	Best = 2.379707e+03	FES = 81100	Diversity = 1.0437e+01
Generation	820:	Best = 2.379707e+03	FES = 82100	Diversity = 1.0407e+01
Generation	830:	Best = 2.379707e+03	FES = 83100	Diversity = 1.0327e+01
Generation	840:	Best = 2.379707e+03	FES = 84100	Diversity = 1.0305e+01
Generation	850:	Best = 2.379707e+03	FES = 85100	Diversity = 1.0300e+01
Generation	860:	Best = 2.379216e+03	FES = 86100	Diversity = 1.0289e+01
Generation	870:	Best = 2.379125e+03	FES = 87100	Diversity = 1.0290e+01
Generation	880:	Best = 2.377806e+03	FES = 88100	Diversity = 1.0295e+01
Generation	890:	Best = 2.377806e+03	FES = 89100	Diversity = 1.0287e+01
Generation	900:	Best = 2.375521e+03	FES = 90100	Diversity = 1.0161e+01
Generation	910:	Best = 2.374824e+03	FES = 91100	Diversity = 1.0122e+01
Generation	920:	Best = 2.374824e+03	FES = 92100	Diversity = 1.0123e+01
Generation	930:	Best = 2.374387e+03	FES = 93100	Diversity = 1.0123e+01
Generation	940:	Best = 2.374387e+03	FES = 94100	Diversity = 1.0185e+01
Generation	950:	Best = 2.372770e+03	FES = 95100	Diversity = 1.0180e+01
Generation	960:	Best = 2.372770e+03	FES = 96100	Diversity = 1.0175e+01
Generation	970:	Best = 2.372770e+03	FES = 97100	Diversity = 1.0179e+01
Generation	980:	Best = 2.372272e+03	FES = 98100	Diversity = 1.0177e+01

Generation 990:	Best = 2.369773e+03	FES = 99100	Diversity = 1.0178e+01
Generation 1000:	Best = 2.366177e+03	FES = 100100	Diversity = 1.0180e+01
Generation 1010:	Best = 2.366105e+03	FES = 101100	Diversity = 1.0169e+01
Generation 1020:	Best = 2.366100e+03	FES = 102100	Diversity = 1.0166e+01
Generation 1030:	Best = 2.366100e+03	FES = 103100	Diversity = 1.0176e+01
Generation 1040:	Best = 2.366100e+03	FES = 104100	Diversity = 1.0176e+01
Generation 1050:	Best = 2.366077e+03	FES = 105100	Diversity = 1.0176e+01
Generation 1060:	Best = 2.365936e+03	FES = 106100	Diversity = 1.0106e+01
Generation 1070:	Best = 2.365908e+03	FES = 107100	Diversity = 1.0106e+01
Generation 1080:	Best = 2.365751e+03	FES = 108100	Diversity = 1.0110e+01
Generation 1090:	Best = 2.365750e+03	FES = 109100	Diversity = 1.0075e+01
Generation 1100:	Best = 2.365748e+03	FES = 110100	Diversity = 1.0074e+01
Generation 1110:	Best = 2.365748e+03	FES = 111100	Diversity = 1.0076e+01
Generation 1120:	Best = 2.365748e+03	FES = 112100	Diversity = 1.0118e+01
Generation 1130:	Best = 2.365720e+03	FES = 113100	Diversity = 1.0111e+01
Generation 1140:	Best = 2.365194e+03	FES = 114100	Diversity = 1.0079e+01
Generation 1150:	Best = 2.365182e+03	FES = 115100	Diversity = 1.0095e+01
Generation 1160:	Best = 2.364519e+03	FES = 116100	Diversity = 1.0074e+01
Generation 1170:	Best = 2.364519e+03	FES = 117100	Diversity = 1.0079e+01
Generation 1180:	Best = 2.364511e+03	FES = 118100	Diversity = 1.0080e+01
Generation 1190:	Best = 2.364511e+03	FES = 119100	Diversity = 1.0076e+01
Generation 1200:	Best = 2.364429e+03	FES = 120100	Diversity = 1.0041e+01
Generation 1210:	Best = 2.364429e+03	FES = 121100	Diversity = 1.0019e+01
Generation 1220:	Best = 2.364428e+03	FES = 122100	Diversity = 1.0022e+01
Generation 1230:	Best = 2.363679e+03	FES = 123100	Diversity = 1.0019e+01
Generation 1240:	Best = 2.363584e+03	FES = 124100	Diversity = 9.9920e+00
Generation 1250:	Best = 2.363502e+03	FES = 125100	Diversity = 9.9851e+00
Generation 1260:	Best = 2.363476e+03	FES = 126100	Diversity = 9.9851e+00
Generation 1270:	Best = 2.363476e+03	FES = 127100	Diversity = 1.0005e+01
Generation 1280:	Best = 2.363469e+03	FES = 128100	Diversity = 1.0032e+01
Generation 1290:	Best = 2.363469e+03	FES = 129100	Diversity = 1.0032e+01
Generation 1300:	Best = 2.363469e+03	FES = 130100	Diversity = 1.0078e+01
Generation 1310:	Best = 2.363469e+03	FES = 131100	Diversity = 1.0106e+01
Generation 1320:	Best = 2.363469e+03	FES = 132100	Diversity = 1.0106e+01
Generation 1330:	Best = 2.363469e+03	FES = 133100	Diversity = 1.0105e+01
Generation 1340:	Best = 2.363459e+03	FES = 134100	Diversity = 1.0114e+01
Generation 1350:	Best = 2.363457e+03	FES = 135100	Diversity = 1.0112e+01
Generation 1360:	Best = 2.363457e+03	FES = 136100	Diversity = 1.0113e+01
Generation 1370:	Best = 2.363444e+03	FES = 137100	Diversity = 1.0111e+01
Generation 1380:	Best = 2.363444e+03	FES = 138100	Diversity = 1.0105e+01
Generation 1390:	Best = 2.363444e+03	FES = 139100	Diversity = 1.0108e+01
Generation 1400:	Best = 2.363444e+03	FES = 140100	Diversity = 1.0108e+01
Generation 1410:	Best = 2.363417e+03	FES = 141100	Diversity = 1.0108e+01
Generation 1420:	Best = 2.363415e+03	FES = 142100	Diversity = 1.0108e+01
Generation 1430:	Best = 2.363415e+03	FES = 143100	Diversity = 1.0111e+01
Generation 1440:	Best = 2.363415e+03	FES = 144100	Diversity = 1.0112e+01
Generation 1450:	Best = 2.363415e+03	FES = 145100	Diversity = 1.0112e+01
Generation 1460:	Best = 2.363410e+03	FES = 146100	Diversity = 1.0102e+01
Generation 1470:	Best = 2.363166e+03	FES = 147100	Diversity = 1.0045e+01
Generation 1480:	Best = 2.363166e+03	FES = 148100	Diversity = 1.0044e+01
Generation 1490:	Best = 2.363166e+03	FES = 149100	Diversity = 1.0036e+01
Generation 1500:	Best = 2.363163e+03	FES = 150100	Diversity = 1.0037e+01
Generation 1510:	Best = 2.363163e+03	FES = 151100	Diversity = 1.0028e+01
Generation 1520:	Best = 2.363149e+03	FES = 152100	Diversity = 1.0067e+01

Generation 1530:	Best = 2.363130e+03	FES = 153100	Diversity = 1.0054e+01
Generation 1540:	Best = 2.363056e+03	FES = 154100	Diversity = 1.0052e+01
Generation 1550:	Best = 2.363056e+03	FES = 155100	Diversity = 1.0052e+01
Generation 1560:	Best = 2.363056e+03	FES = 156100	Diversity = 1.0052e+01
Generation 1570:	Best = 2.363056e+03	FES = 157100	Diversity = 1.0053e+01
Generation 1580:	Best = 2.363056e+03	FES = 158100	Diversity = 1.0093e+01
Generation 1590:	Best = 2.362768e+03	FES = 159100	Diversity = 1.0097e+01
Generation 1600:	Best = 2.362768e+03	FES = 160100	Diversity = 1.0119e+01
Generation 1610:	Best = 2.362768e+03	FES = 161100	Diversity = 1.0120e+01
Generation 1620:	Best = 2.362768e+03	FES = 162100	Diversity = 1.0122e+01
Generation 1630:	Best = 2.362723e+03	FES = 163100	Diversity = 1.0106e+01
Generation 1640:	Best = 2.362723e+03	FES = 164100	Diversity = 1.0062e+01
Generation 1650:	Best = 2.362711e+03	FES = 165100	Diversity = 1.0064e+01
Generation 1660:	Best = 2.362705e+03	FES = 166100	Diversity = 1.0077e+01
Generation 1670:	Best = 2.362705e+03	FES = 167100	Diversity = 1.0123e+01
Generation 1680:	Best = 2.362705e+03	FES = 168100	Diversity = 1.0110e+01
Generation 1690:	Best = 2.362699e+03	FES = 169100	Diversity = 1.0112e+01
Generation 1700:	Best = 2.362699e+03	FES = 170100	Diversity = 1.0174e+01
Generation 1710:	Best = 2.362692e+03	FES = 171100	Diversity = 1.0171e+01
Generation 1720:	Best = 2.362692e+03	FES = 172100	Diversity = 1.0168e+01
Generation 1730:	Best = 2.362692e+03	FES = 173100	Diversity = 1.0168e+01
Generation 1740:	Best = 2.362692e+03	FES = 174100	Diversity = 1.0166e+01
Generation 1750:	Best = 2.362692e+03	FES = 175100	Diversity = 1.0094e+01
Generation 1760:	Best = 2.362690e+03	FES = 176100	Diversity = 1.0095e+01
Generation 1770:	Best = 2.360535e+03	FES = 177100	Diversity = 1.0068e+01
Generation 1780:	Best = 2.360535e+03	FES = 178100	Diversity = 1.0069e+01
Generation 1790:	Best = 2.360535e+03	FES = 179100	Diversity = 1.0069e+01
Generation 1800:	Best = 2.360203e+03	FES = 180100	Diversity = 1.0068e+01
Generation 1810:	Best = 2.360203e+03	FES = 181100	Diversity = 1.0077e+01
Generation 1820:	Best = 2.360203e+03	FES = 182100	Diversity = 1.0079e+01
Generation 1830:	Best = 2.360203e+03	FES = 183100	Diversity = 1.0064e+01
Generation 1840:	Best = 2.360203e+03	FES = 184100	Diversity = 1.0064e+01
Generation 1850:	Best = 2.360203e+03	FES = 185100	Diversity = 1.0066e+01
Generation 1860:	Best = 2.360203e+03	FES = 186100	Diversity = 1.0065e+01
Generation 1870:	Best = 2.360200e+03	FES = 187100	Diversity = 1.0066e+01
Generation 1880:	Best = 2.360200e+03	FES = 188100	Diversity = 1.0062e+01
Generation 1890:	Best = 2.360041e+03	FES = 189100	Diversity = 1.0052e+01
Generation 1900:	Best = 2.360041e+03	FES = 190100	Diversity = 1.0057e+01
Generation 1910:	Best = 2.360041e+03	FES = 191100	Diversity = 1.0056e+01
Generation 1920:	Best = 2.358892e+03	FES = 192100	Diversity = 1.0054e+01
Generation 1930:	Best = 2.358892e+03	FES = 193100	Diversity = 1.0054e+01
Generation 1940:	Best = 2.358892e+03	FES = 194100	Diversity = 1.0035e+01
Generation 1950:	Best = 2.358892e+03	FES = 195100	Diversity = 1.0034e+01
Generation 1960:	Best = 2.358892e+03	FES = 196100	Diversity = 1.0024e+01
Generation 1970:	Best = 2.358892e+03	FES = 197100	Diversity = 1.0024e+01
Generation 1980:	Best = 2.358892e+03	FES = 198100	Diversity = 1.0022e+01
Generation 1990:	Best = 2.358882e+03	FES = 199100	Diversity = 1.0022e+01
Generation 2000:	Best = 2.358882e+03	FES = 200100	Diversity = 1.0017e+01
Generation 2010:	Best = 2.358882e+03	FES = 201100	Diversity = 9.9796e+00
Generation 2020:	Best = 2.358882e+03	FES = 202100	Diversity = 9.9838e+00
Generation 2030:	Best = 2.358882e+03	FES = 203100	Diversity = 9.9865e+00
Generation 2040:	Best = 2.358882e+03	FES = 204100	Diversity = 9.9882e+00
Generation 2050:	Best = 2.358882e+03	FES = 205100	Diversity = 9.9866e+00
Generation 2060:	Best = 2.358882e+03	FES = 206100	Diversity = 9.9865e+00

Generation 2070:	Best = 2.358756e+03	FES = 207100	Diversity = 9.9859e+00
Generation 2080:	Best = 2.358756e+03	FES = 208100	Diversity = 1.0007e+01
Generation 2090:	Best = 2.358756e+03	FES = 209100	Diversity = 1.0006e+01
Generation 2100:	Best = 2.358756e+03	FES = 210100	Diversity = 1.0006e+01
Generation 2110:	Best = 2.358756e+03	FES = 211100	Diversity = 1.0005e+01
Generation 2120:	Best = 2.358756e+03	FES = 212100	Diversity = 9.9773e+00
Generation 2130:	Best = 2.358756e+03	FES = 213100	Diversity = 9.9961e+00
Generation 2140:	Best = 2.358756e+03	FES = 214100	Diversity = 9.9964e+00
Generation 2150:	Best = 2.358756e+03	FES = 215100	Diversity = 9.9966e+00
Generation 2160:	Best = 2.358756e+03	FES = 216100	Diversity = 9.9966e+00
Generation 2170:	Best = 2.358756e+03	FES = 217100	Diversity = 9.9796e+00
Generation 2180:	Best = 2.358756e+03	FES = 218100	Diversity = 9.9469e+00
Generation 2190:	Best = 2.358756e+03	FES = 219100	Diversity = 9.9779e+00
Generation 2200:	Best = 2.358756e+03	FES = 220100	Diversity = 9.9786e+00
Generation 2210:	Best = 2.358755e+03	FES = 221100	Diversity = 9.9791e+00
Generation 2220:	Best = 2.358381e+03	FES = 222100	Diversity = 9.9797e+00
Generation 2230:	Best = 2.358351e+03	FES = 223100	Diversity = 9.9782e+00
Generation 2240:	Best = 2.357386e+03	FES = 224100	Diversity = 9.9634e+00
Generation 2250:	Best = 2.355686e+03	FES = 225100	Diversity = 9.9642e+00
Generation 2260:	Best = 2.355657e+03	FES = 226100	Diversity = 9.9658e+00
Generation 2270:	Best = 2.354867e+03	FES = 227100	Diversity = 9.9466e+00
Generation 2280:	Best = 2.354845e+03	FES = 228100	Diversity = 9.9626e+00
Generation 2290:	Best = 2.354544e+03	FES = 229100	Diversity = 9.9551e+00
Generation 2300:	Best = 2.354544e+03	FES = 230100	Diversity = 9.9783e+00
Generation 2310:	Best = 2.354544e+03	FES = 231100	Diversity = 9.9790e+00
Generation 2320:	Best = 2.354525e+03	FES = 232100	Diversity = 9.9813e+00
Generation 2330:	Best = 2.354443e+03	FES = 233100	Diversity = 9.9819e+00
Generation 2340:	Best = 2.354443e+03	FES = 234100	Diversity = 9.9843e+00
Generation 2350:	Best = 2.354250e+03	FES = 235100	Diversity = 9.9848e+00
Generation 2360:	Best = 2.354249e+03	FES = 236100	Diversity = 9.9857e+00
Generation 2370:	Best = 2.354219e+03	FES = 237100	Diversity = 9.9853e+00
Generation 2380:	Best = 2.354185e+03	FES = 238100	Diversity = 9.9909e+00
Generation 2390:	Best = 2.354185e+03	FES = 239100	Diversity = 9.9520e+00
Generation 2400:	Best = 2.354175e+03	FES = 240100	Diversity = 9.9523e+00
Generation 2410:	Best = 2.354175e+03	FES = 241100	Diversity = 9.9510e+00
Generation 2420:	Best = 2.354165e+03	FES = 242100	Diversity = 9.9528e+00
Generation 2430:	Best = 2.354165e+03	FES = 243100	Diversity = 9.9529e+00
Generation 2440:	Best = 2.354164e+03	FES = 244100	Diversity = 9.9501e+00
Generation 2450:	Best = 2.354164e+03	FES = 245100	Diversity = 9.9494e+00
Generation 2460:	Best = 2.354162e+03	FES = 246100	Diversity = 9.9502e+00
Generation 2470:	Best = 2.354162e+03	FES = 247100	Diversity = 9.9497e+00
Generation 2480:	Best = 2.354162e+03	FES = 248100	Diversity = 9.9483e+00
Generation 2490:	Best = 2.354131e+03	FES = 249100	Diversity = 9.9485e+00
Generation 2500:	Best = 2.354107e+03	FES = 250100	Diversity = 9.9487e+00
Generation 2510:	Best = 2.354095e+03	FES = 251100	Diversity = 9.9496e+00
Generation 2520:	Best = 2.354094e+03	FES = 252100	Diversity = 9.9490e+00
Generation 2530:	Best = 2.354094e+03	FES = 253100	Diversity = 9.9488e+00
Generation 2540:	Best = 2.354094e+03	FES = 254100	Diversity = 9.9465e+00
Generation 2550:	Best = 2.354093e+03	FES = 255100	Diversity = 9.9467e+00
Generation 2560:	Best = 2.354087e+03	FES = 256100	Diversity = 9.9463e+00
Generation 2570:	Best = 2.354087e+03	FES = 257100	Diversity = 9.9434e+00
Generation 2580:	Best = 2.354083e+03	FES = 258100	Diversity = 9.9434e+00
Generation 2590:	Best = 2.354082e+03	FES = 259100	Diversity = 9.9439e+00
Generation 2600:	Best = 2.354082e+03	FES = 260100	Diversity = 9.9004e+00

Generation 2610:	Best = 2.354080e+03	FES = 261100	Diversity = 9.8763e+00
Generation 2620:	Best = 2.354080e+03	FES = 262100	Diversity = 9.8764e+00
Generation 2630:	Best = 2.354080e+03	FES = 263100	Diversity = 9.8765e+00
Generation 2640:	Best = 2.353968e+03	FES = 264100	Diversity = 9.8648e+00
Generation 2650:	Best = 2.353968e+03	FES = 265100	Diversity = 9.9124e+00
Generation 2660:	Best = 2.353966e+03	FES = 266100	Diversity = 9.9124e+00
Generation 2670:	Best = 2.353965e+03	FES = 267100	Diversity = 9.9129e+00
Generation 2680:	Best = 2.353965e+03	FES = 268100	Diversity = 9.9181e+00
Generation 2690:	Best = 2.353965e+03	FES = 269100	Diversity = 9.9193e+00
Generation 2700:	Best = 2.353964e+03	FES = 270100	Diversity = 9.8074e+00
Generation 2710:	Best = 2.353964e+03	FES = 271100	Diversity = 9.8046e+00
Generation 2720:	Best = 2.353963e+03	FES = 272100	Diversity = 9.8040e+00
Generation 2730:	Best = 2.353963e+03	FES = 273100	Diversity = 9.8038e+00
Generation 2740:	Best = 2.353962e+03	FES = 274100	Diversity = 9.8311e+00
Generation 2750:	Best = 2.353956e+03	FES = 275100	Diversity = 9.8328e+00
Generation 2760:	Best = 2.353954e+03	FES = 276100	Diversity = 9.8324e+00
Generation 2770:	Best = 2.353952e+03	FES = 277100	Diversity = 9.7843e+00
Generation 2780:	Best = 2.353931e+03	FES = 278100	Diversity = 9.7802e+00
Generation 2790:	Best = 2.353931e+03	FES = 279100	Diversity = 9.7725e+00
Generation 2800:	Best = 2.353931e+03	FES = 280100	Diversity = 9.7743e+00
Generation 2810:	Best = 2.353931e+03	FES = 281100	Diversity = 9.7674e+00
Generation 2820:	Best = 2.353931e+03	FES = 282100	Diversity = 9.7420e+00
Generation 2830:	Best = 2.353931e+03	FES = 283100	Diversity = 9.6559e+00
Generation 2840:	Best = 2.353931e+03	FES = 284100	Diversity = 9.6535e+00
Generation 2850:	Best = 2.353931e+03	FES = 285100	Diversity = 9.6887e+00
Generation 2860:	Best = 2.353931e+03	FES = 286100	Diversity = 9.6696e+00
Generation 2870:	Best = 2.353931e+03	FES = 287100	Diversity = 9.6699e+00
Generation 2880:	Best = 2.353931e+03	FES = 288100	Diversity = 9.6700e+00
Generation 2890:	Best = 2.353931e+03	FES = 289100	Diversity = 9.5539e+00
Generation 2900:	Best = 2.353931e+03	FES = 290100	Diversity = 9.5553e+00
Generation 2910:	Best = 2.353720e+03	FES = 291100	Diversity = 9.4488e+00
Generation 2920:	Best = 2.353587e+03	FES = 292100	Diversity = 9.4491e+00
Generation 2930:	Best = 2.353550e+03	FES = 293100	Diversity = 9.4476e+00
Generation 2940:	Best = 2.353537e+03	FES = 294100	Diversity = 9.4478e+00
Generation 2950:	Best = 2.353532e+03	FES = 295100	Diversity = 9.4453e+00
Generation 2960:	Best = 2.353531e+03	FES = 296100	Diversity = 9.4524e+00
Generation 2970:	Best = 2.353531e+03	FES = 297100	Diversity = 9.4824e+00
Generation 2980:	Best = 2.353531e+03	FES = 298100	Diversity = 9.5016e+00
Generation 2990:	Best = 2.353531e+03	FES = 299100	Diversity = 9.5201e+00

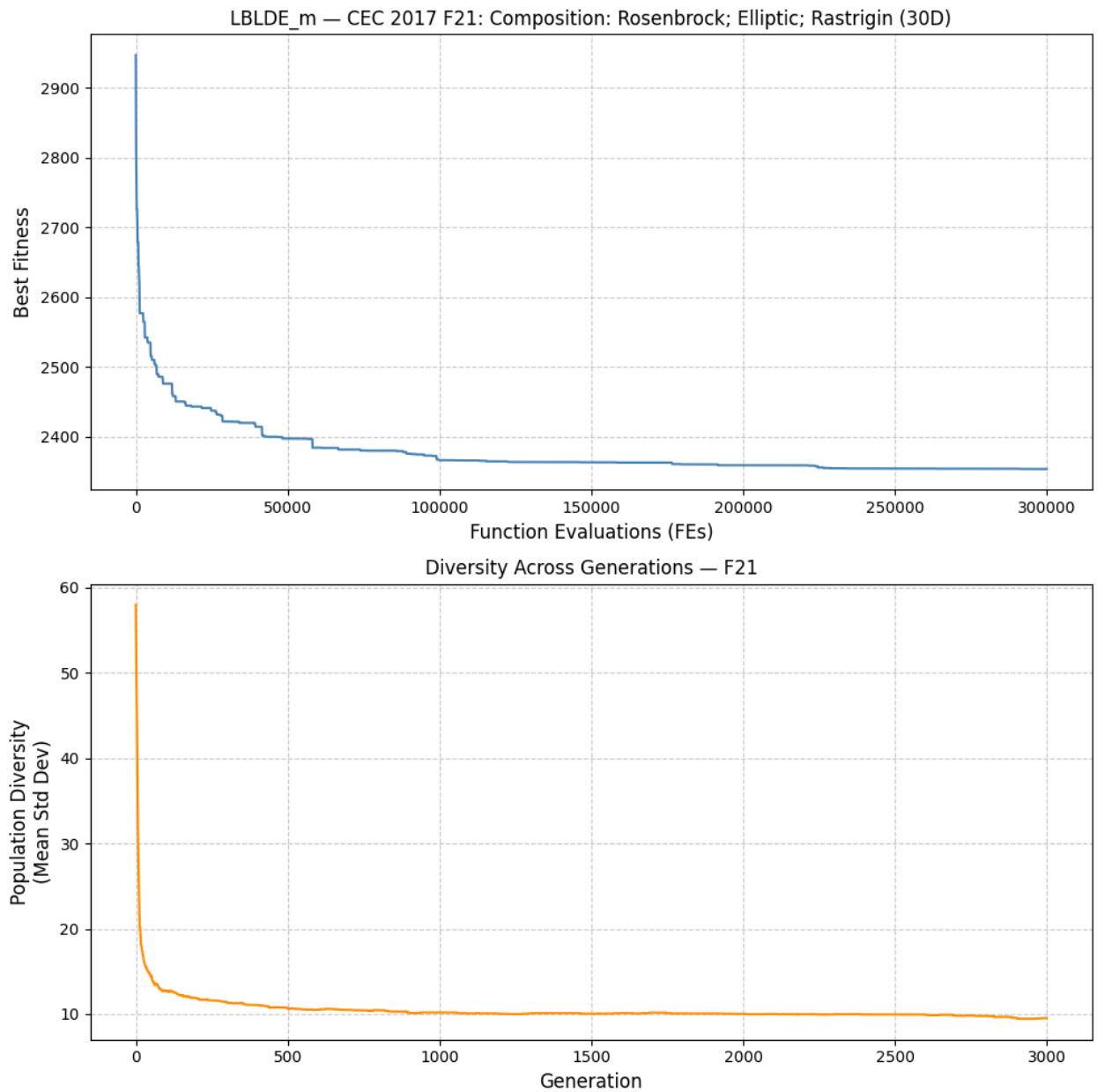
Optimization completed!

Final Best Fitness = 2.353531e+03

Total Function Evaluations = 300000

```
=====
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Results:
  Best Fitness           : 2.353531e+03
  Global Optimum         : 2.100000e+03
  Error (|best-optimum|) : 2.535308e+02
  Best Solution (5 dim)  : [ -5.17926562 -25.08900334 -33.38889536 15.94474178
34.56542983]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F21.png

```
In [48]: # =====  
# F22 – Composition Function 2 (Rastrigin; Griewank; Schwefel)  
# =====  
run_and_plot('F22', make_wrapper(composition.f22), 2200.0, 'Composition: Rastr
```

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LBLDE_m on CEC 2017 F22 – Single Run

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Function : Composition: Rastrigin; Griewank; Schwefel (30D)

Bounds : [-100, 100]

Global optimum : 2200.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 1.151639e+04 | Diversity = 5.7976e+01

Generation	10:	Best = 5.204290e+03	FES = 1100	Diversity = 3.9055e+01
Generation	20:	Best = 3.636578e+03	FES = 2100	Diversity = 8.2276e+00
Generation	30:	Best = 3.076656e+03	FES = 3100	Diversity = 3.7569e+00
Generation	40:	Best = 2.887574e+03	FES = 4100	Diversity = 1.9152e+00
Generation	50:	Best = 2.757439e+03	FES = 5100	Diversity = 1.1377e+00
Generation	60:	Best = 2.715828e+03	FES = 6100	Diversity = 7.2252e-01
Generation	70:	Best = 2.701644e+03	FES = 7100	Diversity = 4.1276e-01
Generation	80:	Best = 2.690515e+03	FES = 8100	Diversity = 4.7196e-01
Generation	90:	Best = 2.674946e+03	FES = 9100	Diversity = 3.8085e-01
Generation	100:	Best = 2.668554e+03	FES = 10100	Diversity = 2.3842e-01
Generation	110:	Best = 2.665516e+03	FES = 11100	Diversity = 1.3589e-01
Generation	120:	Best = 2.663966e+03	FES = 12100	Diversity = 9.3020e-02
Generation	130:	Best = 2.662116e+03	FES = 13100	Diversity = 8.0892e-02
Generation	140:	Best = 2.661410e+03	FES = 14100	Diversity = 5.6853e-02
Generation	150:	Best = 2.660637e+03	FES = 15100	Diversity = 4.7064e-02
Generation	160:	Best = 2.660161e+03	FES = 16100	Diversity = 3.9321e-02
Generation	170:	Best = 2.659751e+03	FES = 17100	Diversity = 3.7445e-02
Generation	180:	Best = 2.659378e+03	FES = 18100	Diversity = 3.1839e-02
Generation	190:	Best = 2.659168e+03	FES = 19100	Diversity = 2.2116e-02
Generation	200:	Best = 2.659049e+03	FES = 20100	Diversity = 1.6652e-02
Generation	210:	Best = 2.658923e+03	FES = 21100	Diversity = 1.7306e-02
Generation	220:	Best = 2.658789e+03	FES = 22100	Diversity = 1.3698e-02
Generation	230:	Best = 2.658749e+03	FES = 23100	Diversity = 1.0618e-02
Generation	240:	Best = 2.658699e+03	FES = 24100	Diversity = 7.7849e-03
Generation	250:	Best = 2.658673e+03	FES = 25100	Diversity = 6.8616e-03
Generation	260:	Best = 2.658645e+03	FES = 26100	Diversity = 6.9375e-03
Generation	270:	Best = 2.658630e+03	FES = 27100	Diversity = 3.6515e-03
Generation	280:	Best = 2.658619e+03	FES = 28100	Diversity = 3.7130e-03
Generation	290:	Best = 2.658606e+03	FES = 29100	Diversity = 3.8624e-03
Generation	300:	Best = 2.658594e+03	FES = 30100	Diversity = 3.2676e-03
Generation	310:	Best = 2.658585e+03	FES = 31100	Diversity = 2.8889e-03
Generation	320:	Best = 2.658582e+03	FES = 32100	Diversity = 2.0418e-03
Generation	330:	Best = 2.658578e+03	FES = 33100	Diversity = 1.7546e-03
Generation	340:	Best = 2.658575e+03	FES = 34100	Diversity = 1.5219e-03
Generation	350:	Best = 2.658573e+03	FES = 35100	Diversity = 1.2193e-03
Generation	360:	Best = 2.658571e+03	FES = 36100	Diversity = 1.1943e-03
Generation	370:	Best = 2.658570e+03	FES = 37100	Diversity = 1.0604e-03
Generation	380:	Best = 2.658569e+03	FES = 38100	Diversity = 9.8937e-04
Generation	390:	Best = 2.658568e+03	FES = 39100	Diversity = 6.7263e-04
Generation	400:	Best = 2.658568e+03	FES = 40100	Diversity = 6.4535e-04
Generation	410:	Best = 2.658567e+03	FES = 41100	Diversity = 5.7147e-04
Generation	420:	Best = 2.658567e+03	FES = 42100	Diversity = 4.5265e-04
Generation	430:	Best = 2.658567e+03	FES = 43100	Diversity = 3.7329e-04
Generation	440:	Best = 2.658567e+03	FES = 44100	Diversity = 2.1145e-04

Generation	450:	Best = 2.658567e+03	FES = 45100	Diversity = 2.0442e-04
Generation	460:	Best = 2.658567e+03	FES = 46100	Diversity = 2.0617e-04
Generation	470:	Best = 2.658567e+03	FES = 47100	Diversity = 1.2324e-04
Generation	480:	Best = 2.658567e+03	FES = 48100	Diversity = 1.1852e-04
Generation	490:	Best = 2.658567e+03	FES = 49100	Diversity = 8.2644e-05
Generation	500:	Best = 2.658567e+03	FES = 50100	Diversity = 7.6920e-05
Generation	510:	Best = 2.658567e+03	FES = 51100	Diversity = 7.2791e-05
Generation	520:	Best = 2.658567e+03	FES = 52100	Diversity = 6.4176e-05
Generation	530:	Best = 2.658567e+03	FES = 53100	Diversity = 6.2812e-05
Generation	540:	Best = 2.658567e+03	FES = 54100	Diversity = 5.0260e-05
Generation	550:	Best = 2.658567e+03	FES = 55100	Diversity = 2.7848e-05
Generation	560:	Best = 2.658567e+03	FES = 56100	Diversity = 3.0612e-05
Generation	570:	Best = 2.658567e+03	FES = 57100	Diversity = 1.8015e-05
Generation	580:	Best = 2.658567e+03	FES = 58100	Diversity = 1.4012e-05
Generation	590:	Best = 2.658567e+03	FES = 59100	Diversity = 1.5019e-05
Generation	600:	Best = 2.658567e+03	FES = 60100	Diversity = 1.1339e-05
Generation	610:	Best = 2.658567e+03	FES = 61100	Diversity = 7.3826e-06
Generation	620:	Best = 2.658567e+03	FES = 62100	Diversity = 8.1209e-06
Generation	630:	Best = 2.658567e+03	FES = 63100	Diversity = 4.9042e-06
Generation	640:	Best = 2.658567e+03	FES = 64100	Diversity = 3.4160e-06
Generation	650:	Best = 2.658567e+03	FES = 65100	Diversity = 3.8024e-06
Generation	660:	Best = 2.658567e+03	FES = 66100	Diversity = 2.7822e-06
Generation	670:	Best = 2.658567e+03	FES = 67100	Diversity = 1.9119e-06
Generation	680:	Best = 2.658567e+03	FES = 68100	Diversity = 1.1023e-06
Generation	690:	Best = 2.658567e+03	FES = 69100	Diversity = 1.1533e-06
Generation	700:	Best = 2.658567e+03	FES = 70100	Diversity = 6.8158e-07
Generation	710:	Best = 2.658567e+03	FES = 71100	Diversity = 4.7662e-07
Generation	720:	Best = 2.658567e+03	FES = 72100	Diversity = 3.2474e-07
Generation	730:	Best = 2.658567e+03	FES = 73100	Diversity = 3.6999e-07
Generation	740:	Best = 2.658567e+03	FES = 74100	Diversity = 3.3275e-07
Generation	750:	Best = 2.658567e+03	FES = 75100	Diversity = 2.8932e-07
Generation	760:	Best = 2.658567e+03	FES = 76100	Diversity = 1.8768e-07
Generation	770:	Best = 2.658567e+03	FES = 77100	Diversity = 9.3084e-08
Generation	780:	Best = 2.658567e+03	FES = 78100	Diversity = 5.2809e-08
Generation	790:	Best = 2.658567e+03	FES = 79100	Diversity = 3.3765e-08
Generation	800:	Best = 2.658567e+03	FES = 80100	Diversity = 3.6875e-08
Generation	810:	Best = 2.658567e+03	FES = 81100	Diversity = 2.0875e-08
Generation	820:	Best = 2.658567e+03	FES = 82100	Diversity = 1.3198e-08
Generation	830:	Best = 2.658567e+03	FES = 83100	Diversity = 1.3904e-08
Generation	840:	Best = 2.658567e+03	FES = 84100	Diversity = 7.7901e-09
Generation	850:	Best = 2.658567e+03	FES = 85100	Diversity = 3.7572e-09
Generation	860:	Best = 2.658567e+03	FES = 86100	Diversity = 1.0759e-09
Generation	870:	Best = 2.658567e+03	FES = 87100	Diversity = 5.2493e-11
Generation	880:	Best = 2.658567e+03	FES = 88100	Diversity = 4.0644e-14
Generation	890:	Best = 2.658567e+03	FES = 89100	Diversity = 4.0451e-14
Generation	900:	Best = 2.658567e+03	FES = 90100	Diversity = 4.0423e-14
Generation	910:	Best = 2.658567e+03	FES = 91100	Diversity = 4.0423e-14
Generation	920:	Best = 2.658567e+03	FES = 92100	Diversity = 4.0423e-14
Generation	930:	Best = 2.658567e+03	FES = 93100	Diversity = 4.0423e-14
Generation	940:	Best = 2.658567e+03	FES = 94100	Diversity = 4.0423e-14
Generation	950:	Best = 2.658567e+03	FES = 95100	Diversity = 4.0423e-14
Generation	960:	Best = 2.658567e+03	FES = 96100	Diversity = 4.0423e-14
Generation	970:	Best = 2.658567e+03	FES = 97100	Diversity = 4.0423e-14
Generation	980:	Best = 2.658567e+03	FES = 98100	Diversity = 4.0423e-14

[illegible]

[illegible]

[illegible]

Generation 2610:	Best = 2.658567e+03	FES = 261100	Diversity = 4.0423e-14
Generation 2620:	Best = 2.658567e+03	FES = 262100	Diversity = 4.0423e-14
Generation 2630:	Best = 2.658567e+03	FES = 263100	Diversity = 4.0423e-14
Generation 2640:	Best = 2.658567e+03	FES = 264100	Diversity = 4.0423e-14
Generation 2650:	Best = 2.658567e+03	FES = 265100	Diversity = 4.0423e-14
Generation 2660:	Best = 2.658567e+03	FES = 266100	Diversity = 4.0423e-14
Generation 2670:	Best = 2.658567e+03	FES = 267100	Diversity = 4.0423e-14
Generation 2680:	Best = 2.658567e+03	FES = 268100	Diversity = 4.0423e-14
Generation 2690:	Best = 2.658567e+03	FES = 269100	Diversity = 4.0423e-14
Generation 2700:	Best = 2.658567e+03	FES = 270100	Diversity = 4.0423e-14
Generation 2710:	Best = 2.658567e+03	FES = 271100	Diversity = 4.0423e-14
Generation 2720:	Best = 2.658567e+03	FES = 272100	Diversity = 4.0423e-14
Generation 2730:	Best = 2.658567e+03	FES = 273100	Diversity = 4.0423e-14
Generation 2740:	Best = 2.658567e+03	FES = 274100	Diversity = 4.0423e-14
Generation 2750:	Best = 2.658567e+03	FES = 275100	Diversity = 4.0423e-14
Generation 2760:	Best = 2.658567e+03	FES = 276100	Diversity = 4.0423e-14
Generation 2770:	Best = 2.658567e+03	FES = 277100	Diversity = 4.0423e-14
Generation 2780:	Best = 2.658567e+03	FES = 278100	Diversity = 4.0423e-14
Generation 2790:	Best = 2.658567e+03	FES = 279100	Diversity = 4.0423e-14
Generation 2800:	Best = 2.658567e+03	FES = 280100	Diversity = 4.0423e-14
Generation 2810:	Best = 2.658567e+03	FES = 281100	Diversity = 4.0423e-14
Generation 2820:	Best = 2.658567e+03	FES = 282100	Diversity = 4.0423e-14
Generation 2830:	Best = 2.658567e+03	FES = 283100	Diversity = 4.0423e-14
Generation 2840:	Best = 2.658567e+03	FES = 284100	Diversity = 4.0423e-14
Generation 2850:	Best = 2.658567e+03	FES = 285100	Diversity = 4.0423e-14
Generation 2860:	Best = 2.658567e+03	FES = 286100	Diversity = 4.0423e-14
Generation 2870:	Best = 2.658567e+03	FES = 287100	Diversity = 4.0423e-14
Generation 2880:	Best = 2.658567e+03	FES = 288100	Diversity = 4.0423e-14
Generation 2890:	Best = 2.658567e+03	FES = 289100	Diversity = 4.0423e-14
Generation 2900:	Best = 2.658567e+03	FES = 290100	Diversity = 4.0423e-14
Generation 2910:	Best = 2.658567e+03	FES = 291100	Diversity = 4.0423e-14
Generation 2920:	Best = 2.658567e+03	FES = 292100	Diversity = 4.0423e-14
Generation 2930:	Best = 2.658567e+03	FES = 293100	Diversity = 4.0423e-14
Generation 2940:	Best = 2.658567e+03	FES = 294100	Diversity = 4.0423e-14
Generation 2950:	Best = 2.658567e+03	FES = 295100	Diversity = 4.0423e-14
Generation 2960:	Best = 2.658567e+03	FES = 296100	Diversity = 4.0423e-14
Generation 2970:	Best = 2.658567e+03	FES = 297100	Diversity = 4.0423e-14
Generation 2980:	Best = 2.658567e+03	FES = 298100	Diversity = 4.0423e-14
Generation 2990:	Best = 2.658567e+03	FES = 299100	Diversity = 4.0423e-14

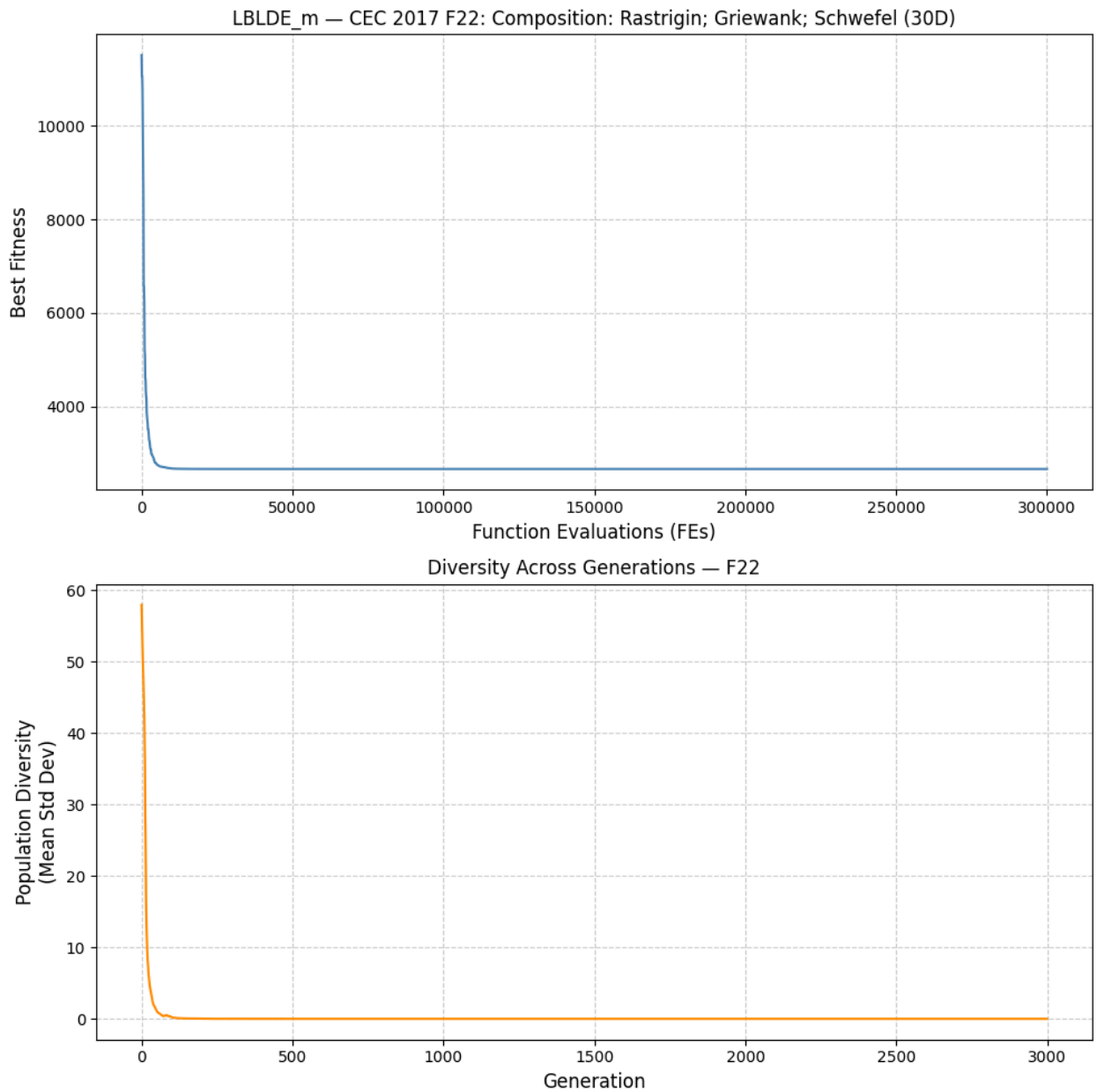
Optimization completed!

Final Best Fitness = 2.658567e+03

Total Function Evaluations = 300000

```
=====
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Results:
  Best Fitness           : 2.658567e+03
  Global Optimum         : 2.200000e+03
  Error (|best-optimum|) : 4.585665e+02
  Best Solution (5 dim)  : [-48.77549064   1.52274714  24.81810481  34.34543449
46.79301546]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F22.png

```
In [49]: # =====  
# F23 — Composition Function 3 (Rosenbrock; Ackley; Schwefel; Rastrigin)  
# =====  
run_and_plot('F23', make_wrapper(composition.f23), 2300.0, 'Composition: Rosen
```

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LBLDE_m on CEC 2017 F23 – Single Run

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Function : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin (30D)
Bounds : [-100, 100]
Global optimum : 2300.0
Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 4.071208e+03 | Diversity = 5.7976e+01

Generation	10:	Best = 3.272313e+03	FES = 1100	Diversity = 3.1946e+01
Generation	20:	Best = 3.020540e+03	FES = 2100	Diversity = 1.7362e+01
Generation	30:	Best = 2.964741e+03	FES = 3100	Diversity = 1.4359e+01
Generation	40:	Best = 2.955529e+03	FES = 4100	Diversity = 1.2536e+01
Generation	50:	Best = 2.901737e+03	FES = 5100	Diversity = 1.0873e+01
Generation	60:	Best = 2.878661e+03	FES = 6100	Diversity = 1.0072e+01
Generation	70:	Best = 2.868192e+03	FES = 7100	Diversity = 8.9005e+00
Generation	80:	Best = 2.843787e+03	FES = 8100	Diversity = 8.2708e+00
Generation	90:	Best = 2.806377e+03	FES = 9100	Diversity = 6.3883e+00
Generation	100:	Best = 2.803818e+03	FES = 10100	Diversity = 3.0619e+00
Generation	110:	Best = 2.781808e+03	FES = 11100	Diversity = 1.9982e+00
Generation	120:	Best = 2.766610e+03	FES = 12100	Diversity = 1.5946e+00
Generation	130:	Best = 2.755489e+03	FES = 13100	Diversity = 1.1061e+00
Generation	140:	Best = 2.752126e+03	FES = 14100	Diversity = 2.6080e-01
Generation	150:	Best = 2.749925e+03	FES = 15100	Diversity = 1.6231e-01
Generation	160:	Best = 2.748527e+03	FES = 16100	Diversity = 1.3942e-01
Generation	170:	Best = 2.747243e+03	FES = 17100	Diversity = 1.1719e-01
Generation	180:	Best = 2.745899e+03	FES = 18100	Diversity = 9.7049e-02
Generation	190:	Best = 2.745202e+03	FES = 19100	Diversity = 6.9733e-02
Generation	200:	Best = 2.744870e+03	FES = 20100	Diversity = 5.3366e-02
Generation	210:	Best = 2.744516e+03	FES = 21100	Diversity = 4.0465e-02
Generation	220:	Best = 2.744335e+03	FES = 22100	Diversity = 3.1682e-02
Generation	230:	Best = 2.744184e+03	FES = 23100	Diversity = 2.4398e-02
Generation	240:	Best = 2.744111e+03	FES = 24100	Diversity = 1.6080e-02
Generation	250:	Best = 2.744023e+03	FES = 25100	Diversity = 1.5249e-02
Generation	260:	Best = 2.743986e+03	FES = 26100	Diversity = 1.1107e-02
Generation	270:	Best = 2.743952e+03	FES = 27100	Diversity = 1.0213e-02
Generation	280:	Best = 2.743921e+03	FES = 28100	Diversity = 8.2219e-03
Generation	290:	Best = 2.743897e+03	FES = 29100	Diversity = 6.8887e-03
Generation	300:	Best = 2.743886e+03	FES = 30100	Diversity = 4.9756e-03
Generation	310:	Best = 2.743875e+03	FES = 31100	Diversity = 4.0694e-03
Generation	320:	Best = 2.743864e+03	FES = 32100	Diversity = 3.7575e-03
Generation	330:	Best = 2.743855e+03	FES = 33100	Diversity = 4.1024e-03
Generation	340:	Best = 2.743848e+03	FES = 34100	Diversity = 3.4446e-03
Generation	350:	Best = 2.743842e+03	FES = 35100	Diversity = 3.1830e-03
Generation	360:	Best = 2.743837e+03	FES = 36100	Diversity = 2.9741e-03
Generation	370:	Best = 2.743832e+03	FES = 37100	Diversity = 2.4117e-03
Generation	380:	Best = 2.743829e+03	FES = 38100	Diversity = 2.4782e-03
Generation	390:	Best = 2.743826e+03	FES = 39100	Diversity = 1.9170e-03
Generation	400:	Best = 2.743823e+03	FES = 40100	Diversity = 1.9181e-03
Generation	410:	Best = 2.743821e+03	FES = 41100	Diversity = 1.7159e-03
Generation	420:	Best = 2.743819e+03	FES = 42100	Diversity = 1.4498e-03
Generation	430:	Best = 2.743818e+03	FES = 43100	Diversity = 1.1114e-03
Generation	440:	Best = 2.743817e+03	FES = 44100	Diversity = 1.0712e-03

Generation	450:	Best = 2.743815e+03	FES = 45100	Diversity = 9.6876e-04
Generation	460:	Best = 2.743814e+03	FES = 46100	Diversity = 1.0008e-03
Generation	470:	Best = 2.743813e+03	FES = 47100	Diversity = 1.0501e-03
Generation	480:	Best = 2.743812e+03	FES = 48100	Diversity = 8.3442e-04
Generation	490:	Best = 2.743812e+03	FES = 49100	Diversity = 8.0810e-04
Generation	500:	Best = 2.743811e+03	FES = 50100	Diversity = 5.7580e-04
Generation	510:	Best = 2.743811e+03	FES = 51100	Diversity = 4.5634e-04
Generation	520:	Best = 2.743811e+03	FES = 52100	Diversity = 4.9305e-04
Generation	530:	Best = 2.743811e+03	FES = 53100	Diversity = 3.4394e-04
Generation	540:	Best = 2.743810e+03	FES = 54100	Diversity = 3.2388e-04
Generation	550:	Best = 2.743810e+03	FES = 55100	Diversity = 2.1400e-04
Generation	560:	Best = 2.743810e+03	FES = 56100	Diversity = 2.0569e-04
Generation	570:	Best = 2.743810e+03	FES = 57100	Diversity = 1.4503e-04
Generation	580:	Best = 2.743810e+03	FES = 58100	Diversity = 1.3070e-04
Generation	590:	Best = 2.743810e+03	FES = 59100	Diversity = 7.6128e-05
Generation	600:	Best = 2.743810e+03	FES = 60100	Diversity = 7.1563e-05
Generation	610:	Best = 2.743810e+03	FES = 61100	Diversity = 6.0927e-05
Generation	620:	Best = 2.743810e+03	FES = 62100	Diversity = 4.6263e-05
Generation	630:	Best = 2.743810e+03	FES = 63100	Diversity = 4.7888e-05
Generation	640:	Best = 2.743810e+03	FES = 64100	Diversity = 4.2078e-05
Generation	650:	Best = 2.743810e+03	FES = 65100	Diversity = 4.3333e-05
Generation	660:	Best = 2.743810e+03	FES = 66100	Diversity = 3.0696e-05
Generation	670:	Best = 2.743810e+03	FES = 67100	Diversity = 1.9093e-05
Generation	680:	Best = 2.743810e+03	FES = 68100	Diversity = 2.0951e-05
Generation	690:	Best = 2.743810e+03	FES = 69100	Diversity = 1.8129e-05
Generation	700:	Best = 2.743810e+03	FES = 70100	Diversity = 1.0171e-05
Generation	710:	Best = 2.743810e+03	FES = 71100	Diversity = 1.0591e-05
Generation	720:	Best = 2.743810e+03	FES = 72100	Diversity = 1.0553e-05
Generation	730:	Best = 2.743810e+03	FES = 73100	Diversity = 6.4821e-06
Generation	740:	Best = 2.743810e+03	FES = 74100	Diversity = 5.6043e-06
Generation	750:	Best = 2.743810e+03	FES = 75100	Diversity = 4.2421e-06
Generation	760:	Best = 2.743810e+03	FES = 76100	Diversity = 2.3152e-06
Generation	770:	Best = 2.743810e+03	FES = 77100	Diversity = 2.1363e-06
Generation	780:	Best = 2.743810e+03	FES = 78100	Diversity = 1.8208e-06
Generation	790:	Best = 2.743810e+03	FES = 79100	Diversity = 1.6447e-06
Generation	800:	Best = 2.743810e+03	FES = 80100	Diversity = 1.4981e-06
Generation	810:	Best = 2.743810e+03	FES = 81100	Diversity = 8.8463e-07
Generation	820:	Best = 2.743810e+03	FES = 82100	Diversity = 9.3593e-07
Generation	830:	Best = 2.743810e+03	FES = 83100	Diversity = 7.0033e-07
Generation	840:	Best = 2.743810e+03	FES = 84100	Diversity = 4.8677e-07
Generation	850:	Best = 2.743810e+03	FES = 85100	Diversity = 3.6787e-07
Generation	860:	Best = 2.743810e+03	FES = 86100	Diversity = 2.5396e-07
Generation	870:	Best = 2.743810e+03	FES = 87100	Diversity = 3.0082e-07
Generation	880:	Best = 2.743810e+03	FES = 88100	Diversity = 1.9420e-07
Generation	890:	Best = 2.743810e+03	FES = 89100	Diversity = 1.1916e-07
Generation	900:	Best = 2.743810e+03	FES = 90100	Diversity = 1.0934e-07
Generation	910:	Best = 2.743810e+03	FES = 91100	Diversity = 7.1066e-08
Generation	920:	Best = 2.743810e+03	FES = 92100	Diversity = 4.1025e-08
Generation	930:	Best = 2.743810e+03	FES = 93100	Diversity = 4.3670e-08
Generation	940:	Best = 2.743810e+03	FES = 94100	Diversity = 2.1469e-08
Generation	950:	Best = 2.743810e+03	FES = 95100	Diversity = 7.3900e-09
Generation	960:	Best = 2.743810e+03	FES = 96100	Diversity = 7.9978e-11
Generation	970:	Best = 2.743810e+03	FES = 97100	Diversity = 1.6668e-12
Generation	980:	Best = 2.743810e+03	FES = 98100	Diversity = 7.6180e-14

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Generation 2610:	Best = 2.743810e+03	FES = 261100	Diversity = 5.9752e-14
Generation 2620:	Best = 2.743810e+03	FES = 262100	Diversity = 5.9752e-14
Generation 2630:	Best = 2.743810e+03	FES = 263100	Diversity = 5.9752e-14
Generation 2640:	Best = 2.743810e+03	FES = 264100	Diversity = 5.9752e-14
Generation 2650:	Best = 2.743810e+03	FES = 265100	Diversity = 5.9752e-14
Generation 2660:	Best = 2.743810e+03	FES = 266100	Diversity = 5.9752e-14
Generation 2670:	Best = 2.743810e+03	FES = 267100	Diversity = 5.9752e-14
Generation 2680:	Best = 2.743810e+03	FES = 268100	Diversity = 5.9752e-14
Generation 2690:	Best = 2.743810e+03	FES = 269100	Diversity = 5.9752e-14
Generation 2700:	Best = 2.743810e+03	FES = 270100	Diversity = 5.9752e-14
Generation 2710:	Best = 2.743810e+03	FES = 271100	Diversity = 5.9752e-14
Generation 2720:	Best = 2.743810e+03	FES = 272100	Diversity = 5.9752e-14
Generation 2730:	Best = 2.743810e+03	FES = 273100	Diversity = 5.9752e-14
Generation 2740:	Best = 2.743810e+03	FES = 274100	Diversity = 5.9752e-14
Generation 2750:	Best = 2.743810e+03	FES = 275100	Diversity = 5.9752e-14
Generation 2760:	Best = 2.743810e+03	FES = 276100	Diversity = 5.9752e-14
Generation 2770:	Best = 2.743810e+03	FES = 277100	Diversity = 5.9752e-14
Generation 2780:	Best = 2.743810e+03	FES = 278100	Diversity = 5.9752e-14
Generation 2790:	Best = 2.743810e+03	FES = 279100	Diversity = 5.9752e-14
Generation 2800:	Best = 2.743810e+03	FES = 280100	Diversity = 5.9752e-14
Generation 2810:	Best = 2.743810e+03	FES = 281100	Diversity = 5.9752e-14
Generation 2820:	Best = 2.743810e+03	FES = 282100	Diversity = 5.9752e-14
Generation 2830:	Best = 2.743810e+03	FES = 283100	Diversity = 5.9752e-14
Generation 2840:	Best = 2.743810e+03	FES = 284100	Diversity = 5.9752e-14
Generation 2850:	Best = 2.743810e+03	FES = 285100	Diversity = 5.9752e-14
Generation 2860:	Best = 2.743810e+03	FES = 286100	Diversity = 5.9752e-14
Generation 2870:	Best = 2.743810e+03	FES = 287100	Diversity = 5.9752e-14
Generation 2880:	Best = 2.743810e+03	FES = 288100	Diversity = 5.9752e-14
Generation 2890:	Best = 2.743810e+03	FES = 289100	Diversity = 5.9752e-14
Generation 2900:	Best = 2.743810e+03	FES = 290100	Diversity = 5.9752e-14
Generation 2910:	Best = 2.743810e+03	FES = 291100	Diversity = 5.9752e-14
Generation 2920:	Best = 2.743810e+03	FES = 292100	Diversity = 5.9752e-14
Generation 2930:	Best = 2.743810e+03	FES = 293100	Diversity = 5.9752e-14
Generation 2940:	Best = 2.743810e+03	FES = 294100	Diversity = 5.9752e-14
Generation 2950:	Best = 2.743810e+03	FES = 295100	Diversity = 5.9752e-14
Generation 2960:	Best = 2.743810e+03	FES = 296100	Diversity = 5.9752e-14
Generation 2970:	Best = 2.743810e+03	FES = 297100	Diversity = 5.9752e-14
Generation 2980:	Best = 2.743810e+03	FES = 298100	Diversity = 5.9752e-14
Generation 2990:	Best = 2.743810e+03	FES = 299100	Diversity = 5.9752e-14

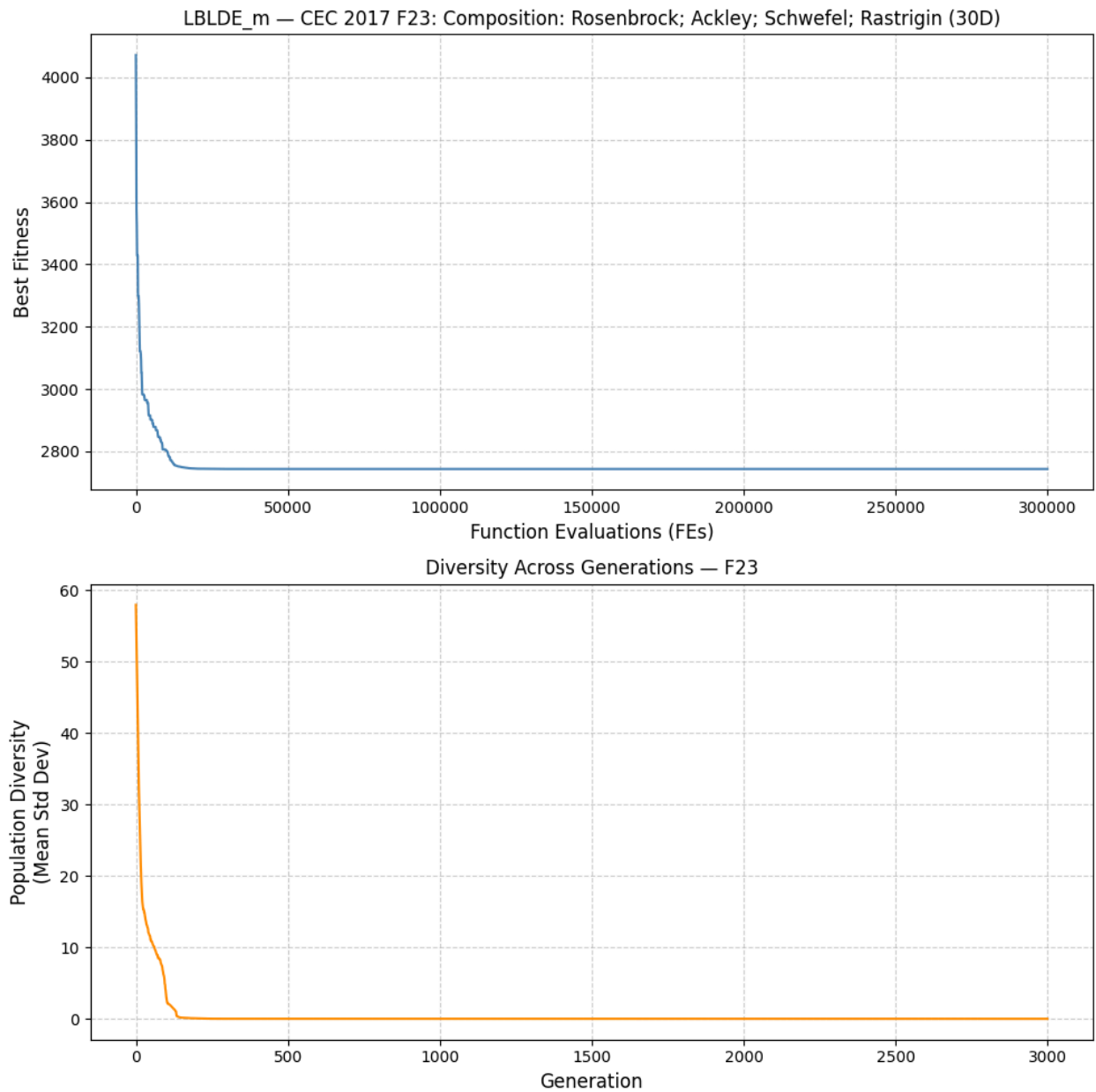
Optimization completed!

Final Best Fitness = 2.743810e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 2.743810e+03
  Global Optimum         : 2.300000e+03
  Error (|best-optimum|) : 4.438101e+02
  Best Solution (5 dim)  : [-63.95988933 -81.02027033 -70.26941227  34.64144197
-80.31982044]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F23.png

```
In [50]: # =====  
# F24 — Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)  
# =====  
run_and_plot('F24', make_wrapper(composition.f24), 2400.0, 'Composition: Ackle
```

=====

=

LBLDE_m on CEC 2017 F24 – Single Run

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=

Function : Composition: Ackley; Elliptic; Griewank; Rastrigin (30D)
Bounds : [-100, 100]
Global optimum : 2400.0
Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 3.931320e+03 | Diversity = 5.7976e+01

Generation 10:	Best = 3.352139e+03	FES = 1100	Diversity = 2.6901e+01
Generation 20:	Best = 3.239145e+03	FES = 2100	Diversity = 1.3783e+01
Generation 30:	Best = 3.129845e+03	FES = 3100	Diversity = 1.0291e+01
Generation 40:	Best = 3.104230e+03	FES = 4100	Diversity = 8.6682e+00
Generation 50:	Best = 3.083423e+03	FES = 5100	Diversity = 7.6522e+00
Generation 60:	Best = 3.038402e+03	FES = 6100	Diversity = 6.8076e+00
Generation 70:	Best = 3.003582e+03	FES = 7100	Diversity = 5.9151e+00
Generation 80:	Best = 2.973573e+03	FES = 8100	Diversity = 4.2999e+00
Generation 90:	Best = 2.952112e+03	FES = 9100	Diversity = 1.3141e+00
Generation 100:	Best = 2.941689e+03	FES = 10100	Diversity = 4.8740e-01
Generation 110:	Best = 2.936410e+03	FES = 11100	Diversity = 2.8247e-01
Generation 120:	Best = 2.933147e+03	FES = 12100	Diversity = 1.9038e-01
Generation 130:	Best = 2.930095e+03	FES = 13100	Diversity = 1.3673e-01
Generation 140:	Best = 2.929030e+03	FES = 14100	Diversity = 9.9262e-02
Generation 150:	Best = 2.928376e+03	FES = 15100	Diversity = 6.4842e-02
Generation 160:	Best = 2.927860e+03	FES = 16100	Diversity = 4.8786e-02
Generation 170:	Best = 2.927443e+03	FES = 17100	Diversity = 4.2901e-02
Generation 180:	Best = 2.927176e+03	FES = 18100	Diversity = 3.8820e-02
Generation 190:	Best = 2.926903e+03	FES = 19100	Diversity = 3.2295e-02
Generation 200:	Best = 2.926768e+03	FES = 20100	Diversity = 2.4849e-02
Generation 210:	Best = 2.926667e+03	FES = 21100	Diversity = 2.0523e-02
Generation 220:	Best = 2.926580e+03	FES = 22100	Diversity = 1.6174e-02
Generation 230:	Best = 2.926517e+03	FES = 23100	Diversity = 1.4134e-02
Generation 240:	Best = 2.926466e+03	FES = 24100	Diversity = 1.0562e-02
Generation 250:	Best = 2.926419e+03	FES = 25100	Diversity = 1.2295e-02
Generation 260:	Best = 2.926363e+03	FES = 26100	Diversity = 1.0867e-02
Generation 270:	Best = 2.926340e+03	FES = 27100	Diversity = 8.7457e-03
Generation 280:	Best = 2.926293e+03	FES = 28100	Diversity = 1.0322e-02
Generation 290:	Best = 2.926272e+03	FES = 29100	Diversity = 6.6375e-03
Generation 300:	Best = 2.926264e+03	FES = 30100	Diversity = 4.1935e-03
Generation 310:	Best = 2.926254e+03	FES = 31100	Diversity = 3.8955e-03
Generation 320:	Best = 2.926248e+03	FES = 32100	Diversity = 3.0544e-03
Generation 330:	Best = 2.926243e+03	FES = 33100	Diversity = 2.9308e-03
Generation 340:	Best = 2.926236e+03	FES = 34100	Diversity = 2.7340e-03
Generation 350:	Best = 2.926231e+03	FES = 35100	Diversity = 2.9948e-03
Generation 360:	Best = 2.926227e+03	FES = 36100	Diversity = 2.9017e-03
Generation 370:	Best = 2.926225e+03	FES = 37100	Diversity = 2.2425e-03
Generation 380:	Best = 2.926223e+03	FES = 38100	Diversity = 1.7068e-03
Generation 390:	Best = 2.926222e+03	FES = 39100	Diversity = 8.6220e-04
Generation 400:	Best = 2.926221e+03	FES = 40100	Diversity = 9.1620e-04
Generation 410:	Best = 2.926220e+03	FES = 41100	Diversity = 7.6636e-04
Generation 420:	Best = 2.926220e+03	FES = 42100	Diversity = 5.5222e-04
Generation 430:	Best = 2.926220e+03	FES = 43100	Diversity = 4.0996e-04
Generation 440:	Best = 2.926219e+03	FES = 44100	Diversity = 3.0441e-04

Generation	450:	Best = 2.926219e+03	FES = 45100	Diversity = 2.9638e-04
Generation	460:	Best = 2.926219e+03	FES = 46100	Diversity = 3.0719e-04
Generation	470:	Best = 2.926219e+03	FES = 47100	Diversity = 2.9323e-04
Generation	480:	Best = 2.926219e+03	FES = 48100	Diversity = 2.4404e-04
Generation	490:	Best = 2.926219e+03	FES = 49100	Diversity = 1.7641e-04
Generation	500:	Best = 2.926219e+03	FES = 50100	Diversity = 1.2050e-04
Generation	510:	Best = 2.926219e+03	FES = 51100	Diversity = 1.2943e-04
Generation	520:	Best = 2.926219e+03	FES = 52100	Diversity = 9.7688e-05
Generation	530:	Best = 2.926219e+03	FES = 53100	Diversity = 1.0268e-04
Generation	540:	Best = 2.926219e+03	FES = 54100	Diversity = 7.9925e-05
Generation	550:	Best = 2.926219e+03	FES = 55100	Diversity = 6.1004e-05
Generation	560:	Best = 2.926219e+03	FES = 56100	Diversity = 5.5493e-05
Generation	570:	Best = 2.926218e+03	FES = 57100	Diversity = 4.6863e-05
Generation	580:	Best = 2.926218e+03	FES = 58100	Diversity = 3.2096e-05
Generation	590:	Best = 2.926218e+03	FES = 59100	Diversity = 2.2063e-05
Generation	600:	Best = 2.926218e+03	FES = 60100	Diversity = 2.4647e-05
Generation	610:	Best = 2.926218e+03	FES = 61100	Diversity = 2.5570e-05
Generation	620:	Best = 2.926218e+03	FES = 62100	Diversity = 2.0751e-05
Generation	630:	Best = 2.926218e+03	FES = 63100	Diversity = 2.2835e-05
Generation	640:	Best = 2.926218e+03	FES = 64100	Diversity = 1.3722e-05
Generation	650:	Best = 2.926218e+03	FES = 65100	Diversity = 1.1483e-05
Generation	660:	Best = 2.926218e+03	FES = 66100	Diversity = 7.9045e-06
Generation	670:	Best = 2.926218e+03	FES = 67100	Diversity = 6.0954e-06
Generation	680:	Best = 2.926218e+03	FES = 68100	Diversity = 3.8138e-06
Generation	690:	Best = 2.926218e+03	FES = 69100	Diversity = 3.7818e-06
Generation	700:	Best = 2.926218e+03	FES = 70100	Diversity = 2.6463e-06
Generation	710:	Best = 2.926218e+03	FES = 71100	Diversity = 2.6465e-06
Generation	720:	Best = 2.926218e+03	FES = 72100	Diversity = 2.0265e-06
Generation	730:	Best = 2.926218e+03	FES = 73100	Diversity = 2.0226e-06
Generation	740:	Best = 2.926218e+03	FES = 74100	Diversity = 1.3060e-06
Generation	750:	Best = 2.926218e+03	FES = 75100	Diversity = 1.1964e-06
Generation	760:	Best = 2.926218e+03	FES = 76100	Diversity = 1.3064e-06
Generation	770:	Best = 2.926218e+03	FES = 77100	Diversity = 8.4801e-07
Generation	780:	Best = 2.926218e+03	FES = 78100	Diversity = 7.1008e-07
Generation	790:	Best = 2.926218e+03	FES = 79100	Diversity = 5.6479e-07
Generation	800:	Best = 2.926218e+03	FES = 80100	Diversity = 3.6154e-07
Generation	810:	Best = 2.926218e+03	FES = 81100	Diversity = 3.0611e-07
Generation	820:	Best = 2.926218e+03	FES = 82100	Diversity = 2.3173e-07
Generation	830:	Best = 2.926218e+03	FES = 83100	Diversity = 1.5669e-07
Generation	840:	Best = 2.926218e+03	FES = 84100	Diversity = 1.1270e-07
Generation	850:	Best = 2.926218e+03	FES = 85100	Diversity = 9.3232e-08
Generation	860:	Best = 2.926218e+03	FES = 86100	Diversity = 7.7942e-08
Generation	870:	Best = 2.926218e+03	FES = 87100	Diversity = 7.4856e-08
Generation	880:	Best = 2.926218e+03	FES = 88100	Diversity = 4.2440e-08
Generation	890:	Best = 2.926218e+03	FES = 89100	Diversity = 4.1703e-08
Generation	900:	Best = 2.926218e+03	FES = 90100	Diversity = 2.8916e-08
Generation	910:	Best = 2.926218e+03	FES = 91100	Diversity = 1.9170e-08
Generation	920:	Best = 2.926218e+03	FES = 92100	Diversity = 1.4140e-08
Generation	930:	Best = 2.926218e+03	FES = 93100	Diversity = 4.6524e-09
Generation	940:	Best = 2.926218e+03	FES = 94100	Diversity = 2.9981e-09
Generation	950:	Best = 2.926218e+03	FES = 95100	Diversity = 4.1768e-09
Generation	960:	Best = 2.926218e+03	FES = 96100	Diversity = 5.5151e-09
Generation	970:	Best = 2.926218e+03	FES = 97100	Diversity = 4.8193e-09
Generation	980:	Best = 2.926218e+03	FES = 98100	Diversity = 4.7133e-09

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[illegible]

[illegible]

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Generation 2610: Best = 2.926218e+03 FES = 261100 Diversity = 5.7865e-14
Generation 2620: Best = 2.926218e+03 FES = 262100 Diversity = 5.7865e-14
Generation 2630: Best = 2.926218e+03 FES = 263100 Diversity = 5.7865e-14
Generation 2640: Best = 2.926218e+03 FES = 264100 Diversity = 5.7865e-14
Generation 2650: Best = 2.926218e+03 FES = 265100 Diversity = 5.7865e-14
Generation 2660: Best = 2.926218e+03 FES = 266100 Diversity = 5.7865e-14
Generation 2670: Best = 2.926218e+03 FES = 267100 Diversity = 5.7865e-14
Generation 2680: Best = 2.926218e+03 FES = 268100 Diversity = 5.7865e-14
Generation 2690: Best = 2.926218e+03 FES = 269100 Diversity = 5.7865e-14
Generation 2700: Best = 2.926218e+03 FES = 270100 Diversity = 5.7865e-14
Generation 2710: Best = 2.926218e+03 FES = 271100 Diversity = 5.7865e-14
Generation 2720: Best = 2.926218e+03 FES = 272100 Diversity = 5.7865e-14
Generation 2730: Best = 2.926218e+03 FES = 273100 Diversity = 5.7865e-14
Generation 2740: Best = 2.926218e+03 FES = 274100 Diversity = 5.7865e-14
Generation 2750: Best = 2.926218e+03 FES = 275100 Diversity = 5.7865e-14
Generation 2760: Best = 2.926218e+03 FES = 276100 Diversity = 5.7865e-14
Generation 2770: Best = 2.926218e+03 FES = 277100 Diversity = 5.7865e-14
Generation 2780: Best = 2.926218e+03 FES = 278100 Diversity = 5.7865e-14
Generation 2790: Best = 2.926218e+03 FES = 279100 Diversity = 5.7865e-14
Generation 2800: Best = 2.926218e+03 FES = 280100 Diversity = 5.7865e-14
Generation 2810: Best = 2.926218e+03 FES = 281100 Diversity = 5.7865e-14
Generation 2820: Best = 2.926218e+03 FES = 282100 Diversity = 5.7865e-14
Generation 2830: Best = 2.926218e+03 FES = 283100 Diversity = 5.7865e-14
Generation 2840: Best = 2.926218e+03 FES = 284100 Diversity = 5.7865e-14
Generation 2850: Best = 2.926218e+03 FES = 285100 Diversity = 5.7865e-14
Generation 2860: Best = 2.926218e+03 FES = 286100 Diversity = 5.7865e-14
Generation 2870: Best = 2.926218e+03 FES = 287100 Diversity = 5.7865e-14
Generation 2880: Best = 2.926218e+03 FES = 288100 Diversity = 5.7865e-14
Generation 2890: Best = 2.926218e+03 FES = 289100 Diversity = 5.7865e-14
Generation 2900: Best = 2.926218e+03 FES = 290100 Diversity = 5.7865e-14
Generation 2910: Best = 2.926218e+03 FES = 291100 Diversity = 5.7865e-14
Generation 2920: Best = 2.926218e+03 FES = 292100 Diversity = 5.7865e-14
Generation 2930: Best = 2.926218e+03 FES = 293100 Diversity = 5.7865e-14
Generation 2940: Best = 2.926218e+03 FES = 294100 Diversity = 5.7865e-14
Generation 2950: Best = 2.926218e+03 FES = 295100 Diversity = 5.7865e-14
Generation 2960: Best = 2.926218e+03 FES = 296100 Diversity = 5.7865e-14
Generation 2970: Best = 2.926218e+03 FES = 297100 Diversity = 5.7865e-14
Generation 2980: Best = 2.926218e+03 FES = 298100 Diversity = 5.7865e-14
Generation 2990: Best = 2.926218e+03 FES = 299100 Diversity = 5.7865e-14

```

Optimization completed!

Final Best Fitness = 2.926218e+03

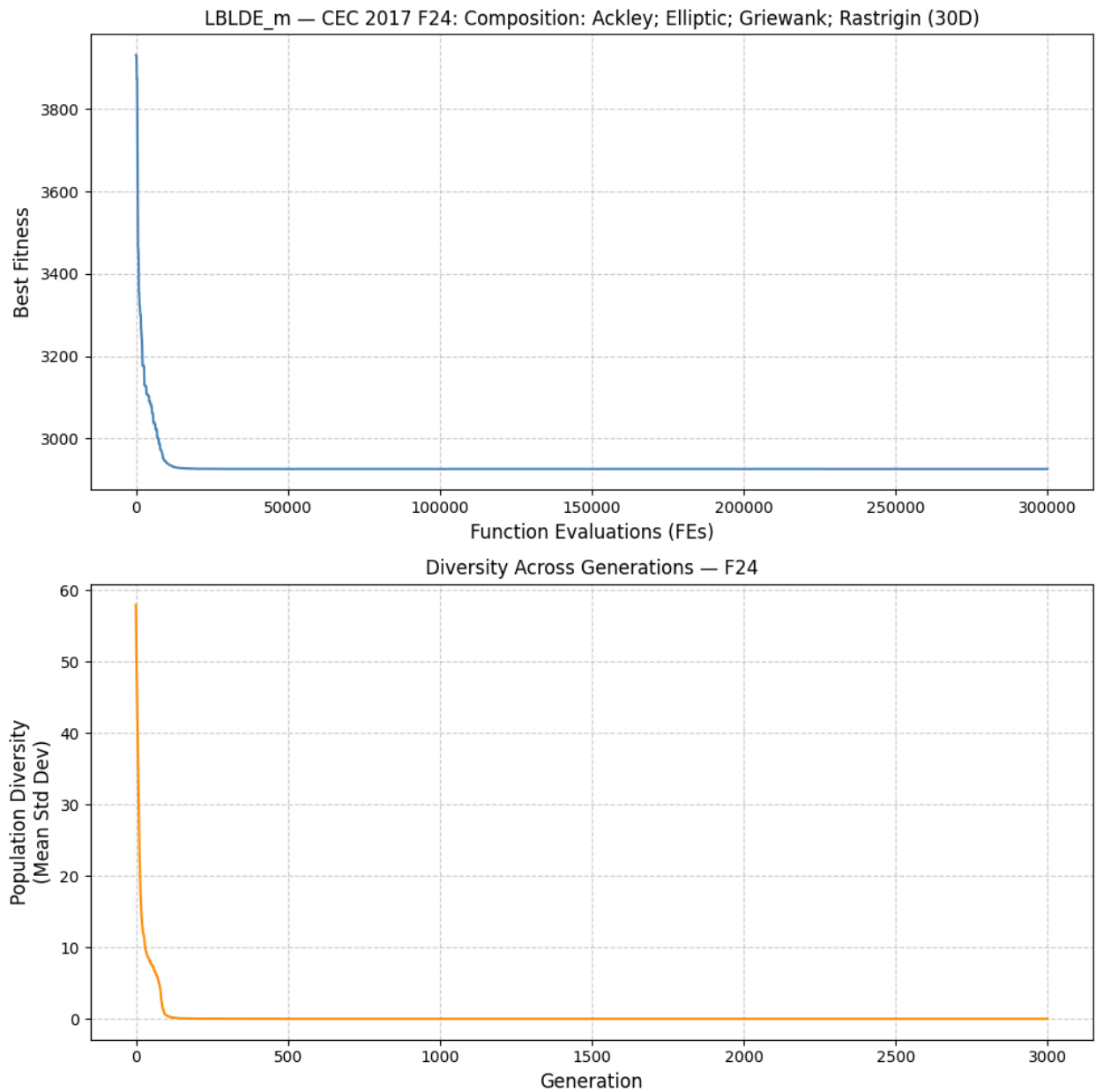
Total Function Evaluations = 300000

```

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Results:
  Best Fitness           : 2.926218e+03
  Global Optimum         : 2.400000e+03
  Error (|best-optimum|) : 5.262185e+02
  Best Solution (5 dim)  : [ 52.91592202  76.09884889 -43.7632307  -61.42119891
34.38827378]
  D^(0) (init. diversity): 5.7976e+01
=====

```

II



Graph saved as convergence_F24.png

```
In [51]: # =====  
# F25 – Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus; Rosenbrock)  
# =====  
run_and_plot('F25', make_wrapper(composition.f25), 2500.0, 'Composition: Rastr
```

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LBLDE_m on CEC 2017 F25 – Single Run

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Function : Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock (30D)

Bounds : [-100, 100]

Global optimum : 2500.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 2.562141e+04 | Diversity = 5.7976e+01

Generation 10: Best = 4.036198e+03 FES = 1100 Diversity = 2.0556e+01

Generation 20: Best = 3.374639e+03 FES = 2100 Diversity = 6.8684e+00

Generation 30: Best = 3.243024e+03 FES = 3100 Diversity = 3.1838e+00

Generation 40: Best = 3.200114e+03 FES = 4100 Diversity = 1.6870e+00

Generation 50: Best = 3.183526e+03 FES = 5100 Diversity = 9.1697e-01

Generation 60: Best = 3.166103e+03 FES = 6100 Diversity = 8.5967e-01

Generation 70: Best = 3.152695e+03 FES = 7100 Diversity = 8.0152e-01

Generation 80: Best = 3.141463e+03 FES = 8100 Diversity = 7.0427e-01

Generation 90: Best = 3.134867e+03 FES = 9100 Diversity = 5.4721e-01

Generation 100: Best = 3.130426e+03 FES = 10100 Diversity = 4.9929e-01

Generation 110: Best = 3.126484e+03 FES = 11100 Diversity = 4.7113e-01

Generation 120: Best = 3.123419e+03 FES = 12100 Diversity = 4.1033e-01

Generation 130: Best = 3.120690e+03 FES = 13100 Diversity = 3.6905e-01

Generation 140: Best = 3.118522e+03 FES = 14100 Diversity = 3.3376e-01

Generation 150: Best = 3.117231e+03 FES = 15100 Diversity = 2.5854e-01

Generation 160: Best = 3.116267e+03 FES = 16100 Diversity = 1.6786e-01

Generation 170: Best = 3.115553e+03 FES = 17100 Diversity = 1.5005e-01

Generation 180: Best = 3.115014e+03 FES = 18100 Diversity = 1.3857e-01

Generation 190: Best = 3.114720e+03 FES = 19100 Diversity = 8.4145e-02

Generation 200: Best = 3.114534e+03 FES = 20100 Diversity = 5.9635e-02

Generation 210: Best = 3.114317e+03 FES = 21100 Diversity = 1.4906e-02

Generation 220: Best = 3.113827e+03 FES = 22100 Diversity = 4.6526e-02

Generation 230: Best = 3.112644e+03 FES = 23100 Diversity = 1.3107e-01

Generation 240: Best = 3.110690e+03 FES = 24100 Diversity = 1.8898e-01

Generation 250: Best = 3.109636e+03 FES = 25100 Diversity = 1.5691e-01

Generation 260: Best = 3.109296e+03 FES = 26100 Diversity = 7.9092e-02

Generation 270: Best = 3.109002e+03 FES = 27100 Diversity = 9.8497e-02

Generation 280: Best = 3.108680e+03 FES = 28100 Diversity = 8.9893e-02

Generation 290: Best = 3.108507e+03 FES = 29100 Diversity = 6.3755e-02

Generation 300: Best = 3.108392e+03 FES = 30100 Diversity = 3.9928e-02

Generation 310: Best = 3.108307e+03 FES = 31100 Diversity = 3.6013e-02

Generation 320: Best = 3.108243e+03 FES = 32100 Diversity = 2.8057e-02

Generation 330: Best = 3.108211e+03 FES = 33100 Diversity = 2.1610e-02

Generation 340: Best = 3.108186e+03 FES = 34100 Diversity = 1.2326e-02

Generation 350: Best = 3.108164e+03 FES = 35100 Diversity = 1.2044e-02

Generation 360: Best = 3.108149e+03 FES = 36100 Diversity = 8.6256e-03

Generation 370: Best = 3.108140e+03 FES = 37100 Diversity = 8.1104e-03

Generation 380: Best = 3.108133e+03 FES = 38100 Diversity = 7.0173e-03

Generation 390: Best = 3.108128e+03 FES = 39100 Diversity = 5.6170e-03

Generation 400: Best = 3.108123e+03 FES = 40100 Diversity = 5.5506e-03

Generation 410: Best = 3.108120e+03 FES = 41100 Diversity = 3.9891e-03

Generation 420: Best = 3.108117e+03 FES = 42100 Diversity = 3.2935e-03

Generation 430: Best = 3.108116e+03 FES = 43100 Diversity = 2.5397e-03

Generation	440:	Best = 3.108115e+03	FES = 44100	Diversity = 1.9886e-03
Generation	450:	Best = 3.108114e+03	FES = 45100	Diversity = 1.4130e-03
Generation	460:	Best = 3.108114e+03	FES = 46100	Diversity = 1.3563e-03
Generation	470:	Best = 3.108113e+03	FES = 47100	Diversity = 1.4722e-03
Generation	480:	Best = 3.108113e+03	FES = 48100	Diversity = 9.9520e-04
Generation	490:	Best = 3.108113e+03	FES = 49100	Diversity = 7.7368e-04
Generation	500:	Best = 3.108113e+03	FES = 50100	Diversity = 6.1614e-04
Generation	510:	Best = 3.108113e+03	FES = 51100	Diversity = 5.7124e-04
Generation	520:	Best = 3.108113e+03	FES = 52100	Diversity = 6.4378e-04
Generation	530:	Best = 3.108113e+03	FES = 53100	Diversity = 3.8046e-04
Generation	540:	Best = 3.108112e+03	FES = 54100	Diversity = 4.1651e-04
Generation	550:	Best = 3.108112e+03	FES = 55100	Diversity = 3.4614e-04
Generation	560:	Best = 3.108112e+03	FES = 56100	Diversity = 2.5633e-04
Generation	570:	Best = 3.108112e+03	FES = 57100	Diversity = 1.1096e-04
Generation	580:	Best = 3.108112e+03	FES = 58100	Diversity = 1.3901e-04
Generation	590:	Best = 3.108112e+03	FES = 59100	Diversity = 7.9959e-05
Generation	600:	Best = 3.108112e+03	FES = 60100	Diversity = 7.3854e-05
Generation	610:	Best = 3.108112e+03	FES = 61100	Diversity = 6.2409e-05
Generation	620:	Best = 3.108112e+03	FES = 62100	Diversity = 4.6679e-05
Generation	630:	Best = 3.108112e+03	FES = 63100	Diversity = 4.0672e-05
Generation	640:	Best = 3.108112e+03	FES = 64100	Diversity = 3.8721e-05
Generation	650:	Best = 3.108112e+03	FES = 65100	Diversity = 3.0204e-05
Generation	660:	Best = 3.108112e+03	FES = 66100	Diversity = 2.2767e-05
Generation	670:	Best = 3.108112e+03	FES = 67100	Diversity = 1.3842e-05
Generation	680:	Best = 3.108112e+03	FES = 68100	Diversity = 1.5926e-05
Generation	690:	Best = 3.108112e+03	FES = 69100	Diversity = 1.5654e-05
Generation	700:	Best = 3.108112e+03	FES = 70100	Diversity = 3.9622e-06
Generation	710:	Best = 3.108112e+03	FES = 71100	Diversity = 5.0041e-06
Generation	720:	Best = 3.108112e+03	FES = 72100	Diversity = 3.1421e-06
Generation	730:	Best = 3.108112e+03	FES = 73100	Diversity = 3.0560e-06
Generation	740:	Best = 3.108112e+03	FES = 74100	Diversity = 2.3188e-06
Generation	750:	Best = 3.108112e+03	FES = 75100	Diversity = 1.7843e-06
Generation	760:	Best = 3.108112e+03	FES = 76100	Diversity = 1.2512e-06
Generation	770:	Best = 3.108112e+03	FES = 77100	Diversity = 1.2244e-06
Generation	780:	Best = 3.108112e+03	FES = 78100	Diversity = 1.0395e-06
Generation	790:	Best = 3.108112e+03	FES = 79100	Diversity = 8.4335e-07
Generation	800:	Best = 3.108112e+03	FES = 80100	Diversity = 4.9585e-07
Generation	810:	Best = 3.108112e+03	FES = 81100	Diversity = 3.9517e-07
Generation	820:	Best = 3.108112e+03	FES = 82100	Diversity = 3.4227e-07
Generation	830:	Best = 3.108112e+03	FES = 83100	Diversity = 2.2067e-07
Generation	840:	Best = 3.108112e+03	FES = 84100	Diversity = 1.1792e-07
Generation	850:	Best = 3.108112e+03	FES = 85100	Diversity = 8.5649e-08
Generation	860:	Best = 3.108112e+03	FES = 86100	Diversity = 6.6565e-09
Generation	870:	Best = 3.108112e+03	FES = 87100	Diversity = 1.4872e-10
Generation	880:	Best = 3.108112e+03	FES = 88100	Diversity = 4.1547e-13
Generation	890:	Best = 3.108112e+03	FES = 89100	Diversity = 7.8278e-14
Generation	900:	Best = 3.108112e+03	FES = 90100	Diversity = 2.7936e-14
Generation	910:	Best = 3.108112e+03	FES = 91100	Diversity = 2.7883e-14
Generation	920:	Best = 3.108112e+03	FES = 92100	Diversity = 2.7857e-14
Generation	930:	Best = 3.108112e+03	FES = 93100	Diversity = 2.7857e-14
Generation	940:	Best = 3.108112e+03	FES = 94100	Diversity = 2.7857e-14
Generation	950:	Best = 3.108112e+03	FES = 95100	Diversity = 2.7857e-14
Generation	960:	Best = 3.108112e+03	FES = 96100	Diversity = 2.7857e-14
Generation	970:	Best = 3.108112e+03	FES = 97100	Diversity = 2.7857e-14

[illegible]

[illegible]

[illegible]

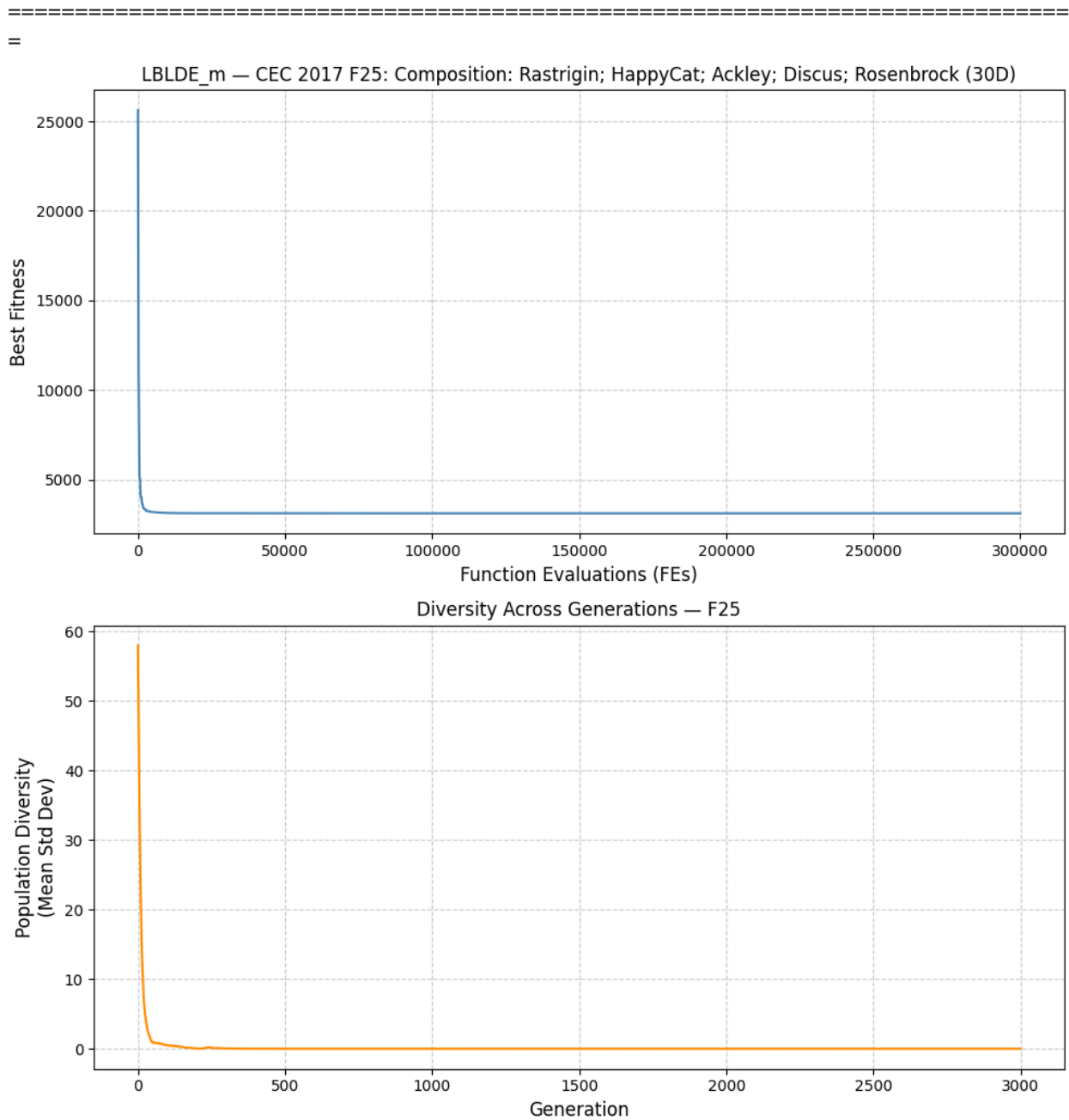
Generation 2600:	Best = 3.108112e+03	FES = 260100	Diversity = 2.7857e-14
Generation 2610:	Best = 3.108112e+03	FES = 261100	Diversity = 2.7857e-14
Generation 2620:	Best = 3.108112e+03	FES = 262100	Diversity = 2.7857e-14
Generation 2630:	Best = 3.108112e+03	FES = 263100	Diversity = 2.7857e-14
Generation 2640:	Best = 3.108112e+03	FES = 264100	Diversity = 2.7857e-14
Generation 2650:	Best = 3.108112e+03	FES = 265100	Diversity = 2.7857e-14
Generation 2660:	Best = 3.108112e+03	FES = 266100	Diversity = 2.7857e-14
Generation 2670:	Best = 3.108112e+03	FES = 267100	Diversity = 2.7857e-14
Generation 2680:	Best = 3.108112e+03	FES = 268100	Diversity = 2.7857e-14
Generation 2690:	Best = 3.108112e+03	FES = 269100	Diversity = 2.7857e-14
Generation 2700:	Best = 3.108112e+03	FES = 270100	Diversity = 2.7857e-14
Generation 2710:	Best = 3.108112e+03	FES = 271100	Diversity = 2.7857e-14
Generation 2720:	Best = 3.108112e+03	FES = 272100	Diversity = 2.7857e-14
Generation 2730:	Best = 3.108112e+03	FES = 273100	Diversity = 2.7857e-14
Generation 2740:	Best = 3.108112e+03	FES = 274100	Diversity = 2.7857e-14
Generation 2750:	Best = 3.108112e+03	FES = 275100	Diversity = 2.7857e-14
Generation 2760:	Best = 3.108112e+03	FES = 276100	Diversity = 2.7857e-14
Generation 2770:	Best = 3.108112e+03	FES = 277100	Diversity = 2.7857e-14
Generation 2780:	Best = 3.108112e+03	FES = 278100	Diversity = 2.7857e-14
Generation 2790:	Best = 3.108112e+03	FES = 279100	Diversity = 2.7857e-14
Generation 2800:	Best = 3.108112e+03	FES = 280100	Diversity = 2.7857e-14
Generation 2810:	Best = 3.108112e+03	FES = 281100	Diversity = 2.7857e-14
Generation 2820:	Best = 3.108112e+03	FES = 282100	Diversity = 2.7857e-14
Generation 2830:	Best = 3.108112e+03	FES = 283100	Diversity = 2.7857e-14
Generation 2840:	Best = 3.108112e+03	FES = 284100	Diversity = 2.7857e-14
Generation 2850:	Best = 3.108112e+03	FES = 285100	Diversity = 2.7857e-14
Generation 2860:	Best = 3.108112e+03	FES = 286100	Diversity = 2.7857e-14
Generation 2870:	Best = 3.108112e+03	FES = 287100	Diversity = 2.7857e-14
Generation 2880:	Best = 3.108112e+03	FES = 288100	Diversity = 2.7857e-14
Generation 2890:	Best = 3.108112e+03	FES = 289100	Diversity = 2.7857e-14
Generation 2900:	Best = 3.108112e+03	FES = 290100	Diversity = 2.7857e-14
Generation 2910:	Best = 3.108112e+03	FES = 291100	Diversity = 2.7857e-14
Generation 2920:	Best = 3.108112e+03	FES = 292100	Diversity = 2.7857e-14
Generation 2930:	Best = 3.108112e+03	FES = 293100	Diversity = 2.7857e-14
Generation 2940:	Best = 3.108112e+03	FES = 294100	Diversity = 2.7857e-14
Generation 2950:	Best = 3.108112e+03	FES = 295100	Diversity = 2.7857e-14
Generation 2960:	Best = 3.108112e+03	FES = 296100	Diversity = 2.7857e-14
Generation 2970:	Best = 3.108112e+03	FES = 297100	Diversity = 2.7857e-14
Generation 2980:	Best = 3.108112e+03	FES = 298100	Diversity = 2.7857e-14
Generation 2990:	Best = 3.108112e+03	FES = 299100	Diversity = 2.7857e-14

Optimization completed!

Final Best Fitness = 3.108112e+03

Total Function Evaluations = 300000

```
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Results:
  Best Fitness           : 3.108112e+03
  Global Optimum         : 2.500000e+03
  Error (|best-optimum|) : 6.081124e+02
  Best Solution (5 dim)  : [ 27.26955868  -0.29450565   4.66419652   0.39162716
-15.43329882]
  D^(0) (init. diversity): 5.7976e+01
```



Graph saved as convergence_F25.png

```
In [52]: # =====
# F26 — Composition Function 6 (Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin)
# =====
run_and_plot('F26', make_wrapper(composition.f26), 2600.0, 'Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin')
```

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LBLDE_m on CEC 2017 F26 – Single Run

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Function : Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrig
in (30D)

Bounds : [-100, 100]

Global optimum : 2600.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 1.514240e+04 | Diversity = 5.7976e+01

Generation 10: Best = 9.182518e+03 FES = 1100 Diversity = 2.5650e+01

Generation 20: Best = 7.897287e+03 FES = 2100 Diversity = 1.1273e+01

Generation 30: Best = 7.332936e+03 FES = 3100 Diversity = 7.3406e+00

Generation 40: Best = 7.038958e+03 FES = 4100 Diversity = 5.5375e+00

Generation 50: Best = 6.733613e+03 FES = 5100 Diversity = 2.3418e+00

Generation 60: Best = 6.627655e+03 FES = 6100 Diversity = 6.7076e-01

Generation 70: Best = 6.575484e+03 FES = 7100 Diversity = 4.1899e-01

Generation 80: Best = 6.560238e+03 FES = 8100 Diversity = 2.4766e-01

Generation 90: Best = 6.545780e+03 FES = 9100 Diversity = 1.5914e-01

Generation 100: Best = 6.536788e+03 FES = 10100 Diversity = 1.2483e-01

Generation 110: Best = 6.533291e+03 FES = 11100 Diversity = 7.0415e-02

Generation 120: Best = 6.531630e+03 FES = 12100 Diversity = 4.3911e-02

Generation 130: Best = 6.530906e+03 FES = 13100 Diversity = 3.1721e-02

Generation 140: Best = 6.530341e+03 FES = 14100 Diversity = 2.1111e-02

Generation 150: Best = 6.529784e+03 FES = 15100 Diversity = 2.0331e-02

Generation 160: Best = 6.529441e+03 FES = 16100 Diversity = 1.5241e-02

Generation 170: Best = 6.529087e+03 FES = 17100 Diversity = 1.3133e-02

Generation 180: Best = 6.528842e+03 FES = 18100 Diversity = 1.0647e-02

Generation 190: Best = 6.528697e+03 FES = 19100 Diversity = 7.4554e-03

Generation 200: Best = 6.528577e+03 FES = 20100 Diversity = 6.4705e-03

Generation 210: Best = 6.528513e+03 FES = 21100 Diversity = 4.8086e-03

Generation 220: Best = 6.528473e+03 FES = 22100 Diversity = 3.4820e-03

Generation 230: Best = 6.528435e+03 FES = 23100 Diversity = 3.4216e-03

Generation 240: Best = 6.528420e+03 FES = 24100 Diversity = 2.1903e-03

Generation 250: Best = 6.528406e+03 FES = 25100 Diversity = 1.9109e-03

Generation 260: Best = 6.528391e+03 FES = 26100 Diversity = 1.9422e-03

Generation 270: Best = 6.528379e+03 FES = 27100 Diversity = 1.5622e-03

Generation 280: Best = 6.528371e+03 FES = 28100 Diversity = 1.2318e-03

Generation 290: Best = 6.528364e+03 FES = 29100 Diversity = 1.2371e-03

Generation 300: Best = 6.528357e+03 FES = 30100 Diversity = 1.1672e-03

Generation 310: Best = 6.528353e+03 FES = 31100 Diversity = 1.0310e-03

Generation 320: Best = 6.528348e+03 FES = 32100 Diversity = 9.5015e-04

Generation 330: Best = 6.528345e+03 FES = 33100 Diversity = 6.6189e-04

Generation 340: Best = 6.528344e+03 FES = 34100 Diversity = 5.7257e-04

Generation 350: Best = 6.528342e+03 FES = 35100 Diversity = 4.9143e-04

Generation 360: Best = 6.528341e+03 FES = 36100 Diversity = 3.8400e-04

Generation 370: Best = 6.528341e+03 FES = 37100 Diversity = 2.4184e-04

Generation 380: Best = 6.528340e+03 FES = 38100 Diversity = 2.5790e-04

Generation 390: Best = 6.528340e+03 FES = 39100 Diversity = 1.9013e-04

Generation 400: Best = 6.528340e+03 FES = 40100 Diversity = 1.5150e-04

Generation 410: Best = 6.528340e+03 FES = 41100 Diversity = 1.2378e-04

Generation 420: Best = 6.528340e+03 FES = 42100 Diversity = 9.8460e-05

Generation 430: Best = 6.528340e+03 FES = 43100 Diversity = 7.7500e-05

Generation	440:	Best = 6.528340e+03	FES = 44100	Diversity = 7.2370e-05
Generation	450:	Best = 6.528340e+03	FES = 45100	Diversity = 6.3164e-05
Generation	460:	Best = 6.528340e+03	FES = 46100	Diversity = 4.8520e-05
Generation	470:	Best = 6.528340e+03	FES = 47100	Diversity = 3.3116e-05
Generation	480:	Best = 6.528340e+03	FES = 48100	Diversity = 2.8225e-05
Generation	490:	Best = 6.528340e+03	FES = 49100	Diversity = 2.2225e-05
Generation	500:	Best = 6.528340e+03	FES = 50100	Diversity = 1.6119e-05
Generation	510:	Best = 6.528340e+03	FES = 51100	Diversity = 1.2559e-05
Generation	520:	Best = 6.528340e+03	FES = 52100	Diversity = 1.3958e-05
Generation	530:	Best = 6.528340e+03	FES = 53100	Diversity = 7.2884e-06
Generation	540:	Best = 6.528340e+03	FES = 54100	Diversity = 7.8333e-06
Generation	550:	Best = 6.528340e+03	FES = 55100	Diversity = 6.4118e-06
Generation	560:	Best = 6.528340e+03	FES = 56100	Diversity = 4.8717e-06
Generation	570:	Best = 6.528340e+03	FES = 57100	Diversity = 2.4658e-06
Generation	580:	Best = 6.528340e+03	FES = 58100	Diversity = 1.8098e-06
Generation	590:	Best = 6.528340e+03	FES = 59100	Diversity = 1.6680e-06
Generation	600:	Best = 6.528340e+03	FES = 60100	Diversity = 1.3214e-06
Generation	610:	Best = 6.528340e+03	FES = 61100	Diversity = 1.1022e-06
Generation	620:	Best = 6.528340e+03	FES = 62100	Diversity = 8.8690e-07
Generation	630:	Best = 6.528340e+03	FES = 63100	Diversity = 8.2761e-07
Generation	640:	Best = 6.528340e+03	FES = 64100	Diversity = 7.0111e-07
Generation	650:	Best = 6.528340e+03	FES = 65100	Diversity = 6.7040e-07
Generation	660:	Best = 6.528340e+03	FES = 66100	Diversity = 4.6430e-07
Generation	670:	Best = 6.528340e+03	FES = 67100	Diversity = 5.3520e-07
Generation	680:	Best = 6.528340e+03	FES = 68100	Diversity = 3.9333e-07
Generation	690:	Best = 6.528340e+03	FES = 69100	Diversity = 2.8028e-07
Generation	700:	Best = 6.528340e+03	FES = 70100	Diversity = 1.6955e-07
Generation	710:	Best = 6.528340e+03	FES = 71100	Diversity = 1.5032e-07
Generation	720:	Best = 6.528340e+03	FES = 72100	Diversity = 1.8900e-07
Generation	730:	Best = 6.528340e+03	FES = 73100	Diversity = 9.7361e-08
Generation	740:	Best = 6.528340e+03	FES = 74100	Diversity = 8.8128e-08
Generation	750:	Best = 6.528340e+03	FES = 75100	Diversity = 8.3612e-08
Generation	760:	Best = 6.528340e+03	FES = 76100	Diversity = 5.0646e-08
Generation	770:	Best = 6.528340e+03	FES = 77100	Diversity = 6.0015e-08
Generation	780:	Best = 6.528340e+03	FES = 78100	Diversity = 3.2637e-08
Generation	790:	Best = 6.528340e+03	FES = 79100	Diversity = 2.1902e-08
Generation	800:	Best = 6.528340e+03	FES = 80100	Diversity = 1.8392e-08
Generation	810:	Best = 6.528340e+03	FES = 81100	Diversity = 1.5503e-08
Generation	820:	Best = 6.528340e+03	FES = 82100	Diversity = 1.5959e-08
Generation	830:	Best = 6.528340e+03	FES = 83100	Diversity = 1.4087e-08
Generation	840:	Best = 6.528340e+03	FES = 84100	Diversity = 1.3565e-08
Generation	850:	Best = 6.528340e+03	FES = 85100	Diversity = 1.1914e-08
Generation	860:	Best = 6.528340e+03	FES = 86100	Diversity = 1.0181e-08
Generation	870:	Best = 6.528340e+03	FES = 87100	Diversity = 1.1242e-08
Generation	880:	Best = 6.528340e+03	FES = 88100	Diversity = 1.0276e-08
Generation	890:	Best = 6.528340e+03	FES = 89100	Diversity = 1.0488e-08
Generation	900:	Best = 6.528340e+03	FES = 90100	Diversity = 8.7830e-09
Generation	910:	Best = 6.528340e+03	FES = 91100	Diversity = 8.7241e-09
Generation	920:	Best = 6.528340e+03	FES = 92100	Diversity = 7.0266e-09
Generation	930:	Best = 6.528340e+03	FES = 93100	Diversity = 6.8201e-09
Generation	940:	Best = 6.528340e+03	FES = 94100	Diversity = 4.2274e-09
Generation	950:	Best = 6.528340e+03	FES = 95100	Diversity = 3.3106e-09
Generation	960:	Best = 6.528340e+03	FES = 96100	Diversity = 1.9776e-09
Generation	970:	Best = 6.528340e+03	FES = 97100	Diversity = 1.1172e-09

[illegible]

[illegible]

[illegible]

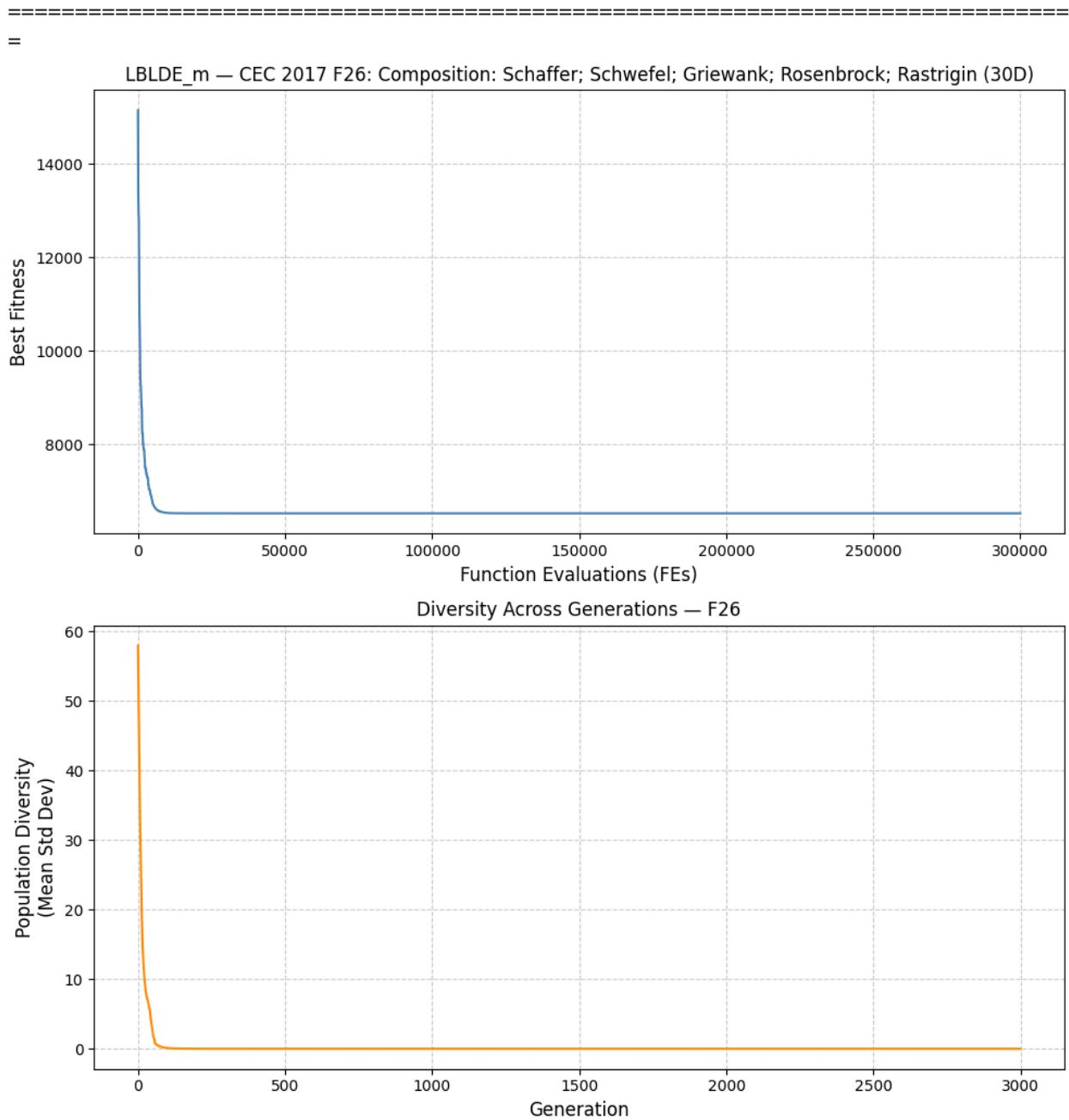
Generation 2600:	Best = 6.528340e+03	FES = 260100	Diversity = 3.3951e-14
Generation 2610:	Best = 6.528340e+03	FES = 261100	Diversity = 3.3951e-14
Generation 2620:	Best = 6.528340e+03	FES = 262100	Diversity = 3.3951e-14
Generation 2630:	Best = 6.528340e+03	FES = 263100	Diversity = 3.3951e-14
Generation 2640:	Best = 6.528340e+03	FES = 264100	Diversity = 3.3951e-14
Generation 2650:	Best = 6.528340e+03	FES = 265100	Diversity = 3.3951e-14
Generation 2660:	Best = 6.528340e+03	FES = 266100	Diversity = 3.3951e-14
Generation 2670:	Best = 6.528340e+03	FES = 267100	Diversity = 3.3951e-14
Generation 2680:	Best = 6.528340e+03	FES = 268100	Diversity = 3.3951e-14
Generation 2690:	Best = 6.528340e+03	FES = 269100	Diversity = 3.3951e-14
Generation 2700:	Best = 6.528340e+03	FES = 270100	Diversity = 3.3951e-14
Generation 2710:	Best = 6.528340e+03	FES = 271100	Diversity = 3.3951e-14
Generation 2720:	Best = 6.528340e+03	FES = 272100	Diversity = 3.3951e-14
Generation 2730:	Best = 6.528340e+03	FES = 273100	Diversity = 3.3951e-14
Generation 2740:	Best = 6.528340e+03	FES = 274100	Diversity = 3.3951e-14
Generation 2750:	Best = 6.528340e+03	FES = 275100	Diversity = 3.3951e-14
Generation 2760:	Best = 6.528340e+03	FES = 276100	Diversity = 3.3951e-14
Generation 2770:	Best = 6.528340e+03	FES = 277100	Diversity = 3.3951e-14
Generation 2780:	Best = 6.528340e+03	FES = 278100	Diversity = 3.3951e-14
Generation 2790:	Best = 6.528340e+03	FES = 279100	Diversity = 3.3951e-14
Generation 2800:	Best = 6.528340e+03	FES = 280100	Diversity = 3.3951e-14
Generation 2810:	Best = 6.528340e+03	FES = 281100	Diversity = 3.3951e-14
Generation 2820:	Best = 6.528340e+03	FES = 282100	Diversity = 3.3951e-14
Generation 2830:	Best = 6.528340e+03	FES = 283100	Diversity = 3.3951e-14
Generation 2840:	Best = 6.528340e+03	FES = 284100	Diversity = 3.3951e-14
Generation 2850:	Best = 6.528340e+03	FES = 285100	Diversity = 3.3951e-14
Generation 2860:	Best = 6.528340e+03	FES = 286100	Diversity = 3.3951e-14
Generation 2870:	Best = 6.528340e+03	FES = 287100	Diversity = 3.3951e-14
Generation 2880:	Best = 6.528340e+03	FES = 288100	Diversity = 3.3951e-14
Generation 2890:	Best = 6.528340e+03	FES = 289100	Diversity = 3.3951e-14
Generation 2900:	Best = 6.528340e+03	FES = 290100	Diversity = 3.3951e-14
Generation 2910:	Best = 6.528340e+03	FES = 291100	Diversity = 3.3951e-14
Generation 2920:	Best = 6.528340e+03	FES = 292100	Diversity = 3.3951e-14
Generation 2930:	Best = 6.528340e+03	FES = 293100	Diversity = 3.3951e-14
Generation 2940:	Best = 6.528340e+03	FES = 294100	Diversity = 3.3951e-14
Generation 2950:	Best = 6.528340e+03	FES = 295100	Diversity = 3.3951e-14
Generation 2960:	Best = 6.528340e+03	FES = 296100	Diversity = 3.3951e-14
Generation 2970:	Best = 6.528340e+03	FES = 297100	Diversity = 3.3951e-14
Generation 2980:	Best = 6.528340e+03	FES = 298100	Diversity = 3.3951e-14
Generation 2990:	Best = 6.528340e+03	FES = 299100	Diversity = 3.3951e-14

Optimization completed!

Final Best Fitness = 6.528340e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 6.528340e+03
  Global Optimum         : 2.600000e+03
  Error (|best-optimum|) : 3.928340e+03
  Best Solution (5 dim)  : [ 1.40223755 -64.52918112  31.4936235  -34.26303419
-34.89608754]
  D^(0) (init. diversity): 5.7976e+01
```



Graph saved as convergence_F26.png

```
In [53]: # =====
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent Cigar; Ellipt
# =====
run_and_plot('F27', make_wrapper(composition.f27), 2700.0, 'Composition: HGBat
```

=====

=

LBLDE_m on CEC 2017 F27 – Single Run

=====

=

Function : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic;
Schaffer (30D)

Bounds : [-100, 100]

Global optimum : 2700.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 4.906147e+03 | Diversity = 5.7976e+01

Generation 10:	Best = 3.726776e+03	FES = 1100	Diversity = 3.5239e+01
Generation 20:	Best = 3.463435e+03	FES = 2100	Diversity = 1.9804e+01
Generation 30:	Best = 3.376332e+03	FES = 3100	Diversity = 1.2924e+01
Generation 40:	Best = 3.327067e+03	FES = 4100	Diversity = 8.4638e+00
Generation 50:	Best = 3.288784e+03	FES = 5100	Diversity = 6.2001e+00
Generation 60:	Best = 3.277520e+03	FES = 6100	Diversity = 3.5232e+00
Generation 70:	Best = 3.268605e+03	FES = 7100	Diversity = 2.5545e+00
Generation 80:	Best = 3.262879e+03	FES = 8100	Diversity = 1.2405e+00
Generation 90:	Best = 3.261982e+03	FES = 9100	Diversity = 9.9628e-01
Generation 100:	Best = 3.260571e+03	FES = 10100	Diversity = 9.2076e-01
Generation 110:	Best = 3.260130e+03	FES = 11100	Diversity = 8.3504e-01
Generation 120:	Best = 3.259518e+03	FES = 12100	Diversity = 6.9457e-01
Generation 130:	Best = 3.258680e+03	FES = 13100	Diversity = 6.0023e-01
Generation 140:	Best = 3.258259e+03	FES = 14100	Diversity = 5.2606e-01
Generation 150:	Best = 3.257673e+03	FES = 15100	Diversity = 4.2345e-01
Generation 160:	Best = 3.257590e+03	FES = 16100	Diversity = 3.1597e-01
Generation 170:	Best = 3.257341e+03	FES = 17100	Diversity = 2.5254e-01
Generation 180:	Best = 3.257141e+03	FES = 18100	Diversity = 2.1965e-01
Generation 190:	Best = 3.257022e+03	FES = 19100	Diversity = 2.0080e-01
Generation 200:	Best = 3.256886e+03	FES = 20100	Diversity = 1.9292e-01
Generation 210:	Best = 3.256810e+03	FES = 21100	Diversity = 1.9340e-01
Generation 220:	Best = 3.256760e+03	FES = 22100	Diversity = 1.9052e-01
Generation 230:	Best = 3.256728e+03	FES = 23100	Diversity = 4.2580e-02
Generation 240:	Best = 3.256693e+03	FES = 24100	Diversity = 4.0213e-02
Generation 250:	Best = 3.256667e+03	FES = 25100	Diversity = 3.5311e-02
Generation 260:	Best = 3.256649e+03	FES = 26100	Diversity = 3.0393e-02
Generation 270:	Best = 3.256632e+03	FES = 27100	Diversity = 2.7037e-02
Generation 280:	Best = 3.256628e+03	FES = 28100	Diversity = 2.3988e-02
Generation 290:	Best = 3.256618e+03	FES = 29100	Diversity = 2.0968e-02
Generation 300:	Best = 3.256611e+03	FES = 30100	Diversity = 1.9386e-02
Generation 310:	Best = 3.256605e+03	FES = 31100	Diversity = 1.9565e-02
Generation 320:	Best = 3.256593e+03	FES = 32100	Diversity = 1.8147e-02
Generation 330:	Best = 3.256589e+03	FES = 33100	Diversity = 1.7340e-02
Generation 340:	Best = 3.256585e+03	FES = 34100	Diversity = 1.5719e-02
Generation 350:	Best = 3.256582e+03	FES = 35100	Diversity = 1.3438e-02
Generation 360:	Best = 3.256579e+03	FES = 36100	Diversity = 1.1984e-02
Generation 370:	Best = 3.256575e+03	FES = 37100	Diversity = 1.0969e-02
Generation 380:	Best = 3.256570e+03	FES = 38100	Diversity = 1.2139e-02
Generation 390:	Best = 3.256567e+03	FES = 39100	Diversity = 1.2098e-02
Generation 400:	Best = 3.256566e+03	FES = 40100	Diversity = 1.0578e-02
Generation 410:	Best = 3.256563e+03	FES = 41100	Diversity = 9.4129e-03
Generation 420:	Best = 3.256561e+03	FES = 42100	Diversity = 8.8791e-03
Generation 430:	Best = 3.256558e+03	FES = 43100	Diversity = 9.1567e-03

Generation	440:	Best = 3.256556e+03	FES = 44100	Diversity = 9.2632e-03
Generation	450:	Best = 3.256553e+03	FES = 45100	Diversity = 9.2672e-03
Generation	460:	Best = 3.256551e+03	FES = 46100	Diversity = 9.8677e-03
Generation	470:	Best = 3.256547e+03	FES = 47100	Diversity = 9.7731e-03
Generation	480:	Best = 3.256544e+03	FES = 48100	Diversity = 1.0748e-02
Generation	490:	Best = 3.256541e+03	FES = 49100	Diversity = 9.8832e-03
Generation	500:	Best = 3.256539e+03	FES = 50100	Diversity = 1.0508e-02
Generation	510:	Best = 3.256538e+03	FES = 51100	Diversity = 1.0403e-02
Generation	520:	Best = 3.256535e+03	FES = 52100	Diversity = 9.2194e-03
Generation	530:	Best = 3.256532e+03	FES = 53100	Diversity = 9.1898e-03
Generation	540:	Best = 3.256528e+03	FES = 54100	Diversity = 1.0647e-02
Generation	550:	Best = 3.256526e+03	FES = 55100	Diversity = 1.0611e-02
Generation	560:	Best = 3.256523e+03	FES = 56100	Diversity = 1.0355e-02
Generation	570:	Best = 3.256521e+03	FES = 57100	Diversity = 1.0017e-02
Generation	580:	Best = 3.256519e+03	FES = 58100	Diversity = 1.0166e-02
Generation	590:	Best = 3.256517e+03	FES = 59100	Diversity = 9.7643e-03
Generation	600:	Best = 3.256512e+03	FES = 60100	Diversity = 9.3017e-03
Generation	610:	Best = 3.256508e+03	FES = 61100	Diversity = 1.1167e-02
Generation	620:	Best = 3.256504e+03	FES = 62100	Diversity = 1.1092e-02
Generation	630:	Best = 3.256503e+03	FES = 63100	Diversity = 1.0098e-02
Generation	640:	Best = 3.256501e+03	FES = 64100	Diversity = 1.0201e-02
Generation	650:	Best = 3.256498e+03	FES = 65100	Diversity = 1.0255e-02
Generation	660:	Best = 3.256496e+03	FES = 66100	Diversity = 9.6642e-03
Generation	670:	Best = 3.256495e+03	FES = 67100	Diversity = 8.4600e-03
Generation	680:	Best = 3.256492e+03	FES = 68100	Diversity = 8.9463e-03
Generation	690:	Best = 3.256491e+03	FES = 69100	Diversity = 8.0836e-03
Generation	700:	Best = 3.256489e+03	FES = 70100	Diversity = 8.0328e-03
Generation	710:	Best = 3.256486e+03	FES = 71100	Diversity = 8.8492e-03
Generation	720:	Best = 3.256483e+03	FES = 72100	Diversity = 9.0192e-03
Generation	730:	Best = 3.256482e+03	FES = 73100	Diversity = 9.0693e-03
Generation	740:	Best = 3.256478e+03	FES = 74100	Diversity = 9.3475e-03
Generation	750:	Best = 3.256475e+03	FES = 75100	Diversity = 9.6380e-03
Generation	760:	Best = 3.256472e+03	FES = 76100	Diversity = 9.8467e-03
Generation	770:	Best = 3.256469e+03	FES = 77100	Diversity = 1.0618e-02
Generation	780:	Best = 3.256466e+03	FES = 78100	Diversity = 1.0128e-02
Generation	790:	Best = 3.256463e+03	FES = 79100	Diversity = 9.9377e-03
Generation	800:	Best = 3.256459e+03	FES = 80100	Diversity = 1.1160e-02
Generation	810:	Best = 3.256456e+03	FES = 81100	Diversity = 1.1485e-02
Generation	820:	Best = 3.256453e+03	FES = 82100	Diversity = 1.0904e-02
Generation	830:	Best = 3.256451e+03	FES = 83100	Diversity = 1.0509e-02
Generation	840:	Best = 3.256447e+03	FES = 84100	Diversity = 1.0661e-02
Generation	850:	Best = 3.256443e+03	FES = 85100	Diversity = 1.1516e-02
Generation	860:	Best = 3.256442e+03	FES = 86100	Diversity = 1.0740e-02
Generation	870:	Best = 3.256440e+03	FES = 87100	Diversity = 1.0064e-02
Generation	880:	Best = 3.256438e+03	FES = 88100	Diversity = 1.0045e-02
Generation	890:	Best = 3.256434e+03	FES = 89100	Diversity = 9.3663e-03
Generation	900:	Best = 3.256433e+03	FES = 90100	Diversity = 9.7598e-03
Generation	910:	Best = 3.256431e+03	FES = 91100	Diversity = 1.0492e-02
Generation	920:	Best = 3.256429e+03	FES = 92100	Diversity = 9.0662e-03
Generation	930:	Best = 3.256428e+03	FES = 93100	Diversity = 8.8652e-03
Generation	940:	Best = 3.256425e+03	FES = 94100	Diversity = 9.1366e-03
Generation	950:	Best = 3.256422e+03	FES = 95100	Diversity = 9.7048e-03
Generation	960:	Best = 3.256420e+03	FES = 96100	Diversity = 9.1663e-03
Generation	970:	Best = 3.256418e+03	FES = 97100	Diversity = 9.6390e-03

Generation 980:	Best = 3.256415e+03	FES = 98100	Diversity = 1.0120e-02
Generation 990:	Best = 3.256413e+03	FES = 99100	Diversity = 9.7031e-03
Generation 1000:	Best = 3.256412e+03	FES = 100100	Diversity = 9.4072e-03
Generation 1010:	Best = 3.256409e+03	FES = 101100	Diversity = 8.7895e-03
Generation 1020:	Best = 3.256406e+03	FES = 102100	Diversity = 9.2872e-03
Generation 1030:	Best = 3.256403e+03	FES = 103100	Diversity = 1.0147e-02
Generation 1040:	Best = 3.256401e+03	FES = 104100	Diversity = 9.8346e-03
Generation 1050:	Best = 3.256396e+03	FES = 105100	Diversity = 9.5470e-03
Generation 1060:	Best = 3.256391e+03	FES = 106100	Diversity = 1.1213e-02
Generation 1070:	Best = 3.256390e+03	FES = 107100	Diversity = 1.1163e-02
Generation 1080:	Best = 3.256387e+03	FES = 108100	Diversity = 1.0175e-02
Generation 1090:	Best = 3.256384e+03	FES = 109100	Diversity = 1.0055e-02
Generation 1100:	Best = 3.256382e+03	FES = 110100	Diversity = 1.0649e-02
Generation 1110:	Best = 3.256379e+03	FES = 111100	Diversity = 9.8689e-03
Generation 1120:	Best = 3.256376e+03	FES = 112100	Diversity = 1.0946e-02
Generation 1130:	Best = 3.256372e+03	FES = 113100	Diversity = 1.2029e-02
Generation 1140:	Best = 3.256367e+03	FES = 114100	Diversity = 1.2399e-02
Generation 1150:	Best = 3.256363e+03	FES = 115100	Diversity = 1.2797e-02
Generation 1160:	Best = 3.256358e+03	FES = 116100	Diversity = 1.1222e-02
Generation 1170:	Best = 3.256355e+03	FES = 117100	Diversity = 1.1012e-02
Generation 1180:	Best = 3.256352e+03	FES = 118100	Diversity = 1.0610e-02
Generation 1190:	Best = 3.256351e+03	FES = 119100	Diversity = 9.6533e-03
Generation 1200:	Best = 3.256349e+03	FES = 120100	Diversity = 9.4913e-03
Generation 1210:	Best = 3.256347e+03	FES = 121100	Diversity = 8.6132e-03
Generation 1220:	Best = 3.256345e+03	FES = 122100	Diversity = 9.8531e-03
Generation 1230:	Best = 3.256342e+03	FES = 123100	Diversity = 9.8949e-03
Generation 1240:	Best = 3.256339e+03	FES = 124100	Diversity = 1.0580e-02
Generation 1250:	Best = 3.256337e+03	FES = 125100	Diversity = 9.4052e-03
Generation 1260:	Best = 3.256335e+03	FES = 126100	Diversity = 9.8971e-03
Generation 1270:	Best = 3.256333e+03	FES = 127100	Diversity = 8.8016e-03
Generation 1280:	Best = 3.256331e+03	FES = 128100	Diversity = 9.1229e-03
Generation 1290:	Best = 3.256328e+03	FES = 129100	Diversity = 9.4185e-03
Generation 1300:	Best = 3.256324e+03	FES = 130100	Diversity = 1.1244e-02
Generation 1310:	Best = 3.256322e+03	FES = 131100	Diversity = 1.0235e-02
Generation 1320:	Best = 3.256317e+03	FES = 132100	Diversity = 1.0441e-02
Generation 1330:	Best = 3.256316e+03	FES = 133100	Diversity = 1.1835e-02
Generation 1340:	Best = 3.256314e+03	FES = 134100	Diversity = 1.1464e-02
Generation 1350:	Best = 3.256311e+03	FES = 135100	Diversity = 1.0836e-02
Generation 1360:	Best = 3.256308e+03	FES = 136100	Diversity = 1.0965e-02
Generation 1370:	Best = 3.256306e+03	FES = 137100	Diversity = 1.0756e-02
Generation 1380:	Best = 3.256303e+03	FES = 138100	Diversity = 1.0508e-02
Generation 1390:	Best = 3.256302e+03	FES = 139100	Diversity = 1.0168e-02
Generation 1400:	Best = 3.256301e+03	FES = 140100	Diversity = 1.0186e-02
Generation 1410:	Best = 3.256299e+03	FES = 141100	Diversity = 9.4851e-03
Generation 1420:	Best = 3.256298e+03	FES = 142100	Diversity = 8.8377e-03
Generation 1430:	Best = 3.256297e+03	FES = 143100	Diversity = 8.6583e-03
Generation 1440:	Best = 3.256296e+03	FES = 144100	Diversity = 8.3574e-03
Generation 1450:	Best = 3.256294e+03	FES = 145100	Diversity = 7.5934e-03
Generation 1460:	Best = 3.256293e+03	FES = 146100	Diversity = 6.8392e-03
Generation 1470:	Best = 3.256292e+03	FES = 147100	Diversity = 6.6145e-03
Generation 1480:	Best = 3.256292e+03	FES = 148100	Diversity = 6.6893e-03
Generation 1490:	Best = 3.256291e+03	FES = 149100	Diversity = 6.0019e-03
Generation 1500:	Best = 3.256290e+03	FES = 150100	Diversity = 6.0850e-03
Generation 1510:	Best = 3.256290e+03	FES = 151100	Diversity = 5.4432e-03

Generation 1520:	Best = 3.256289e+03	FES = 152100	Diversity = 5.2048e-03
Generation 1530:	Best = 3.256289e+03	FES = 153100	Diversity = 5.0921e-03
Generation 1540:	Best = 3.256288e+03	FES = 154100	Diversity = 5.0127e-03
Generation 1550:	Best = 3.256288e+03	FES = 155100	Diversity = 5.1371e-03
Generation 1560:	Best = 3.256288e+03	FES = 156100	Diversity = 5.2821e-03
Generation 1570:	Best = 3.256288e+03	FES = 157100	Diversity = 4.9060e-03
Generation 1580:	Best = 3.256287e+03	FES = 158100	Diversity = 4.2419e-03
Generation 1590:	Best = 3.256287e+03	FES = 159100	Diversity = 4.2391e-03
Generation 1600:	Best = 3.256287e+03	FES = 160100	Diversity = 3.7794e-03
Generation 1610:	Best = 3.256287e+03	FES = 161100	Diversity = 3.4137e-03
Generation 1620:	Best = 3.256286e+03	FES = 162100	Diversity = 3.4579e-03
Generation 1630:	Best = 3.256286e+03	FES = 163100	Diversity = 3.4518e-03
Generation 1640:	Best = 3.256286e+03	FES = 164100	Diversity = 3.2256e-03
Generation 1650:	Best = 3.256286e+03	FES = 165100	Diversity = 3.2618e-03
Generation 1660:	Best = 3.256286e+03	FES = 166100	Diversity = 3.1588e-03
Generation 1670:	Best = 3.256286e+03	FES = 167100	Diversity = 3.0577e-03
Generation 1680:	Best = 3.256285e+03	FES = 168100	Diversity = 3.2270e-03
Generation 1690:	Best = 3.256285e+03	FES = 169100	Diversity = 3.1123e-03
Generation 1700:	Best = 3.256285e+03	FES = 170100	Diversity = 2.9314e-03
Generation 1710:	Best = 3.256285e+03	FES = 171100	Diversity = 3.3266e-03
Generation 1720:	Best = 3.256284e+03	FES = 172100	Diversity = 3.5923e-03
Generation 1730:	Best = 3.256284e+03	FES = 173100	Diversity = 3.5574e-03
Generation 1740:	Best = 3.256283e+03	FES = 174100	Diversity = 3.7787e-03
Generation 1750:	Best = 3.256283e+03	FES = 175100	Diversity = 4.0272e-03
Generation 1760:	Best = 3.256283e+03	FES = 176100	Diversity = 3.8737e-03
Generation 1770:	Best = 3.256282e+03	FES = 177100	Diversity = 3.7051e-03
Generation 1780:	Best = 3.256282e+03	FES = 178100	Diversity = 3.5964e-03
Generation 1790:	Best = 3.256282e+03	FES = 179100	Diversity = 3.5178e-03
Generation 1800:	Best = 3.256281e+03	FES = 180100	Diversity = 3.9561e-03
Generation 1810:	Best = 3.256280e+03	FES = 181100	Diversity = 4.8425e-03
Generation 1820:	Best = 3.256280e+03	FES = 182100	Diversity = 5.1231e-03
Generation 1830:	Best = 3.256279e+03	FES = 183100	Diversity = 4.7974e-03
Generation 1840:	Best = 3.256278e+03	FES = 184100	Diversity = 5.1069e-03
Generation 1850:	Best = 3.256278e+03	FES = 185100	Diversity = 5.6868e-03
Generation 1860:	Best = 3.256277e+03	FES = 186100	Diversity = 4.8618e-03
Generation 1870:	Best = 3.256277e+03	FES = 187100	Diversity = 4.9262e-03
Generation 1880:	Best = 3.256276e+03	FES = 188100	Diversity = 4.8906e-03
Generation 1890:	Best = 3.256276e+03	FES = 189100	Diversity = 4.6137e-03
Generation 1900:	Best = 3.256276e+03	FES = 190100	Diversity = 4.2701e-03
Generation 1910:	Best = 3.256275e+03	FES = 191100	Diversity = 4.6552e-03
Generation 1920:	Best = 3.256275e+03	FES = 192100	Diversity = 5.2487e-03
Generation 1930:	Best = 3.256275e+03	FES = 193100	Diversity = 4.5034e-03
Generation 1940:	Best = 3.256274e+03	FES = 194100	Diversity = 4.3015e-03
Generation 1950:	Best = 3.256274e+03	FES = 195100	Diversity = 4.2972e-03
Generation 1960:	Best = 3.256274e+03	FES = 196100	Diversity = 3.7415e-03
Generation 1970:	Best = 3.256273e+03	FES = 197100	Diversity = 3.7245e-03
Generation 1980:	Best = 3.256273e+03	FES = 198100	Diversity = 3.8514e-03
Generation 1990:	Best = 3.256272e+03	FES = 199100	Diversity = 4.3825e-03
Generation 2000:	Best = 3.256272e+03	FES = 200100	Diversity = 4.1134e-03
Generation 2010:	Best = 3.256272e+03	FES = 201100	Diversity = 3.5507e-03
Generation 2020:	Best = 3.256271e+03	FES = 202100	Diversity = 4.0396e-03
Generation 2030:	Best = 3.256271e+03	FES = 203100	Diversity = 3.4218e-03
Generation 2040:	Best = 3.256271e+03	FES = 204100	Diversity = 3.6892e-03
Generation 2050:	Best = 3.256270e+03	FES = 205100	Diversity = 3.8871e-03

Generation 2060:	Best = 3.256270e+03	FES = 206100	Diversity = 3.7665e-03
Generation 2070:	Best = 3.256269e+03	FES = 207100	Diversity = 4.1121e-03
Generation 2080:	Best = 3.256269e+03	FES = 208100	Diversity = 4.2691e-03
Generation 2090:	Best = 3.256269e+03	FES = 209100	Diversity = 4.2237e-03
Generation 2100:	Best = 3.256268e+03	FES = 210100	Diversity = 4.0905e-03
Generation 2110:	Best = 3.256268e+03	FES = 211100	Diversity = 4.5092e-03
Generation 2120:	Best = 3.256267e+03	FES = 212100	Diversity = 4.3643e-03
Generation 2130:	Best = 3.256267e+03	FES = 213100	Diversity = 5.0098e-03
Generation 2140:	Best = 3.256266e+03	FES = 214100	Diversity = 5.0595e-03
Generation 2150:	Best = 3.256266e+03	FES = 215100	Diversity = 4.8462e-03
Generation 2160:	Best = 3.256266e+03	FES = 216100	Diversity = 4.7794e-03
Generation 2170:	Best = 3.256265e+03	FES = 217100	Diversity = 4.8332e-03
Generation 2180:	Best = 3.256264e+03	FES = 218100	Diversity = 4.6729e-03
Generation 2190:	Best = 3.256263e+03	FES = 219100	Diversity = 5.1705e-03
Generation 2200:	Best = 3.256263e+03	FES = 220100	Diversity = 4.9554e-03
Generation 2210:	Best = 3.256263e+03	FES = 221100	Diversity = 4.6107e-03
Generation 2220:	Best = 3.256262e+03	FES = 222100	Diversity = 4.5310e-03
Generation 2230:	Best = 3.256262e+03	FES = 223100	Diversity = 4.2802e-03
Generation 2240:	Best = 3.256262e+03	FES = 224100	Diversity = 4.2454e-03
Generation 2250:	Best = 3.256261e+03	FES = 225100	Diversity = 4.8989e-03
Generation 2260:	Best = 3.256260e+03	FES = 226100	Diversity = 4.6638e-03
Generation 2270:	Best = 3.256260e+03	FES = 227100	Diversity = 4.4255e-03
Generation 2280:	Best = 3.256260e+03	FES = 228100	Diversity = 4.5080e-03
Generation 2290:	Best = 3.256259e+03	FES = 229100	Diversity = 5.0013e-03
Generation 2300:	Best = 3.256259e+03	FES = 230100	Diversity = 5.2127e-03
Generation 2310:	Best = 3.256258e+03	FES = 231100	Diversity = 4.6926e-03
Generation 2320:	Best = 3.256258e+03	FES = 232100	Diversity = 5.1475e-03
Generation 2330:	Best = 3.256257e+03	FES = 233100	Diversity = 4.9927e-03
Generation 2340:	Best = 3.256257e+03	FES = 234100	Diversity = 5.5253e-03
Generation 2350:	Best = 3.256256e+03	FES = 235100	Diversity = 5.3321e-03
Generation 2360:	Best = 3.256255e+03	FES = 236100	Diversity = 5.1980e-03
Generation 2370:	Best = 3.256255e+03	FES = 237100	Diversity = 4.6571e-03
Generation 2380:	Best = 3.256254e+03	FES = 238100	Diversity = 4.8152e-03
Generation 2390:	Best = 3.256254e+03	FES = 239100	Diversity = 4.4378e-03
Generation 2400:	Best = 3.256253e+03	FES = 240100	Diversity = 4.6436e-03
Generation 2410:	Best = 3.256253e+03	FES = 241100	Diversity = 4.4563e-03
Generation 2420:	Best = 3.256252e+03	FES = 242100	Diversity = 4.5088e-03
Generation 2430:	Best = 3.256252e+03	FES = 243100	Diversity = 4.7521e-03
Generation 2440:	Best = 3.256251e+03	FES = 244100	Diversity = 4.7476e-03
Generation 2450:	Best = 3.256251e+03	FES = 245100	Diversity = 4.5550e-03
Generation 2460:	Best = 3.256250e+03	FES = 246100	Diversity = 4.6576e-03
Generation 2470:	Best = 3.256250e+03	FES = 247100	Diversity = 5.1061e-03
Generation 2480:	Best = 3.256250e+03	FES = 248100	Diversity = 4.3565e-03
Generation 2490:	Best = 3.256249e+03	FES = 249100	Diversity = 4.5290e-03
Generation 2500:	Best = 3.256249e+03	FES = 250100	Diversity = 4.1966e-03
Generation 2510:	Best = 3.256249e+03	FES = 251100	Diversity = 4.0623e-03
Generation 2520:	Best = 3.256248e+03	FES = 252100	Diversity = 4.3209e-03
Generation 2530:	Best = 3.256248e+03	FES = 253100	Diversity = 4.4685e-03
Generation 2540:	Best = 3.256247e+03	FES = 254100	Diversity = 4.3689e-03
Generation 2550:	Best = 3.256247e+03	FES = 255100	Diversity = 4.2348e-03
Generation 2560:	Best = 3.256247e+03	FES = 256100	Diversity = 4.2156e-03
Generation 2570:	Best = 3.256246e+03	FES = 257100	Diversity = 4.6372e-03
Generation 2580:	Best = 3.256246e+03	FES = 258100	Diversity = 4.4168e-03
Generation 2590:	Best = 3.256245e+03	FES = 259100	Diversity = 4.6282e-03

Generation 2600:	Best = 3.256245e+03	FES = 260100	Diversity = 4.5171e-03
Generation 2610:	Best = 3.256244e+03	FES = 261100	Diversity = 4.7568e-03
Generation 2620:	Best = 3.256244e+03	FES = 262100	Diversity = 4.6022e-03
Generation 2630:	Best = 3.256244e+03	FES = 263100	Diversity = 4.8528e-03
Generation 2640:	Best = 3.256243e+03	FES = 264100	Diversity = 4.6822e-03
Generation 2650:	Best = 3.256243e+03	FES = 265100	Diversity = 4.6432e-03
Generation 2660:	Best = 3.256242e+03	FES = 266100	Diversity = 5.5173e-03
Generation 2670:	Best = 3.256241e+03	FES = 267100	Diversity = 5.5844e-03
Generation 2680:	Best = 3.256241e+03	FES = 268100	Diversity = 5.9599e-03
Generation 2690:	Best = 3.256240e+03	FES = 269100	Diversity = 5.6497e-03
Generation 2700:	Best = 3.256239e+03	FES = 270100	Diversity = 5.6597e-03
Generation 2710:	Best = 3.256237e+03	FES = 271100	Diversity = 6.2983e-03
Generation 2720:	Best = 3.256237e+03	FES = 272100	Diversity = 6.3977e-03
Generation 2730:	Best = 3.256236e+03	FES = 273100	Diversity = 6.6367e-03
Generation 2740:	Best = 3.256236e+03	FES = 274100	Diversity = 6.8613e-03
Generation 2750:	Best = 3.256235e+03	FES = 275100	Diversity = 6.1238e-03
Generation 2760:	Best = 3.256234e+03	FES = 276100	Diversity = 5.9088e-03
Generation 2770:	Best = 3.256234e+03	FES = 277100	Diversity = 5.5433e-03
Generation 2780:	Best = 3.256233e+03	FES = 278100	Diversity = 5.2725e-03
Generation 2790:	Best = 3.256233e+03	FES = 279100	Diversity = 5.3390e-03
Generation 2800:	Best = 3.256232e+03	FES = 280100	Diversity = 4.8955e-03
Generation 2810:	Best = 3.256232e+03	FES = 281100	Diversity = 5.3111e-03
Generation 2820:	Best = 3.256232e+03	FES = 282100	Diversity = 5.3365e-03
Generation 2830:	Best = 3.256231e+03	FES = 283100	Diversity = 4.6590e-03
Generation 2840:	Best = 3.256231e+03	FES = 284100	Diversity = 4.9952e-03
Generation 2850:	Best = 3.256230e+03	FES = 285100	Diversity = 5.0823e-03
Generation 2860:	Best = 3.256230e+03	FES = 286100	Diversity = 4.8767e-03
Generation 2870:	Best = 3.256230e+03	FES = 287100	Diversity = 4.9023e-03
Generation 2880:	Best = 3.256229e+03	FES = 288100	Diversity = 4.3457e-03
Generation 2890:	Best = 3.256229e+03	FES = 289100	Diversity = 4.3224e-03
Generation 2900:	Best = 3.256228e+03	FES = 290100	Diversity = 4.5634e-03
Generation 2910:	Best = 3.256228e+03	FES = 291100	Diversity = 4.6395e-03
Generation 2920:	Best = 3.256228e+03	FES = 292100	Diversity = 4.4362e-03
Generation 2930:	Best = 3.256227e+03	FES = 293100	Diversity = 4.3058e-03
Generation 2940:	Best = 3.256227e+03	FES = 294100	Diversity = 4.6317e-03
Generation 2950:	Best = 3.256227e+03	FES = 295100	Diversity = 4.7300e-03
Generation 2960:	Best = 3.256226e+03	FES = 296100	Diversity = 4.4305e-03
Generation 2970:	Best = 3.256226e+03	FES = 297100	Diversity = 4.4028e-03
Generation 2980:	Best = 3.256225e+03	FES = 298100	Diversity = 4.5243e-03
Generation 2990:	Best = 3.256225e+03	FES = 299100	Diversity = 4.3458e-03

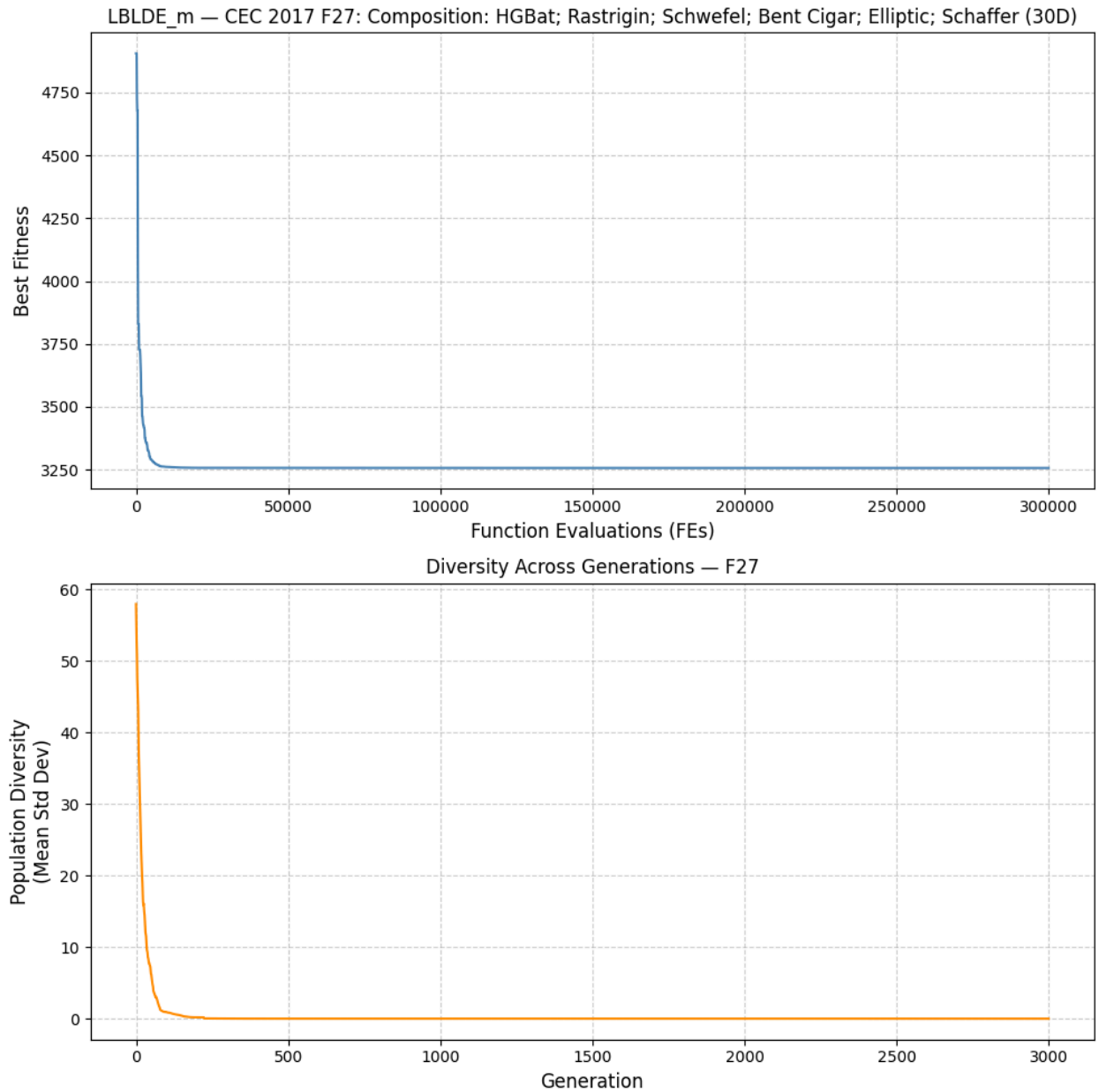
Optimization completed!

Final Best Fitness = 3.256224e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 3.256224e+03
  Global Optimum         : 2.700000e+03
  Error (|best-optimum|) : 5.562245e+02
  Best Solution (5 dim)  : [-94.94495663 -72.2582117 -99.47902863 -58.25401183
-24.91160631]
  D^(0) (init. diversity): 5.7976e+01
```

II



Graph saved as convergence_F27.png

```
In [54]: # =====  
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock; HappyCat  
# =====  
run_and_plot('F28', make_wrapper(composition.f28), 2800.0, 'Composition: Ackle
```

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LBLDE_m on CEC 2017 F28 – Single Run

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Function : Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat; S
chaffer (30D)

Bounds : [-100, 100]

Global optimum : 2800.0

Initial $D^*(0) = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 1.208847e+04 | Diversity = 5.7976e+01

Generation 10: Best = 5.821938e+03 FES = 1100 Diversity = 2.2581e+01

Generation 20: Best = 4.741516e+03 FES = 2100 Diversity = 8.7796e+00

Generation 30: Best = 4.389447e+03 FES = 3100 Diversity = 4.5585e+00

Generation 40: Best = 4.243347e+03 FES = 4100 Diversity = 2.4449e+00

Generation 50: Best = 4.171381e+03 FES = 5100 Diversity = 1.3605e+00

Generation 60: Best = 4.122833e+03 FES = 6100 Diversity = 1.0262e+00

Generation 70: Best = 4.088328e+03 FES = 7100 Diversity = 9.9001e-01

Generation 80: Best = 4.056328e+03 FES = 8100 Diversity = 9.7164e-01

Generation 90: Best = 4.022483e+03 FES = 9100 Diversity = 1.3523e+00

Generation 100: Best = 3.997023e+03 FES = 10100 Diversity = 1.0238e+00

Generation 110: Best = 3.976933e+03 FES = 11100 Diversity = 1.1631e+00

Generation 120: Best = 3.957763e+03 FES = 12100 Diversity = 1.2702e+00

Generation 130: Best = 3.937180e+03 FES = 13100 Diversity = 8.6497e-01

Generation 140: Best = 3.921009e+03 FES = 14100 Diversity = 7.4128e-01

Generation 150: Best = 3.905160e+03 FES = 15100 Diversity = 6.9455e-01

Generation 160: Best = 3.893730e+03 FES = 16100 Diversity = 5.9430e-01

Generation 170: Best = 3.885140e+03 FES = 17100 Diversity = 5.3288e-01

Generation 180: Best = 3.874733e+03 FES = 18100 Diversity = 4.8828e-01

Generation 190: Best = 3.867997e+03 FES = 19100 Diversity = 3.6913e-01

Generation 200: Best = 3.862253e+03 FES = 20100 Diversity = 3.2894e-01

Generation 210: Best = 3.859361e+03 FES = 21100 Diversity = 3.0207e-01

Generation 220: Best = 3.856834e+03 FES = 22100 Diversity = 2.5150e-01

Generation 230: Best = 3.855096e+03 FES = 23100 Diversity = 2.0474e-01

Generation 240: Best = 3.853472e+03 FES = 24100 Diversity = 1.3999e-01

Generation 250: Best = 3.851791e+03 FES = 25100 Diversity = 1.3538e-01

Generation 260: Best = 3.850408e+03 FES = 26100 Diversity = 1.3058e-01

Generation 270: Best = 3.849742e+03 FES = 27100 Diversity = 9.1159e-02

Generation 280: Best = 3.849222e+03 FES = 28100 Diversity = 8.6549e-02

Generation 290: Best = 3.848708e+03 FES = 29100 Diversity = 8.2752e-02

Generation 300: Best = 3.848358e+03 FES = 30100 Diversity = 6.2174e-02

Generation 310: Best = 3.848005e+03 FES = 31100 Diversity = 5.3619e-02

Generation 320: Best = 3.847812e+03 FES = 32100 Diversity = 4.8110e-02

Generation 330: Best = 3.847647e+03 FES = 33100 Diversity = 3.7351e-02

Generation 340: Best = 3.847462e+03 FES = 34100 Diversity = 3.3176e-02

Generation 350: Best = 3.847338e+03 FES = 35100 Diversity = 3.1120e-02

Generation 360: Best = 3.847246e+03 FES = 36100 Diversity = 2.6117e-02

Generation 370: Best = 3.847172e+03 FES = 37100 Diversity = 1.7542e-02

Generation 380: Best = 3.847138e+03 FES = 38100 Diversity = 1.1854e-02

Generation 390: Best = 3.847108e+03 FES = 39100 Diversity = 1.0130e-02

Generation 400: Best = 3.847079e+03 FES = 40100 Diversity = 9.4100e-03

Generation 410: Best = 3.847060e+03 FES = 41100 Diversity = 8.1528e-03

Generation 420: Best = 3.847039e+03 FES = 42100 Diversity = 7.8591e-03

Generation 430: Best = 3.847020e+03 FES = 43100 Diversity = 6.9450e-03

Generation	440:	Best = 3.847005e+03	FES = 44100	Diversity = 5.8390e-03
Generation	450:	Best = 3.847000e+03	FES = 45100	Diversity = 4.6760e-03
Generation	460:	Best = 3.846995e+03	FES = 46100	Diversity = 3.3114e-03
Generation	470:	Best = 3.846992e+03	FES = 47100	Diversity = 2.9691e-03
Generation	480:	Best = 3.846991e+03	FES = 48100	Diversity = 2.1993e-03
Generation	490:	Best = 3.846989e+03	FES = 49100	Diversity = 1.8609e-03
Generation	500:	Best = 3.846988e+03	FES = 50100	Diversity = 1.2611e-03
Generation	510:	Best = 3.846987e+03	FES = 51100	Diversity = 1.4164e-03
Generation	520:	Best = 3.846986e+03	FES = 52100	Diversity = 1.0767e-03
Generation	530:	Best = 3.846986e+03	FES = 53100	Diversity = 1.0589e-03
Generation	540:	Best = 3.846985e+03	FES = 54100	Diversity = 8.6946e-04
Generation	550:	Best = 3.846985e+03	FES = 55100	Diversity = 6.4722e-04
Generation	560:	Best = 3.846985e+03	FES = 56100	Diversity = 4.1766e-04
Generation	570:	Best = 3.846985e+03	FES = 57100	Diversity = 3.6175e-04
Generation	580:	Best = 3.846985e+03	FES = 58100	Diversity = 2.8889e-04
Generation	590:	Best = 3.846985e+03	FES = 59100	Diversity = 1.6847e-04
Generation	600:	Best = 3.846985e+03	FES = 60100	Diversity = 1.9211e-04
Generation	610:	Best = 3.846985e+03	FES = 61100	Diversity = 1.9045e-04
Generation	620:	Best = 3.846985e+03	FES = 62100	Diversity = 1.5368e-04
Generation	630:	Best = 3.846985e+03	FES = 63100	Diversity = 8.9020e-05
Generation	640:	Best = 3.846985e+03	FES = 64100	Diversity = 6.9749e-05
Generation	650:	Best = 3.846985e+03	FES = 65100	Diversity = 6.0223e-05
Generation	660:	Best = 3.846985e+03	FES = 66100	Diversity = 6.2445e-05
Generation	670:	Best = 3.846985e+03	FES = 67100	Diversity = 3.7110e-05
Generation	680:	Best = 3.846985e+03	FES = 68100	Diversity = 3.1461e-05
Generation	690:	Best = 3.846985e+03	FES = 69100	Diversity = 3.0435e-05
Generation	700:	Best = 3.846985e+03	FES = 70100	Diversity = 1.9405e-05
Generation	710:	Best = 3.846985e+03	FES = 71100	Diversity = 1.3563e-05
Generation	720:	Best = 3.846985e+03	FES = 72100	Diversity = 1.2843e-05
Generation	730:	Best = 3.846985e+03	FES = 73100	Diversity = 9.9839e-06
Generation	740:	Best = 3.846985e+03	FES = 74100	Diversity = 7.0167e-06
Generation	750:	Best = 3.846985e+03	FES = 75100	Diversity = 7.1226e-06
Generation	760:	Best = 3.846985e+03	FES = 76100	Diversity = 4.2070e-06
Generation	770:	Best = 3.846985e+03	FES = 77100	Diversity = 4.1284e-06
Generation	780:	Best = 3.846985e+03	FES = 78100	Diversity = 3.2802e-06
Generation	790:	Best = 3.846985e+03	FES = 79100	Diversity = 2.7586e-06
Generation	800:	Best = 3.846985e+03	FES = 80100	Diversity = 1.7010e-06
Generation	810:	Best = 3.846985e+03	FES = 81100	Diversity = 1.2929e-06
Generation	820:	Best = 3.846985e+03	FES = 82100	Diversity = 8.3750e-07
Generation	830:	Best = 3.846985e+03	FES = 83100	Diversity = 8.7461e-07
Generation	840:	Best = 3.846985e+03	FES = 84100	Diversity = 7.4364e-07
Generation	850:	Best = 3.846985e+03	FES = 85100	Diversity = 5.1863e-07
Generation	860:	Best = 3.846985e+03	FES = 86100	Diversity = 3.8327e-07
Generation	870:	Best = 3.846985e+03	FES = 87100	Diversity = 2.9434e-07
Generation	880:	Best = 3.846985e+03	FES = 88100	Diversity = 2.4604e-07
Generation	890:	Best = 3.846985e+03	FES = 89100	Diversity = 2.1967e-07
Generation	900:	Best = 3.846985e+03	FES = 90100	Diversity = 8.4235e-08
Generation	910:	Best = 3.846985e+03	FES = 91100	Diversity = 8.7947e-08
Generation	920:	Best = 3.846985e+03	FES = 92100	Diversity = 7.4826e-08
Generation	930:	Best = 3.846985e+03	FES = 93100	Diversity = 5.2697e-08
Generation	940:	Best = 3.846985e+03	FES = 94100	Diversity = 4.0234e-08
Generation	950:	Best = 3.846985e+03	FES = 95100	Diversity = 2.1975e-08
Generation	960:	Best = 3.846985e+03	FES = 96100	Diversity = 5.3318e-09
Generation	970:	Best = 3.846985e+03	FES = 97100	Diversity = 7.6439e-10

[illegible]

[illegible]

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Generation 2600:	Best = 3.846985e+03	FES = 260100	Diversity = 3.5135e-14
Generation 2610:	Best = 3.846985e+03	FES = 261100	Diversity = 3.5135e-14
Generation 2620:	Best = 3.846985e+03	FES = 262100	Diversity = 3.5135e-14
Generation 2630:	Best = 3.846985e+03	FES = 263100	Diversity = 3.5135e-14
Generation 2640:	Best = 3.846985e+03	FES = 264100	Diversity = 3.5135e-14
Generation 2650:	Best = 3.846985e+03	FES = 265100	Diversity = 3.5135e-14
Generation 2660:	Best = 3.846985e+03	FES = 266100	Diversity = 3.5135e-14
Generation 2670:	Best = 3.846985e+03	FES = 267100	Diversity = 3.5135e-14
Generation 2680:	Best = 3.846985e+03	FES = 268100	Diversity = 3.5135e-14
Generation 2690:	Best = 3.846985e+03	FES = 269100	Diversity = 3.5135e-14
Generation 2700:	Best = 3.846985e+03	FES = 270100	Diversity = 3.5135e-14
Generation 2710:	Best = 3.846985e+03	FES = 271100	Diversity = 3.5135e-14
Generation 2720:	Best = 3.846985e+03	FES = 272100	Diversity = 3.5135e-14
Generation 2730:	Best = 3.846985e+03	FES = 273100	Diversity = 3.5135e-14
Generation 2740:	Best = 3.846985e+03	FES = 274100	Diversity = 3.5135e-14
Generation 2750:	Best = 3.846985e+03	FES = 275100	Diversity = 3.5135e-14
Generation 2760:	Best = 3.846985e+03	FES = 276100	Diversity = 3.5135e-14
Generation 2770:	Best = 3.846985e+03	FES = 277100	Diversity = 3.5135e-14
Generation 2780:	Best = 3.846985e+03	FES = 278100	Diversity = 3.5135e-14
Generation 2790:	Best = 3.846985e+03	FES = 279100	Diversity = 3.5135e-14
Generation 2800:	Best = 3.846985e+03	FES = 280100	Diversity = 3.5135e-14
Generation 2810:	Best = 3.846985e+03	FES = 281100	Diversity = 3.5135e-14
Generation 2820:	Best = 3.846985e+03	FES = 282100	Diversity = 3.5135e-14
Generation 2830:	Best = 3.846985e+03	FES = 283100	Diversity = 3.5135e-14
Generation 2840:	Best = 3.846985e+03	FES = 284100	Diversity = 3.5135e-14
Generation 2850:	Best = 3.846985e+03	FES = 285100	Diversity = 3.5135e-14
Generation 2860:	Best = 3.846985e+03	FES = 286100	Diversity = 3.5135e-14
Generation 2870:	Best = 3.846985e+03	FES = 287100	Diversity = 3.5135e-14
Generation 2880:	Best = 3.846985e+03	FES = 288100	Diversity = 3.5135e-14
Generation 2890:	Best = 3.846985e+03	FES = 289100	Diversity = 3.5135e-14
Generation 2900:	Best = 3.846985e+03	FES = 290100	Diversity = 3.5135e-14
Generation 2910:	Best = 3.846985e+03	FES = 291100	Diversity = 3.5135e-14
Generation 2920:	Best = 3.846985e+03	FES = 292100	Diversity = 3.5135e-14
Generation 2930:	Best = 3.846985e+03	FES = 293100	Diversity = 3.5135e-14
Generation 2940:	Best = 3.846985e+03	FES = 294100	Diversity = 3.5135e-14
Generation 2950:	Best = 3.846985e+03	FES = 295100	Diversity = 3.5135e-14
Generation 2960:	Best = 3.846985e+03	FES = 296100	Diversity = 3.5135e-14
Generation 2970:	Best = 3.846985e+03	FES = 297100	Diversity = 3.5135e-14
Generation 2980:	Best = 3.846985e+03	FES = 298100	Diversity = 3.5135e-14
Generation 2990:	Best = 3.846985e+03	FES = 299100	Diversity = 3.5135e-14

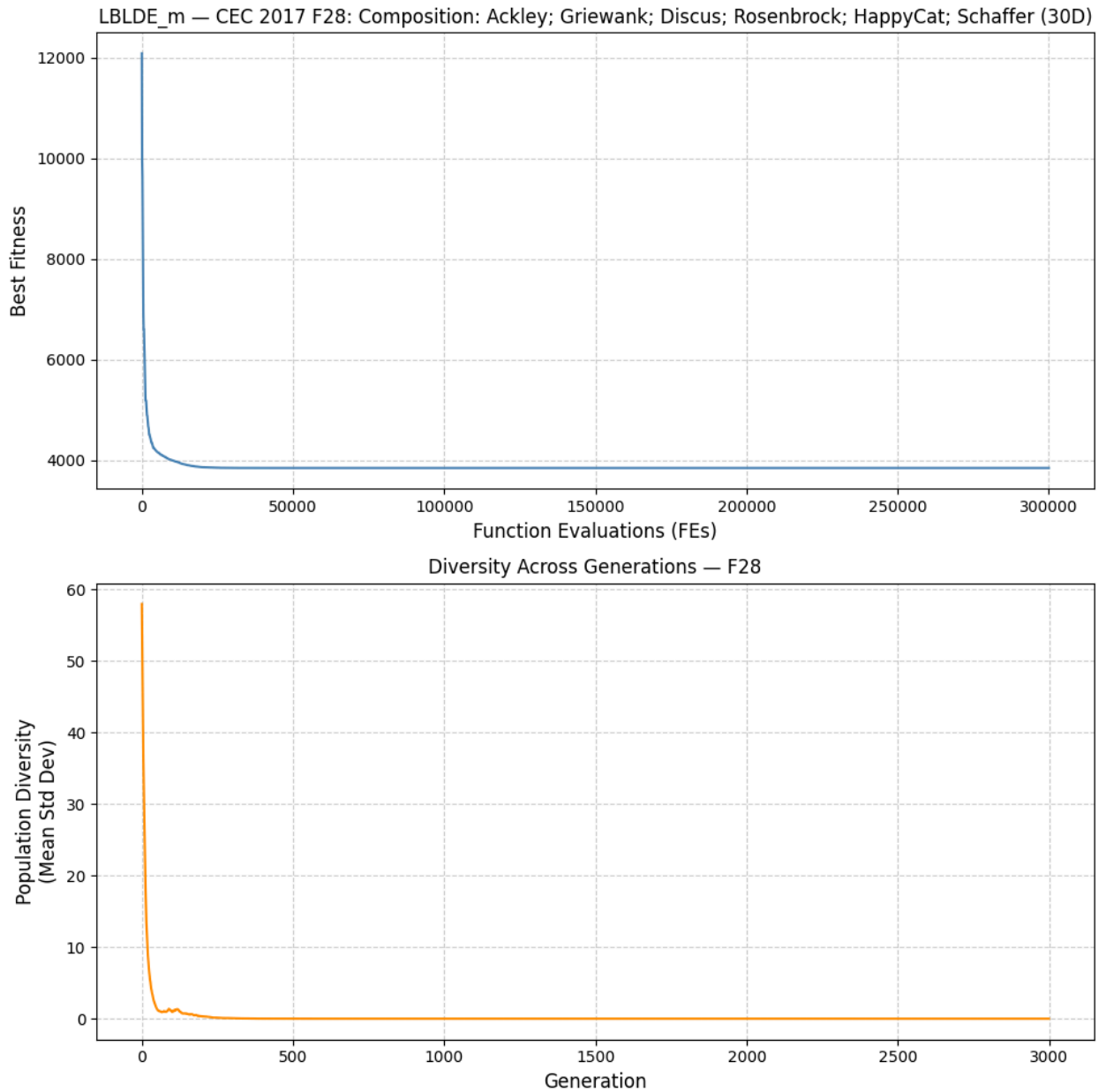
Optimization completed!

Final Best Fitness = 3.846985e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness           : 3.846985e+03
  Global Optimum         : 2.800000e+03
  Error (|best-optimum|) : 1.046985e+03
  Best Solution (5 dim)  : [-38.22329487  10.83150092  18.25059028  55.05074297
-23.03579385]
  D^(0) (init. diversity): 5.7976e+01
```

II



Graph saved as convergence_F28.png

```
In [55]: # =====  
# F29 — Composition Function 9 (F15; F16; F17)  
# =====  
run_and_plot('F29', make_wrapper(composition.f29), 2900.0, 'Composition: F15;
```

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LBLDE_m on CEC 2017 F29 – Single Run

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Function : Composition: F15; F16; F17 (30D)
Bounds : [-100, 100]
Global optimum : 2900.0
Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00
Generation 0: Best Fitness = 8.977257e+03 | Diversity = 5.7976e+01

Generation	10:	Best = 6.129356e+03	FES = 1100	Diversity = 2.8628e+01
Generation	20:	Best = 5.204913e+03	FES = 2100	Diversity = 2.2896e+01
Generation	30:	Best = 4.763572e+03	FES = 3100	Diversity = 1.8452e+01
Generation	40:	Best = 4.452465e+03	FES = 4100	Diversity = 1.7027e+01
Generation	50:	Best = 4.260058e+03	FES = 5100	Diversity = 1.5746e+01
Generation	60:	Best = 4.260058e+03	FES = 6100	Diversity = 1.5128e+01
Generation	70:	Best = 4.240666e+03	FES = 7100	Diversity = 1.4747e+01
Generation	80:	Best = 4.121100e+03	FES = 8100	Diversity = 1.4594e+01
Generation	90:	Best = 3.943475e+03	FES = 9100	Diversity = 1.4366e+01
Generation	100:	Best = 3.943475e+03	FES = 10100	Diversity = 1.4101e+01
Generation	110:	Best = 3.917491e+03	FES = 11100	Diversity = 1.3617e+01
Generation	120:	Best = 3.917491e+03	FES = 12100	Diversity = 1.3184e+01
Generation	130:	Best = 3.876786e+03	FES = 13100	Diversity = 1.2988e+01
Generation	140:	Best = 3.867790e+03	FES = 14100	Diversity = 1.3132e+01
Generation	150:	Best = 3.854659e+03	FES = 15100	Diversity = 1.2862e+01
Generation	160:	Best = 3.854659e+03	FES = 16100	Diversity = 1.2804e+01
Generation	170:	Best = 3.794983e+03	FES = 17100	Diversity = 1.2608e+01
Generation	180:	Best = 3.756990e+03	FES = 18100	Diversity = 1.2141e+01
Generation	190:	Best = 3.713313e+03	FES = 19100	Diversity = 1.2085e+01
Generation	200:	Best = 3.713313e+03	FES = 20100	Diversity = 1.1907e+01
Generation	210:	Best = 3.637671e+03	FES = 21100	Diversity = 1.1713e+01
Generation	220:	Best = 3.637671e+03	FES = 22100	Diversity = 1.1720e+01
Generation	230:	Best = 3.618810e+03	FES = 23100	Diversity = 1.1557e+01
Generation	240:	Best = 3.616322e+03	FES = 24100	Diversity = 1.1637e+01
Generation	250:	Best = 3.575438e+03	FES = 25100	Diversity = 1.1403e+01
Generation	260:	Best = 3.575438e+03	FES = 26100	Diversity = 1.1272e+01
Generation	270:	Best = 3.532264e+03	FES = 27100	Diversity = 1.0944e+01
Generation	280:	Best = 3.532264e+03	FES = 28100	Diversity = 1.0532e+01
Generation	290:	Best = 3.521778e+03	FES = 29100	Diversity = 1.0458e+01
Generation	300:	Best = 3.521778e+03	FES = 30100	Diversity = 1.0452e+01
Generation	310:	Best = 3.521778e+03	FES = 31100	Diversity = 1.0323e+01
Generation	320:	Best = 3.521778e+03	FES = 32100	Diversity = 1.0394e+01
Generation	330:	Best = 3.509489e+03	FES = 33100	Diversity = 1.0393e+01
Generation	340:	Best = 3.497185e+03	FES = 34100	Diversity = 1.0204e+01
Generation	350:	Best = 3.486313e+03	FES = 35100	Diversity = 1.0116e+01
Generation	360:	Best = 3.485284e+03	FES = 36100	Diversity = 9.7089e+00
Generation	370:	Best = 3.480767e+03	FES = 37100	Diversity = 9.7041e+00
Generation	380:	Best = 3.480767e+03	FES = 38100	Diversity = 9.7221e+00
Generation	390:	Best = 3.477784e+03	FES = 39100	Diversity = 9.6035e+00
Generation	400:	Best = 3.469647e+03	FES = 40100	Diversity = 9.3475e+00
Generation	410:	Best = 3.457529e+03	FES = 41100	Diversity = 9.3304e+00
Generation	420:	Best = 3.457529e+03	FES = 42100	Diversity = 9.1277e+00
Generation	430:	Best = 3.457529e+03	FES = 43100	Diversity = 9.0916e+00
Generation	440:	Best = 3.457529e+03	FES = 44100	Diversity = 9.0424e+00

Generation	450:	Best = 3.445149e+03	FES = 45100	Diversity = 8.9728e+00
Generation	460:	Best = 3.440613e+03	FES = 46100	Diversity = 8.9537e+00
Generation	470:	Best = 3.440613e+03	FES = 47100	Diversity = 8.9244e+00
Generation	480:	Best = 3.438457e+03	FES = 48100	Diversity = 8.8291e+00
Generation	490:	Best = 3.438457e+03	FES = 49100	Diversity = 8.8063e+00
Generation	500:	Best = 3.438457e+03	FES = 50100	Diversity = 8.7714e+00
Generation	510:	Best = 3.430887e+03	FES = 51100	Diversity = 8.7819e+00
Generation	520:	Best = 3.430887e+03	FES = 52100	Diversity = 8.6575e+00
Generation	530:	Best = 3.430690e+03	FES = 53100	Diversity = 8.4439e+00
Generation	540:	Best = 3.430690e+03	FES = 54100	Diversity = 8.3823e+00
Generation	550:	Best = 3.430690e+03	FES = 55100	Diversity = 8.4270e+00
Generation	560:	Best = 3.430682e+03	FES = 56100	Diversity = 8.2583e+00
Generation	570:	Best = 3.430457e+03	FES = 57100	Diversity = 8.2336e+00
Generation	580:	Best = 3.430457e+03	FES = 58100	Diversity = 8.1851e+00
Generation	590:	Best = 3.429856e+03	FES = 59100	Diversity = 8.1902e+00
Generation	600:	Best = 3.429856e+03	FES = 60100	Diversity = 7.9882e+00
Generation	610:	Best = 3.429856e+03	FES = 61100	Diversity = 7.9678e+00
Generation	620:	Best = 3.429856e+03	FES = 62100	Diversity = 7.9623e+00
Generation	630:	Best = 3.429856e+03	FES = 63100	Diversity = 7.9996e+00
Generation	640:	Best = 3.429855e+03	FES = 64100	Diversity = 7.9919e+00
Generation	650:	Best = 3.429855e+03	FES = 65100	Diversity = 7.9386e+00
Generation	660:	Best = 3.429855e+03	FES = 66100	Diversity = 7.9354e+00
Generation	670:	Best = 3.416658e+03	FES = 67100	Diversity = 7.8734e+00
Generation	680:	Best = 3.413262e+03	FES = 68100	Diversity = 7.8576e+00
Generation	690:	Best = 3.413262e+03	FES = 69100	Diversity = 7.8485e+00
Generation	700:	Best = 3.413262e+03	FES = 70100	Diversity = 7.8317e+00
Generation	710:	Best = 3.413262e+03	FES = 71100	Diversity = 7.7600e+00
Generation	720:	Best = 3.412249e+03	FES = 72100	Diversity = 7.6169e+00
Generation	730:	Best = 3.412249e+03	FES = 73100	Diversity = 7.6661e+00
Generation	740:	Best = 3.412249e+03	FES = 74100	Diversity = 7.6388e+00
Generation	750:	Best = 3.412249e+03	FES = 75100	Diversity = 7.6189e+00
Generation	760:	Best = 3.412249e+03	FES = 76100	Diversity = 7.6453e+00
Generation	770:	Best = 3.412249e+03	FES = 77100	Diversity = 7.6764e+00
Generation	780:	Best = 3.412061e+03	FES = 78100	Diversity = 7.5607e+00
Generation	790:	Best = 3.412061e+03	FES = 79100	Diversity = 7.4257e+00
Generation	800:	Best = 3.412061e+03	FES = 80100	Diversity = 7.4222e+00
Generation	810:	Best = 3.412061e+03	FES = 81100	Diversity = 7.3322e+00
Generation	820:	Best = 3.411889e+03	FES = 82100	Diversity = 7.2529e+00
Generation	830:	Best = 3.411889e+03	FES = 83100	Diversity = 7.2076e+00
Generation	840:	Best = 3.411889e+03	FES = 84100	Diversity = 7.2094e+00
Generation	850:	Best = 3.411889e+03	FES = 85100	Diversity = 7.1824e+00
Generation	860:	Best = 3.411621e+03	FES = 86100	Diversity = 7.2213e+00
Generation	870:	Best = 3.411621e+03	FES = 87100	Diversity = 7.2420e+00
Generation	880:	Best = 3.410313e+03	FES = 88100	Diversity = 7.2768e+00
Generation	890:	Best = 3.409832e+03	FES = 89100	Diversity = 7.2681e+00
Generation	900:	Best = 3.409832e+03	FES = 90100	Diversity = 7.2594e+00
Generation	910:	Best = 3.409143e+03	FES = 91100	Diversity = 7.2820e+00
Generation	920:	Best = 3.409143e+03	FES = 92100	Diversity = 7.2265e+00
Generation	930:	Best = 3.409143e+03	FES = 93100	Diversity = 7.1924e+00
Generation	940:	Best = 3.409143e+03	FES = 94100	Diversity = 7.1874e+00
Generation	950:	Best = 3.409143e+03	FES = 95100	Diversity = 7.2114e+00
Generation	960:	Best = 3.409143e+03	FES = 96100	Diversity = 7.2629e+00
Generation	970:	Best = 3.406123e+03	FES = 97100	Diversity = 7.2541e+00
Generation	980:	Best = 3.406123e+03	FES = 98100	Diversity = 7.1522e+00

Generation 990:	Best = 3.406123e+03	FES = 99100	Diversity = 7.1435e+00
Generation 1000:	Best = 3.404968e+03	FES = 100100	Diversity = 7.1242e+00
Generation 1010:	Best = 3.404968e+03	FES = 101100	Diversity = 7.1180e+00
Generation 1020:	Best = 3.404968e+03	FES = 102100	Diversity = 7.0033e+00
Generation 1030:	Best = 3.404968e+03	FES = 103100	Diversity = 6.9890e+00
Generation 1040:	Best = 3.404968e+03	FES = 104100	Diversity = 6.9932e+00
Generation 1050:	Best = 3.404968e+03	FES = 105100	Diversity = 6.9008e+00
Generation 1060:	Best = 3.404968e+03	FES = 106100	Diversity = 6.8808e+00
Generation 1070:	Best = 3.404968e+03	FES = 107100	Diversity = 6.8612e+00
Generation 1080:	Best = 3.404968e+03	FES = 108100	Diversity = 6.9048e+00
Generation 1090:	Best = 3.404968e+03	FES = 109100	Diversity = 6.9021e+00
Generation 1100:	Best = 3.404827e+03	FES = 110100	Diversity = 6.9175e+00
Generation 1110:	Best = 3.404538e+03	FES = 111100	Diversity = 6.9247e+00
Generation 1120:	Best = 3.404523e+03	FES = 112100	Diversity = 6.8944e+00
Generation 1130:	Best = 3.404475e+03	FES = 113100	Diversity = 6.9082e+00
Generation 1140:	Best = 3.404422e+03	FES = 114100	Diversity = 6.9090e+00
Generation 1150:	Best = 3.403759e+03	FES = 115100	Diversity = 6.9081e+00
Generation 1160:	Best = 3.403759e+03	FES = 116100	Diversity = 6.9099e+00
Generation 1170:	Best = 3.403759e+03	FES = 117100	Diversity = 6.9077e+00
Generation 1180:	Best = 3.403759e+03	FES = 118100	Diversity = 6.8786e+00
Generation 1190:	Best = 3.403759e+03	FES = 119100	Diversity = 6.8616e+00
Generation 1200:	Best = 3.403759e+03	FES = 120100	Diversity = 6.8481e+00
Generation 1210:	Best = 3.403054e+03	FES = 121100	Diversity = 6.8437e+00
Generation 1220:	Best = 3.402421e+03	FES = 122100	Diversity = 6.8533e+00
Generation 1230:	Best = 3.401356e+03	FES = 123100	Diversity = 6.8239e+00
Generation 1240:	Best = 3.401356e+03	FES = 124100	Diversity = 6.7973e+00
Generation 1250:	Best = 3.401356e+03	FES = 125100	Diversity = 6.7867e+00
Generation 1260:	Best = 3.401199e+03	FES = 126100	Diversity = 6.7788e+00
Generation 1270:	Best = 3.401199e+03	FES = 127100	Diversity = 6.7771e+00
Generation 1280:	Best = 3.400682e+03	FES = 128100	Diversity = 6.7263e+00
Generation 1290:	Best = 3.400434e+03	FES = 129100	Diversity = 6.7265e+00
Generation 1300:	Best = 3.400299e+03	FES = 130100	Diversity = 6.7215e+00
Generation 1310:	Best = 3.400299e+03	FES = 131100	Diversity = 6.7075e+00
Generation 1320:	Best = 3.397060e+03	FES = 132100	Diversity = 6.7079e+00
Generation 1330:	Best = 3.394555e+03	FES = 133100	Diversity = 6.7090e+00
Generation 1340:	Best = 3.394555e+03	FES = 134100	Diversity = 6.7074e+00
Generation 1350:	Best = 3.394529e+03	FES = 135100	Diversity = 6.7091e+00
Generation 1360:	Best = 3.394456e+03	FES = 136100	Diversity = 6.7057e+00
Generation 1370:	Best = 3.394456e+03	FES = 137100	Diversity = 6.6725e+00
Generation 1380:	Best = 3.394438e+03	FES = 138100	Diversity = 6.6730e+00
Generation 1390:	Best = 3.394438e+03	FES = 139100	Diversity = 6.6752e+00
Generation 1400:	Best = 3.394438e+03	FES = 140100	Diversity = 6.6912e+00
Generation 1410:	Best = 3.394438e+03	FES = 141100	Diversity = 6.6909e+00
Generation 1420:	Best = 3.394438e+03	FES = 142100	Diversity = 6.6841e+00
Generation 1430:	Best = 3.394438e+03	FES = 143100	Diversity = 6.6811e+00
Generation 1440:	Best = 3.394438e+03	FES = 144100	Diversity = 6.6822e+00
Generation 1450:	Best = 3.394438e+03	FES = 145100	Diversity = 6.6814e+00
Generation 1460:	Best = 3.394438e+03	FES = 146100	Diversity = 6.6075e+00
Generation 1470:	Best = 3.394438e+03	FES = 147100	Diversity = 6.6016e+00
Generation 1480:	Best = 3.394429e+03	FES = 148100	Diversity = 6.5959e+00
Generation 1490:	Best = 3.394429e+03	FES = 149100	Diversity = 6.5977e+00
Generation 1500:	Best = 3.394429e+03	FES = 150100	Diversity = 6.5522e+00
Generation 1510:	Best = 3.394429e+03	FES = 151100	Diversity = 6.5466e+00
Generation 1520:	Best = 3.394429e+03	FES = 152100	Diversity = 6.5451e+00

Generation 1530:	Best = 3.394429e+03	FES = 153100	Diversity = 6.5396e+00
Generation 1540:	Best = 3.394429e+03	FES = 154100	Diversity = 6.5406e+00
Generation 1550:	Best = 3.394429e+03	FES = 155100	Diversity = 6.4409e+00
Generation 1560:	Best = 3.394383e+03	FES = 156100	Diversity = 6.3975e+00
Generation 1570:	Best = 3.394383e+03	FES = 157100	Diversity = 6.3860e+00
Generation 1580:	Best = 3.394383e+03	FES = 158100	Diversity = 6.3873e+00
Generation 1590:	Best = 3.394383e+03	FES = 159100	Diversity = 6.3856e+00
Generation 1600:	Best = 3.382738e+03	FES = 160100	Diversity = 6.3559e+00
Generation 1610:	Best = 3.382738e+03	FES = 161100	Diversity = 6.3544e+00
Generation 1620:	Best = 3.382738e+03	FES = 162100	Diversity = 6.3536e+00
Generation 1630:	Best = 3.382738e+03	FES = 163100	Diversity = 6.3539e+00
Generation 1640:	Best = 3.378315e+03	FES = 164100	Diversity = 6.2928e+00
Generation 1650:	Best = 3.378315e+03	FES = 165100	Diversity = 6.2914e+00
Generation 1660:	Best = 3.376540e+03	FES = 166100	Diversity = 6.2917e+00
Generation 1670:	Best = 3.376540e+03	FES = 167100	Diversity = 6.2903e+00
Generation 1680:	Best = 3.376540e+03	FES = 168100	Diversity = 6.2258e+00
Generation 1690:	Best = 3.376540e+03	FES = 169100	Diversity = 6.2168e+00
Generation 1700:	Best = 3.376540e+03	FES = 170100	Diversity = 6.2279e+00
Generation 1710:	Best = 3.376540e+03	FES = 171100	Diversity = 6.2278e+00
Generation 1720:	Best = 3.376540e+03	FES = 172100	Diversity = 6.2134e+00
Generation 1730:	Best = 3.376156e+03	FES = 173100	Diversity = 6.2110e+00
Generation 1740:	Best = 3.376156e+03	FES = 174100	Diversity = 6.2090e+00
Generation 1750:	Best = 3.376156e+03	FES = 175100	Diversity = 6.2066e+00
Generation 1760:	Best = 3.376156e+03	FES = 176100	Diversity = 6.2053e+00
Generation 1770:	Best = 3.375982e+03	FES = 177100	Diversity = 6.2057e+00
Generation 1780:	Best = 3.375777e+03	FES = 178100	Diversity = 6.1734e+00
Generation 1790:	Best = 3.375659e+03	FES = 179100	Diversity = 6.1757e+00
Generation 1800:	Best = 3.375659e+03	FES = 180100	Diversity = 6.1755e+00
Generation 1810:	Best = 3.375659e+03	FES = 181100	Diversity = 6.1741e+00
Generation 1820:	Best = 3.375659e+03	FES = 182100	Diversity = 6.1809e+00
Generation 1830:	Best = 3.375374e+03	FES = 183100	Diversity = 6.1687e+00
Generation 1840:	Best = 3.375374e+03	FES = 184100	Diversity = 6.1673e+00
Generation 1850:	Best = 3.375374e+03	FES = 185100	Diversity = 6.1650e+00
Generation 1860:	Best = 3.375374e+03	FES = 186100	Diversity = 6.1664e+00
Generation 1870:	Best = 3.375374e+03	FES = 187100	Diversity = 6.1709e+00
Generation 1880:	Best = 3.375374e+03	FES = 188100	Diversity = 6.1761e+00
Generation 1890:	Best = 3.375374e+03	FES = 189100	Diversity = 6.1818e+00
Generation 1900:	Best = 3.375242e+03	FES = 190100	Diversity = 6.1803e+00
Generation 1910:	Best = 3.375043e+03	FES = 191100	Diversity = 6.1792e+00
Generation 1920:	Best = 3.375043e+03	FES = 192100	Diversity = 6.1482e+00
Generation 1930:	Best = 3.375043e+03	FES = 193100	Diversity = 6.1574e+00
Generation 1940:	Best = 3.375043e+03	FES = 194100	Diversity = 6.1327e+00
Generation 1950:	Best = 3.375043e+03	FES = 195100	Diversity = 6.1256e+00
Generation 1960:	Best = 3.375043e+03	FES = 196100	Diversity = 6.1289e+00
Generation 1970:	Best = 3.375043e+03	FES = 197100	Diversity = 6.1291e+00
Generation 1980:	Best = 3.375043e+03	FES = 198100	Diversity = 6.1126e+00
Generation 1990:	Best = 3.375043e+03	FES = 199100	Diversity = 6.0464e+00
Generation 2000:	Best = 3.375043e+03	FES = 200100	Diversity = 6.0422e+00
Generation 2010:	Best = 3.375043e+03	FES = 201100	Diversity = 6.0421e+00
Generation 2020:	Best = 3.375043e+03	FES = 202100	Diversity = 6.0403e+00
Generation 2030:	Best = 3.375043e+03	FES = 203100	Diversity = 6.0381e+00
Generation 2040:	Best = 3.375043e+03	FES = 204100	Diversity = 6.0039e+00
Generation 2050:	Best = 3.375043e+03	FES = 205100	Diversity = 6.0052e+00
Generation 2060:	Best = 3.375043e+03	FES = 206100	Diversity = 6.0057e+00

Generation 2070:	Best = 3.375043e+03	FES = 207100	Diversity = 5.9990e+00
Generation 2080:	Best = 3.375043e+03	FES = 208100	Diversity = 6.0024e+00
Generation 2090:	Best = 3.375043e+03	FES = 209100	Diversity = 5.9855e+00
Generation 2100:	Best = 3.375043e+03	FES = 210100	Diversity = 5.9874e+00
Generation 2110:	Best = 3.375043e+03	FES = 211100	Diversity = 5.9880e+00
Generation 2120:	Best = 3.375043e+03	FES = 212100	Diversity = 5.9875e+00
Generation 2130:	Best = 3.375043e+03	FES = 213100	Diversity = 6.0038e+00
Generation 2140:	Best = 3.375043e+03	FES = 214100	Diversity = 6.0033e+00
Generation 2150:	Best = 3.375043e+03	FES = 215100	Diversity = 5.9602e+00
Generation 2160:	Best = 3.375043e+03	FES = 216100	Diversity = 5.9628e+00
Generation 2170:	Best = 3.374311e+03	FES = 217100	Diversity = 5.9611e+00
Generation 2180:	Best = 3.374311e+03	FES = 218100	Diversity = 5.9579e+00
Generation 2190:	Best = 3.374311e+03	FES = 219100	Diversity = 5.9577e+00
Generation 2200:	Best = 3.374311e+03	FES = 220100	Diversity = 5.9554e+00
Generation 2210:	Best = 3.374311e+03	FES = 221100	Diversity = 5.9537e+00
Generation 2220:	Best = 3.374311e+03	FES = 222100	Diversity = 5.9272e+00
Generation 2230:	Best = 3.374228e+03	FES = 223100	Diversity = 5.9266e+00
Generation 2240:	Best = 3.374228e+03	FES = 224100	Diversity = 5.9418e+00
Generation 2250:	Best = 3.374228e+03	FES = 225100	Diversity = 5.9443e+00
Generation 2260:	Best = 3.374228e+03	FES = 226100	Diversity = 5.9440e+00
Generation 2270:	Best = 3.374228e+03	FES = 227100	Diversity = 5.9367e+00
Generation 2280:	Best = 3.374228e+03	FES = 228100	Diversity = 5.9320e+00
Generation 2290:	Best = 3.374206e+03	FES = 229100	Diversity = 5.9310e+00
Generation 2300:	Best = 3.373955e+03	FES = 230100	Diversity = 5.9363e+00
Generation 2310:	Best = 3.373955e+03	FES = 231100	Diversity = 5.9345e+00
Generation 2320:	Best = 3.373955e+03	FES = 232100	Diversity = 5.9342e+00
Generation 2330:	Best = 3.373955e+03	FES = 233100	Diversity = 5.9338e+00
Generation 2340:	Best = 3.373955e+03	FES = 234100	Diversity = 5.9316e+00
Generation 2350:	Best = 3.373955e+03	FES = 235100	Diversity = 5.9321e+00
Generation 2360:	Best = 3.373955e+03	FES = 236100	Diversity = 5.9204e+00
Generation 2370:	Best = 3.373955e+03	FES = 237100	Diversity = 5.9201e+00
Generation 2380:	Best = 3.373955e+03	FES = 238100	Diversity = 5.9249e+00
Generation 2390:	Best = 3.373955e+03	FES = 239100	Diversity = 5.8946e+00
Generation 2400:	Best = 3.373955e+03	FES = 240100	Diversity = 5.8943e+00
Generation 2410:	Best = 3.373955e+03	FES = 241100	Diversity = 5.8926e+00
Generation 2420:	Best = 3.373955e+03	FES = 242100	Diversity = 5.9075e+00
Generation 2430:	Best = 3.373955e+03	FES = 243100	Diversity = 5.8709e+00
Generation 2440:	Best = 3.373955e+03	FES = 244100	Diversity = 5.7755e+00
Generation 2450:	Best = 3.373944e+03	FES = 245100	Diversity = 5.7187e+00
Generation 2460:	Best = 3.373944e+03	FES = 246100	Diversity = 5.7103e+00
Generation 2470:	Best = 3.373944e+03	FES = 247100	Diversity = 5.7063e+00
Generation 2480:	Best = 3.373944e+03	FES = 248100	Diversity = 5.7050e+00
Generation 2490:	Best = 3.373944e+03	FES = 249100	Diversity = 5.7062e+00
Generation 2500:	Best = 3.373944e+03	FES = 250100	Diversity = 5.7177e+00
Generation 2510:	Best = 3.373944e+03	FES = 251100	Diversity = 5.7167e+00
Generation 2520:	Best = 3.373944e+03	FES = 252100	Diversity = 5.7213e+00
Generation 2530:	Best = 3.373944e+03	FES = 253100	Diversity = 5.7260e+00
Generation 2540:	Best = 3.373944e+03	FES = 254100	Diversity = 5.7254e+00
Generation 2550:	Best = 3.373944e+03	FES = 255100	Diversity = 5.7348e+00
Generation 2560:	Best = 3.373940e+03	FES = 256100	Diversity = 5.7300e+00
Generation 2570:	Best = 3.373940e+03	FES = 257100	Diversity = 5.7142e+00
Generation 2580:	Best = 3.373940e+03	FES = 258100	Diversity = 5.6986e+00
Generation 2590:	Best = 3.373940e+03	FES = 259100	Diversity = 5.6983e+00
Generation 2600:	Best = 3.373934e+03	FES = 260100	Diversity = 5.7072e+00

Generation 2610:	Best = 3.373934e+03	FES = 261100	Diversity = 5.7065e+00
Generation 2620:	Best = 3.373231e+03	FES = 262100	Diversity = 5.7065e+00
Generation 2630:	Best = 3.373231e+03	FES = 263100	Diversity = 5.7086e+00
Generation 2640:	Best = 3.373231e+03	FES = 264100	Diversity = 5.7088e+00
Generation 2650:	Best = 3.373145e+03	FES = 265100	Diversity = 5.7077e+00
Generation 2660:	Best = 3.373140e+03	FES = 266100	Diversity = 5.7085e+00
Generation 2670:	Best = 3.373140e+03	FES = 267100	Diversity = 5.6752e+00
Generation 2680:	Best = 3.373140e+03	FES = 268100	Diversity = 5.6691e+00
Generation 2690:	Best = 3.373140e+03	FES = 269100	Diversity = 5.6677e+00
Generation 2700:	Best = 3.373140e+03	FES = 270100	Diversity = 5.6547e+00
Generation 2710:	Best = 3.373140e+03	FES = 271100	Diversity = 5.6792e+00
Generation 2720:	Best = 3.373140e+03	FES = 272100	Diversity = 5.6917e+00
Generation 2730:	Best = 3.373140e+03	FES = 273100	Diversity = 5.7247e+00
Generation 2740:	Best = 3.373140e+03	FES = 274100	Diversity = 5.7242e+00
Generation 2750:	Best = 3.373140e+03	FES = 275100	Diversity = 5.7218e+00
Generation 2760:	Best = 3.373140e+03	FES = 276100	Diversity = 5.7099e+00
Generation 2770:	Best = 3.373140e+03	FES = 277100	Diversity = 5.6965e+00
Generation 2780:	Best = 3.373119e+03	FES = 278100	Diversity = 5.6944e+00
Generation 2790:	Best = 3.373119e+03	FES = 279100	Diversity = 5.7077e+00
Generation 2800:	Best = 3.373119e+03	FES = 280100	Diversity = 5.7080e+00
Generation 2810:	Best = 3.373119e+03	FES = 281100	Diversity = 5.7075e+00
Generation 2820:	Best = 3.373119e+03	FES = 282100	Diversity = 5.7083e+00
Generation 2830:	Best = 3.373119e+03	FES = 283100	Diversity = 5.7072e+00
Generation 2840:	Best = 3.373119e+03	FES = 284100	Diversity = 5.6906e+00
Generation 2850:	Best = 3.373119e+03	FES = 285100	Diversity = 5.6905e+00
Generation 2860:	Best = 3.373119e+03	FES = 286100	Diversity = 5.7075e+00
Generation 2870:	Best = 3.373119e+03	FES = 287100	Diversity = 5.7067e+00
Generation 2880:	Best = 3.373119e+03	FES = 288100	Diversity = 5.7070e+00
Generation 2890:	Best = 3.373119e+03	FES = 289100	Diversity = 5.7061e+00
Generation 2900:	Best = 3.373119e+03	FES = 290100	Diversity = 5.7061e+00
Generation 2910:	Best = 3.373119e+03	FES = 291100	Diversity = 5.7047e+00
Generation 2920:	Best = 3.373119e+03	FES = 292100	Diversity = 5.7042e+00
Generation 2930:	Best = 3.372907e+03	FES = 293100	Diversity = 5.7041e+00
Generation 2940:	Best = 3.372907e+03	FES = 294100	Diversity = 5.7061e+00
Generation 2950:	Best = 3.372907e+03	FES = 295100	Diversity = 5.7049e+00
Generation 2960:	Best = 3.372907e+03	FES = 296100	Diversity = 5.7036e+00
Generation 2970:	Best = 3.372907e+03	FES = 297100	Diversity = 5.7034e+00
Generation 2980:	Best = 3.372907e+03	FES = 298100	Diversity = 5.7024e+00
Generation 2990:	Best = 3.372907e+03	FES = 299100	Diversity = 5.7019e+00

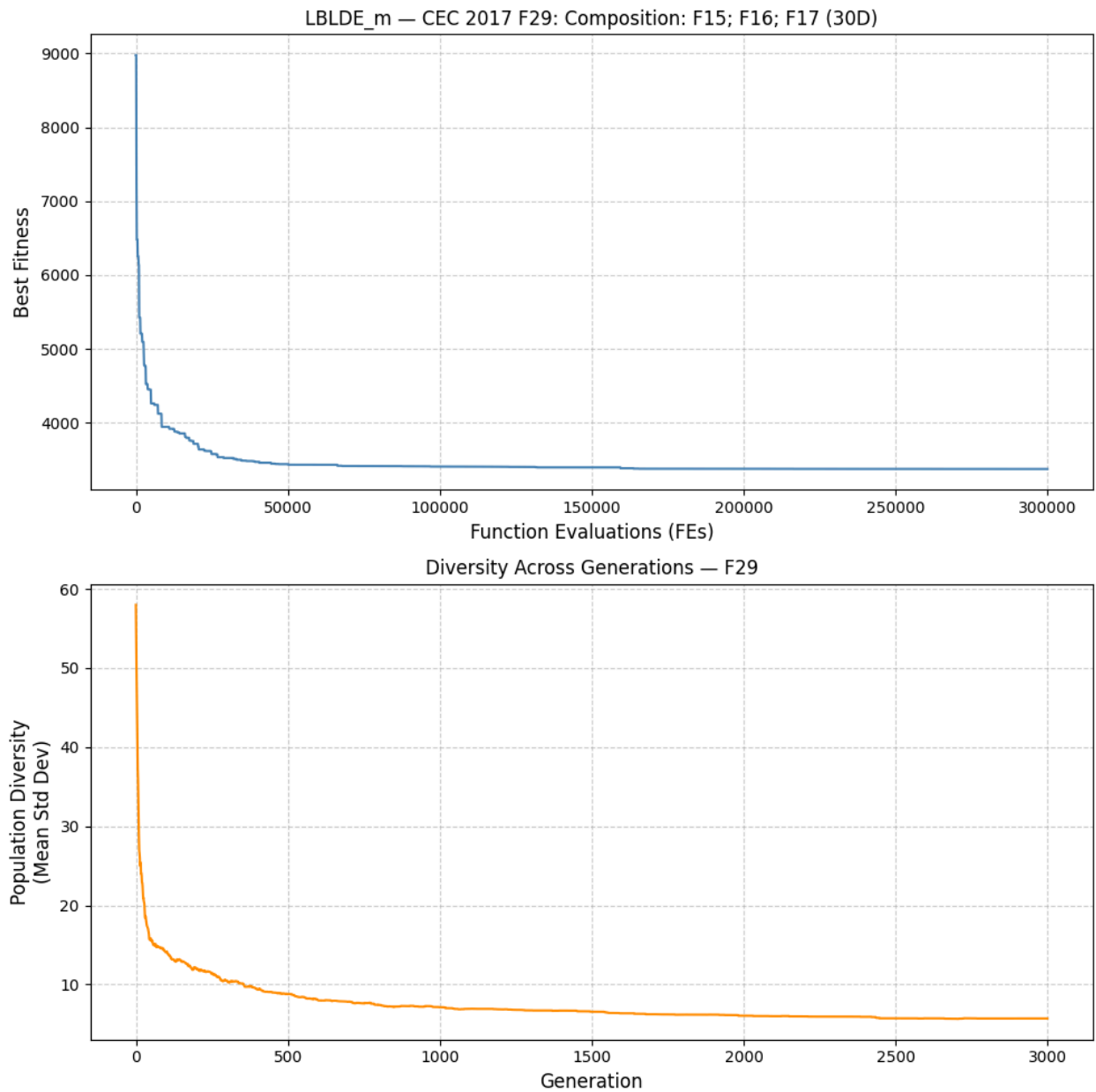
Optimization completed!

Final Best Fitness = 3.372907e+03

Total Function Evaluations = 300000

```
=====
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Results:
  Best Fitness           : 3.372907e+03
  Global Optimum         : 2.900000e+03
  Error (|best-optimum|) : 4.729068e+02
  Best Solution (5 dim)  : [-41.99809585 -37.52103473 -22.8117433  98.58871835
36.27083764]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F29.png

```
In [56]: # =====  
# F30 — Composition Function 10 (F15; F18; F19)  
# =====  
run_and_plot('F30', make_wrapper(composition.f30), 3000.0, 'Composition: F15;
```

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LBLDE_m on CEC 2017 F30 – Single Run

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Function : Composition: F15; F18; F19 (30D)

Bounds : [-100, 100]

Global optimum : 3000.0

Initial $D^{(0)} = 5.7976e+01$ | Diversity threshold = 5.7976e+00

Generation 0: Best Fitness = 3.426498e+10 | Diversity = 5.7976e+01

Generation	10:	Best = 1.176319e+09	FES = 1100	Diversity = 2.7703e+01
Generation	20:	Best = 1.427063e+08	FES = 2100	Diversity = 1.5958e+01
Generation	30:	Best = 3.966096e+07	FES = 3100	Diversity = 1.2156e+01
Generation	40:	Best = 2.417788e+07	FES = 4100	Diversity = 8.9410e+00
Generation	50:	Best = 1.860363e+07	FES = 5100	Diversity = 7.0480e+00
Generation	60:	Best = 1.253034e+07	FES = 6100	Diversity = 6.5769e+00
Generation	70:	Best = 1.123589e+07	FES = 7100	Diversity = 7.3551e+00
Generation	80:	Best = 8.653035e+06	FES = 8100	Diversity = 7.0879e+00
Generation	90:	Best = 6.596329e+06	FES = 9100	Diversity = 7.8409e+00
Generation	100:	Best = 5.073371e+06	FES = 10100	Diversity = 8.7492e+00
Generation	110:	Best = 3.556282e+06	FES = 11100	Diversity = 9.8823e+00
Generation	120:	Best = 2.758083e+06	FES = 12100	Diversity = 1.1994e+01
Generation	130:	Best = 2.098440e+06	FES = 13100	Diversity = 1.4087e+01
Generation	140:	Best = 1.675161e+06	FES = 14100	Diversity = 1.5139e+01
Generation	150:	Best = 9.276348e+05	FES = 15100	Diversity = 1.6193e+01
Generation	160:	Best = 6.466235e+05	FES = 16100	Diversity = 1.8292e+01
Generation	170:	Best = 3.870975e+05	FES = 17100	Diversity = 2.1385e+01
Generation	180:	Best = 3.219870e+05	FES = 18100	Diversity = 1.9858e+01
Generation	190:	Best = 2.174362e+05	FES = 19100	Diversity = 1.6342e+01
Generation	200:	Best = 1.388460e+05	FES = 20100	Diversity = 1.4152e+01
Generation	210:	Best = 6.369670e+04	FES = 21100	Diversity = 1.2161e+01
Generation	220:	Best = 5.418617e+04	FES = 22100	Diversity = 1.0270e+01
Generation	230:	Best = 3.984730e+04	FES = 23100	Diversity = 1.0068e+01
Generation	240:	Best = 3.149127e+04	FES = 24100	Diversity = 8.5225e+00
Generation	250:	Best = 2.905659e+04	FES = 25100	Diversity = 7.6191e+00
Generation	260:	Best = 2.381872e+04	FES = 26100	Diversity = 6.2192e+00
Generation	270:	Best = 2.074706e+04	FES = 27100	Diversity = 5.1059e+00
Generation	280:	Best = 1.901243e+04	FES = 28100	Diversity = 5.7733e+00
Generation	290:	Best = 1.766198e+04	FES = 29100	Diversity = 5.7362e+00
Generation	300:	Best = 1.456946e+04	FES = 30100	Diversity = 6.0468e+00
Generation	310:	Best = 1.413065e+04	FES = 31100	Diversity = 4.8143e+00
Generation	320:	Best = 1.330878e+04	FES = 32100	Diversity = 3.7274e+00
Generation	330:	Best = 1.280770e+04	FES = 33100	Diversity = 3.5353e+00
Generation	340:	Best = 1.173194e+04	FES = 34100	Diversity = 3.2979e+00
Generation	350:	Best = 1.108194e+04	FES = 35100	Diversity = 3.5028e+00
Generation	360:	Best = 1.075180e+04	FES = 36100	Diversity = 3.2791e+00
Generation	370:	Best = 1.024287e+04	FES = 37100	Diversity = 2.9267e+00
Generation	380:	Best = 9.960936e+03	FES = 38100	Diversity = 3.2678e+00
Generation	390:	Best = 9.546153e+03	FES = 39100	Diversity = 3.0747e+00
Generation	400:	Best = 9.037743e+03	FES = 40100	Diversity = 3.2800e+00
Generation	410:	Best = 8.950720e+03	FES = 41100	Diversity = 3.3240e+00
Generation	420:	Best = 8.671666e+03	FES = 42100	Diversity = 2.9579e+00
Generation	430:	Best = 8.486425e+03	FES = 43100	Diversity = 2.7522e+00
Generation	440:	Best = 8.292385e+03	FES = 44100	Diversity = 2.6130e+00

Generation	450:	Best = 8.167082e+03	FES = 45100	Diversity = 2.5489e+00
Generation	460:	Best = 8.167082e+03	FES = 46100	Diversity = 2.3412e+00
Generation	470:	Best = 7.956789e+03	FES = 47100	Diversity = 2.5561e+00
Generation	480:	Best = 7.893545e+03	FES = 48100	Diversity = 2.3014e+00
Generation	490:	Best = 7.675456e+03	FES = 49100	Diversity = 2.2366e+00
Generation	500:	Best = 7.564547e+03	FES = 50100	Diversity = 2.1802e+00
Generation	510:	Best = 7.480597e+03	FES = 51100	Diversity = 2.0492e+00
Generation	520:	Best = 7.425618e+03	FES = 52100	Diversity = 1.9515e+00
Generation	530:	Best = 7.364770e+03	FES = 53100	Diversity = 1.8740e+00
Generation	540:	Best = 7.293931e+03	FES = 54100	Diversity = 1.5615e+00
Generation	550:	Best = 7.245521e+03	FES = 55100	Diversity = 1.5366e+00
Generation	560:	Best = 7.192259e+03	FES = 56100	Diversity = 1.4926e+00
Generation	570:	Best = 7.167973e+03	FES = 57100	Diversity = 1.3382e+00
Generation	580:	Best = 7.140099e+03	FES = 58100	Diversity = 1.2742e+00
Generation	590:	Best = 7.103493e+03	FES = 59100	Diversity = 1.1937e+00
Generation	600:	Best = 7.039866e+03	FES = 60100	Diversity = 1.2175e+00
Generation	610:	Best = 7.020809e+03	FES = 61100	Diversity = 1.1932e+00
Generation	620:	Best = 6.971335e+03	FES = 62100	Diversity = 1.1199e+00
Generation	630:	Best = 6.923351e+03	FES = 63100	Diversity = 1.2353e+00
Generation	640:	Best = 6.860606e+03	FES = 64100	Diversity = 1.2149e+00
Generation	650:	Best = 6.828432e+03	FES = 65100	Diversity = 1.6431e+00
Generation	660:	Best = 6.755723e+03	FES = 66100	Diversity = 1.9998e+00
Generation	670:	Best = 6.645762e+03	FES = 67100	Diversity = 1.8589e+00
Generation	680:	Best = 6.625203e+03	FES = 68100	Diversity = 1.5170e+00
Generation	690:	Best = 6.541292e+03	FES = 69100	Diversity = 1.1112e+00
Generation	700:	Best = 6.456383e+03	FES = 70100	Diversity = 1.2168e+00
Generation	710:	Best = 6.425569e+03	FES = 71100	Diversity = 1.1655e+00
Generation	720:	Best = 6.383036e+03	FES = 72100	Diversity = 1.2120e+00
Generation	730:	Best = 6.295790e+03	FES = 73100	Diversity = 1.1526e+00
Generation	740:	Best = 6.285805e+03	FES = 74100	Diversity = 1.1120e+00
Generation	750:	Best = 6.247254e+03	FES = 75100	Diversity = 1.0109e+00
Generation	760:	Best = 6.190024e+03	FES = 76100	Diversity = 1.1646e+00
Generation	770:	Best = 6.168417e+03	FES = 77100	Diversity = 1.2162e+00
Generation	780:	Best = 6.105112e+03	FES = 78100	Diversity = 1.2238e+00
Generation	790:	Best = 6.062674e+03	FES = 79100	Diversity = 1.0712e+00
Generation	800:	Best = 6.051344e+03	FES = 80100	Diversity = 1.0221e+00
Generation	810:	Best = 6.034864e+03	FES = 81100	Diversity = 9.6323e-01
Generation	820:	Best = 5.995434e+03	FES = 82100	Diversity = 9.2832e-01
Generation	830:	Best = 5.986760e+03	FES = 83100	Diversity = 9.7820e-01
Generation	840:	Best = 5.960323e+03	FES = 84100	Diversity = 9.7970e-01
Generation	850:	Best = 5.942528e+03	FES = 85100	Diversity = 9.3570e-01
Generation	860:	Best = 5.930778e+03	FES = 86100	Diversity = 8.5241e-01
Generation	870:	Best = 5.921120e+03	FES = 87100	Diversity = 7.7055e-01
Generation	880:	Best = 5.907910e+03	FES = 88100	Diversity = 8.2231e-01
Generation	890:	Best = 5.899092e+03	FES = 89100	Diversity = 8.0048e-01
Generation	900:	Best = 5.892512e+03	FES = 90100	Diversity = 9.3345e-01
Generation	910:	Best = 5.873979e+03	FES = 91100	Diversity = 8.3987e-01
Generation	920:	Best = 5.873323e+03	FES = 92100	Diversity = 6.5188e-01
Generation	930:	Best = 5.864108e+03	FES = 93100	Diversity = 5.0469e-01
Generation	940:	Best = 5.859154e+03	FES = 94100	Diversity = 4.7059e-01
Generation	950:	Best = 5.851993e+03	FES = 95100	Diversity = 4.3685e-01
Generation	960:	Best = 5.850849e+03	FES = 96100	Diversity = 4.2360e-01
Generation	970:	Best = 5.849985e+03	FES = 97100	Diversity = 4.1777e-01
Generation	980:	Best = 5.838384e+03	FES = 98100	Diversity = 3.7161e-01

Generation 990:	Best = 5.833935e+03	FES = 99100	Diversity = 4.0406e-01
Generation 1000:	Best = 5.828218e+03	FES = 100100	Diversity = 3.7777e-01
Generation 1010:	Best = 5.824324e+03	FES = 101100	Diversity = 3.4657e-01
Generation 1020:	Best = 5.821453e+03	FES = 102100	Diversity = 3.3850e-01
Generation 1030:	Best = 5.810695e+03	FES = 103100	Diversity = 3.2204e-01
Generation 1040:	Best = 5.810695e+03	FES = 104100	Diversity = 3.1945e-01
Generation 1050:	Best = 5.809030e+03	FES = 105100	Diversity = 3.0749e-01
Generation 1060:	Best = 5.806290e+03	FES = 106100	Diversity = 2.9166e-01
Generation 1070:	Best = 5.806290e+03	FES = 107100	Diversity = 2.7064e-01
Generation 1080:	Best = 5.805315e+03	FES = 108100	Diversity = 2.6393e-01
Generation 1090:	Best = 5.805261e+03	FES = 109100	Diversity = 2.6209e-01
Generation 1100:	Best = 5.803067e+03	FES = 110100	Diversity = 2.4773e-01
Generation 1110:	Best = 5.803014e+03	FES = 111100	Diversity = 2.3422e-01
Generation 1120:	Best = 5.802822e+03	FES = 112100	Diversity = 2.3103e-01
Generation 1130:	Best = 5.802447e+03	FES = 113100	Diversity = 2.2614e-01
Generation 1140:	Best = 5.800504e+03	FES = 114100	Diversity = 2.2214e-01
Generation 1150:	Best = 5.796413e+03	FES = 115100	Diversity = 2.2439e-01
Generation 1160:	Best = 5.796413e+03	FES = 116100	Diversity = 2.1711e-01
Generation 1170:	Best = 5.796413e+03	FES = 117100	Diversity = 2.0736e-01
Generation 1180:	Best = 5.796413e+03	FES = 118100	Diversity = 2.0877e-01
Generation 1190:	Best = 5.796413e+03	FES = 119100	Diversity = 2.0047e-01
Generation 1200:	Best = 5.796413e+03	FES = 120100	Diversity = 2.0253e-01
Generation 1210:	Best = 5.796413e+03	FES = 121100	Diversity = 1.9678e-01
Generation 1220:	Best = 5.796413e+03	FES = 122100	Diversity = 1.9960e-01
Generation 1230:	Best = 5.795899e+03	FES = 123100	Diversity = 1.9788e-01
Generation 1240:	Best = 5.794255e+03	FES = 124100	Diversity = 1.9364e-01
Generation 1250:	Best = 5.793654e+03	FES = 125100	Diversity = 1.9656e-01
Generation 1260:	Best = 5.793178e+03	FES = 126100	Diversity = 1.9837e-01
Generation 1270:	Best = 5.791967e+03	FES = 127100	Diversity = 1.9491e-01
Generation 1280:	Best = 5.791691e+03	FES = 128100	Diversity = 1.9847e-01
Generation 1290:	Best = 5.791691e+03	FES = 129100	Diversity = 2.0937e-01
Generation 1300:	Best = 5.790526e+03	FES = 130100	Diversity = 2.0460e-01
Generation 1310:	Best = 5.790526e+03	FES = 131100	Diversity = 2.0544e-01
Generation 1320:	Best = 5.790447e+03	FES = 132100	Diversity = 2.0177e-01
Generation 1330:	Best = 5.789678e+03	FES = 133100	Diversity = 2.0635e-01
Generation 1340:	Best = 5.788757e+03	FES = 134100	Diversity = 2.0599e-01
Generation 1350:	Best = 5.788757e+03	FES = 135100	Diversity = 2.0229e-01
Generation 1360:	Best = 5.788757e+03	FES = 136100	Diversity = 1.9231e-01
Generation 1370:	Best = 5.788728e+03	FES = 137100	Diversity = 1.9495e-01
Generation 1380:	Best = 5.788051e+03	FES = 138100	Diversity = 1.9480e-01
Generation 1390:	Best = 5.787532e+03	FES = 139100	Diversity = 1.9166e-01
Generation 1400:	Best = 5.787457e+03	FES = 140100	Diversity = 1.9567e-01
Generation 1410:	Best = 5.787457e+03	FES = 141100	Diversity = 1.9634e-01
Generation 1420:	Best = 5.786978e+03	FES = 142100	Diversity = 1.9513e-01
Generation 1430:	Best = 5.786518e+03	FES = 143100	Diversity = 1.8909e-01
Generation 1440:	Best = 5.786050e+03	FES = 144100	Diversity = 1.8417e-01
Generation 1450:	Best = 5.786014e+03	FES = 145100	Diversity = 1.7969e-01
Generation 1460:	Best = 5.785863e+03	FES = 146100	Diversity = 1.8675e-01
Generation 1470:	Best = 5.785548e+03	FES = 147100	Diversity = 1.8430e-01
Generation 1480:	Best = 5.785508e+03	FES = 148100	Diversity = 1.7361e-01
Generation 1490:	Best = 5.785479e+03	FES = 149100	Diversity = 1.7320e-01
Generation 1500:	Best = 5.784752e+03	FES = 150100	Diversity = 1.8408e-01
Generation 1510:	Best = 5.784730e+03	FES = 151100	Diversity = 1.9298e-01
Generation 1520:	Best = 5.784730e+03	FES = 152100	Diversity = 1.9359e-01

Generation 1530:	Best = 5.783471e+03	FES = 153100	Diversity = 1.9594e-01
Generation 1540:	Best = 5.783471e+03	FES = 154100	Diversity = 1.9709e-01
Generation 1550:	Best = 5.783471e+03	FES = 155100	Diversity = 2.1300e-01
Generation 1560:	Best = 5.783168e+03	FES = 156100	Diversity = 2.1373e-01
Generation 1570:	Best = 5.783094e+03	FES = 157100	Diversity = 2.0670e-01
Generation 1580:	Best = 5.781103e+03	FES = 158100	Diversity = 2.0756e-01
Generation 1590:	Best = 5.781103e+03	FES = 159100	Diversity = 2.0218e-01
Generation 1600:	Best = 5.781103e+03	FES = 160100	Diversity = 2.0135e-01
Generation 1610:	Best = 5.780546e+03	FES = 161100	Diversity = 2.0030e-01
Generation 1620:	Best = 5.780546e+03	FES = 162100	Diversity = 2.0226e-01
Generation 1630:	Best = 5.780349e+03	FES = 163100	Diversity = 2.0980e-01
Generation 1640:	Best = 5.780021e+03	FES = 164100	Diversity = 2.1447e-01
Generation 1650:	Best = 5.778909e+03	FES = 165100	Diversity = 2.1359e-01
Generation 1660:	Best = 5.778896e+03	FES = 166100	Diversity = 2.1027e-01
Generation 1670:	Best = 5.778896e+03	FES = 167100	Diversity = 2.1930e-01
Generation 1680:	Best = 5.778067e+03	FES = 168100	Diversity = 2.2288e-01
Generation 1690:	Best = 5.778067e+03	FES = 169100	Diversity = 2.2943e-01
Generation 1700:	Best = 5.778067e+03	FES = 170100	Diversity = 2.3477e-01
Generation 1710:	Best = 5.778067e+03	FES = 171100	Diversity = 2.3441e-01
Generation 1720:	Best = 5.778022e+03	FES = 172100	Diversity = 2.2604e-01
Generation 1730:	Best = 5.778022e+03	FES = 173100	Diversity = 2.2221e-01
Generation 1740:	Best = 5.777820e+03	FES = 174100	Diversity = 2.3554e-01
Generation 1750:	Best = 5.777074e+03	FES = 175100	Diversity = 2.3625e-01
Generation 1760:	Best = 5.777074e+03	FES = 176100	Diversity = 2.2954e-01
Generation 1770:	Best = 5.777074e+03	FES = 177100	Diversity = 2.2203e-01
Generation 1780:	Best = 5.777074e+03	FES = 178100	Diversity = 2.1801e-01
Generation 1790:	Best = 5.777074e+03	FES = 179100	Diversity = 2.1668e-01
Generation 1800:	Best = 5.775649e+03	FES = 180100	Diversity = 2.1685e-01
Generation 1810:	Best = 5.775649e+03	FES = 181100	Diversity = 2.1828e-01
Generation 1820:	Best = 5.775289e+03	FES = 182100	Diversity = 2.1682e-01
Generation 1830:	Best = 5.775289e+03	FES = 183100	Diversity = 2.2476e-01
Generation 1840:	Best = 5.772384e+03	FES = 184100	Diversity = 2.2510e-01
Generation 1850:	Best = 5.771961e+03	FES = 185100	Diversity = 2.3129e-01
Generation 1860:	Best = 5.771862e+03	FES = 186100	Diversity = 2.2841e-01
Generation 1870:	Best = 5.771842e+03	FES = 187100	Diversity = 2.2166e-01
Generation 1880:	Best = 5.771694e+03	FES = 188100	Diversity = 2.2653e-01
Generation 1890:	Best = 5.771694e+03	FES = 189100	Diversity = 2.1837e-01
Generation 1900:	Best = 5.771638e+03	FES = 190100	Diversity = 2.2547e-01
Generation 1910:	Best = 5.771638e+03	FES = 191100	Diversity = 2.2189e-01
Generation 1920:	Best = 5.771018e+03	FES = 192100	Diversity = 2.3310e-01
Generation 1930:	Best = 5.768617e+03	FES = 193100	Diversity = 2.3837e-01
Generation 1940:	Best = 5.768585e+03	FES = 194100	Diversity = 2.4234e-01
Generation 1950:	Best = 5.764882e+03	FES = 195100	Diversity = 2.3016e-01
Generation 1960:	Best = 5.764871e+03	FES = 196100	Diversity = 2.4035e-01
Generation 1970:	Best = 5.764818e+03	FES = 197100	Diversity = 2.2927e-01
Generation 1980:	Best = 5.764754e+03	FES = 198100	Diversity = 2.2082e-01
Generation 1990:	Best = 5.764754e+03	FES = 199100	Diversity = 2.1971e-01
Generation 2000:	Best = 5.763137e+03	FES = 200100	Diversity = 1.8705e-01
Generation 2010:	Best = 5.761951e+03	FES = 201100	Diversity = 1.8639e-01
Generation 2020:	Best = 5.761951e+03	FES = 202100	Diversity = 1.7921e-01
Generation 2030:	Best = 5.761546e+03	FES = 203100	Diversity = 1.7614e-01
Generation 2040:	Best = 5.761546e+03	FES = 204100	Diversity = 1.6565e-01
Generation 2050:	Best = 5.761546e+03	FES = 205100	Diversity = 1.5993e-01
Generation 2060:	Best = 5.761546e+03	FES = 206100	Diversity = 1.4664e-01

Generation 2070:	Best = 5.761546e+03	FES = 207100	Diversity = 1.4571e-01
Generation 2080:	Best = 5.761546e+03	FES = 208100	Diversity = 1.4011e-01
Generation 2090:	Best = 5.760986e+03	FES = 209100	Diversity = 1.4625e-01
Generation 2100:	Best = 5.760986e+03	FES = 210100	Diversity = 1.4317e-01
Generation 2110:	Best = 5.760943e+03	FES = 211100	Diversity = 1.4145e-01
Generation 2120:	Best = 5.760893e+03	FES = 212100	Diversity = 1.4478e-01
Generation 2130:	Best = 5.760893e+03	FES = 213100	Diversity = 1.3722e-01
Generation 2140:	Best = 5.760893e+03	FES = 214100	Diversity = 1.3091e-01
Generation 2150:	Best = 5.760718e+03	FES = 215100	Diversity = 1.3137e-01
Generation 2160:	Best = 5.760718e+03	FES = 216100	Diversity = 1.3047e-01
Generation 2170:	Best = 5.760192e+03	FES = 217100	Diversity = 1.3245e-01
Generation 2180:	Best = 5.760137e+03	FES = 218100	Diversity = 1.3051e-01
Generation 2190:	Best = 5.759882e+03	FES = 219100	Diversity = 1.2629e-01
Generation 2200:	Best = 5.759882e+03	FES = 220100	Diversity = 1.2385e-01
Generation 2210:	Best = 5.758445e+03	FES = 221100	Diversity = 1.2197e-01
Generation 2220:	Best = 5.758059e+03	FES = 222100	Diversity = 1.1929e-01
Generation 2230:	Best = 5.758059e+03	FES = 223100	Diversity = 1.1928e-01
Generation 2240:	Best = 5.758059e+03	FES = 224100	Diversity = 1.1943e-01
Generation 2250:	Best = 5.758059e+03	FES = 225100	Diversity = 1.1648e-01
Generation 2260:	Best = 5.758059e+03	FES = 226100	Diversity = 1.1636e-01
Generation 2270:	Best = 5.758059e+03	FES = 227100	Diversity = 1.1542e-01
Generation 2280:	Best = 5.758059e+03	FES = 228100	Diversity = 1.1432e-01
Generation 2290:	Best = 5.758035e+03	FES = 229100	Diversity = 1.1317e-01
Generation 2300:	Best = 5.758022e+03	FES = 230100	Diversity = 1.1232e-01
Generation 2310:	Best = 5.758022e+03	FES = 231100	Diversity = 1.1129e-01
Generation 2320:	Best = 5.757657e+03	FES = 232100	Diversity = 1.1142e-01
Generation 2330:	Best = 5.757657e+03	FES = 233100	Diversity = 1.1135e-01
Generation 2340:	Best = 5.757657e+03	FES = 234100	Diversity = 1.1163e-01
Generation 2350:	Best = 5.757656e+03	FES = 235100	Diversity = 1.0526e-01
Generation 2360:	Best = 5.757646e+03	FES = 236100	Diversity = 9.4793e-02
Generation 2370:	Best = 5.757641e+03	FES = 237100	Diversity = 9.3192e-02
Generation 2380:	Best = 5.757641e+03	FES = 238100	Diversity = 9.3229e-02
Generation 2390:	Best = 5.757635e+03	FES = 239100	Diversity = 9.2794e-02
Generation 2400:	Best = 5.757635e+03	FES = 240100	Diversity = 9.2185e-02
Generation 2410:	Best = 5.757631e+03	FES = 241100	Diversity = 9.2182e-02
Generation 2420:	Best = 5.757535e+03	FES = 242100	Diversity = 9.2020e-02
Generation 2430:	Best = 5.757535e+03	FES = 243100	Diversity = 9.1036e-02
Generation 2440:	Best = 5.757367e+03	FES = 244100	Diversity = 9.0999e-02
Generation 2450:	Best = 5.757367e+03	FES = 245100	Diversity = 8.9605e-02
Generation 2460:	Best = 5.757367e+03	FES = 246100	Diversity = 8.8961e-02
Generation 2470:	Best = 5.757367e+03	FES = 247100	Diversity = 8.7221e-02
Generation 2480:	Best = 5.757367e+03	FES = 248100	Diversity = 8.7540e-02
Generation 2490:	Best = 5.757367e+03	FES = 249100	Diversity = 8.4757e-02
Generation 2500:	Best = 5.757367e+03	FES = 250100	Diversity = 8.4947e-02
Generation 2510:	Best = 5.757252e+03	FES = 251100	Diversity = 8.3678e-02
Generation 2520:	Best = 5.757252e+03	FES = 252100	Diversity = 8.2046e-02
Generation 2530:	Best = 5.756782e+03	FES = 253100	Diversity = 8.1983e-02
Generation 2540:	Best = 5.756781e+03	FES = 254100	Diversity = 8.1981e-02
Generation 2550:	Best = 5.756781e+03	FES = 255100	Diversity = 8.2166e-02
Generation 2560:	Best = 5.756670e+03	FES = 256100	Diversity = 8.2363e-02
Generation 2570:	Best = 5.756670e+03	FES = 257100	Diversity = 8.2327e-02
Generation 2580:	Best = 5.756670e+03	FES = 258100	Diversity = 8.2289e-02
Generation 2590:	Best = 5.756667e+03	FES = 259100	Diversity = 8.2253e-02
Generation 2600:	Best = 5.756653e+03	FES = 260100	Diversity = 8.2252e-02

Generation 2610:	Best = 5.756653e+03	FES = 261100	Diversity = 8.0964e-02
Generation 2620:	Best = 5.756593e+03	FES = 262100	Diversity = 8.0397e-02
Generation 2630:	Best = 5.756593e+03	FES = 263100	Diversity = 8.1007e-02
Generation 2640:	Best = 5.756588e+03	FES = 264100	Diversity = 8.0481e-02
Generation 2650:	Best = 5.756588e+03	FES = 265100	Diversity = 8.0193e-02
Generation 2660:	Best = 5.756587e+03	FES = 266100	Diversity = 7.9596e-02
Generation 2670:	Best = 5.756587e+03	FES = 267100	Diversity = 8.0065e-02
Generation 2680:	Best = 5.756578e+03	FES = 268100	Diversity = 8.1847e-02
Generation 2690:	Best = 5.756577e+03	FES = 269100	Diversity = 8.1910e-02
Generation 2700:	Best = 5.756571e+03	FES = 270100	Diversity = 8.1825e-02
Generation 2710:	Best = 5.756518e+03	FES = 271100	Diversity = 8.1386e-02
Generation 2720:	Best = 5.756518e+03	FES = 272100	Diversity = 8.1553e-02
Generation 2730:	Best = 5.756471e+03	FES = 273100	Diversity = 8.1512e-02
Generation 2740:	Best = 5.756435e+03	FES = 274100	Diversity = 8.0079e-02
Generation 2750:	Best = 5.756425e+03	FES = 275100	Diversity = 8.0045e-02
Generation 2760:	Best = 5.756424e+03	FES = 276100	Diversity = 8.0057e-02
Generation 2770:	Best = 5.756415e+03	FES = 277100	Diversity = 8.0004e-02
Generation 2780:	Best = 5.756342e+03	FES = 278100	Diversity = 7.9389e-02
Generation 2790:	Best = 5.756296e+03	FES = 279100	Diversity = 7.9388e-02
Generation 2800:	Best = 5.756281e+03	FES = 280100	Diversity = 7.8153e-02
Generation 2810:	Best = 5.756267e+03	FES = 281100	Diversity = 7.8142e-02
Generation 2820:	Best = 5.756242e+03	FES = 282100	Diversity = 7.8778e-02
Generation 2830:	Best = 5.756242e+03	FES = 283100	Diversity = 7.8733e-02
Generation 2840:	Best = 5.756242e+03	FES = 284100	Diversity = 7.8201e-02
Generation 2850:	Best = 5.756235e+03	FES = 285100	Diversity = 7.8224e-02
Generation 2860:	Best = 5.756220e+03	FES = 286100	Diversity = 7.8306e-02
Generation 2870:	Best = 5.756177e+03	FES = 287100	Diversity = 7.8351e-02
Generation 2880:	Best = 5.756177e+03	FES = 288100	Diversity = 7.8320e-02
Generation 2890:	Best = 5.756177e+03	FES = 289100	Diversity = 7.8088e-02
Generation 2900:	Best = 5.756176e+03	FES = 290100	Diversity = 7.8602e-02
Generation 2910:	Best = 5.756176e+03	FES = 291100	Diversity = 7.8409e-02
Generation 2920:	Best = 5.756175e+03	FES = 292100	Diversity = 7.8138e-02
Generation 2930:	Best = 5.756171e+03	FES = 293100	Diversity = 7.8069e-02
Generation 2940:	Best = 5.755975e+03	FES = 294100	Diversity = 7.7854e-02
Generation 2950:	Best = 5.755975e+03	FES = 295100	Diversity = 7.7746e-02
Generation 2960:	Best = 5.755971e+03	FES = 296100	Diversity = 7.8181e-02
Generation 2970:	Best = 5.755961e+03	FES = 297100	Diversity = 7.7098e-02
Generation 2980:	Best = 5.755961e+03	FES = 298100	Diversity = 7.6137e-02
Generation 2990:	Best = 5.755956e+03	FES = 299100	Diversity = 7.6099e-02

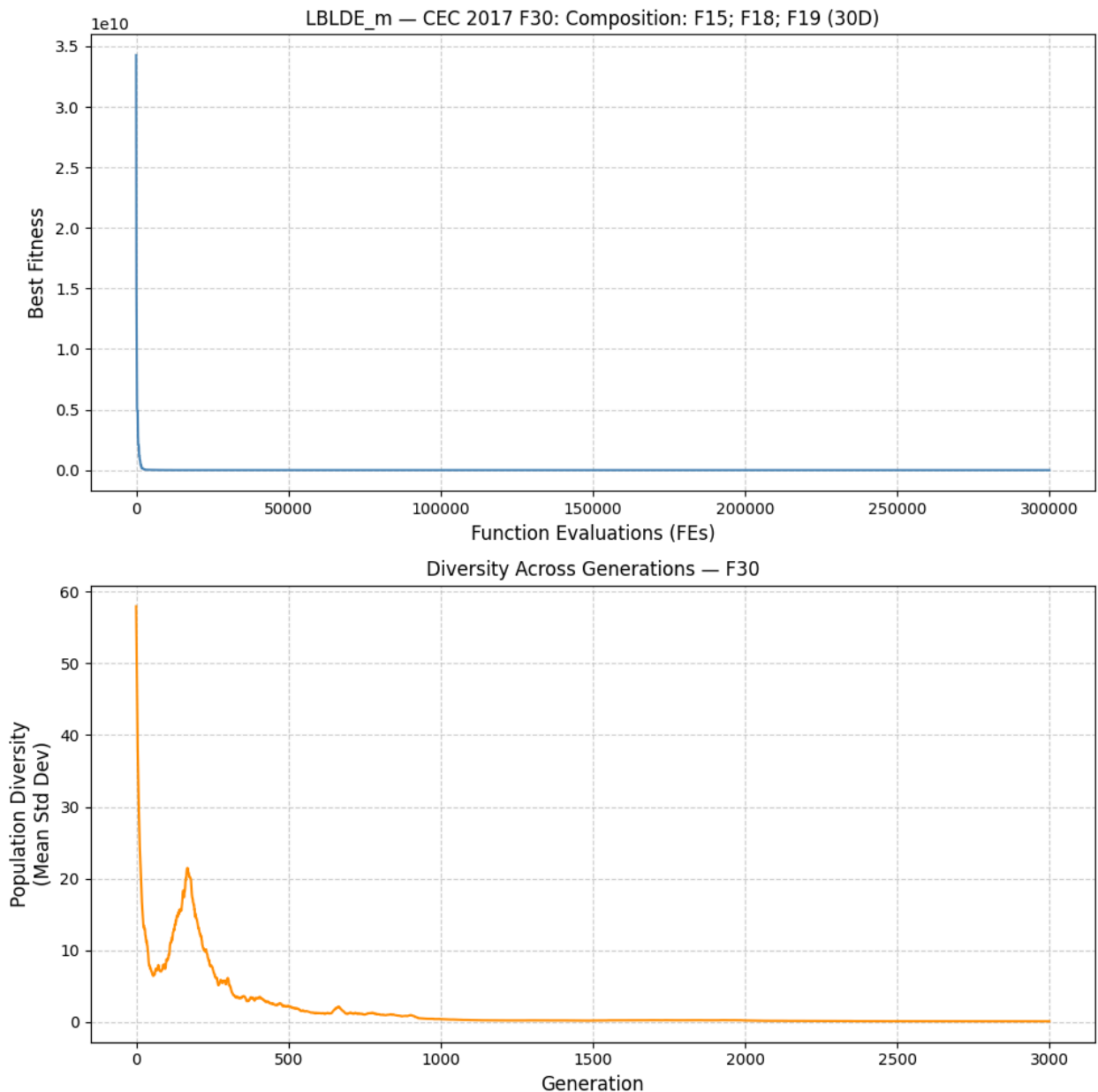
Optimization completed!

Final Best Fitness = 5.755954e+03

Total Function Evaluations = 300000

```
=====
=
Results:
  Best Fitness      : 5.755954e+03
  Global Optimum    : 3.000000e+03
  Error (|best-optimum|) : 2.755954e+03
  Best Solution (5 dim) : [ -2.88339392 -21.93397308  89.52557742  62.14511643
3.21135087]
  D^(0) (init. diversity): 5.7976e+01
=====
```

II



Graph saved as convergence_F30.png

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
In [57]: print('=' * 80)
print('SUMMARY — LBLDE_m on CEC 2017 (10D), Single Run, Seed=42')
print('=' * 80)
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |
print('-' * 70)
```

```

for fn in [f'F{i}' for i in range(1, 31)]:
    if fn not in all_results:
        continue
    r = all_results[fn]
    if r['best_fit'] is None:
        print(f'{fn:<10} {"N/A (deprecated)":>18} {r["f_global"]:>18.2f} {"N/A"}')
    else:
        print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f} {r["error"]:>18.2f}')
print('=' * 80)

```

```

=====
=
SUMMARY – LBLDE_m on CEC 2017 (10D), Single Run, Seed=42
=====
=
Function          Best Fitness          Global Optimum      Error |best-opt|
-----
F1                1.620461e+10          100.00             1.620461e+10
F2                3.723569e+28          200.00             3.723569e+28
F3                1.302988e+04          300.00             1.272988e+04
F4                1.178556e+03          400.00             7.785557e+02
F5                5.412613e+02          500.00             4.126126e+01
F6                6.038764e+02          600.00             3.876356e+00
F7                8.075230e+02          700.00             1.075230e+02
F8                8.378620e+02          800.00             3.786196e+01
F9                9.009706e+02          900.00             9.705759e-01
F10               4.595022e+03          1000.00            3.595022e+03
F11               1.374674e+03          1100.00            2.746740e+02
F12               2.677164e+09          1200.00            2.677163e+09
F13               1.962182e+04          1300.00            1.832182e+04
F14               1.441525e+03          1400.00            4.152461e+01
F15               1.543183e+03          1500.00            4.318287e+01
F16               2.251600e+03          1600.00            6.515999e+02
F17               1.756774e+03          1700.00            5.677415e+01
F18               5.479425e+03          1800.00            3.679425e+03
F19               1.962082e+03          1900.00            6.208153e+01
F20               2.161276e+03          2000.00            1.612756e+02
F21               2.353531e+03          2100.00            2.535308e+02
F22               2.658567e+03          2200.00            4.585665e+02
F23               2.743810e+03          2300.00            4.438101e+02
F24               2.926218e+03          2400.00            5.262185e+02
F25               3.108112e+03          2500.00            6.081124e+02
F26               6.528340e+03          2600.00            3.928340e+03
F27               3.256224e+03          2700.00            5.562245e+02
F28               3.846985e+03          2800.00            1.046985e+03
F29               3.372907e+03          2900.00            4.729068e+02
F30               5.755954e+03          3000.00            2.755954e+03
=====
=

```

In [58]: !zip plots.zip convergence_F{1..30}.png

adding: convergence_F1.png (deflated 23%)
adding: convergence_F2.png (deflated 23%)
adding: convergence_F3.png (deflated 23%)
adding: convergence_F4.png (deflated 24%)
adding: convergence_F5.png (deflated 20%)
adding: convergence_F6.png (deflated 21%)
adding: convergence_F7.png (deflated 22%)
adding: convergence_F8.png (deflated 21%)
adding: convergence_F9.png (deflated 23%)
adding: convergence_F10.png (deflated 19%)
adding: convergence_F11.png (deflated 22%)
adding: convergence_F12.png (deflated 22%)
adding: convergence_F13.png (deflated 23%)
adding: convergence_F14.png (deflated 21%)
adding: convergence_F15.png (deflated 22%)
adding: convergence_F16.png (deflated 21%)
adding: convergence_F17.png (deflated 19%)
adding: convergence_F18.png (deflated 17%)
adding: convergence_F19.png (deflated 17%)
adding: convergence_F20.png (deflated 18%)
adding: convergence_F21.png (deflated 20%)
adding: convergence_F22.png (deflated 23%)
adding: convergence_F23.png (deflated 21%)
adding: convergence_F24.png (deflated 22%)
adding: convergence_F25.png (deflated 23%)
adding: convergence_F26.png (deflated 22%)
adding: convergence_F27.png (deflated 22%)
adding: convergence_F28.png (deflated 22%)
adding: convergence_F29.png (deflated 21%)
adding: convergence_F30.png (deflated 22%)